

mielke-geometroynamics: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometroynamics_F00001	7	■ 1.000	A	A	potential 1-form A via electron interference.
mielke-geometroynamics_F00002	7	■ 0.980	$\operatorname{SL}(5, \mathbb{R})$	$\mathrm{SL}(5, \mathbb{R})$	model with a primordial $\mathrm{SL}(5, \mathbb{R})$ gauge group is presented. After a tiny
mielke-geometroynamics_F00003	33	■ 0.985	$\mathbb{R}$	$\mathbb{R}$	tiabile functions f to the space $\mathbb{R}$ of real numbers, defines the <i>tangent vector</i> $e(m)$ or
mielke-geometroynamics_F00004	17	■ 1.000	β	β	and Fermi’s theory of weak interactions, being the cause for the β -decay of certain
mielke-geometroynamics_F00005	17	■ 0.965	$\mathrm{W}^{\pm}$	$W^{\pm}$	empirical evidence by the detection of the mediating $W^{\pm}$ and Z^0 bosons as the
mielke-geometroynamics_F00006	17	■ 0.965	Z°	Z^0	empirical evidence by the detection of the mediating $W^{\pm}$ and Z^0 bosons as the
mielke-geometroynamics_F00007	19	■ 1.000	1×10^{-12}	1×10^{-12}	mass is measured nowadays with a precision better than 1×10^{-12} ; this accuracy
mielke-geometroynamics_F00008	20	■ 1.000	∂_j	∂_j	to the electromagnetic fields if the partial derivatives ∂_j in the Dirac equation are
mielke-geometroynamics_F00009	21	■ 1.000	$D_j := \partial_j + i A_j$	$D_j := \partial_j + i A_j$	replaced by covariant derivatives $D_j := \partial_j + i A_j$. In the latter concept, a “minimal”
mielke-geometroynamics_F00010	21	■ 1.000	A_j	A_j	substitution of the electromagnetic vector potential A_j is taken into consideration.
mielke-geometroynamics_F00011	21	■ 1.000	$\psi \rightarrow e^{i\theta} \psi$	$\psi \rightarrow e^{i\theta} \psi$	remains invariant under the <i>local</i> transformation $\psi \rightarrow e^{i\theta} \psi$ of the phase of the
mielke-geometroynamics_F00012	21	■ 1.000	ψ	ψ	wave function ψ and the simultaneous transformation $A_j \rightarrow A_j - \partial_j \theta$ of the four-
mielke-geometroynamics_F00013	21	■ 1.000	$A_j \rightarrow A_j - \partial_j \theta$	$A_j \rightarrow A_j - \partial_j \theta$	wave function ψ and the simultaneous transformation $A_j \rightarrow A_j - \partial_j \theta$ of the four-
mielke-geometroynamics_F00014	22	■ 0.526	$n > 4$	$n > 4$	($n > 4$) spaces and was, to a certain extent, anticipated by the pioneering works of
mielke-geometroynamics_F00015	23	■ 0.343	$\{ \}^{+}$	$+$	an SU(3) nonet that consists of the massive spin-2 ⁺ mesons $f, f', A_2, K^*(1420)$.
mielke-geometroynamics_F00016	23	■ 0.343	$A_2, K^*(1420)$	$A_2, K^*(1420)$	an SU(3) nonet that consists of the massive spin-2 ⁺ mesons $f, f', A_2, K^*(1420)$.
mielke-geometroynamics_F00017	23	■ 0.987	$\operatorname{SL}(2, \mathbb{C})$	$\mathrm{SL}(2, \mathbb{C})$	$\mathrm{SL}(2, \mathbb{C})$, respectively. If we apply these findings to <i>strong gravity</i> , then it is obvious
mielke-geometroynamics_F00018	23	■ 0.697	$\operatorname{SU}(3)$	$\mathrm{SU}(3)$	to tie together the internal symmetries, such as the SU(3) one, with the spacetime
mielke-geometroynamics_F00019	23	■ 0.997	$\operatorname{SL}(2, \mathbb{C}) \subset \operatorname{SL}(2, \mathbb{C}) \otimes \operatorname{SU}(3) \subset \operatorname{SL}(6, \mathbb{C})$	$\mathrm{SL}(2, \mathbb{C}) \subset \mathrm{SL}(2, \mathbb{C}) \otimes \mathrm{SU}(3) \subset \mathrm{SL}(6, \mathbb{C})$	chain $\mathrm{SL}(2, \mathbb{C}) \subset \mathrm{SL}(2, \mathbb{C}) \otimes \mathrm{SU}(3) \subset \mathrm{SL}(6, \mathbb{C})$. If these symmetries are regarded
mielke-geometroynamics_F00020	24	■ 0.834	$\operatorname{SL}(6, \mathbb{C})$	$\mathrm{SL}(6, \mathbb{C})$	as local ones, we get an $\mathrm{SL}(6, \mathbb{C})$ gauge theory of strong interaction (SALAM 1973)
mielke-geometroynamics_F00021	24	■ 1.000	$\ell := \ell^* / \sqrt{\kappa}$	$\ell := \ell^* / \sqrt{\kappa}$	a further length $\ell := \ell^* / \sqrt{\kappa}$ is introduced into particle physics that corresponds to
mielke-geometroynamics_F00022	24	■ 0.995	$G_S = G_N / \kappa$	$G_S = G_N / \kappa$	the scaled “Newtonian” coupling constant $G_S = G_N / \kappa$ of strong gravity. SALAM &

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mielke-geometrodynamics_F00023	25	■ 1.000	$\sqrt{s}=540 \text{ GeV}$	$\sqrt{s} = 540\text{GeV}$	associated missing energy at $\sqrt{s} = 540 \text{ GeV}$. Phys Lett B 122:103–116
mielke-geometrodynamics_F00024	30	■ 1.000	$\mathrm{x}: \mathrm{M} \rightarrow \mathbb{R}^{\mathrm{n}}$	$x : M \rightarrow \mathbb{R}^n$	achieve this, we consider injective (reversible) mappings $\boxed{x : M \rightarrow \mathbb{R}^n}$, whose range
mielke-geometrodynamics_F00025	30	■ 1.000	E^{n}	E^n	extends to an open subset of the usual Euclidean space E^n . Such a map together with
mielke-geometrodynamics_F00026	30	■ 0.993	U_{i}	U_i	its domain U_i is called an n-dimensional chart. By the projection π_i onto the single
mielke-geometrodynamics_F00027	30	■ 0.993	π_{i}	π_i	its domain U_i is called an n-dimensional chart. By the projection π_i onto the single
mielke-geometrodynamics_F00028	30	■ 1.000	$\mathbb{R}^{\mathrm{n}}$	$\mathbb{R}^n$	axes of the coordinate system of $\mathbb{R}^n$, the <i>local coordinates</i> $x_i = \pi_i \circ x, i = 1, \dots, n$
mielke-geometrodynamics_F00029	30	■ 1.000	$\mathrm{x}_{\mathrm{i}}=\pi_{\mathrm{i}} \circ \mathrm{x}, \mathrm{i}=1, \ldots, \mathrm{n}$	$x_i = \pi_i \circ x, i = 1, \dots, n$	axes of the coordinate system of $\mathbb{R}^n$, the <i>local coordinates</i> $x_i = \pi_i \circ x, i = 1, \dots, n$
mielke-geometrodynamics_F00030	30	■ 1.000	S^{n}	S^n	sphere S^n). However, it is possible to construct a collection of charts of identical
mielke-geometrodynamics_F00031	31	■ 0.996	$\mathrm{U}_{\mathrm{i}} \subset \mathrm{M}$	$U_i \subset M$	dimensions so that the set-theoretic union of their domains $U_i \subset M$ cover M com-
mielke-geometrodynamics_F00032	31	■ 0.999	$\mathscr{D}(\mathrm{M})$	$\mathscr{D}(M)$	form an infinite-dimensional group with respect to composition, i.e., the group $\mathscr{D}(M)$
mielke-geometrodynamics_F00033	31	■ 1.000	$\left(\mathrm{U}_{\mathrm{i}}, \mathrm{x}_{\mathrm{i}}\right)$	(U_i, x_i)	A collection (U_i, x_i) of compatible charts is called a C^∞ -atlas of M . (The common
mielke-geometrodynamics_F00034	31	■ 1.000	C^{∞}	C^∞	A collection (U_i, x_i) of compatible charts is called a C^∞ -atlas of M . (The common
mielke-geometrodynamics_F00035	31	■ 0.997	C^{r}	C^r	of class C^r). This atlas again can be extended unequivocally to a <i>complete</i> one, which
mielke-geometrodynamics_F00036	32	■ 1.000	$\mathrm{E}^{\mathrm{n}}(\mathrm{n}+1) / 2$	$E^{n(n+1)/2}$	locally isometric embeddings of analytic Riemannian manifolds into $E^{n(n+1)/2}$ are
mielke-geometrodynamics_F00037	32	■ 0.936	$N=n(3 \ n+11) / 2$	$N = n(3n + 11)/2$	dimension $\boxed{N = n(3n + 11)/2}$ was proven by NASH (1956); for a noncompact one,
mielke-geometrodynamics_F00038	32	■ 1.000	$N=n(n+1)(3 \ n+11) / 2$	$N = n(n + 1)(3n + 11)/2$	the extravagant dimension $\boxed{N = n(n + 1)(3n + 11)/2}$ results; cf. CHEN (2000) for a
mielke-geometrodynamics_F00039	32	■ 1.000	$\mathbb{E}^4$	$\mathbb{E}^4$	surface of a rotating Kerr–Newman black hole into E^4 .
mielke-geometrodynamics_F00040	32	■ 0.997	$\mathrm{T}_{\mathrm{m}}(\mathrm{M})$	$T_m(M)$	<i>space</i> $T_m(M)$ attached to a point $m \in M$. Contrary to the given illustration (see
mielke-geometrodynamics_F00041	32	■ 0.997	$\mathrm{m} \in \mathrm{M}$	$m \in M$	<i>space</i> $T_m(M)$ attached to a point $m \in M$. Contrary to the given illustration (see
mielke-geometrodynamics_F00042	32	■ 0.573	$\mathrm{s}(\mathrm{t})$	$s(t)$	to achieve this, a curve $s(t)$ on the manifold is chosen that passes through a point
mielke-geometrodynamics_F00043	33	■ 0.985	$\mathrm{e}(\mathrm{m})$	$e(m)$	tiabile functions f to the space $\mathbb{R}$ of real numbers, defines the <i>tangent vector</i> $e(m)$ or
mielke-geometrodynamics_F00044	33	■ 0.999	$\mathrm{e}_{\alpha}(\mathrm{m}), \alpha=1, \ldots, \mathrm{n}$	$e_\alpha(m), \alpha = 1, \dots, n$	independent vectors $\boxed{e_\alpha(m), \alpha = 1, \dots, n}$. It follows from (2.1.2) that the partial
mielke-geometrodynamics_F00045	33	■ 1.000	$\partial_{\mathrm{i}}:=\partial / \partial \mathrm{x}^{\mathrm{i}}$	$\partial_i := \partial / \partial x^i$	derivates $\partial_i := \partial / \partial x^i$ constitute a local basis for $T_m(M)$ with respect to a <i>holonomic</i>
mielke-geometrodynamics_F00046	33	■ 1.000	$\left[\mathrm{e}_{\mathrm{i}}(\mathrm{~m}), \mathrm{e}_{\mathrm{j}}(\mathrm{~m})\right]=\left[\partial_{\mathrm{i}}, \partial_{\mathrm{j}}\right]=0$	$[e_i(\ m), e_j(\ m)] = [\partial_i, \partial_j] = 0$	coordinate system. By definition, the relation $[e_i(m), e_j(m)] = [\partial_i, \partial_j] = 0$ is satis-
mielke-geometrodynamics_F00047	33	■ 1.000	$\mathrm{e}_{\alpha}^{\mathrm{i}}(\mathrm{m})$	$e_\alpha^i(m)$	The occurring nonsingular matrices $e_\alpha^i(m)$ depend on the point in question and
mielke-geometrodynamics_F00048	33	■ 0.628	$\mathrm{T}(\mathrm{M})$	$T(M)$	In addition to the tangent bundle $T(M)$, a so-called <i>dual</i> bundle $T^*(M) = \bigcup_{m \in M} T_m^*(M)$
mielke-geometrodynamics_F00049	33	■ 0.628	$\mathrm{T}^{*}(\mathrm{M})=\bigcup_{\mathrm{m} \in \mathrm{M}} \mathrm{T}_{\mathrm{m}}^{*}(\mathrm{M})$	$T^*(M) = \bigcup_{m \in M} T_m^*(M)$	In addition to the tangent bundle $T(M)$, a so-called <i>dual</i> bundle $T^*(M) = \bigcup_{m \in M} T_m^*(M)$

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mielke-geometrodynamics_F00050	33	■ 1.000	$\mathrm{T}_{\mathrm{m}}^*(\mathrm{M})$	$T_m^*(M)$	consisting of all possible cotangent spaces $T_m^*(M)$ on the manifold can be constructed.
mielke-geometrodynamics_F00051	33	■ 1.000	$\vartheta^\alpha, \alpha=1, \ldots, n$	$\vartheta^\alpha, \alpha = 1, \dots, n$	Let $\vartheta^\alpha, \alpha = 1, \dots, n$, be a basis of $T_m^*(M)$. Then the “duality” ³ requires that ϑ^α
mielke-geometrodynamics_F00052	33	■ 1.000	ϑ^α	ϑ^α	Let $\vartheta^\alpha, \alpha = 1, \dots, n$, be a basis of $T_m^*(M)$. Then the “duality” ³ requires that ϑ^α
mielke-geometrodynamics_F00053	33	■ 0.999	$\mathrm{d}x^i$	dx^i	by dx^i . In terms of the rigid basis of E^n , the 1-forms
mielke-geometrodynamics_F00054	34	■ 0.998	$\mathrm{d} \vartheta^\alpha$	$d\vartheta^\alpha$	derivative $d\vartheta^\alpha$, in contrast to that of dx^j , is generally different from zero. Following
mielke-geometrodynamics_F00055	34	■ 0.998	$\mathrm{d}x^j$	dx^j	derivative $d\vartheta^\alpha$, in contrast to that of dx^j , is generally different from zero. Following
mielke-geometrodynamics_F00056	34	■ 0.599	$\mathrm{E}_{\cdot j}^\alpha$	$E_{\cdot j}^\alpha$	(2.2.1), $E_{\cdot j}^\alpha$ can be understood as the dual tetrad field “reciprocal” to e_β^i (cf. (2.1.3)).
mielke-geometrodynamics_F00057	34	■ 0.599	e_β^i	e_β^i	(2.2.1), $E_{\cdot j}^\alpha$ can be understood as the dual tetrad field “reciprocal” to e_β^i (cf. (2.1.3)).
mielke-geometrodynamics_F00058	34	■ 0.979	$\otimes_{\mathrm{s}, \mathrm{a}}$	$\otimes_{s,a}$	Completely symmetric or antisymmetric products are denoted by $\otimes_{s,a}$ or, even more
mielke-geometrodynamics_F00059	34	■ 1.000	$\vee$	$\vee$	commonly, by the symbols $\vee$ and $\wedge$, respectively. Note that a tensor is a geometric
mielke-geometrodynamics_F00060	34	■ 1.000	$\wedge$	$\wedge$	commonly, by the symbols $\vee$ and $\wedge$, respectively. Note that a tensor is a geometric
mielke-geometrodynamics_F00061	34	■ 0.962	$\mathrm{T}_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(\mathrm{m})$	$T_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(m)$	The quantities $T_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(m)$ are called its p covariant and q contravariant components,
mielke-geometrodynamics_F00062	34	■ 0.919	$\mathrm{T}_p^q(\mathrm{M})$	$T_p^q(M)$	and $T_p^q(M)$ may be regarded as a bundle associated with $L(M)$ having the manifold
mielke-geometrodynamics_F00063	34	■ 0.919	$\mathrm{L}(\mathrm{M})$	$L(M)$	and $T_p^q(M)$ may be regarded as a bundle associated with $L(M)$ having the manifold
mielke-geometrodynamics_F00064	34	■ 0.708	$\mathrm{D}^{(\mathrm{p}, \mathrm{q})}(\mathrm{p}, \mathrm{q})$	$D^{(p,q)}$	M as a base, the tensor representation $D^{(p,q)}$ of $GL(n, \mathbb{R})$, i.e., $\rho^{(p,q)}(GL(n, \mathbb{R}))$ as a
mielke-geometrodynamics_F00065	34	■ 0.708	$\rho^{(\mathrm{p}, \mathrm{q})}(\mathrm{GL}(\mathrm{n}, \mathbb{R}), \mathbb{R})$	$\rho^{(p,q)}(GL(n, \mathbb{R}))$	M as a base, the tensor representation $D^{(p,q)}$ of $GL(n, \mathbb{R})$, i.e., $\rho^{(p,q)}(GL(n, \mathbb{R}))$ as a
mielke-geometrodynamics_F00066	34	■ 0.907	$\otimes^{\mathrm{p}*} \mathrm{R}^n \otimes^{\mathrm{q}} \mathbb{R}^n$	$\otimes^{p*} \mathbb{R}^n \otimes^q \mathbb{R}^n$	structure group and the (Cartesian) product $\otimes^{p*} \mathbb{R}^n \otimes^q \mathbb{R}^n$ as a typical fiber.
mielke-geometrodynamics_F00067	34	■ 1.000	$\mathrm{g}_{\mu \nu}$	$g_{\mu\nu}$	the scale of the manifold. Due to the postulated symmetry of the components $g_{\mu \nu}$,
mielke-geometrodynamics_F00068	35	■ 0.651	$(\mathrm{p}, 0)$	$(p, 0)$	$(p, 0)$. Usually, $\wedge := \otimes_a$ is an abbreviation for the antisymmetrized tensor product,
mielke-geometrodynamics_F00069	35	■ 0.651	$\wedge := \otimes_{\mathrm{a}}$	$\wedge := \otimes_a$	$(p, 0)$. Usually, $\wedge := \otimes_a$ is an abbreviation for the antisymmetrized tensor product,
mielke-geometrodynamics_F00070	35	■ 1.000	$(1939, 1957)$	$(1939, 1957)$	(1939, 1957) of the representations of the transformation group in flat spacetime, i.e.,
mielke-geometrodynamics_F00071	36	■ 1.000	$\mathscr{H}$	$\mathcal{H}$	rays in a Hilbert space $\mathcal{H}$, which is to be transformed according to an irreducible
mielke-geometrodynamics_F00072	36	■ 0.997	φ	φ	assumed that φ at a point m is equal to the value $\varphi(m)$ in a complex vector space
mielke-geometrodynamics_F00073	36	■ 0.997	$\varphi(\mathrm{m})$	$\varphi(m)$	assumed that φ at a point m is equal to the value $\varphi(m)$ in a complex vector space
mielke-geometrodynamics_F00074	36	■ 1.000	V_{m}	V_m	V_m . Additionally, it can be postulated that a vector space of this kind is attached
mielke-geometrodynamics_F00075	36	■ 0.941	π	π	Definition A fiber bundle (F, M, π) consists of two C^∞ -differentiable manifolds F
mielke-geometrodynamics_F00076	36	■ 0.362	$\mathrm{F}_{\mathrm{m}} := -\pi^{-1}(\mathrm{m})$	$F_m := -\pi^{-1}(m)$	and π the <i>projection</i> . The closed submanifold $F_m := \pi^{-1}(m)$ will be called a <i>typical</i>

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mielke-geometrodynamics_F00077	37	■ 0.980	$\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}_m$ F_m	$\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}_m$	point $m \in \mathrm{M}$ has a neighborhood U such that $\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}_m$ is isomorphic to the
mielke-geometrodynamics_F00078	37	■ 0.686	$\left(\mathrm{U} \times \mathrm{F}_m, \pi_u\right)$ U, π_u	$(\mathrm{U} \times \mathrm{F}_m, \mathrm{U}, \pi_u)$	product bundle $(\mathrm{U} \times \mathrm{F}_m, \mathrm{U}, \pi_u)$. The mapping
mielke-geometrodynamics_F00079	37	■ 0.637	$\mathrm{P}(\mathrm{M}, \mathrm{G}, \pi, \delta)$	$P(\mathrm{M}, \mathrm{G}, \pi, \delta)$	group, then the collection $P(\mathrm{M}, \mathrm{G}, \pi, \delta)$ is called a <i>principal fiber bundle</i> if
mielke-geometrodynamics_F00080	37	■ 0.790	$(\mathrm{P}, \mathrm{M}, \pi)$	$(\mathrm{P}, \mathrm{M}, \pi)$	(i) $(\mathrm{P}, \mathrm{M}, \pi)$ is a fiber bundle with typical fiber G ;
mielke-geometrodynamics_F00081	37	■ 0.866	$\mathrm{P}, \mathrm{G}, \delta$	$\mathrm{P}, \mathrm{G}, \delta$	(ii) $(\mathrm{P}, \mathrm{G}, \delta)$ is a G -manifold (whereby G is acting on P from the right);
mielke-geometrodynamics_F00082	37	■ 0.662	$\mathrm{U} \subset$	$\mathrm{U} \subset$	(iii) P is <i>locally</i> trivial, which means that every point of M has a neighborhood $\mathrm{U} \subset$
mielke-geometrodynamics_F00083	37	■ 0.941	$\iota: \mathrm{U} \times \mathrm{G} \rightarrow \pi^{-1}(\mathrm{U})$ $\pi^{-1}(\mathrm{U})$	$\iota: \mathrm{U} \times \mathrm{G} \rightarrow \pi^{-1}(\mathrm{U})$	M together with an isomorphism $\iota: \mathrm{U} \times \mathrm{G} \rightarrow \pi^{-1}(\mathrm{U})$ for which the following
mielke-geometrodynamics_F00084	37	■ 0.857	G	G	A manifold is called a <i>G-manifold</i> $(\mathrm{P}, \mathrm{G}, \delta)$ if G acts on it as a free transformation
mielke-geometrodynamics_F00085	37	■ 0.970	$\mathrm{p}_2 = \mathrm{g} \mathrm{p}_1$	$p_2 = g p_1$	there always existed a g that would provide a relation $p_2 = g p_1$ for given p_1
mielke-geometrodynamics_F00086	37	■ 0.970	p_1	p_1	there always existed a g that would provide a relation $p_2 = g p_1$ for given p_1
mielke-geometrodynamics_F00087	37	■ 0.666	p_2	p_2	and p_2 , then G considered as a manifold would be isomorphic to the base M .)
mielke-geometrodynamics_F00088	38	■ 0.985	π, δ	π, δ	this purpose, the fiber bundles $V(\mathrm{M}, \mathrm{F}; \mathrm{G}, \mathrm{P})$ associated with $P(\mathrm{M}, \mathrm{G}, \pi, \delta)$ are
mielke-geometrodynamics_F00089	38	■ 0.999	$\mathrm{P} \times \mathrm{F}$	$\mathrm{P} \times \mathrm{F}$	i.e., on the product space $\mathrm{P} \times \mathrm{F}$, an action from the right-hand side is defined by
mielke-geometrodynamics_F00090	38	■ 0.678	$\mathrm{V} = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$	$V = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$	by $V = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$. The isomorphism $\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$ concerning the original domain
mielke-geometrodynamics_F00091	38	■ 0.678	$\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$	$\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$	by $V = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$. The isomorphism $\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$ concerning the original domain
mielke-geometrodynamics_F00092	38	■ 0.307	$\pi_v^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}$	$\pi_v^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}$	induce the isomorphism $\pi_v^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}$ in the associated bundle. Provided that
mielke-geometrodynamics_F00093	38	■ 0.907	$\pi_v^{-1}(\mathrm{U})$	$\pi_v^{-1}(\mathrm{U})$	$\pi_v^{-1}(\mathrm{U})$ is an open submanifold of V , then V can be equipped with a differentiable
mielke-geometrodynamics_F00094	38	■ 0.998	$\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	$\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	occurs, but its linear representation $\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$, i.e., the Lie homomorphism
mielke-geometrodynamics_F00095	38	■ 0.852	$\mathrm{N} \times \mathrm{N}$	$\mathrm{N} \times \mathrm{N}$	of G into the general linear group of complex $\mathrm{N} \times \mathrm{N}$ matrices. If their representation
mielke-geometrodynamics_F00096	38	■ 0.780	$\mathbb{C}^{\mathrm{N}}$	$\mathbb{C}^{\mathrm{N}}$	space, the N -dimensional vector space $\mathbb{C}^{\mathrm{N}}$ over the field of complex numbers, is
mielke-geometrodynamics_F00097	38	■ 1.000	$T_{\mathbb{C}}^*(\mathrm{M})$	$T_{\mathbb{C}}^*(\mathrm{M})$	with the complexified ⁴ cotangent bundle $T_{\mathbb{C}}^*(\mathrm{M})$, further vector bundles will come
mielke-geometrodynamics_F00098	38	■ 1.000	ρ	ρ	ρ , i.e., the cross section
mielke-geometrodynamics_F00099	38	■ 0.986	$\mathrm{N} \geq 1$	$\mathrm{N} \geq 1$	For $\mathrm{N} \geq 1$, these fields form an infinite-dimensional vector space over $\mathbb{C}^{\mathrm{N}}$. In
mielke-geometrodynamics_F00100	38	■ 1.000	$\mathrm{b} = (b_A): \mathrm{U} \subset \mathrm{M} \rightarrow$	$b = (b_A): \mathrm{U} \subset \mathrm{M} \rightarrow$	(2.4.3), the bundle coordinates relative to the local basis fields $b = (b_A): \mathrm{U} \subset \mathrm{M} \rightarrow$
mielke-geometrodynamics_F00101	38	■ 1.000	V^{ρ}	V^{ρ}	V^{ρ} are denoted by
mielke-geometrodynamics_F00102	39	■ 1.000	$\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	$\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	Let the Lie homomorphism $\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$ be explicitly given by the nonsin-
mielke-geometrodynamics_F00103	39	■ 0.964	$\rho_A^B(g(\xi))$	$\rho_A^B(g(\xi))$	gular complex $\mathrm{N} \times \mathrm{N}$ matrices $\rho_A^B(g(\xi))$, where ξ^j ($j = 1, \dots, \dim \mathrm{G}$) prescribes a

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mielke-geometrodynamics_F00104	39	■ 0.964	$\xi^j(\mathrm{j}=1, \ldots, \operatorname{dim} \mathrm{G})$	$\xi^j(j = 1, \dots, \dim G)$	gular complex $N \times N$ matrices $\rho_A^B(g(\xi))$, where $\xi(j = 1, \dots, \dim G)$ prescribes a
mielke-geometrodynamics_F00105	39	■ 1.000	$\rho=\mathrm{id}$	$\rho = \mathrm{id}$	identical representation, i.e., for $\rho = \mathrm{id}$, the infinitesimal generators I_j form a basis
mielke-geometrodynamics_F00106	39	■ 1.000	I_{j}	I_j	identical representation, i.e., for $\rho = \mathrm{id}$, the infinitesimal generators I_j form a basis
mielke-geometrodynamics_F00107	39	■ 0.938	$\mathfrak{g}$	$\mathfrak{g}$	of the Lie algebra $\mathfrak{g}$ of G . As a representation of I_j , the infinitesimal operators inherit
mielke-geometrodynamics_F00108	39	■ 1.000	$\mathrm{c}_{ij}^{\mathrm{k}}$	c_{ij}^k	These algebraic relations are determined by the structure constants c_{ij}^k of the Lie
mielke-geometrodynamics_F00109	39	■ 0.960	V^{Ad}	V^{Ad}	vector bundle V^{Ad} respecting the adjoint representation of G as a structure group.
mielke-geometrodynamics_F00110	39	■ 0.983	$\mathrm{s}(\mathrm{t}) \subset \mathrm{M}$	$s(t) \subset M$	in a differentiable manner along a given curve $s(t) \subset M$ in the base manifold. In order
mielke-geometrodynamics_F00111	39	■ 1.000	$\mathrm{T}_{\mathrm{p}}^*(\mathrm{P})$	$T_p^*(P)$	cotangent bundle $T_p^*(P)$ at the point $p \in P$ (see KN I, Chap. 1). Then decompose it
mielke-geometrodynamics_F00112	39	■ 1.000	$\mathrm{p} \in \mathrm{P}$	$p \in P$	cotangent bundle $T_p^*(P)$ at the point $p \in P$ (see KN I, Chap. 1). Then decompose it
mielke-geometrodynamics_F00113	40	■ 1.000	$\mathrm{T}_{\mathrm{p}}^*(\mathrm{P})=\mathrm{H}_{\mathrm{p}} \oplus \mathrm{V}_{\mathrm{p}}$	$T_p^*(P) = H_p \oplus V_p$	into the direct sum of horizontal and the vertical subspaces, i.e., $T_p^*(P) = H_p \oplus V_p$.
mielke-geometrodynamics_F00114	40	■ 1.000	$\mathrm{F}_{\mathrm{m}}(\mathrm{M})$	$F_m(M)$	i.e., in the cotangent space of the bundle $F_m(M)$ of typical fibers on M .
mielke-geometrodynamics_F00115	40	■ 0.999	$\omega(\mathrm{e}) \in \mathscr{C}$	$\omega(e) \in \mathcal{C}$	This will be achieved by the assertion of a 1-form $\omega(e) \in \mathcal{C}$ with values in the
mielke-geometrodynamics_F00116	40	■ 1.000	$\mathrm{e}_{\mathrm{v}}=\mathrm{pdg} \in \mathrm{V}_{\mathrm{p}}$	$e_v = \mathrm{pdg} \in V_p$	(ii) For vertical elements $e_v = \mathrm{pdg} \in V_p$, it will be mapped to the left-invariant
mielke-geometrodynamics_F00117	40	■ 1.000	$\bigwedge^{\mathrm{p}} \mathrm{T}_{\mathbb{C}}^*(\mathrm{M}) \otimes V^{\rho}$	$\bigwedge^p T_{\mathbb{C}}^*(M) \otimes V^\rho$	connection in an associated vector bundle $\bigwedge^p T_{\mathbb{C}}^*(M) \otimes V^\rho$ is defined via the <i>covari-</i>
mielke-geometrodynamics_F00118	40	■ 1.000	$\mathrm{T}_{\mathbb{C}}^*(\mathrm{M}) \otimes V^{\rho}$	$T_{\mathbb{C}}^*(M) \otimes V^\rho$	onto those of the product bundle $T_{\mathbb{C}}^*(M) \otimes V^\rho$. This operator will be defined in such
mielke-geometrodynamics_F00119	41	■ 1.000	V^{ρ}	V^ρ	expressed as a linear transformation of the selfsame basis of V^ρ , i.e.,
mielke-geometrodynamics_F00120	41	■ 0.856	b	b	The image of Db will be referred to as the <i>(G)-covariant exterior derivative</i> of b .
mielke-geometrodynamics_F00121	41	■ 0.641	$\mathrm{b}_{\mathrm{A}}, \mathrm{A}=1, \ldots, \mathrm{N}$	$b_A, A = 1, \dots, N$	restriction to a neighborhood U of M . Let $b_A, A = 1, \dots, N$ be a local basis for the
mielke-geometrodynamics_F00122	41	■ 0.996	$\mathrm{V}_{\mathbb{U}}^{\rho}$	V_U^ρ	cross section V_U^ρ restricted to U . With respect to this local system of reference, the
mielke-geometrodynamics_F00123	41	■ 0.998	b_{A}	b_A	resulting effect of D on b_A can be expanded as follows:
mielke-geometrodynamics_F00124	41	■ 0.998	$\left[\omega_{\mathrm{A}}^{\mathrm{B}}\right]$	$[\omega_A^B]$	matrix $[\omega_A^B]$ acquires values in $T_{\mathbb{C}}^*(M)$ and will consequently be called a <i>connection</i>
mielke-geometrodynamics_F00125	41	■ 1.000	$\Gamma_{\alpha}^j(\mathrm{m})$	$\Gamma_\alpha^j(m)$	The coefficients $\Gamma_\alpha^j(m)$ will turn out to be generalized gauge potentials as they
mielke-geometrodynamics_F00126	42	■ 0.997	$\mathrm{d} d \equiv 0$	$dd \equiv 0$	For differential forms in general, the identity $dd \equiv 0$ holds, which may be regarded as
mielke-geometrodynamics_F00127	42	■ 1.000	$\partial \partial \equiv 0$	$\partial\partial \equiv 0$	a counterpart to the homological relation $\partial\partial \equiv 0$. Figuratively speaking, this means
mielke-geometrodynamics_F00128	42	■ 0.905	Ω	Ω	that the curvature 2-form Ω satisfies the <i>second structure equation</i> of É. CARTAN:
mielke-geometrodynamics_F00129	42	■ 1.000	$\mathrm{G}=\mathrm{U}(1)$	$G = U(1)$	which can be formulated in a fiber bundle with abelian structure group $G = U(1)$,
mielke-geometrodynamics_F00130	43	■ 1.000	$\bigcirc=\partial U \subset M$	$\bigcirc = \partial U \subset M$	along an <i>infinitesimal</i> closed curve $\bigcirc = \partial U \subset M$. The very concept of a connection

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mielke-geometrodynamics_F00131	43	■ 1.000	$\partial \mathrm{U}$	∂U	closed curve ∂U . Figure 2.2 shows this for a parallel displacement of a pair of tangent
mielke-geometrodynamics_F00132	43	■ 1.000	$\mathrm{e}(\mathrm{m}) \in \mathrm{T}_{\mathrm{m}}(\mathrm{M})$	$e(m) \in T_m(M)$	$T_m(M)$ onto itself. These are generated by displacements of $\mathfrak{e}(m) \in T_m(M)$ along
mielke-geometrodynamics_F00133	44	■ 0.996	$\mathrm{G}(\mathrm{p})$	$G(p)$	(i) $G(p)$ is equivariant, i.e.,
mielke-geometrodynamics_F00134	44	■ 0.897	$\mathrm{F}_{\mathrm{m}} = \pi^{-1}(\mathrm{m})$	$F_m = \pi^{-1}(m)$	(ii) $G(p)$ preserves each fiber $F_m = \pi^{-1}(m)$, i.e., acts trivially on the base space,
mielke-geometrodynamics_F00135	44	■ 1.000	$\mathscr{G}_{\mathrm{P}}$	$\mathcal{G}_P$	position, they constitute an infinite-dimensional group, the group $\mathcal{G}_P$ of <i>local</i>
mielke-geometrodynamics_F00136	45	■ 0.513	$\exp : \mathfrak{g} \rightarrow \mathrm{G}$	$\exp : \mathfrak{g} \rightarrow G$	The exponential mapping $\exp : \mathfrak{g} \rightarrow G$ from the Lie algebra $\mathfrak{g}$ into the structure
mielke-geometrodynamics_F00137	45	■ 0.614	$P \times_{\mathrm{Ad}} \mathrm{G}$	$P \times_{\mathrm{Ad}} G$	group G of P induces a corresponding mapping within $P \times_{\mathrm{Ad}} G$. As a result,
mielke-geometrodynamics_F00138	45	■ 1.000	$\theta^k(m) \in C^\infty(M)$	$\theta^k(m) \in C^\infty(M)$	Here $\mathfrak{\theta}^k(m) \in C^\infty(M)$ denote real functions on the base space. Gauge trans-
mielke-geometrodynamics_F00139	45	■ 0.984	$\mathrm{V}^\rho(\mathrm{M}, \mathbb{C}^N, \mathrm{GL}(N, \mathbb{C}), \mathrm{P})$	$V^\rho(M, \mathbb{C}^N, \mathrm{GL}(N, \mathbb{C}), P)$	formations within an associate vector bundle $V^\rho(M, \mathbb{C}^N, \mathrm{GL}(N, \mathbb{C}), P)$, i.e.,
mielke-geometrodynamics_F00140	45	■ 1.000	$\mathscr{G}_{\mathrm{V}}$	$\mathcal{G}_V$	elements of $\mathcal{G}_V$, are represented by the same expression to the extent that L^k
mielke-geometrodynamics_F00141	45	■ 1.000	L^k	L^k	elements of $\mathcal{G}_V$, are represented by the same expression to the extent that L^k
mielke-geometrodynamics_F00142	45	■ 0.482	$\rho : \mathrm{G} \rightarrow \mathrm{GL}(N, \mathbb{C})$	$\rho : G \rightarrow \mathrm{GL}(N, \mathbb{C})$	$\varrho : G \rightarrow \mathrm{GL}(N, \mathbb{C})$.
mielke-geometrodynamics_F00143	45	■ 1.000	$\bar{\varphi}$	$\bar{\varphi}$	It has to be emphasized that the bundle coordinates $\bar{\varphi}$ of the vector bundle constructed
mielke-geometrodynamics_F00144	45	■ 1.000	$\bar{\rho}$	$\bar{\rho}$	on the conjugate (“Dirac adjoint”) representation $\bar{\rho}$ transforms according to
mielke-geometrodynamics_F00145	46	■ 0.798	$\mathrm{V}(\mathrm{M}, \mathbb{C}^N, \mathrm{GL}(N, \mathbb{C}), \mathrm{P})$	$V(M, \mathbb{C}^N, \mathrm{GL}(N, \mathbb{C}), P)$	Let $V(M, \mathbb{C}^N, \mathrm{GL}(N, \mathbb{C}), P)$ be a complex vector bundle that is associated with
mielke-geometrodynamics_F00146	46	■ 0.438	γ_k	γ_k	2k-forms γ_k on P whose inverse images after the projection on the base space M read
mielke-geometrodynamics_F00147	46	■ 1.000	$k \leq n/2$	$k \leq n/2$	The number of curvature 2-forms Ω in the exterior product is $k \leq n/2$. From the
mielke-geometrodynamics_F00148	46	■ 1.000	ω	ω	of the latter is not determined by the particular choice of the connection ω , but depends
mielke-geometrodynamics_F00149	46	■ 0.959	k^{th}	k^{th}	precisely (see: KN II, Theorem 3.1), it can be stated that the abstractly defined k^{th}
mielke-geometrodynamics_F00150	46	■ 0.570	$c_k(\mathrm{V})$	$c_k(V)$	Chern class $c_k(V)$ of a complex vector bundle V is represented by the closed 2k-form
mielke-geometrodynamics_F00151	46	■ 0.546	$V^{\mathbb{R}}$	$V^{\mathbb{R}}$	In order to obtain the characteristic classes of a real vector bundle $V^{\mathbb{R}}$ over M with
mielke-geometrodynamics_F00152	46	■ 0.983	$\mathbb{R}^N$	$\mathbb{R}^N$	the typical fiber $\mathbb{R}^N$, it will be enlarged to a complex vector bundle V with the typical
mielke-geometrodynamics_F00153	46	■ 0.753	$\mathbb{C}^N$	$\mathbb{C}^N$	fiber $\mathbb{C}^N$. The latter arises from the complexification of each fiber of $V^{\mathbb{R}}$. Then the
mielke-geometrodynamics_F00154	46	■ 0.753	$V^{\mathbb{R}}$	$V^{\mathbb{R}}$	fiber $\mathbb{C}^N$. The latter arises from the complexification of each fiber of $V^{\mathbb{R}}$. Then the
mielke-geometrodynamics_F00155	47	■ 1.000	$s=0$	$s = 0$	the Euclidean signature $s = 0$, we find locally, cf. DANIEL & VIALLET (1980),
mielke-geometrodynamics_F00156	47	■ 1.000	$\mathrm{P}(\mathrm{S}^4, \mathrm{SU}(f), \pi, \delta)$	$P(S^4, \mathrm{SU}(f), \pi, \delta)$	It is typical for the principal fiber bundles $P(S^4, \mathrm{SU}(f), \pi, \delta)$, which are to be found in
mielke-geometrodynamics_F00157	47	■ 1.000	$\det \mathrm{G}=1$	$\det G = 1$	for which $\det G = 1$ holds. Inasmuch as the trace of the Lie-algebra-valued curvature

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mielke-geometrodynamics_F00158	47	■ 0.565	<code>\operatorname{Tr}\{\backslash 0\omega \ \wedge \ 0\omega\}</code>	$\mathrm{Tr}(\Omega \wedge \Omega)$	2-form Ω vanishes in such instances, the gauge-invariant 4-form $\mathrm{Tr}(\Omega \wedge \Omega)$ suffices
mielke-geometrodynamics_F00159	47	■ 0.892	<code>\stackrel{*}{\pi}\left(\gamma_2\right)</code>	$\overset{*}{\pi}(\gamma_2)$	The integration of the second Chern class $\overset{*}{\pi}(\gamma_2)$ over the base space yields a
mielke-geometrodynamics_F00160	47	■ 0.992	<code>\mathrm{G}=\mathrm{SU}\left(\mathrm{f}\right)</code>	$G = \mathrm{SU}(f)$	index”; see JACKIW (1977). Concerning bundles with structure group $G = \mathrm{SU}(f)$,
mielke-geometrodynamics_F00161	47	■ 0.827	<code>\mathrm{SU}\left(2\right)</code>	$\mathrm{SU}(2)$	et al. (1975) is correct, since in their paper, the isospin group $\mathrm{SU}(2)$ has been enlarged
mielke-geometrodynamics_F00162	47	■ 0.749	<code>\widetilde{\mathrm{SO}}\left(4\right) \approx \mathrm{SU}\left(2\right) \otimes \mathrm{SU}\left(2\right)</code>	$\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)$	to the real structure group $\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)$, due to a special isomorphism
mielke-geometrodynamics_F00163	47	■ 0.995	<code>\gamma_2</code>	γ_2	of the closed form γ_2 onto M is even an exact one. This is a property that it shares
mielke-geometrodynamics_F00164	48	■ 1.000	<code>\mathrm{F}_{\alpha \beta}^{\mathrm{j}}</code>	$F_{\alpha\beta}^{\mathrm{j}}$	that the field strengths $F_{\alpha\beta}^{\mathrm{j}}$ (components of the curvature form) have to vanish faster
mielke-geometrodynamics_F00165	48	■ 1.000	<code> \mathrm{x} ^{-2}</code>	$ \mathrm{x} ^{-2}$	than $ \mathrm{x} ^{-2}$ at infinity, since otherwise, the integral (2.8.13) would not exist. In order to
mielke-geometrodynamics_F00166	49	■ 1.000	<code> \mathrm{x} \rightarrow \infty</code>	$ \mathrm{x} \rightarrow \infty$	should hold for $ \mathrm{x} \rightarrow \infty$. Here $\overset{\infty}{\omega}$ denotes a Maurer–Cartan connection, i.e., a left-
mielke-geometrodynamics_F00167	49	■ 1.000	<code>\stackrel{\infty}{\omega}</code>	$\overset{\infty}{\omega}$	should hold for $ \mathrm{x} \rightarrow \infty$. Here $\overset{\infty}{\omega}$ denotes a Maurer–Cartan connection, i.e., a left-
mielke-geometrodynamics_F00168	49	■ 0.978	<code>\partial \mathrm{M}</code>	∂M	If the boundary ∂M is chosen in such a way that only a pure gauge connection $\overset{\infty}{\omega}$
mielke-geometrodynamics_F00169	49	■ 0.999	<code>\mathrm{G}=\mathrm{SU}\left(2\right)</code>	$G = \mathrm{SU}(2)$	in full compliance with (2.8.15); cf. JACKIW (1980). If $G = \mathrm{SU}(2)$ serves as a struc-
mielke-geometrodynamics_F00170	49	■ 1.000	<code>\mathrm{S}^3</code>	S^3	S^3 topologically, then it can be shown that $c_2(M)$ determines the “mapping degree”
mielke-geometrodynamics_F00171	49	■ 1.000	<code>\mathrm{c}_2\left(\mathrm{M}\right)</code>	$c_2(M)$	S^3 topologically, then it can be shown that $c_2(M)$ determines the “mapping degree”
mielke-geometrodynamics_F00172	49	■ 0.954	<code>\mathrm{S}^3 \approx \mathrm{SU}\left(2\right)</code>	$S^3 \approx \mathrm{SU}(2)$	or the <i>winding number</i> ⁶ of the mapping of S^3 in $S^3 \approx \mathrm{SU}(2)$. These considerations
mielke-geometrodynamics_F00173	49	■ 0.995	<code>\mathrm{n}^{\mathrm{th}}</code>	n^{th}	⁶ Speaking more generally, the n^{th} -homotopy group of the n -dimensional sphere S^n is given by the
mielke-geometrodynamics_F00174	49	■ 1.000	<code>\mathbb{Z}</code>	$\mathbb{Z}$	group $\mathbb{Z}$ of integers, i.e., by $\pi_n(S^n) = \mathbb{Z}$. As such, it determines the winding number of the mapping
mielke-geometrodynamics_F00175	49	■ 1.000	<code>\pi_n\left(S^n\right)=\mathbb{Z}</code>	$\pi_n\left(S^n\right) = \mathbb{Z}$	group $\mathbb{Z}$ of integers, i.e., by $\pi_n(S^n) = \mathbb{Z}$. As such, it determines the winding number of the mapping
mielke-geometrodynamics_F00176	49	■ 1.000	<code>\mathrm{S}^n \rightarrow \mathrm{S}^n</code>	$S^n \rightarrow S^n$	$S^n \rightarrow S^n$.
mielke-geometrodynamics_F00177	52	■ 1.000	<code>\delta \varphi</code>	$\delta\varphi$	The operator $\delta\varphi$ of <i>small variations</i> $\varphi + \delta\varphi$ of the fields φ is satisfying the Leibniz
mielke-geometrodynamics_F00178	52	■ 1.000	<code>\varphi+\delta \varphi</code>	$\varphi + \delta\varphi$	The operator $\delta\varphi$ of <i>small variations</i> $\varphi + \delta\varphi$ of the fields φ is satisfying the Leibniz
mielke-geometrodynamics_F00179	53	■ 0.937	<code>(n-1)</code>	$(n-1)$	be the $(n-1)$ -form of the <i>field momenta</i> which are <i>canonically conjugated</i> to $d\varphi$. It
mielke-geometrodynamics_F00180	53	■ 0.937	<code>d \varphi</code>	$d\varphi$	be the $(n-1)$ -form of the <i>field momenta</i> which are <i>canonically conjugated</i> to $d\varphi$. It
mielke-geometrodynamics_F00181	53	■ 1.000	<code>\delta</code>	δ	after δ has been interchanged with the integration. The last term is an exact form.
mielke-geometrodynamics_F00182	53	■ 1.000	<code>s</code>	s	where s is the signature of the pseudo-Riemannian manifold.
mielke-geometrodynamics_F00183	53	■ 1.000	<code>d \, d \equiv 0</code>	$dd \equiv 0$	where the last term vanishes identically, due to $dd \equiv 0$. Following the physically
mielke-geometrodynamics_F00184	53	■ 0.913	<code>\mathrm{x}^i</code>	x^i	quently does not depend upon the coordinates x^i explicitly, we get the well-known

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mielke-geometrodynamics_F00185	54	■ 1.000	$\varphi \rightarrow \psi := \left(\varphi, \Theta^{(1)} := d\varphi \right)$	$\varphi \rightarrow \psi := \left(\varphi, \Theta^{(1)} := d\varphi \right)$	via the substitution $\varphi \rightarrow \psi := (\varphi, \Theta^{(1)} := d\varphi)$ into a first order system of the
mielke-geometrodynamics_F00186	54	■ 1.000	$\mathscr{D}(M)$	$\mathscr{D}(M)$	respect to the group $\mathscr{D}(M)$ of general coordinate-transformations of the spacetime
mielke-geometrodynamics_F00187	54	■ 1.000	$\mathscr{G}$	$\mathscr{G}$	labels, but as well with respect to the “internal covariance group” $\mathscr{G}$ of gauge trans-
mielke-geometrodynamics_F00188	54	■ 0.989	L_{mat}	L_{mat}	The invariance of L_{mat} with respect to $\mathscr{D}(M)$ is already guaranteed by the use
mielke-geometrodynamics_F00189	55	■ 1.000	$\bar{\varphi} \mathscr{O} \varphi$	$\bar{\varphi} \mathscr{O} \varphi$	depend upon $\bar{\varphi} \mathscr{O} \varphi$, where $\mathscr{O}$ denotes a gauge-covariant operator
mielke-geometrodynamics_F00190	55	■ 1.000	$\mathscr{O}$	$\mathscr{O}$	depend upon $\bar{\varphi} \mathscr{O} \varphi$, where $\mathscr{O}$ denotes a gauge-covariant operator
mielke-geometrodynamics_F00191	55	■ 0.996	ω'	ω'	by means of a 1-form ω' in such a way that the term GdG^{-1} resulting from gauge
mielke-geometrodynamics_F00192	55	■ 0.996	GdG^{-1}	GdG^{-1}	by means of a 1-form ω' in such a way that the term GdG^{-1} resulting from gauge
mielke-geometrodynamics_F00193	55	■ 1.000	$G \in \mathscr{G}_V$	$G \in \mathscr{G}_V$	concerning the “actively” applied gauge transformation $G \in \mathscr{G}_V$. For the derivation
mielke-geometrodynamics_F00194	55	■ 1.000	${}^G\omega$	${}^G\omega$	matrix ${}^G\omega$ of the connection relative to a basis ${}^G\mathbf{b}$, which is locally “rotated” by
mielke-geometrodynamics_F00195	55	■ 1.000	${}^G\mathbf{b}$	${}^G\mathbf{b}$	matrix ${}^G\omega$ of the connection relative to a basis ${}^G\mathbf{b}$, which is locally “rotated” by
mielke-geometrodynamics_F00196	55	■ 1.000	G^{-1}	G^{-1}	If G ist substituted by G^{-1} exactly the relation (3.2.3) comes into existence, which
mielke-geometrodynamics_F00197	56	■ 0.749	$\sigma_{\alpha}^* \omega$	$\sigma_{\alpha}^* \omega$	image or pull-back $\sigma_{\alpha}^* \omega$ due to the defining properties of a connection (see: KN I,
mielke-geometrodynamics_F00198	56	■ 0.818	$G^{-1} \omega$	$G^{-1} \omega$	The transformed connection $G^{-1} \omega$ deviates from the original one solely by a gauge-
mielke-geometrodynamics_F00199	56	■ 1.000	$\bar{\omega}$	$\bar{\omega}$	For a so-called “copy” $\bar{\omega}$ of a connection ω , this expression is to be taken as different
mielke-geometrodynamics_F00200	56	■ 1.000	$\bar{\Omega} = G^{-1} \Omega G$	$\bar{\Omega} = G^{-1} \Omega G$	from zero, although - according to the definition - its curvature $\bar{\Omega} = G^{-1} \Omega G$ is
mielke-geometrodynamics_F00201	56	■ 1.000	$d^* \omega \simeq 0$	$d^* \omega \simeq 0$	transversal or Coulomb condition $d^* \omega \simeq 0$ does not fix uniquely (large) gauge trans-
mielke-geometrodynamics_F00202	56	■ 1.000	$A = A_i dx^i$	$A = A_i dx^i$	tions of Maxwell, the electromagnetic four-potential $A = A_i dx^i$, a one-form, is the
mielke-geometrodynamics_F00203	56	■ 1.000	F	F	fundamental variable, whereas Faraday’s field strength F is a derived concept and
mielke-geometrodynamics_F00204	57	■ 0.992	$dd\Phi \equiv 0$	$dd\Phi \equiv 0$	On the other hand, $dd\Phi \equiv 0$ for any exterior p-form Φ , due to the Poincaré lemma.
mielke-geometrodynamics_F00205	57	■ 0.992	Φ	Φ	On the other hand, $dd\Phi \equiv 0$ for any exterior p-form Φ , due to the Poincaré lemma.
mielke-geometrodynamics_F00206	57	■ 1.000	$dF \equiv 0$	$dF \equiv 0$	Thus if $dF \equiv 0$, there exists locally a one-form A such that $F = dA$. However, the
mielke-geometrodynamics_F00207	57	■ 1.000	$F = dA$	$F = dA$	Thus if $dF \equiv 0$, there exists locally a one-form A such that $F = dA$. However, the
mielke-geometrodynamics_F00208	57	■ 0.981	$U(1)$	$U(1)$	local $U(1)$ gauge transformation
mielke-geometrodynamics_F00209	57	■ 1.000	$D \rightarrow \mathscr{D} := D + ieA$	$D \rightarrow \mathscr{D} := D + ieA$	Although, A is needed for the standard minimal coupling $D \rightarrow \mathscr{D} := D + ieA$ to
mielke-geometrodynamics_F00210	57	■ 1.000	L	L	For a Lagrangian formulation, let us depart from the four-form L which consist of
mielke-geometrodynamics_F00211	57	■ 0.980	Ψ	Ψ	the gauge field part and L_{mat} governing the matter field Ψ and its minimal coupling

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mielke-geometrodynamics_F00212	57	■ 1.000	$S=\int L$	$S = \int L$	usually be assumed to be of first order in the fields. Stationarity of the action $S = \int L$
mielke-geometrodynamics_F00213	57	■ 1.000	H	H	The excitation H is the field momentum conjugated to A , i.e.
mielke-geometrodynamics_F00214	58	■ 1.000	$d\,d\,H \equiv 0$	$ddH \equiv 0$	$ddH \equiv 0$, the field equations imposes ‘on shell’ that the electric current j is con-
mielke-geometrodynamics_F00215	58	■ 1.000	j	j	$ddH \equiv 0$, the field equations imposes ‘on shell’ that the electric current j is con-
mielke-geometrodynamics_F00216	58	■ 1.000	$C:=A \wedge F=A \wedge d\,A$	$C := A \wedge F = A \wedge dA$	where $C:=A \wedge F = A \wedge dA$ is the abelian Chern–Simons term, known to violate
mielke-geometrodynamics_F00217	58	■ 1.000	P	P	parity P .
mielke-geometrodynamics_F00218	58	■ 0.885	$\stackrel{\theta}{L}=-\left(\cos \theta F \wedge F+\sin \theta F \wedge F\right) / 2$	$\stackrel{\theta}{L} = -(\cos \theta F \wedge *F + \sin \theta F \wedge F)/2$	$\stackrel{\theta}{L} = -(\cos \theta F \wedge *F + \sin \theta F \wedge F)/2$, where θ is the ‘vacuum’ angle of duality rotations or
mielke-geometrodynamics_F00219	58	■ 0.885	θ	θ	$\stackrel{\theta}{L} = -(\cos \theta F \wedge *F + \sin \theta F \wedge F)/2$, where θ is the ‘vacuum’ angle of duality rotations or
mielke-geometrodynamics_F00220	59	■ 0.950	$*$	$*$	The Hodge dual $*$ depends on the metric g and on the orientation of the manifold, as
mielke-geometrodynamics_F00221	59	■ 0.950	g	g	The Hodge dual $*$ depends on the metric g and on the orientation of the manifold, as
mielke-geometrodynamics_F00222	59	■ 1.000	η	η	where η is the standard volume four-form. The Pontryagin four-form can be written
mielke-geometrodynamics_F00223	59	■ 1.000	$\partial \mathfrak{D} / \partial t$	$\partial \mathfrak{D} / \partial t$	³ The displacement current $\partial \mathfrak{D} / \partial t$, where $\mathfrak{D} = \varepsilon \mathbf{E}$, was anticipated 1839 by James Mac Cullagh, cf.
mielke-geometrodynamics_F00224	59	■ 1.000	$\mathfrak{D}=\varepsilon \mathbf{E}$	$\mathfrak{D} = \varepsilon \mathbf{E}$	³ The displacement current $\partial \mathfrak{D} / \partial t$, where $\mathfrak{D} = \varepsilon \mathbf{E}$, was anticipated 1839 by James Mac Cullagh, cf.
mielke-geometrodynamics_F00225	60	■ 1.000	$\vartheta_{[\alpha} \wedge \Sigma_{\beta]}=0$	$\vartheta_{[\alpha} \wedge \Sigma_{\beta]} = 0$	and turns out to be symmetric, i.e., $\vartheta_{[\alpha} \wedge \Sigma_{\beta]} \equiv 0$. For non-vanishing charge current
mielke-geometrodynamics_F00226	60	■ 1.000	$B=\frac{1}{2} B_{i j} d x^i \wedge d x^j$	$B = \frac{1}{2} B_{ij} dx^i \wedge dx^j$	two-form $B = \frac{1}{2} B_{ij} dx^i \wedge dx^j$ are varied <i>independently</i> , slightly reminiscent of the
mielke-geometrodynamics_F00227	60	■ 1.000	B	B	Independent variations with respect to A and B lead to $dB \cong 0$ and to the <i>constraint</i> of
mielke-geometrodynamics_F00228	60	■ 1.000	$d\,B \cong 0$	$dB \cong 0$	Independent variations with respect to A and B lead to $dB \cong 0$ and to the <i>constraint</i> of
mielke-geometrodynamics_F00229	60	■ 1.000	$F:=d\,A \cong 0$	$F := dA \cong 0$	vanishing field strength $F := dA \cong 0$. This implies, however, that such a primordial
mielke-geometrodynamics_F00230	60	■ 1.000	$d\,C$	dC	(3.3.22) only by a boundary term dC derived from a <i>Chern–Simons</i> (CS) three-form
mielke-geometrodynamics_F00231	60	■ 1.000	$H=d\,B$	$H = dB$	excitation $H = dB$, the Kalb–Ramond <i>axion</i> three-form.
mielke-geometrodynamics_F00232	61	■ 1.000	$\delta d\,C / \delta A=d\,F \equiv 0$	$\delta dC / \delta A = dF \equiv 0$	can be recovered via the variation of the Pontryagin term, e.g., $\delta dC / \delta A = dF \equiv 0$
mielke-geometrodynamics_F00233	61	■ 0.998	$\psi=\psi_i d x^i$	$\psi = \psi_i dx^i$	where $\psi = \psi_i dx^i$ is a one-form or a covector; cf. LUCCHESI et al. (1993). In a
mielke-geometrodynamics_F00234	61	■ 1.000	$s\,\psi=D\,\phi$	$s\psi = D\phi$	BRST quantization, due to the zero mode $s\psi = D\phi$ with $s\phi = 0$, the “topological”
mielke-geometrodynamics_F00235	61	■ 1.000	$s\,\phi=0$	$s\phi = 0$	BRST quantization, due to the zero mode $s\psi = D\phi$ with $s\phi = 0$, the “topological”
mielke-geometrodynamics_F00236	61	■ 1.000	$B\,F$	BF	to topological field theory is the BF formalism which, in the case of gravity, was
mielke-geometrodynamics_F00237	61	■ 1.000	$F=\left\{ \right\}^* B$	$F = *B$	the definition of the field strength (3.3.24), but as a bonus, from the relation $F = *B$
mielke-geometrodynamics_F00238	61	■ 0.998	$j:=\delta L_{\text {matter }} / \delta A$	$j := \delta L_{\text{matter}} / \delta A$	where $j := \delta L_{\text{matter}} / \delta A$ is the matter current.

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mielke-geometrodynamics_F00239	61	■ 1.000	$d\,j\,\cong\,0$	$dj \cong 0$	which “on shell,” is conserved classically, i.e., $dj \cong 0$. In general, this includes
mielke-geometrodynamics_F00240	61	■ 0.999	$d\,A$	dA	<i>Pauli-type terms</i> generated by the variation of the Lagrangian with respect to dA ; cf.
mielke-geometrodynamics_F00241	62	■ 1.000	$V(B)$	$V(B)$	The quadratic term in (3.3.28) can be generalized to a four-form potential $V(B)$.
mielke-geometrodynamics_F00242	62	■ 1.000	$d\,B+j\,\cong\,0$	$dB + j \cong 0$	Then the field equation $dB + j \cong 0$ together with the relation $F \cong \partial V/\partial B$ for
mielke-geometrodynamics_F00243	62	■ 1.000	$F\,\cong\,\partial V/\partial B$	$F \cong \partial V/\partial B$	Then the field equation $dB + j \cong 0$ together with the relation $F \cong \partial V/\partial B$ for
mielke-geometrodynamics_F00244	62	■ 1.000	$-B\,\wedge\,F+B\,\wedge\,B\,\wedge\,B\,/\,3$	$-B \wedge F + B \wedge B \wedge B/3$	systems with a cubic term $-B \wedge F + B \wedge B \wedge B/3$ are directly related to Chern–Simons
mielke-geometrodynamics_F00245	62	■ 0.999	$C:=A\,\wedge\,F-A\,\wedge\,A\,\wedge\,A\,/\,3$	$C := A \wedge F - A \wedge A \wedge A/3$	theories departing from the noninvariant three-form $C := A \wedge F - A \wedge A \wedge A/3$; cf.
mielke-geometrodynamics_F00246	62	■ 1.000	$F=-B\,\wedge\,B$	$F = -B \wedge B$	$F = -B \wedge B$ together with the Bianchi identity $DB = DF \equiv 0$ correspond, in 3D
mielke-geometrodynamics_F00247	62	■ 1.000	$D\,B=D\,F\,\equiv\,0$	$DB = DF \equiv 0$	$F = -B \wedge B$ together with the Bianchi identity $DB = DF \equiv 0$ correspond, in 3D
mielke-geometrodynamics_F00248	62	■ 0.999	$\Gamma_{\alpha}^j(\mathrm{m})$	$\Gamma_{\alpha}^j(\mathrm{m})$	Speaking a priori, the connection ω and the corresponding gauge potentials $\Gamma_{\alpha}^j(\mathrm{m})$
mielke-geometrodynamics_F00249	62	■ 1.000	$d\,\omega$	$d\omega$	only of its first derivative $d\omega$. The only <i>gauge-covariant</i> object of this kind that meets
mielke-geometrodynamics_F00250	62	■ 0.668	$\mathscr{F}$	$\mathscr{F}$	4-forms into 0-forms, i.e., scalar functions. In quantized gauge theories, $\mathscr{F}$ and $^*\mathscr{F}$
mielke-geometrodynamics_F00251	62	■ 0.668	$\{ \, \}^{*} \mathscr{F}$	$^*\mathscr{F}$	4-forms into 0-forms, i.e., scalar functions. In quantized gauge theories, $\mathscr{F}$ and $^*\mathscr{F}$
mielke-geometrodynamics_F00252	63	■ 0.731	Π^{Ω}	Π^{Ω}	The 2-form Π^{Ω} of the <i>field momenta</i> being <i>canonically conjugated</i> to Ω are formally
mielke-geometrodynamics_F00253	63	■ 0.894	$n-1$	$n - 1$	$(n - 1)$ -form τ of the matter current as a source. The latter is canonically conjugate
mielke-geometrodynamics_F00254	63	■ 0.894	τ	τ	$(n - 1)$ -form τ of the matter current as a source. The latter is canonically conjugate
mielke-geometrodynamics_F00255	64	■ 1.000	$\delta L\,/\,\delta\omega$	$\delta L/\delta\omega$	the variation $\delta L/\delta\omega$ of the total system. They are complemented by the Bianchi
mielke-geometrodynamics_F00256	64	■ 1.000	(1978,1980)	(1978, 1980)	it has been shown by HORNDESKI (1978, 1980) that a field theory of Yang–Mills
mielke-geometrodynamics_F00257	65	■ 1.000	H^3	H^3	hypersurface H^3 or its boundary $\partial\mathrm{H}^3$, respectively. Note that ι involves not only
mielke-geometrodynamics_F00258	65	■ 1.000	$\partial\,\mathrm{H}^3$	$\partial\mathrm{H}^3$	hypersurface H^3 or its boundary $\partial\mathrm{H}^3$, respectively. Note that ι involves not only
mielke-geometrodynamics_F00259	65	■ 1.000	ι	ι	hypersurface H^3 or its boundary $\partial\mathrm{H}^3$, respectively. Note that ι involves not only
mielke-geometrodynamics_F00260	65	■ 0.879	$\mathrm{SU}(\mathrm{N})$	$\mathrm{SU}(\mathrm{N})$	In more generalized versions, these theories make use of $\mathrm{SU}(\mathrm{N})$ as a structure group,
mielke-geometrodynamics_F00261	66	■ 0.971	$L\!\left(\mathscr{F},\{ \, \}^{*} \mathscr{F}\right)$	$L(\mathscr{F}, ^*\mathscr{F})$	$L(\mathscr{F}, ^*\mathscr{F})$ is of course a mere empirical one and can be answered only by means of
mielke-geometrodynamics_F00262	66	■ 0.435	$\mathrm{I}^j=\mathrm{i}\,\sigma^j$	$\mathbb{I}^j = \mathrm{i}\sigma^j$	with the infinitesimal generators $\mathbb{I}^j = \mathrm{i}\sigma^j$, thus obtaining the group $\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$
mielke-geometrodynamics_F00263	66	■ 0.435	$\mathrm{SU}(2)\,\times\,\mathrm{SU}(2)\,\approx$	$\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$	with the infinitesimal generators $\mathbb{I}^j = \mathrm{i}\sigma^j$, thus obtaining the group $\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$
mielke-geometrodynamics_F00264	66	■ 0.765	$\mathrm{SO}(4)$	$\mathrm{SO}(4)$	$\mathrm{SO}(4)$ by proceeding from the array of matrices
mielke-geometrodynamics_F00265	67	■ 0.998	$\mathrm{h}=\mathrm{h}(\mathrm{x})$	$\mathrm{h} = \mathrm{h}(\mathrm{x})$	i.e., $\mathrm{h} = \mathrm{h}(\mathrm{x})$ in the spherically symmetric case. A similar nonlinear equation occurs

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mielke-geometrodynamics_F00266	67	■ 0.499	M^4	M^4	cf. JACKIW et al. (1977). In a flat space M^4 with Euclidean signature $s = 0$, this is a
mielke-geometrodynamics_F00267	67	■ 0.999	$\mathrm{k}=1$	$k = 1$	in the case of $k = 1$ (T HOOFT 1977). The configuration as given above depends
mielke-geometrodynamics_F00268	67	■ 0.512	$5 \mathrm{k}+4$	$5k + 4$	on $5k + 4$ parameters. From studies of related objects in algebraic topology it is
mielke-geometrodynamics_F00269	67	■ 1.000	$\mathbb{R}^4 \cup \{\infty\} \approx \mathrm{S}^4$	$\mathbb{R}^4 \cup \{\infty\} \approx S^4$	of the Yang–Mills equation in a Euclidean space $\mathbb{R}^4 \cup \{\infty\} \approx S^4$, which is confor-
mielke-geometrodynamics_F00270	68	■ 1.000	$ \!-\!\infty\rangle$	$ \!-\!\infty\rangle$	transition amplitude between the asymptotic states $ \!-\!\infty\rangle$ and $\langle\infty $. This ampli-
mielke-geometrodynamics_F00271	68	■ 1.000	$\langle\infty $	$\langle\infty $	transition amplitude between the asymptotic states $ \!-\!\infty\rangle$ and $\langle\infty $. This ampli-
mielke-geometrodynamics_F00272	68	■ 1.000	$\mathscr{M}_{\omega}$	$\mathcal{M}_{\omega}$	configuration space $\mathcal{M}_{\omega}$ of all inequivalent gauge fields, i.e., of all possible connec-
mielke-geometrodynamics_F00273	68	■ 0.991	$\mathscr{M}_{\omega}=\mathscr{C} / \mathscr{G}_p$	$\mathcal{M}_{\omega} = \mathcal{C} / \mathcal{G}_p$	$\mathcal{M}_{\omega} = \mathcal{C} / \mathcal{G}_p$ has to be taken as the configuration space proper. Under not too stringent
mielke-geometrodynamics_F00274	68	■ 0.999	$\mu[\omega]$	$\mu[\omega]$	can be shown that the “measure” $\mu[\omega]$ for the functional integration, as given by the
mielke-geometrodynamics_F00275	69	■ 0.976	S	S	regime of gauge fields, the so-called <i>S-duality</i> .
mielke-geometrodynamics_F00276	69	■ 1.000	$\operatorname{Spin}^c(4)$	$\operatorname{Spin}^c(4)$	(2006) coupled to spinors ψ regarded as a lift from the frame bundle to $\operatorname{Spin}^c(4)$.
mielke-geometrodynamics_F00277	69	■ 0.988	$\psi=\psi_L+\psi_R=P_{-}\psi+P_{+}\psi$	$\psi = \psi_L + \psi_R = P_{-}\psi + P_{+}\psi$	cf. JOST et al. (1996), where here the decomposition $\overline{\psi} = \overline{\psi}_L + \overline{\psi}_R = P_{-}\overline{\psi} + P_{+}\overline{\psi}$
mielke-geometrodynamics_F00278	70	■ 1.000	$\bar{\psi}$	$\bar{\psi}$	The variation with respect to $\bar{\psi}$ and A leads to the <i>convective</i> -type spinor equation
mielke-geometrodynamics_F00279	70	■ 1.000	R	R	(3.6.1). For a nonnegative scalar curvature R in the Weizenböck formula, all solutions
mielke-geometrodynamics_F00280	70	■ 1.000	$\psi=0$	$\psi = 0$	satisfy $\overline{\psi} \equiv 0$ and then are $U(1)$ instantons.
mielke-geometrodynamics_F00281	70	■ 0.996	$F^{\pm}$	$F^{\pm}$	$F^{\pm}$ resembles (MIELKE 1984) the modified (double) duality ansatz
mielke-geometrodynamics_F00282	71	■ 1.000	A_{α}^j	A_{α}^j	gauge potentials A_{α}^j in the Lagrangian n -form. However, there is only one massless
mielke-geometrodynamics_F00283	71	■ 1.000	φ^j	φ^j	fields φ^j with f components but also the vector fields $\Theta^{(1)j}$ will be coupled to the
mielke-geometrodynamics_F00284	71	■ 1.000	$\Theta^{(1) j}$	$\Theta^{(1)j}$	fields φ^j with f components but also the vector fields $\Theta^{(1)j}$ will be coupled to the
mielke-geometrodynamics_F00285	71	■ 0.902	$\phi^{\{0\}}$	$\phi^{(0)}$	the 0-form $\phi^{(0)}$ and the 1-form $\Theta^{(1)}$ with values in the adjoint representation of G .
mielke-geometrodynamics_F00286	71	■ 0.902	$\Theta^{\{1\}}$	$\Theta^{(1)}$	the 0-form $\phi^{(0)}$ and the 1-form $\Theta^{(1)}$ with values in the adjoint representation of G .
mielke-geometrodynamics_F00287	71	■ 1.000	$\mathrm{A}_{\alpha}^j(\mathrm{m})$	$A_{\alpha}^j(m)$	gies to the Higgs model. The coupling of the gauge potentials $A_{\alpha}^j(m)$ to the Higgs
mielke-geometrodynamics_F00288	71	■ 1.000	$\delta L_{\mathrm{H}} / \delta \bar{\phi}^{\{0\}}$	$\delta L_H / \delta \bar{\phi}^{(0)}$	the G -equivalence principle. From the variations and $\delta L_H / \delta \bar{\phi}^{(0)}$ and $\delta L_H / \delta \bar{\Theta}^{(1)}$, the
mielke-geometrodynamics_F00289	71	■ 1.000	$\delta L_{\mathrm{H}} / \delta \bar{\Theta}^{\{1\}}$	$\delta L_H / \delta \bar{\Theta}^{(1)}$	the G -equivalence principle. From the variations and $\delta L_H / \delta \bar{\phi}^{(0)}$ and $\delta L_H / \delta \bar{\Theta}^{(1)}$, the
mielke-geometrodynamics_F00290	72	■ 1.000	μ	μ	It should be noted that the parameter μ in (3.7.5) complies formally with an imaginary
mielke-geometrodynamics_F00291	72	■ 1.000	$\varphi=0$	$\varphi = 0$	mass of the Klein–Gordon-type field. However, the trivial field configuration $\varphi = 0$
mielke-geometrodynamics_F00292	72	■ 1.000	$ \varphi $	$ \varphi $	This is a consequence of the <i>nonlinear term</i> in (3.7.1), which is quartic in $ \varphi $.

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mielke-geometrodynamics_F00293	72	■ 1.000	$\{ \}^{\lambda} \varphi$	${}^{\lambda}\varphi$	quantum-mechanical state ${}^{\lambda}\varphi$, for which the vacuum expectation value
mielke-geometrodynamics_F00294	72	■ 1.000	$\mathrm{\varphi}_{\mathrm{H}}:=\mathrm{\varphi}-\mu/\sqrt{\lambda}$	$\varphi_{\mathrm{H}}:=\varphi-\mu/\sqrt{\lambda}$	$\varphi_{\mathrm{H}}:=\varphi-\mu/\sqrt{\lambda}$ along with the gauge fields A_{α}^j acquire a real, i.e., physical, mass
mielke-geometrodynamics_F00295	72	■ 1.000	$\mathrm{m}_{\mathrm{H}}=\sqrt{2}\,\mu$	$m_{\mathrm{H}}=\sqrt{2}\mu$	that is determined by the curvature of the potential at the bottom, i.e., by $m_{\mathrm{H}}=\sqrt{2}\mu$,
mielke-geometrodynamics_F00296	72	■ 0.882	G/H	G/H	field φ such that its range is an orbit G/H of G in V^{ρ} with respect to a subgroup H ,
mielke-geometrodynamics_F00297	72	■ 1.000	$\mathrm{v}_{\{0\}}\in V^{\rho}$	$v_0\in V^{\rho}$	Let $v_0\in V^{\rho}$ be a fixed vector for which the cross section $\overset{\circ}{\varphi}_{\min}:=\sigma(v_0)\in C^{\infty}(V^{\rho})$
mielke-geometrodynamics_F00298	72	■ 1.000	$\stackrel{\circ}{\mathrm{\varphi}}_{\mathrm{min}}:=\sigma\left(\mathrm{v}_{\{0\}}\right)\in\mathrm{C}^{\infty}\left(\mathrm{V}^{\rho}\right)$	$\overset{\circ}{\varphi}_{\min}:=\sigma(v_0)\in C^{\infty}(V^{\rho})$	Let $v_0\in V^{\rho}$ be a fixed vector for which the cross section $\overset{\circ}{\varphi}_{\min}:=\sigma(v_0)\in C^{\infty}(V^{\rho})$
mielke-geometrodynamics_F00299	72	■ 0.683	$\mathrm{a}\,\mathrm{g}\in\mathrm{G}$	$ag\in G$	to find a $g\in G$ that connects any two vectors v and v_0 from V^{ρ} via the relation
mielke-geometrodynamics_F00300	72	■ 0.683	$\mathrm{v}_{\{0\}}$	v_0	to find a $g\in G$ that connects any two vectors v and v_0 from V^{ρ} via the relation
mielke-geometrodynamics_F00301	72	■ 1.000	$\mathrm{v}=\rho(\mathrm{g})\cdot\mathrm{v}_{\{0\}}$	$v=\rho(g)\cdot v_0$	$v=\rho(g)\cdot v_0$. An invariant subgroup of G , the so-called isotropy group
mielke-geometrodynamics_F00302	73	■ 0.996	$\stackrel{\circ}{\mathrm{\varphi}}_{\mathrm{min}}:=\sigma\left(\mathrm{v}_{\{0\}}\right)$	$\overset{\circ}{\varphi}_{\min}=\sigma(v_0)$	can be <i>uniquely</i> adjoined to the fixed $\overset{\circ}{\varphi}_{\min}=\sigma(v_0)$. Then
mielke-geometrodynamics_F00303	73	■ 1.000	$\mathrm{P}(\mathrm{G},\mathrm{M},\pi,\delta)$	$P(G,M,\pi,\delta)$	is a subbundle of $P(G,M,\pi,\delta)$ over the same base space; its structure group is H
mielke-geometrodynamics_F00304	73	■ 0.702	$\mathfrak{h}$	$\mathfrak{h}$	Proposition A connection ω in P is reducible to an $\mathfrak{h}$ -valued connection ω_{H} in Q if
mielke-geometrodynamics_F00305	73	■ 0.702	ω_{H}	ω_{H}	Proposition A connection ω in P is reducible to an $\mathfrak{h}$ -valued connection ω_{H} in Q if
mielke-geometrodynamics_F00306	73	■ 0.702	Q	Q	Proposition A connection ω in P is reducible to an $\mathfrak{h}$ -valued connection ω_{H} in Q if
mielke-geometrodynamics_F00307	73	■ 1.000	$\omega_{\{0\}}$	ω_0	theories of gravity that will be developed later on. Let ω_0 be a timelike component
mielke-geometrodynamics_F00308	74	■ 1.000	ϕ	ϕ	of the connection can be identified with the Higgs field ϕ without self-interaction:
mielke-geometrodynamics_F00309	74	■ 1.000	$\lambda\rightarrow 0$	$\lambda\rightarrow 0$	limit $\lambda\rightarrow 0$ may be obtained by identifying ω_0 in the interaction-free Yang–Mills
mielke-geometrodynamics_F00310	74	■ 1.000	$\stackrel{*}{\sigma}\vec{\omega}$	$\overset{*}{\sigma}\vec{\omega}$	provides the static Yang–Mills potential $\overset{*}{\sigma}\vec{\omega}$ of the monopole. After this identifica-
mielke-geometrodynamics_F00311	74	■ 0.503	(1981,1983)	(1981,1983)	(1977), TCHRAKIAN (1981, 1983), PRASAD & ROSSI (1981a, b). The topological
mielke-geometrodynamics_F00312	75	■ 1.000	σ	σ	In this context, it is worthwhile to mention that in the nonlinear σ -model, a variant
mielke-geometrodynamics_F00313	75	■ 1.000	$\mathbb{R}^2\times\mathrm{S}^2$	$\mathbb{R}^2\times S^2$	connection in a nontrivial $U(1)$ -bundle over $\mathbb{R}^2\times S^2$
mielke-geometrodynamics_F00314	75	■ 0.657	S^4	S^4	classification of $SU(2)$ -bundles over S^4 according to the
mielke-geometrodynamics_F00315	78	■ 1.000	$\mathrm{N}=2$	$N=2$	$N=2$ supersymmetric Yang-Mills theory. Nuclear Phys B 426(1):19–52
mielke-geometrodynamics_F00316	79	■ 1.000	ℓ^*	ℓ^*	diminutive coupling constant given by the Planck length ℓ^* . It may even provide the
mielke-geometrodynamics_F00317	80	■ 0.998	(1980,1981)	(1980,1981)	those theories of gravity, was worked out in more detail by HEHL (1980, 1981) using
mielke-geometrodynamics_F00318	81	■ 0.951	$\mathrm{A}(\mathrm{n},\mathbb{R})$	$A(n,\mathbb{R})$	and to start off deductively from the affine group $A(n,\mathbb{R})$ as structure group. As a
mielke-geometrodynamics_F00319	81	■ 1.000	A^{n}	A^n	Euclidean space E^n , regarded as an affine space A^n .

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mielke-geometrodynamics_F00320	81	■ 0.999	$\mathrm{a}=\left(\mathrm{a}^{\alpha}\right) \in \mathbb{R}^n$	$a = (a^\alpha) \in \mathbb{R}^n$	Let $\underline{a} = (a^\alpha) \in \mathbb{R}^n$ be a row vector. According to its definition, the action of an
mielke-geometrodynamics_F00321	81	■ 0.992	$\mathrm{x} \in \mathrm{A}^{\mathrm{n}}$	$x \in A^n$	element of $A(n, \mathbb{R})$ on a vector $\underline{x} \in A^n$ can expressed as the composition of a general
mielke-geometrodynamics_F00322	81	■ 1.000	$\tilde{x}:=\mathrm{binom}{x}{1} \in \mathbb{R}^{n+1}$	$\tilde{x} := \begin{pmatrix} x \\ 1 \end{pmatrix} \in \mathbb{R}^{n+1}$	In the following, it will be convenient to introduce a vector $\tilde{x} := \begin{pmatrix} x \\ 1 \end{pmatrix} \in \mathbb{R}^{n+1}$ with
mielke-geometrodynamics_F00323	81	■ 1.000	$\mathrm{n}+1$	$n + 1$	$n + 1$ components and to replace the group action (4.1.1) by the abstract transfor-
mielke-geometrodynamics_F00324	81	■ 0.998	$\tilde{x}$	$\tilde{x}$	mations of $\tilde{x}$ via the matrices
mielke-geometrodynamics_F00325	81	■ 0.890	$\mathrm{GL}(\mathrm{n}+1, \mathbb{R})$	$GL(n + 1, \mathbb{R})$	This faithful representation of $A(n, \mathbb{R})$ by a subgroup of $GL(n + 1, \mathbb{R})$ shows clearly
mielke-geometrodynamics_F00326	81	■ 0.623	$\mathrm{a} \in \mathbb{R}^n$	$a \in \mathbb{R}^n$	of the translations $a \in \mathbb{R}^n$ and the linear transformations $g \in GL(n, \mathbb{R})$. The rule for
mielke-geometrodynamics_F00327	81	■ 0.623	$\mathrm{g} \in \mathrm{GL}(\mathrm{n}, \mathbb{R})$	$g \in GL(n, \mathbb{R})$	of the translations $a \in \mathbb{R}^n$ and the linear transformations $\underline{g} \in GL(n, \mathbb{R})$. The rule for
mielke-geometrodynamics_F00328	81	■ 1.000	$0(1, n-1) \subset \mathrm{GL}(\mathrm{n}, \mathbb{R})$	$0(1, n - 1) \subset GL(n, \mathbb{R})$	$0(1, n - 1) \subset GL(n, \mathbb{R})$. The invariance of the (flat) metric ground form of the
mielke-geometrodynamics_F00329	81	■ 1.000	$\mathfrak{a}(\mathrm{n}, \mathbb{R})$	$\mathfrak{a}(n, \mathbb{R})$	$\mathfrak{a}(n, \mathbb{R})$ of the affine group, the analogous decomposition
mielke-geometrodynamics_F00330	82	■ 0.999	$L_{\alpha \beta} \in \mathfrak{gl}(\mathrm{n}, \mathbb{R})$	$L_{\alpha\beta} \in \mathfrak{gl}(n, \mathbb{R})$	In these formulas, $L_{\alpha\beta} \in \mathfrak{gl}(n, \mathbb{R})$ and $P_\alpha \in \mathbb{R}^n$ denote the infinitesimal generators of
mielke-geometrodynamics_F00331	82	■ 0.999	$P_{\alpha} \in \mathbb{R}^n$	$P_\alpha \in \mathbb{R}^n$	In these formulas, $L_{\alpha\beta} \in \mathfrak{gl}(n, \mathbb{R})$ and $P_\alpha \in \mathbb{R}^n$ denote the infinitesimal generators of
mielke-geometrodynamics_F00332	82	■ 1.000	$\mathrm{GL}(\mathrm{n}, \mathbb{R})$	$GL(n, \mathbb{R})$	for the tangent space. The general linear group $GL(n, \mathbb{R})$ acts on this basis as follows:
mielke-geometrodynamics_F00333	82	■ 0.833	$L_m(\mathrm{M})$	$L_m(M)$	It thus generates a space $L_m(M)$ of <i>linear frames of reference</i> at the point m. Let
mielke-geometrodynamics_F00334	82	■ 0.993	$\mathrm{U} \subset \mathrm{M}$	$U \subset M$	remember that in a neighborhood $U \subset M$ of a point $m \in M$ with coordinates x^i ,
mielke-geometrodynamics_F00335	82	■ 0.994	$e_a(\mathrm{m})$	$e_a(m)$	a basis vector $e_a(m)$ of $T_m(M)$ may always be written in the way given above.
mielke-geometrodynamics_F00336	82	■ 0.852	$\mathbb{R}^{n^2}$	$\mathbb{R}^{n^2}$	an open neighborhood of $\mathbb{R}^{n^2}$. Since $e_\alpha^j(m)$ represents a nonsingular $n \times n$ matrix,
mielke-geometrodynamics_F00337	82	■ 0.852	$e_\alpha^j(\mathrm{m})$	$e_\alpha^j(m)$	an open neighborhood of $\mathbb{R}^{n^2}$. Since $e_\alpha^j(m)$ represents a nonsingular $n \times n$ matrix,
mielke-geometrodynamics_F00338	82	■ 0.852	$n \times n$	$n \times n$	an open neighborhood of $\mathbb{R}^{n^2}$. Since $e_\alpha^j(m)$ represents a nonsingular $n \times n$ matrix,
mielke-geometrodynamics_F00339	82	■ 0.987	$\left(x^i, e_{\alpha}^{-j}\right)$	(x^i, e_α^{-j})	the collection (x^i, e_α^j) may be regarded as a choice of coordinates of the Cartesian
mielke-geometrodynamics_F00340	82	■ 0.840	$\mathrm{U} \times \mathrm{GL}(\mathrm{n}, \mathbb{R})$	$U \times GL(n, \mathbb{R})$	product $\underline{U} \times GL(n, \mathbb{R})$. These coordinates arise from the original domain $\pi^{-1}(U)$, for
mielke-geometrodynamics_F00341	82	■ 0.840	$\pi^{-1}(\mathrm{U})$	$\pi^{-1}(U)$	product $U \times GL(n, \mathbb{R})$. These coordinates arise from the original domain $\pi^{-1}(U)$, for
mielke-geometrodynamics_F00342	83	■ 0.858	$\operatorname{GL}(\mathrm{n})$	$GL(n)$	which the projection π of the principal fiber bundle with the structure group $GL(n,$
mielke-geometrodynamics_F00343	83	■ 0.899	ϑ	ϑ	of linear frames. This is the case if an $\mathbb{R}^n$ -valued 1-form ϑ exists that not only is
mielke-geometrodynamics_F00344	83	■ 0.825	$\mathrm{n}=\operatorname{dim} \mathrm{M}$	$n = \dim M$	this close contact and due to the fact that ϑ has rank $n = \dim M$, it is natural to call
mielke-geometrodynamics_F00345	83	■ 0.394	$\mathbb{R}^n$	$\mathbb{R}^n$	associated with $L(M)$. Its standard fiber is $\mathbb{R}^n$, and thus $T(M)$ has the same dimension
mielke-geometrodynamics_F00346	83	■ 0.569	$\mathrm{A}(\mathrm{M}) / \mathbb{R}^n$	$A(M)/\mathbb{R}^n$	The possibility of identifying $L(M)$ with the quotient bundle $A(M)/\mathbb{R}^n$ results from

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mielke-geometrodynamics_F00347	84	■ 0.999	$\mathrm{e}_{\alpha}(\mathrm{m})$	$\mathrm{e}_{\alpha}(\mathrm{m})$	the basis vectors $\mathrm{e}_{\alpha}(\mathrm{m})$ in the tangent space $T_{\mathrm{m}}(\mathrm{M})$ at the point $\mathrm{m} \in \mathrm{M}$ appears to
mielke-geometrodynamics_F00348	84	■ 0.593	$\mathrm{T}=\mathrm{mathbb{R}}^{\mathrm{n}}$	$T = \mathbb{R}^n$	group $T = \mathbb{R}^n$ of translations can be regarded as the quotient space $A(n, \mathbb{R})/\mathrm{GL}(n, \mathbb{R})$,
mielke-geometrodynamics_F00349	84	■ 0.593	$\mathrm{A}(\mathrm{n}, \mathrm{mathbb{R}}) / \mathrm{GL}(\mathrm{n}, \mathrm{mathbb{R}})$	$A(n, \mathbb{R})/\mathrm{GL}(n, \mathbb{R})$	group $T = \mathbb{R}^n$ of translations can be regarded as the quotient space $A(n, \mathbb{R})/\mathrm{GL}(n, \mathbb{R})$,
mielke-geometrodynamics_F00350	84	■ 1.000	$\mathrm{T}=\mathrm{P} / \mathrm{L}$	$T = P/L$	realized by translations acting on the coset space $T = P/L$, which is isomorphic to the
mielke-geometrodynamics_F00351	84	■ 0.550	$\widetilde{\omega}$	$\widetilde{\omega}$	connection. Let $\widetilde{\omega}$ be the 1-form of such a <i>generalized affine connection</i> , and let
mielke-geometrodynamics_F00352	84	■ 0.999	$\chi: \mathrm{L}(\mathrm{M}) \rightarrow 0_{\mathrm{m}} \times \mathrm{L}(\mathrm{M}) \subset \mathrm{A}(\mathrm{M})$	$\chi : L(M) \rightarrow 0_{\mathrm{m}} \times L(M) \subset A(M)$	$\chi : L(M) \rightarrow 0_{\mathrm{m}} \times L(M) \subset A(M)$ be the natural injection of linear frames into the
mielke-geometrodynamics_F00353	85	■ 0.448	$\stackrel{*}{\chi} \widetilde{\omega}$	$\stackrel{*}{\chi} \widetilde{\omega}$	image $\stackrel{*}{\chi} \widetilde{\omega}$ of the connection $\widetilde{\omega}$ is an $\mathfrak{a}(n, \mathbb{R})$ -valued 1-form in $L(M)$ that decomposes
mielke-geometrodynamics_F00354	85	■ 0.507	ω^{L}	ω^{L}	tation (4.1.2) of $A(n, \mathbb{R})$. In the above formula, ω^{L} denotes a $\mathfrak{gl}(n, \mathbb{R})$ -valued 1-form
mielke-geometrodynamics_F00355	85	■ 0.507	$\mathfrak{gl}(\mathrm{n}, \mathrm{mathbb{R}})$	$\mathfrak{gl}(n, \mathbb{R})$	tation (4.1.2) of $A(n, \mathbb{R})$. In the above formula, ω^{L} denotes a $\mathfrak{gl}(n, \mathbb{R})$ -valued 1-form
mielke-geometrodynamics_F00356	85	■ 0.912	ω^{T}	ω^{T}	in M , while ω^{T} is an $\mathbb{R}^n$ -valued 1-form, i.e., a tensorial form of the representation
mielke-geometrodynamics_F00357	85	■ 0.990	$\mathrm{H} \subset \mathrm{G}$	$H \subset G$	structure with respect to a subgroup $H \subset G$.
mielke-geometrodynamics_F00358	85	■ 0.944	$\widetilde{\Omega}$	$\widetilde{\Omega}$	The curvature $\widetilde{\Omega}$ in $A(M)$ can be derived from $\widetilde{\omega}$ by an analogous application of
mielke-geometrodynamics_F00359	85	■ 1.000	$\left[\omega^{\mathrm{T}}, \omega^{\mathrm{T}}\right]$	$[\omega^{\mathrm{T}}, \omega^{\mathrm{T}}]$	The commutator $[\omega^{\mathrm{T}}, \omega^{\mathrm{T}}]$ occurring in the calculation
mielke-geometrodynamics_F00360	85	■ 0.989	$\omega^{\mathrm{T}}, \omega^{\mathrm{L}}$	$\omega^{\mathrm{T}}, \omega^{\mathrm{L}}$	which is induced by ω^{L} . It is of importance to realize that the commutator $[\omega^{\mathrm{T}}, \omega^{\mathrm{L}}]$
mielke-geometrodynamics_F00361	86	■ 1.000	$\mathscr{A}(n, \mathrm{mathbb{R}})$	$\mathscr{A}(n, \mathbb{R})$	of local translations are subgroups of $\mathscr{A}(n, \mathbb{R})$. Taking the cross section in the asso-
mielke-geometrodynamics_F00362	86	■ 0.996	C^{∞}	C^{∞}	ciated bundle is abbreviated by C^{∞} , and Ad denotes the adjoint representation with
mielke-geometrodynamics_F00363	86	■ 0.998	$\mathrm{G} \mathrm{L}(n, \mathrm{mathbb{R}})$	$GL(n, \mathbb{R})$	respect to $GL(n, \mathbb{R})$. Due to its construction, the group of local translations $\mathscr{T}(n, \mathbb{R})$
mielke-geometrodynamics_F00364	86	■ 0.998	$\mathscr{T}(n, \mathrm{mathbb{R}})$	$\mathscr{T}(n, \mathbb{R})$	respect to $GL(n, \mathbb{R})$. Due to its construction, the group of local translations $\mathscr{T}(n, \mathbb{R})$
mielke-geometrodynamics_F00365	86	■ 1.000	$\operatorname{Diff}(n, \mathrm{mathbb{R}})$	$\mathrm{Diff}(n, \mathbb{R})$	is <i>locally</i> isomorphic to the group of active diffeomorphisms $\mathrm{Diff}(n, \mathbb{R})$ of the man-
mielke-geometrodynamics_F00366	86	■ 0.905	$(n, \mathrm{mathbb{R}})$	$(n, \mathbb{R})$	$\mathrm{Diff}(n, \mathbb{R})$ contains the $(n + n^2)$ -dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of <i>holonomic</i> affine
mielke-geometrodynamics_F00367	86	■ 0.905	$\left(n+n^2\right)$	$(n + n^2)$	$\mathrm{Diff}(n, \mathbb{R})$ contains the $(n + n^2)$ -dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of <i>holonomic</i> affine
mielke-geometrodynamics_F00368	86	■ 0.905	$\mathrm{A}(n, \mathrm{mathbb{R}})_{\mathrm{H}}$	$A(n, \mathbb{R})_{\mathrm{H}}$	$\mathrm{Diff}(n, \mathbb{R})$ contains the $(n + n^2)$ -dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of <i>holonomic</i> affine
mielke-geometrodynamics_F00369	86	■ 0.999	$P_{\mathrm{i}}=\partial_{\mathrm{i}}:=\partial / \partial x^{\mathrm{i}}$	$P_i = \partial_i := \partial / \partial x^i$	transformations as a subgroup, generated by the vector fields $P_i = \partial_i := \partial / \partial x^i$ and
mielke-geometrodynamics_F00370	86	■ 1.000	$L^{\mathrm{i}}{ }_{\mathrm{j}}=x^{\mathrm{i}} \partial_{\mathrm{j}}$	$L^i{}_j = x^i \partial_j$	$L^i{}_j = x^i \partial_j$. Note that differentiable coordinate transformations, which leave exterior
mielke-geometrodynamics_F00371	86	■ 1.000	χ	χ	the natural injection χ , this formula decomposes into
mielke-geometrodynamics_F00372	87	■ 1.000	V^{id}	V^{id}	in V^{id} consists of an orbit $\mathbb{R}^n$ of $A(n, \mathbb{R})$. The isotropy group of the zero section

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mielke-geometrodynamics_F00373	87	■ 0.928	$\stackrel{\circ}{\varphi}=\sigma\left(0_{\mathrm{m}}\right)$	$\stackrel{\circ}{\varphi}=\sigma\left(0_{\mathrm{m}}\right)$	$\stackrel{\circ}{\varphi}=\sigma\left(0_{\mathrm{m}}\right)$ is $\mathrm{H}=\mathrm{GL}(\mathrm{n}, \mathbb{R})$, i.e., the general linear group. Then the subbundle
mielke-geometrodynamics_F00374	87	■ 0.928	$\mathrm{H}=\mathrm{GL}(\mathrm{n}, \mathbb{R})$	$\mathrm{H}=\mathrm{GL}(\mathrm{n}, \mathbb{R})$	$\stackrel{\circ}{\varphi}=\sigma\left(0_{\mathrm{m}}\right)$ is $\mathrm{H}=\mathrm{GL}(\mathrm{n}, \mathbb{R})$, i.e., the general linear group. Then the subbundle
mielke-geometrodynamics_F00375	87	■ 0.863	$\widetilde{\vartheta}$	$\widetilde{\vartheta}$	bution $\widetilde{\vartheta}$ transforms under $\mathcal{A}_{\mathrm{p}}$ like
mielke-geometrodynamics_F00376	87	■ 0.863	$\mathscr{A}_{\mathrm{p}}$	$\mathscr{A}_{\mathrm{p}}$	bution $\widetilde{\vartheta}$ transforms under $\mathcal{A}_{\mathrm{p}}$ like
mielke-geometrodynamics_F00377	88	■ 0.968	$\mathrm{D}=(\stackrel{\star}{\chi} \widetilde{\mathrm{D}})$	$\mathrm{D}=(\stackrel{\star}{\chi} \widetilde{\mathrm{D}})$	to $\mathrm{D}=(\stackrel{\star}{\chi} \widetilde{\mathrm{D}})$, it follows that the torsion 2-form transforms like
mielke-geometrodynamics_F00378	88	■ 0.999	$\mathscr{G}_{\mathrm{p}}$	$\mathscr{G}_{\mathrm{p}}$	i.e., as a vector with respect to the subgroup $\mathscr{G}_{\mathrm{p}}$ of linear gauge transformations. As
mielke-geometrodynamics_F00379	89	■ 0.993	$\widetilde{\mathrm{D}}=\mathrm{D}$	$\widetilde{\mathrm{D}}=\mathrm{D}$	$\widetilde{\mathrm{D}}=\mathrm{D}$ it remains valid) the affine connection to the linear $\mathfrak{gl}(\mathrm{n}, \mathbb{R})$ -valued connection
mielke-geometrodynamics_F00380	90	■ 1.000	$\mathscr{C} \subset \mathrm{M}$	$\mathscr{C} \subset \mathrm{M}$	to a smooth curve $\mathscr{C} \subset \mathrm{M}$ that is parametrized by $\mathrm{s}(\mathrm{t})$. In order to satisfy
mielke-geometrodynamics_F00381	90	■ 1.000	$\mathrm{F}(\mathrm{m}, \vartheta): \mathrm{M} \times \mathrm{T}^*(\mathrm{M}) \rightarrow \mathbb{R}^+$	$\mathrm{F}(\mathrm{m}, \vartheta): \mathrm{M} \times \mathrm{T}^*(\mathrm{M}) \rightarrow \mathbb{R}^+$	where $\mathrm{F}(\mathrm{m}, \vartheta): \mathrm{M} \times \mathrm{T}^*(\mathrm{M}) \rightarrow \mathbb{R}^+$ is a smooth, <i>positive</i> , and <i>homogeneous</i> funda-
mielke-geometrodynamics_F00382	91	■ 1.000	$\mathscr{C}$	$\mathscr{C}$	ance of (4.3.4) versus admissible reparametrizations of the curve $\mathscr{C}$. Generally, these
mielke-geometrodynamics_F00383	91	■ 1.000	$(2,0)$	$(2,0)$	Subsequently, a covariant symmetric tensor field of degree (2, 0) is imprinted onto
mielke-geometrodynamics_F00384	91	■ 1.000	$\mathrm{g}_{ij} \in \mathrm{C}^{\infty}(\mathrm{M})$	$\mathrm{g}_{ij} \in \mathrm{C}^{\infty}(\mathrm{M})$	Indeed, the functions $\mathrm{g}_{ij} \in \mathrm{C}^{\infty}(\mathrm{M})$ describe “concerning the chosen, arbitrary coor-
mielke-geometrodynamics_F00385	92	■ 0.689	e_{α}^i	e_{α}^i	1-forms ϑ admitted. Since the e_{α}^i have been considered as local bundle coordinates
mielke-geometrodynamics_F00386	92	■ 1.000	$\mathfrak{D}(\mathrm{s}, \mathrm{n}-\mathrm{s})$	$\mathfrak{D}(\mathrm{s}, \mathrm{n}-\mathrm{s})$	$\mathfrak{D}(\mathrm{s}, \mathrm{n}-\mathrm{s})$ -valued connection $\omega^{\mathfrak{g}}$, the constraint
mielke-geometrodynamics_F00387	92	■ 1.000	$\omega^{\mathfrak{g}}$	$\omega^{\mathfrak{g}}$	$\mathfrak{D}(\mathrm{s}, \mathrm{n}-\mathrm{s})$ -valued connection $\omega^{\mathfrak{g}}$, the constraint
mielke-geometrodynamics_F00388	92	■ 0.532	$\phi^{(\circ)}$	$\phi^{(\circ)}$	is a necessary and sufficient condition. For tensor-valued 0-forms $\phi^{(\circ)}$, the exterior
mielke-geometrodynamics_F00389	92	■ 0.651	$\Gamma_{\mathrm{i}}^{\alpha \beta}$	$\Gamma_{\mathrm{i}}^{\alpha \beta}$	where the $\Gamma_{\mathrm{i}}^{\alpha \beta}$ denote the (local) <i>Ricci rotation coefficients</i> . Due to the fact that
mielke-geometrodynamics_F00390	92	■ 1.000	$L_{\alpha \beta}$	$L_{\alpha \beta}$	the infinitesimal generators $L_{\alpha \beta}$ become skew after the restriction to the orthogonal
mielke-geometrodynamics_F00391	92	■ 1.000	$\omega^{\alpha \beta}=-\omega^{\beta \alpha}$	$\omega^{\alpha \beta}=-\omega^{\beta \alpha}$	subgroup, the constraint (4.3.14) analogously yields $\omega^{\alpha \beta}=-\omega^{\beta \alpha}$, i.e., the skew-
mielke-geometrodynamics_F00392	92	■ 0.988	$\mathrm{m}_{\mathrm{o}} \in \mathrm{M}$	$\mathrm{m}_{\mathrm{o}} \in \mathrm{M}$	neighborhood of a point $\mathrm{m}_{\mathrm{o}} \in \mathrm{M}$ depends entirely on Minkowski’s metric tensor o_{ij}
mielke-geometrodynamics_F00393	92	■ 0.988	o_{ij}	o_{ij}	neighborhood of a point $\mathrm{m}_{\mathrm{o}} \in \mathrm{M}$ depends entirely on Minkowski’s metric tensor o_{ij}
mielke-geometrodynamics_F00394	93	■ 1.000	$\mathrm{H}=\mathrm{O}(\mathrm{s}, \mathrm{n}-\mathrm{s})$	$\mathrm{H}=\mathrm{O}(\mathrm{s}, \mathrm{n}-\mathrm{s})$	gauge theory has only the orthogonal “isotropy group” $\mathrm{H}=\mathrm{O}(\mathrm{s}, \mathrm{n}-\mathrm{s})$ as resid-
mielke-geometrodynamics_F00395	93	■ 1.000	$\mathfrak{o}(\mathrm{s}, \mathrm{n}-\mathrm{s})$	$\mathfrak{o}(\mathrm{s}, \mathrm{n}-\mathrm{s})$	the reduction of the linear connection ω^{L} to an $\mathfrak{o}(\mathrm{s}, \mathrm{n}-\mathrm{s})$ -valued connection $\omega^{\mathfrak{g}}$
mielke-geometrodynamics_F00396	93	■ 0.999	$\mathrm{L}^{\mathfrak{g}}(\mathrm{M})$	$\mathrm{L}^{\mathfrak{g}}(\mathrm{M})$	in $\mathrm{L}^{\mathfrak{g}}(\mathrm{M})$. This consistency does not, as is implied by HANSON & REGGE (1979),
mielke-geometrodynamics_F00397	93	■ 1.000	$\mathrm{D} \vartheta=0$	$\mathrm{D} \vartheta=0$	demand the vanishing of the torsion, i.e., $\mathrm{D} \vartheta=0$. In a Lagrangian formulation of
mielke-geometrodynamics_F00398	93	■ 0.995	Θ	Θ	Higgs field ϑ , i.e., the canonical <i>orthogonal</i> 1-form, ⁴ and the resulting torsion Θ

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mielke-geometrodynamics_F00399	93	■ 0.913	ϑ^{g}	ϑ^{g}	⁴ As for the nomenclature, the 1-form ϑ , which is restricted by (4.3.10), will not be denoted by ϑ^{g}
mielke-geometrodynamics_F00400	93	■ 1.000	Ω^{g}	Ω^{g}	and thus differs from ω^{g} and $\varOmega^{\mathrm{g}}$, since we prefer to stick to (4.3.10) or (4.3.9), respectively, as
mielke-geometrodynamics_F00401	94	■ 0.997	$\omega^{\{\}}$	$\omega^{\{\}}$	Here $\omega^{\{\}}$ denotes the 1-form of the torsion-free Christoffel-type connection for which
mielke-geometrodynamics_F00402	94	■ 0.890	K	K	between the 2-form Θ and the 1-form K of the so-called <i>contortion</i> , having values
mielke-geometrodynamics_F00403	94	■ 1.000	U_{n}	U_{n}	denoted by U_{n} , in contrast to a purely Riemannian manifold, which is symbolized by V_{n} .
mielke-geometrodynamics_F00404	94	■ 1.000	V_{n}	V_{n}	denoted by U_{n} , in contrast to a purely Riemannian manifold, which is symbolized by V_{n} .
mielke-geometrodynamics_F00405	95	■ 0.456	$R_{\mathrm{i} \mathrm{j} \mathrm{c}}^{\{\dots \mathrm{d}\}}$	$R_{ijc}^{\{\dots d\}}$	In this formula, $R_{ijc}^{\{\dots d\}}$ are the mixed components of the Riemannian curvature tensor,
mielke-geometrodynamics_F00406	95	■ 0.999	$R_{\mathrm{i} \mathrm{j} \mathrm{k} \mathrm{l}}^{\{\}\}$	$R_{ijkl}^{\{\}\}$	holonomic components $R_{ijcl}^{\{\}\}$ have the following symmetries with respect to an inter-
mielke-geometrodynamics_F00407	95	■ 0.666	$S \ S \ B$	SSB	cation of a Higgs–Kibble-type mechanism of SSB that resulted in the construction of
mielke-geometrodynamics_F00408	95	■ 1.000	$\mathrm{O}(1, \mathrm{n}-1)$	$\mathrm{O}(1, \mathrm{n}-1)$	$\mathrm{O}(1, \mathrm{n}-1)$ as the “residual” symmetry of the theory via the general linear group
mielke-geometrodynamics_F00409	95	■ 0.778	$\operatorname{GL}(\mathrm{n}, \mathbb{R})$	$\mathrm{GL}(\mathrm{n}, \mathbb{R})$	$\mathrm{GL}(\mathrm{n}, \mathbb{R})$.
mielke-geometrodynamics_F00410	96	■ 1.000	$\mathrm{R}_{\alpha \beta}^{\mathrm{cd}}(\mathrm{m})$	$\mathrm{R}_{\alpha \beta}^{\mathrm{cd}}(\mathrm{m})$	spacetime-dependent “structure constants” $\mathrm{R}_{\alpha \beta}^{\mathrm{cd}}(\mathrm{m})$ occur in the generic case. (The
mielke-geometrodynamics_F00411	96	■ 1.000	$\vartheta, \Theta, \Omega^{\mathrm{g}}$	$\vartheta, \Theta, \Omega^{\mathrm{g}}$	However, contrasting with $\vartheta, \Theta, \varOmega^{\mathrm{g}}$ (see (4.2.20), (4.2.14), and (4.2.13)), the
mielke-geometrodynamics_F00412	97	■ 1.000	L_{g}	L_{g}	only via of the forms Θ and $\varOmega^{\mathrm{g}}$. The variation of L_{g} , regarded as a total differential,
mielke-geometrodynamics_F00413	97	■ 1.000	t	t	defines the translational and (Lorentz-)rotational canonical gauge field momenta Π^{T}
mielke-geometrodynamics_F00414	97	■ 1.000	Π^{T}	Π^{T}	defines the translational and (Lorentz-)rotational canonical gauge field momenta Π^{T}
mielke-geometrodynamics_F00415	97	■ 0.999	Π^{L}	Π^{L}	and Π^{L} . Due to the special role of ϑ , there occurs additionally the (n-1)-form E ,
mielke-geometrodynamics_F00416	97	■ 0.999	E	E	and Π^{L} . Due to the special role of ϑ , there occurs additionally the (n-1)-form E ,
mielke-geometrodynamics_F00417	97	■ 0.998	$\Pi_{\square}^{\mathrm{T}}$	$\Pi_{\square}^{\mathrm{T}}$	where the “contortional” 1-form $\Pi_{\square}^{\mathrm{T}}$, having values in the Lie algebra of the
mielke-geometrodynamics_F00418	97	■ 0.931	$\Pi^{\mathrm{T}} := [\Pi_{\square}^{\mathrm{T}}, \vartheta]$	$\Pi^{\mathrm{T}} := [\Pi_{\square}^{\mathrm{T}}, \vartheta]$	$\Pi^{\mathrm{T}} := [\Pi_{\square}^{\mathrm{T}}, \vartheta]$. This is in complete analogy to the relation (4.3.24) for the contor-
mielke-geometrodynamics_F00419	97	■ 1.000	$\delta \vartheta$	$\delta \vartheta$	The application of Hamilton’s principle to the independent variations of S for $\delta \vartheta$
mielke-geometrodynamics_F00420	97	■ 1.000	$\delta \omega^{\mathrm{g}}$	$\delta \omega^{\mathrm{g}}$	and $\delta \omega^{\mathrm{g}}$ leads to the <i>Euler–Lagrange</i> equations
mielke-geometrodynamics_F00421	98	■ 0.844	Σ	Σ	fied with the 3-form Σ , the energy–momentum current, and the 3-form τ of the spin
mielke-geometrodynamics_F00422	98	■ 0.999	$\lambda^{\varphi} = \mathrm{o}$	$\lambda^{\varphi} = \mathrm{o}$	If the Euler–Lagrange equations $\lambda^{\varphi} = \mathrm{o}$ of the matter fields are satisfied, it will
mielke-geometrodynamics_F00423	98	■ 1.000	τ_{s}	τ_{s}	turn out that Σ and τ_{s} are the field momenta of matter that are <i>canonically conjugate</i>
mielke-geometrodynamics_F00424	98	■ 1.000	Υ	Υ	hypermomentum current Υ of the matter fields in such a generalization (HEHL &
mielke-geometrodynamics_F00425	100	■ 1.000	$\Gamma^{\alpha \beta} = -\Gamma^{\beta \alpha}$	$\Gamma^{\alpha \beta} = -\Gamma^{\beta \alpha}$	symmetric connection 1-form $\Gamma^{\alpha \beta} = -\Gamma^{\beta \alpha}$ is defined by $\delta L = \delta \Gamma^{\alpha \beta} \wedge (\partial L / \partial \Gamma^{\alpha \beta})$.

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mielke-geometrodynamics_F00426	100	■ 1.000	$\delta L = \delta \Gamma^{\alpha\beta} \wedge \left(\frac{\partial L}{\partial \Gamma^{\alpha\beta}} \right)$	$\delta L = \delta \Gamma^{\alpha\beta} \wedge \left(\partial L / \partial \Gamma^{\alpha\beta} \right)$	symmetric connection 1-form $\Gamma^{\alpha\beta} = -\Gamma^{\beta\alpha}$ is defined by $\delta L = \delta \Gamma^{\alpha\beta} \wedge (\partial L / \partial \Gamma^{\alpha\beta})$.
mielke-geometrodynamics_F00427	100	■ 0.629	$S = \int_M L_{\text{mat}}$	$S = \int_M L_{\text{mat}}$	The action $S = \int_M L_{\text{mat}}$ for the matter Lagrangian is, by construction, invari-
mielke-geometrodynamics_F00428	100	■ 1.000	$\operatorname{Diff}(M)$	$\operatorname{Diff}(M)$	ant under the group $\operatorname{Diff}(M)$ of coordinate transformations and local frame rota-
mielke-geometrodynamics_F00429	100	■ 0.924	$\mathscr{T} \subset \operatorname{Diff}(M)$	$\mathscr{T} \subset \operatorname{Diff}(M)$	one-parameter group of <i>local</i> translations $\mathscr{T} \subset \operatorname{Diff}(M)$, we employ the $SO(1, 3)$ -
mielke-geometrodynamics_F00430	100	■ 0.924	$SO(1, 3)$	$SO(1, 3)$	one-parameter group of <i>local</i> translations $\mathscr{T} \subset \operatorname{Diff}(M)$, we employ the $SO(1, 3)$ -
mielke-geometrodynamics_F00431	100	■ 0.590	$\operatorname{L}_{\xi} := \xi \lrcorner D + D\xi \lrcorner$	<i>(not rendered)</i>	covariant Lie derivative $\operatorname{L}_{\xi} := \xi \lrcorner D + D\xi \lrcorner$ on M with respect to an arbitrary vector
mielke-geometrodynamics_F00432	100	■ 0.590	M	M	covariant Lie derivative $\operatorname{L}_{\xi} := \xi \lrcorner D + D\xi \lrcorner$ on M with respect to an arbitrary vector
mielke-geometrodynamics_F00433	100	■ 1.000	ξ	ξ	field ξ . Since $Dg_{\alpha\beta} = 0$, we obtain
mielke-geometrodynamics_F00434	100	■ 1.000	$Dg_{\alpha\beta} = 0$	$Dg_{\alpha\beta} = 0$	field ξ . Since $Dg_{\alpha\beta} = 0$, we obtain
mielke-geometrodynamics_F00435	101	■ 0.779	$\operatorname{L}_{\xi} L = D\xi \lrcorner L$	$\operatorname{L}_{\xi} L = D\xi \lrcorner L$	$\operatorname{L}_{\xi} L = D\xi \lrcorner L$. Comparing the boundary term, we can read off the identity
mielke-geometrodynamics_F00436	101	■ 1.000	$\xi \rightarrow e_{\alpha}$	$\xi \rightarrow e_{\alpha}$	If we replace $\xi \rightarrow e_{\alpha}$ by the basis frame, then (4.5.4) yields directly the explicit
mielke-geometrodynamics_F00437	101	■ 0.995	$\left(F_{\alpha}^{\text{Max}} = \left(e_{\alpha} \lrcorner F \right) \wedge j \right)$	$F_{\alpha}^{\text{Max}} = (e_{\alpha} \lrcorner F) \wedge j$	calculus $F_{\alpha}^{\text{Max}} = (e_{\alpha} \lrcorner F) \wedge j$.
mielke-geometrodynamics_F00438	101	■ 1.000	$\simeq$	$\simeq$	no field equations are invoked. <i>Weak</i> identities, which are denoted by $\simeq$, hold only
mielke-geometrodynamics_F00439	101	■ 0.999	$\delta L / \delta \Psi = 0$	$\delta L / \delta \Psi = 0$	if the matter field equation $\delta L / \delta \Psi = 0$ is satisfied.
mielke-geometrodynamics_F00440	101	■ 0.972	$\delta D\Psi = D\delta\Psi + \delta\Gamma_{\alpha}^{\beta} \wedge I^{\alpha}_{\beta} \Psi$	$\delta D\Psi = D\delta\Psi + \delta\Gamma_{\alpha}^{\beta} \wedge I^{\alpha}_{\beta} \Psi$	Since $\delta D\Psi = D\delta\Psi + \delta\Gamma_{\alpha}^{\beta} \wedge I^{\alpha}_{\beta} \Psi$, this is equivalent to
mielke-geometrodynamics_F00441	102	■ 1.000	$\varepsilon_{\alpha}^{\beta}(x) := \Lambda_{\alpha}^{\beta}(x) - \delta_{\alpha}^{\beta}$	$\varepsilon_{\alpha}^{\beta}(x) := \Lambda_{\alpha}^{\beta}(x) - \delta_{\alpha}^{\beta}$	Under an infinitesimal Lorentz rotation $\varepsilon_{\alpha}^{\beta}(x) := \Lambda_{\alpha}^{\beta}(x) - \delta_{\alpha}^{\beta}$ of the frames, we
mielke-geometrodynamics_F00445	129	■ 1.000	Γ_{α}^{β}	Γ_{α}^{β}	contains the linear connection Γ_{α}^{β} , which is combined with the complex translational
mielke-geometrodynamics_F00449	102	■ 0.986	$\widehat{D}$	$\widehat{D}$	$\widehat{D}$ with respect to the <i>transposed</i> connection $\widehat{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + e_{\alpha} \lrcorner T^{\beta}$.
mielke-geometrodynamics_F00450	102	■ 0.986	$\widehat{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + e_{\alpha} \lrcorner T^{\beta}$	$\widehat{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + e_{\alpha} \lrcorner T^{\beta}$	$\widehat{D}$ with respect to the <i>transposed</i> connection $\widehat{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + e_{\alpha} \lrcorner T^{\beta}$.
mielke-geometrodynamics_F00451	102	■ 0.999	$\tau_{\alpha\beta} = -\tau_{\beta\alpha}$	$\tau_{\alpha\beta} = -\tau_{\beta\alpha}$	identity, consists of an intrinsic or <i>spin</i> part $\tau_{\alpha\beta} = -\tau_{\beta\alpha}$ and an orbital part $x_{[\alpha} \wedge \Sigma_{\beta]}$,
mielke-geometrodynamics_F00452	102	■ 0.999	$x_{[\alpha} \wedge \Sigma_{\beta]}$	$x_{[\alpha} \wedge \Sigma_{\beta]}$	identity, consists of an intrinsic or <i>spin</i> part $\tau_{\alpha\beta} = -\tau_{\beta\alpha}$ and an orbital part $x_{[\alpha} \wedge \Sigma_{\beta]}$,
mielke-geometrodynamics_F00453	102	■ 1.000	x^i	x^i	coordinates x^i with $x^{\alpha} = \delta_i^{\alpha} x^i$.
mielke-geometrodynamics_F00454	102	■ 1.000	$x^{\alpha} = \delta_i^{\alpha} x^i$	$x^{\alpha} = \delta_i^{\alpha} x^i$	coordinates x^i with $x^{\alpha} = \delta_i^{\alpha} x^i$.
mielke-geometrodynamics_F00455	103	■ 1.000	$\xi = \partial / \partial t$	$\xi = \partial / \partial t$	solution, if $\xi = \partial / \partial t$ or $\xi = \partial / \partial \varphi$, respectively, is chosen. Here no “factor of two”
mielke-geometrodynamics_F00456	103	■ 1.000	$\xi = \partial / \partial \varphi$	$\xi = \partial / \partial \varphi$	solution, if $\xi = \partial / \partial t$ or $\xi = \partial / \partial \varphi$, respectively, is chosen. Here no “factor of two”
mielke-geometrodynamics_F00457	104	■ 1.000	$\pm(\Lambda/3)r^3$	$\pm(\Lambda/3)r^3$	complex each contribute <i>one-half</i> of the mass, so that the divergent terms $\pm(\Lambda/3)r^3$
mielke-geometrodynamics_F00458	104	■ 1.000	Λ	Λ	proportional to the cosmological constant Λ exactly cancel each other.

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mielke-geometrodynamics_F00459	104	■ 0.539	$\Gamma^{\{T\}\alpha}$	$\Gamma^{(T)\alpha}$	includes the true translational potential $\Gamma^{(T)\alpha}$ and the $GL(n, \mathbb{R})$ -gauge connection
mielke-geometrodynamics_F00460	104	■ 1.000	$\Gamma_{\alpha}^{\{L\}\beta}$	$\Gamma_{\alpha}^{(L)\beta}$	$\Gamma_{\alpha}^{(L)\beta}$, as can be deduced from the Möbius-type five-dimensional representation of
mielke-geometrodynamics_F00461	104	■ 1.000	$A(n, \mathbb{R}) := \mathbb{R}^n \otimes GL(n, \mathbb{R})$	$A(n, \mathbb{R}) := \mathbb{R}^n \otimes GL(n, \mathbb{R})$	the affine gauge group $A(n, \mathbb{R}) := \mathbb{R}^n \otimes GL(n, \mathbb{R})$.
mielke-geometrodynamics_F00462	104	■ 0.559	$\widetilde{\xi}$	$\widetilde{\xi}$	view: the condition for a parallel transport of an affine vector $\widetilde{\xi}$ around a small closed
mielke-geometrodynamics_F00463	104	■ 0.993	$\widetilde{\Gamma}$	$\widetilde{\Gamma}$	loop by means of $\widetilde{\Gamma}$ reads
mielke-geometrodynamics_F00464	104	■ 0.272	$\mathbb{L}_y := y \rfloor D + Dy$	$\mathbb{L}_y := y \rfloor D + Dy$	where $\mathbb{L}_y := y \rfloor D + Dy \rfloor$ denotes the gauge-covariant Lie derivative and $R^{(T)\alpha}$ the
mielke-geometrodynamics_F00465	104	■ 0.272	$R^{\{T\}\alpha}$	$R^{(T)\alpha}$	where $\mathbb{L}_y := y \rfloor D + Dy \rfloor$ denotes the gauge-covariant Lie derivative and $R^{(T)\alpha}$ the
mielke-geometrodynamics_F00466	104	■ 1.000	y	y	loop parametrized by y yields
mielke-geometrodynamics_F00467	105	■ 1.000	$A = A_i^J \lambda_J dx^i$	$A = A_i^J \lambda_J dx^i$	where $A = A_i^J \lambda_J dx^i$ is a Yang–Mills-type connection and P is the principal value
mielke-geometrodynamics_F00468	105	■ 0.996	γ	γ	of the integral. If the loop γ lies in a field-free region, i.e., one with $U(1)$ field
mielke-geometrodynamics_F00469	105	■ 1.000	$F := dA = 0$	$F := dA = 0$	strength $F := dA = 0$ or Yang–Mills curvature $F = dA + A \wedge A = 0$, but encloses
mielke-geometrodynamics_F00470	105	■ 1.000	$F = dA + A \wedge A = 0$	$F = dA + A \wedge A = 0$	strength $F := dA = 0$ or Yang–Mills curvature $F = dA + A \wedge A = 0$, but encloses
mielke-geometrodynamics_F00471	105	■ 1.000	$F \neq 0$	$F \neq 0$	a “confined” region with nontrivial “magnetic” flux $F \neq 0$, the potential A can still
mielke-geometrodynamics_F00472	105	■ 1.000	$A(4, \mathbb{R})$	$A(4, \mathbb{R})$	arising from a gauge theory of the affine structure group $A(4, \mathbb{R})$.
mielke-geometrodynamics_F00473	105	■ 0.924	$GL(4, \mathbb{R})$	$GL(4, \mathbb{R})$	For a so-called manifold Ψ carrying no $GL(4, \mathbb{R})$ -excitations, i.e., no spin, no
mielke-geometrodynamics_F00474	105	■ 1.000	$\Gamma^{(T)\alpha} \stackrel{*}{=} 0$	$\Gamma^{(T)\alpha} \stackrel{*}{=} 0$	of the cone there is $\Gamma^{(T)\alpha} \stackrel{*}{=} 0$ locally.
mielke-geometrodynamics_F00475	105	■ 0.651	(1993, 1994)	(1993, 1994)	time manifold; cf. ANANDAN (1993, 1994). Because the coframe is nondegenerate
mielke-geometrodynamics_F00476	105	■ 1.000	P_{α}	P_{α}	Since the physical dimension of P_{α} is $2\pi\hbar/\ell$, the gravitational analogue of Dirac’s
mielke-geometrodynamics_F00477	105	■ 1.000	$2\pi\hbar/\ell$	$2\pi\hbar/\ell$	Since the physical dimension of P_{α} is $2\pi\hbar/\ell$, the gravitational analogue of Dirac’s
mielke-geometrodynamics_F00478	105	■ 1.000	$\mathcal{M}_n = nM_{\text{Planck}}$	$\mathcal{M}_n = nM_{\text{Planck}}$	i.e., the mass would turn out to be a multiple $\mathcal{M}_n = nM_{\text{Planck}}$ of the (huge) Planck
mielke-geometrodynamics_F00479	109	■ 0.485	$\Gamma^{\{\alpha\beta\}} = \Gamma^{\{j\alpha\beta\}} - K^{\alpha\beta}$	$\Gamma^{\alpha\beta} = \Gamma^{\{j\alpha\beta\}} - K^{\alpha\beta}$	The Riemann–Cartan (RC) connection $\Gamma^{\alpha\beta} = \Gamma^{\{j\}\alpha\beta} - K^{\alpha\beta}$, a one-form, can be
mielke-geometrodynamics_F00480	109	■ 0.469	$\Gamma^{\{\alpha\beta\}}$	$\Gamma^{\{\alpha\beta\}}$	split into the unique Levi-Civita connection $\Gamma^{\{j\}\alpha\beta}$ of Riemannian geometry and
mielke-geometrodynamics_F00481	109	■ 1.000	$K_{\alpha\beta} = -K_{\beta\alpha}$	$K_{\alpha\beta} = -K_{\beta\alpha}$	a <i>contortion</i> one-form $K_{\alpha\beta} = -K_{\beta\alpha}$, a tensor-valued one-form $K = iK^{\alpha\beta}\sigma_{\alpha\beta}/4$,
mielke-geometrodynamics_F00482	109	■ 1.000	$K = iK^{\alpha\beta}\sigma_{\alpha\beta}/4$	$K = iK^{\alpha\beta}\sigma_{\alpha\beta}/4$	a <i>contortion</i> one-form $K_{\alpha\beta} = -K_{\beta\alpha}$, a tensor-valued one-form $K = iK^{\alpha\beta}\sigma_{\alpha\beta}/4$,
mielke-geometrodynamics_F00483	109	■ 1.000	$T^{\alpha} = K^{\alpha}_{\beta} \wedge \vartheta^{\beta}$	$T^{\alpha} = K^{\alpha}_{\beta} \wedge \vartheta^{\beta}$	implicitly related to torsion via $T^{\alpha} = K^{\alpha}_{\beta} \wedge \vartheta^{\beta}$. Recently, it has been stressed by
mielke-geometrodynamics_F00484	109	■ 1.000	$\Gamma = \Gamma^{\{j\}} - K$	$\Gamma = \Gamma^{\{j\}} - K$	WEINBERG (2005) that this RC connection $\Gamma = \Gamma^{\{j\}} - K$ is just a <i>deformation</i> (or
mielke-geometrodynamics_F00485	109	■ 1.000	$\Gamma^{\{j\}}$	$\Gamma^{\{j\}}$	field redefinition) of the Christoffel connection $\Gamma^{\{j\}}$ by the contortion, at least from

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mielke-geometrodynamics_F00486	110	■ 1.000	D>2	$D > 2$	renormalizable in $D > 2$ without Chern–Simons (CS) terms (ALVES et al. 1999),
mielke-geometrodynamics_F00487	110	■ 1.000	j_{5}	j_5	equation relates torsion to spin, i.e., to the axial current j_5 in the case of Dirac fields,
mielke-geometrodynamics_F00488	110	■ 0.972	\bar{\psi}:=\psi^{\dagger}\gamma_0	$\bar{\psi} := \psi^\dagger \gamma_0$	$\bar{\psi} := \psi^\dagger \gamma_0$ denotes the Dirac adjoint and $D\psi := d\psi + \Gamma \wedge \psi$ is the exterior covari-
mielke-geometrodynamics_F00489	110	■ 0.972	D \psi:=d \psi+\Gamma \wedge \psi	$D\psi := d\psi + \Gamma \wedge \psi$	$\bar{\psi} := \psi^\dagger \gamma_0$ denotes the Dirac adjoint and $\overline{D\psi} := d\bar{\psi} + \Gamma \wedge \bar{\psi}$ is the exterior covari-
mielke-geometrodynamics_F00490	110	■ 1.000	\gamma:=\gamma_{\alpha}\vartheta^{\alpha}	$\gamma := \gamma_\alpha \vartheta^\alpha$	where $\gamma := \gamma_\alpha \vartheta^\alpha$ is the Clifford-algebra-valued coframe satisfying $D\gamma = [\gamma, K]$
mielke-geometrodynamics_F00491	110	■ 1.000	D \gamma=[\gamma, K]	$D\gamma = [\gamma, K]$	where $\gamma := \gamma_\alpha \vartheta^\alpha$ is the Clifford-algebra-valued coframe satisfying $D\gamma = [\gamma, K]$
mielke-geometrodynamics_F00492	110	■ 1.000	=\gamma_{\alpha} T^{\alpha}	$= \gamma_\alpha T^\alpha$	$= \gamma_\alpha T^\alpha$, and $T^\alpha := D\vartheta^\alpha$ is the torsion two-form.
mielke-geometrodynamics_F00493	110	■ 1.000	T^{\alpha}:=D \vartheta^{\alpha}	$T^\alpha := D\vartheta^\alpha$	$= \gamma_\alpha T^\alpha$, and $\overline{T^\alpha} := D\bar{\vartheta}^\alpha$ is the torsion two-form.
mielke-geometrodynamics_F00494	110	■ 1.000	L_{\mathrm{D}}=\bar{L}_{\mathrm{D}}=L_{\mathrm{D}}^{\dagger}	$L_D = \bar{L}_D = L_D^\dagger$	Since $L_D = \bar{L}_D = L_D^\dagger$ even in an unholonomic frame, the minimal coupling pro-
mielke-geometrodynamics_F00495	111	■ 1.000	\tau_{\alpha \beta \gamma}=\tau_{[\alpha \beta \gamma]}	$\tau_{\alpha\beta\gamma} = \tau_{[\alpha\beta\gamma]}$	with <i>totally antisymmetric</i> components $\tau_{\alpha\beta\gamma} = \tau_{[\alpha\beta\gamma]}$. Eventually, torsion merely
mielke-geometrodynamics_F00496	111	■ 0.975	\mu_{\alpha}=\vartheta_{\alpha} \wedge * j_5 / 4	$\mu_\alpha = \vartheta_\alpha \wedge * j_5 / 4$	couples to the <i>spin–energy potential</i> $\mu_\alpha = \vartheta_\alpha \wedge * j_5 / 4$, i.e., to a two-form propor-
mielke-geometrodynamics_F00497	111	■ 0.792	\mathrm{GR}_{\parallel}	$\mathrm{GR}_\parallel$	Transitions from EC theory or GR to $\mathrm{GR}_\parallel$ can be <i>generated</i> via the Chern–Simons-
mielke-geometrodynamics_F00498	111	■ 0.701	C_{\mathrm{TT}^*}:=\vartheta^{\alpha} \wedge * T_{\alpha}	$C_{\mathrm{TT}^*} := \vartheta^\alpha \wedge * T_\alpha$	type term $C_{\mathrm{TT}^*} := \vartheta^\alpha \wedge * T_\alpha$, involving the Hodge dual of the torsion. On the material
mielke-geometrodynamics_F00499	111	■ 1.000	L_{\mathrm{D}} \rightarrow L_{\mathrm{D}}+d U	$L_D \rightarrow L_D + dU$	side, a related change of the Dirac Lagrangian can be induced via $L_D \rightarrow L_D + dU$,
mielke-geometrodynamics_F00500	111	■ 1.000	U=\vartheta^{\alpha} \wedge \mu_{\alpha}	$U = \vartheta^\alpha \wedge \mu_\alpha$	where $\overline{U} := \vartheta^\alpha \wedge \bar{\mu}_\alpha$ features as a superpotential. Then the corresponding boundary
mielke-geometrodynamics_F00501	111	■ 1.000	\hat{\tau}_{\alpha \beta}	$\hat{\tau}_{\alpha\beta}$	sions (R1) and (R2) of MIELKE et al. (1989), such that the relocalized spin $\hat{\tau}_{\alpha\beta}$
mielke-geometrodynamics_F00502	111	■ 1.000	U=\vartheta^{\alpha} \wedge \vartheta_{\alpha} \wedge * j_5 / 4	$U = \vartheta^\alpha \wedge \vartheta_\alpha \wedge * j_5 / 4$	vanishes. Observe that for Dirac spinors, $U = \vartheta^\alpha \wedge \vartheta_\alpha \wedge * j_5 / 4$ is trivial “on shell.”
mielke-geometrodynamics_F00503	111	■ 1.000	\kappa=8 \pi G_{\mathrm{N}}	$\kappa = 8\pi G_N$	and the orthogonal tetrads, whereas $\kappa = 8\pi G_N$ is the gravitational constant in natural
mielke-geometrodynamics_F00504	112	■ 1.000	\Sigma_{\alpha}	Σ_α	coupled to the canonical energy–momentum current Σ_α of matter, is obtained by
mielke-geometrodynamics_F00505	112	■ 1.000	L_{\mathrm{EC}}	L_{EC}	varying the matter coupled Lagrangian L_{EC} with respect to the coframe ϑ^α .
mielke-geometrodynamics_F00506	112	■ 0.973	\left.\widehat{\Gamma}_{\alpha}^{\beta}:=\Gamma_{\alpha}^{\beta}+e_{\alpha}\right\rfloor T^{\beta}	$\widehat{\Gamma}_\alpha{}^\beta := \Gamma_\alpha{}^\beta + e_\alpha \rfloor T^\beta$	with respect to the transposed connection $\widehat{\Gamma}_\alpha{}^\beta := \Gamma_\alpha{}^\beta + e_\alpha \rfloor T^\beta$. Observe that the
mielke-geometrodynamics_F00507	113	■ 1.000	\sigma:=\frac{1}{2} \sigma_{\alpha \beta} \vartheta^{\alpha} \wedge \vartheta^{\beta}=\frac{i}{2} \gamma \wedge \gamma	$\sigma := \frac{1}{2} \sigma_{\alpha\beta} \vartheta^\alpha \wedge \vartheta^\beta = \frac{i}{2} \gamma \wedge \gamma$	involving the unit-curvature two-form $\sigma := \frac{1}{2} \sigma_{\alpha\beta} \vartheta^\alpha \wedge \vartheta^\beta = \frac{i}{2} \gamma \wedge \gamma$, the equiva-
mielke-geometrodynamics_F00508	113	■ 1.000	\Theta=0	$\Theta = 0$	for vanishing torsion, i.e., for $\Theta = 0$ in Einstein’s standard GR.
mielke-geometrodynamics_F00509	113	■ 1.000	\Gamma^{\alpha \beta}	$\Gamma^{\alpha\beta}$	By varying (5.3.1) with respect to the linear connection $\Gamma^{\alpha\beta}$, we obtain the second
mielke-geometrodynamics_F00510	113	■ 0.839	\tilde{\mathrm{T}}_{\mu \nu}	$\tilde{\mathrm{T}}_{\mu\nu}$	$\tilde{\mathrm{T}}_{\mu\nu}$ contains additional terms depending quadratically on the spin angular momen-
mielke-geometrodynamics_F00511	113	■ 1.000	\tau_{\mathrm{s}}=\mathrm{o}	$\tau_s = \mathrm{o}$	tum of matter; cf. HEHL et al. (1976). In the instance that $\tau_s = \mathrm{o}$, the EC equation
mielke-geometrodynamics_F00512	113	■ 1.000	\ell^{* 4}	ℓ^{*4}	terms in (5.3.12) are proportional to ℓ^{*4} and could thus, with a modified Planck

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mielke-geometrodynamics_F00513	114	■ 1.000	$\ell^*=10^{-33} \mathrm{~cm}$	$\ell^* = 10^{-33} \text{ cm}$	length $\ell^* = 10^{-33} \text{ cm}$, play a role on the macroscopic scale only in cosmology; cf.,
mielke-geometrodynamics_F00514	114	■ 1.000	10^{80}	10^{80}	be threatened by a final singularity, since the gravitational collapse of 10^{80} baryons
mielke-geometrodynamics_F00515	114	■ 0.528	$R^{\{\lambda\alpha\beta\}}$	$R^{\{\lambda\alpha\beta\}}$	into the Riemannian curvature $R^{(\lambda\alpha\beta)}$ plus contortion pieces.
mielke-geometrodynamics_F00516	114	■ 1.000	$\{ \}^{(i)} T_{\alpha}$	$^{(i)}T_{\alpha}$	allelism as a consequence. Here $^{(i)}T_{\alpha}$ are the three irreducible pieces of the torsion.
mielke-geometrodynamics_F00517	114	■ 1.000	$\mathscr{A}$	$\mathscr{A}$	When only this axial torsion $\mathscr{A}$ enters algebraically, the EC Lagrangian
mielke-geometrodynamics_F00518	114	■ 1.000	L_{HE}	L_{HE}	generalizes ¹ the metric Hilbert–Einstein Lagrangian L_{HE} to an RC spacetime with
mielke-geometrodynamics_F00519	114	■ 0.647	$T^{\alpha} \wedge^{*} \left(\{ \}^{(1)} T_{\alpha} - 2 \{ \}^{(2)} T_{\alpha} - \frac{1}{2} \{ \}^{(3)} T_{\alpha} \right)$	$T^{\alpha} \wedge^{*} \left(^{(1)}T_{\alpha} - 2 ^{(2)}T_{\alpha} - \frac{1}{2} ^{(3)}T_{\alpha} \right)$	dure, since the particular combination $T^{\alpha} \wedge^{*} \left(^{(1)}T_{\alpha} - 2 ^{(2)}T_{\alpha} - \frac{1}{2} ^{(3)}T_{\alpha} \right)$ of irreducible pieces is
mielke-geometrodynamics_F00520	114	■ 0.558	$d C_{\mathrm{TT}}^{*}$	dC_{TT}^{*}	the space of gravity theories, the term dC_{TT}^{*} interrelates GR with its teleparallelism equivalent
mielke-geometrodynamics_F00521	115	■ 1.000	$G_{\alpha}^{\{\}} = G_{\alpha}^{\beta} \eta_{\beta}$	$G_{\alpha}^{\{\}} = G_{\alpha}^{\beta} \eta_{\beta}$	can be decomposed into the Einstein three-form $G_{\alpha}^{(\lambda)} = G_{\alpha}^{\beta} \eta_{\beta}$ with respect to the
mielke-geometrodynamics_F00522	115	■ 1.000	$\ell = \sqrt{\kappa}$	$\ell = \sqrt{\kappa}$	coupled via the “bare” fundamental length $\ell = \sqrt{\kappa}$. Then “on shell,” EC theory
mielke-geometrodynamics_F00523	115	■ 1.000	$k \rightarrow \infty$	$k \rightarrow \infty$	bounded in the ultraviolet limit $k \rightarrow \infty$. Quite generally, the renormalization flow is
mielke-geometrodynamics_F00524	115	■ 1.000	Γ_k	Γ_k	for the effective action Γ_k , i.e.,
mielke-geometrodynamics_F00525	116	■ 0.999	$k \partial_k$	$k \partial_k$	where $k \partial_k$ is a scale derivative returning the Euler degree of homogeneous functionals.
mielke-geometrodynamics_F00526	116	■ 1.000	k	k	where k is the renormalization group (RG) scale in momentum space and Λ the
mielke-geometrodynamics_F00527	116	■ 1.000	ρ_{Λ}	ρ_{Λ}	cosmological constant related to dark energy (DE) of density ρ_{Λ} ; see also BJORKEN
mielke-geometrodynamics_F00528	116	■ 1.000	d_{N}	d_{N}	where d_{N} is the anomalous dimension of the running Newton coupling g_{N} .
mielke-geometrodynamics_F00529	116	■ 1.000	g_{N}	g_{N}	where d_{N} is the anomalous dimension of the running Newton coupling g_{N} .
mielke-geometrodynamics_F00530	116	■ 1.000	g_{N}^{*}	g_{N}^{*}	stants (5.4.3) run into some nontrivial fixed points g_{N}^{*} and λ_{*} , depending on the
mielke-geometrodynamics_F00531	116	■ 1.000	λ_{*}	λ_{*}	stants (5.4.3) run into some nontrivial fixed points g_{N}^{*} and λ_{*} , depending on the
mielke-geometrodynamics_F00532	116	■ 1.000	R^n	R^n	high-order polynomials R^n of the Ricci scalar, similarly as in the classically bifur-
mielke-geometrodynamics_F00533	116	■ 1.000	$f(R)$	$f(R)$	cating $f(R)$ models (SCHUNCK et al. 2005), but then the issue of physical ghosts or
mielke-geometrodynamics_F00534	117	■ 1.000	κ	κ	also scales with the gravitational constant κ , but inversely compared to the Hilbert–
mielke-geometrodynamics_F00535	117	■ 1.000	$1 / k^4$	$1/k^4$	bilinears is mandatory, including the $1/k^4$ dependence in the functional integral.
mielke-geometrodynamics_F00536	117	■ 1.000	f_{*}^2	f_{*}^2	and the corresponding dimensionless renormalized running coupling f_{*}^2 becomes
mielke-geometrodynamics_F00537	117	■ 1.000	$R_{\alpha\beta} \equiv 0$	$R_{\alpha\beta} \equiv 0$	One way to avoid such anomalies is to employ curvature constraints like $R_{\alpha\beta} \equiv 0$,
mielke-geometrodynamics_F00538	117	■ 1.000	$SL(5, \mathbb{R})$	$SL(5, \mathbb{R})$	from a “minimalist” $SL(5, \mathbb{R})$ gauge model that includes only a “bare” Pontryagin-
mielke-geometrodynamics_F00539	118	■ 1.000	$\Delta c / c \simeq 10^{-17}$	$\Delta c / c \simeq 10^{-17}$	Lorentz-invariant, nowadays measured at the unprecedented $\Delta c / c \simeq 10^{-17}$ level

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mielke-geometrodynamics_F00540	118	■ 1.000	10^{-13}	10^{-13}	et al. 2012) at the 10^{-13} level. Moreover, the upper bound for a violation of the
mielke-geometrodynamics_F00541	118	■ 1.000	10^{-8}	10^{-8}	quarks, is about 10^{-8} ; cf. HUGHES (1993).
mielke-geometrodynamics_F00542	118	■ 0.979	$\bar{\nu}_e$	$\bar{\nu}_e$	Assuming that not only the dominant delayed antineutrinos $\bar{\nu}_e$ but at least one prompt
mielke-geometrodynamics_F00543	118	■ 0.813	ν_e	ν_e	neutrino ν_e was in the pulse, one could infer that the equivalence principle is valid
mielke-geometrodynamics_F00544	118	■ 1.000	10^{-6}	10^{-6}	(GUZZO et al. 2002) for electron–neutrinos ν_e up to 10^{-6} .
mielke-geometrodynamics_F00545	118	■ 0.996	CPT	CPT	Moreover, a test of the CPT theorem for the neutral kaon system indirectly
mielke-geometrodynamics_F00546	118	■ 1.000	K^0	K^0	yielded an upper bound for the relative difference of the <i>inertial masses</i> of the K^0
mielke-geometrodynamics_F00547	118	■ 1.000	$\bar{K}^0$	$\bar{K}^0$	meson and its antiparticle $\bar{K}^0$ of less than 10^{-18} , when employing (MURAYAMA
mielke-geometrodynamics_F00548	118	■ 1.000	10^{-18}	10^{-18}	meson and its antiparticle $\bar{K}^0$ of less than 10^{-18} , when employing (MURAYAMA
mielke-geometrodynamics_F00549	120	■ 0.761	$(5, \mathbb{R})$	$(5, \mathbb{R})$	Mielke EW (2012) Einstein-Weyl gravity from a topological gauge $SL(5, \mathbb{R})$ invariant action. Adv
mielke-geometrodynamics_F00550	122	■ 1.000	(1986, 1991)	(1986, 1991)	On the other hand, Ashtekar’s reformulation ASHTEKAR (1986, 1991) of general
mielke-geometrodynamics_F00551	122	■ 0.996	dC_{TT}	dC_{TT}	ary term dC_{TT} constructed from a <i>translational Chern–Simons term</i> (CS) multiplied
mielke-geometrodynamics_F00552	122	■ 0.987	$(\mathrm{GR}_{\parallel})$	$(\mathrm{GR}_{\parallel})$	as was already suggested by EINSTEIN (1928), a theory with <i>teleparallelism</i> ($\mathrm{GR}_{\parallel}$).
mielke-geometrodynamics_F00553	123	■ 1.000	$\ell := \sqrt{8\pi\hbar G_{\mathrm{N}}/c^3} \simeq 10^{-33} \text{ cm}$	$\ell := \sqrt{8\pi\hbar G_{\mathrm{N}}/c^3} \simeq 10^{-33} \text{ cm}$	scale $\ell := \sqrt{8\pi\hbar G_{\mathrm{N}}/c^3} \simeq 10^{-33} \text{ cm}$, a topologically rich spectrum of dislocations
mielke-geometrodynamics_F00554	123	■ 1.000	q	q	In the one-dimensional harmonic oscillator model (MIELKE 1998) with q as gen-
mielke-geometrodynamics_F00555	123	■ 0.994	$\mathcal{C} = q^2/2$	$\mathcal{C} = q^2/2$	term derived from the <i>Chern–Simons</i> (CS)-type term $\mathcal{C} = q^2/2$ as <i>generating func-</i>
mielke-geometrodynamics_F00556	123	■ 0.999	$\mathcal{S} = \exp(-\mathcal{C}) = \exp(-q^2/2)$	$\mathcal{S} = \exp(-\mathcal{C}) = \exp(-q^2/2)$	<i>tion</i> . After quantization, the corresponding operator $\mathcal{S} = \exp(-\mathcal{C}) = \exp(-q^2/2)$
mielke-geometrodynamics_F00557	123	■ 0.999	$\Psi = N \exp(i\theta \int \mathcal{C})$	$\Psi = N \exp\left(i\theta \int \mathcal{C}\right)$	shown that $\Psi = N \exp(i\theta \int \mathcal{C})$ is, up to an overall factor, the unique solution of the
mielke-geometrodynamics_F00558	123	■ 1.000	$P := \mathbb{R}^4 \otimes SO(1, 3)$	$P := \mathbb{R}^4 \otimes SO(1, 3)$	Since the Poincaré group $P := \mathbb{R}^4 \ltimes SO(1, 3)$ is the semidirect product of trans-
mielke-geometrodynamics_F00559	124	■ 1.000	$\mathcal{A} := *(\vartheta_{\alpha} \wedge T^{\alpha}) =$	$\mathcal{A} := *(\vartheta_{\alpha} \wedge T^{\alpha}) =$	They are <i>translational</i> and Lorentz-rotational CS terms, where $\mathcal{A} := *(\vartheta_{\alpha} \wedge T^{\alpha}) \equiv$
mielke-geometrodynamics_F00560	124	■ 1.000	$\mathcal{A}_i dx^i$	$\mathcal{A}_i dx^i$	$\mathcal{A}_i dx^i$ is the <i>axial torsion</i> one-form. In the first <i>translational</i> CS three-form, there
mielke-geometrodynamics_F00561	124	■ 1.000	ℓ	ℓ	necessarily occurs a fundamental length ℓ for dimensional reasons.
mielke-geometrodynamics_F00562	124	■ 0.907	$SO(1, 4)$	$SO(1, 4)$	the de Sitter group $SO(1, 4)$ or $SO(2, 3)$ as a subgroup, both CS three-forms are
mielke-geometrodynamics_F00563	124	■ 0.907	$SO(2, 3)$	$SO(2, 3)$	the de Sitter group $SO(1, 4)$ or $SO(2, 3)$ as a subgroup, both CS three-forms are
mielke-geometrodynamics_F00564	124	■ 1.000	$g_{\alpha\beta}$	$g_{\alpha\beta}$	free Pontryagin form, in the NY term (6.2.4), a metric $g_{\alpha\beta}$ is needed to raise and
mielke-geometrodynamics_F00565	124	■ 1.000	$T_{\alpha} = g_{\alpha\beta} T^{\beta}$	$T_{\alpha} = g_{\alpha\beta} T^{\beta}$	lower the indices, for instance in $T_{\alpha} = g_{\alpha\beta} T^{\beta}$.
mielke-geometrodynamics_F00566	124	■ 1.000	ϑ_{α}	ϑ_{α}	ϑ_{α} , on the first Bianchi identity

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mielke-geometrodynamics_F00567	125	■ 1.000	$R_{\{\alpha \beta\}}$	$R_{\alpha\beta}^{\{\}}$	of general relativity (GR), where $R_{\alpha\beta}^{\{\}}$ denotes the Riemannian curvature with respect
mielke-geometrodynamics_F00568	125	■ 1.000	$\Gamma_{\{\alpha \beta\}}$	$\Gamma_{\alpha\beta}^{\{\}}$	to the Levi-Civita connection $\Gamma_{\alpha\beta}^{\{\}}$.
mielke-geometrodynamics_F00569	125	■ 1.000	$K_{\{\alpha \beta\}}$	$K_{\alpha\beta}$	is dual to the contortion one-form $K_{\alpha\beta}$, which features in the decomposition $\Gamma_{\alpha\beta} =$
mielke-geometrodynamics_F00570	125	■ 1.000	$\Gamma_{\{\alpha \beta\}} =$	$\Gamma_{\alpha\beta} =$	is dual to the contortion one-form $K_{\alpha\beta}$, which features in the decomposition $\Gamma_{\alpha\beta} =$
mielke-geometrodynamics_F00571	125	■ 0.985	$\left. \left. \left. \Gamma_{\{\beta \alpha\}} = \Gamma_{\{\alpha \beta\}} - K_{\{\alpha \beta\}} = \Gamma_{\{\alpha \beta\}} + e_{\alpha} \right\rfloor T_{\beta} + \left(e_{\alpha} \right\rfloor e_{\beta} \right\rfloor T_{\gamma} \right) \wedge \vartheta^{\gamma}$	$-\Gamma_{\beta\alpha} = \Gamma_{\alpha\beta}^{\{\}} - K_{\alpha\beta} = \Gamma_{\alpha\beta}^{\{\}} + e_{\alpha} \rfloor T_{\beta} + (e_{\alpha} \rfloor e_{\beta}) \rfloor T_{\gamma} \Big) \wedge \vartheta^{\gamma}$	$-\Gamma_{\beta\alpha} = \Gamma_{\alpha\beta}^{\{\}} - K_{\alpha\beta} = \Gamma_{\alpha\beta}^{\{\}} + e_{\alpha} \rfloor T_{\beta} + (e_{\alpha} \rfloor e_{\beta}) \rfloor T_{\gamma} \Big) \wedge \vartheta^{\gamma}$ of the RC connection.
mielke-geometrodynamics_F00572	125	■ 0.971	$C_{\mathrm{TT}*} := \vartheta^{\alpha} \wedge T_{\alpha} / 2 \ell^2$	$C_{\mathrm{TT}*} := \vartheta^{\alpha} \wedge T_{\alpha} / 2 \ell^2$	where $C_{\mathrm{TT}*} := \vartheta^{\alpha} \wedge T_{\alpha} / 2 \ell^2$ is a dual CS term, proper <i>teleparallelism</i> (GR) spec-
mielke-geometrodynamics_F00573	125	■ 1.000	$R^{\{\alpha \beta\}} = 0$	$R^{\alpha\beta} = 0$	the proper constraint $R^{\alpha\beta} = 0$ has to be imposed by subtracting $R^{\alpha\beta} \wedge \lambda_{\alpha\beta}$ from
mielke-geometrodynamics_F00574	125	■ 1.000	$R^{\{\alpha \beta\}} \wedge \lambda_{\{\alpha \beta\}}$	$R^{\alpha\beta} \wedge \lambda_{\alpha\beta}$	the proper constraint $R^{\alpha\beta} = 0$ has to be imposed by subtracting $R^{\alpha\beta} \wedge \lambda_{\alpha\beta}$ from
mielke-geometrodynamics_F00575	125	■ 1.000	$\lambda_{\{\alpha \beta\}} = -\lambda_{\{\beta \alpha\}}$	$\lambda_{\alpha\beta} = -\lambda_{\beta\alpha}$	(6.2.9), where the 2-form $\lambda_{\alpha\beta} = -\lambda_{\beta\alpha}$ is a Lagrange multiplier, see HEHL et al.
mielke-geometrodynamics_F00576	126	■ 1.000	$\vartheta^{\alpha}, \Gamma^{\{\alpha \beta\}}$	$\vartheta^{\alpha}, \Gamma^{\alpha\beta}$	By varying this Lagrangian independently with respect to ϑ^{α} , $\Gamma^{\alpha\beta}$, and the mul-
mielke-geometrodynamics_F00577	126	■ 1.000	$\lambda_{\{\alpha \beta\}}$	$\lambda_{\alpha\beta}$	tiplier $\lambda_{\alpha\beta}$, we obtain as field equations
mielke-geometrodynamics_F00578	126	■ 0.856	$L_{\{\}}$	$L_{ }$	first field Eq. (6.2.13) is the same as that of the unconstrained Lagrangian $L_{ }$. The
mielke-geometrodynamics_F00579	126	■ 1.000	$\lambda_{\{\alpha \beta\}} = \left(\gamma / \ell^2 \right) \eta_{\{\alpha \beta\}}$	$\lambda_{\alpha\beta} = (\gamma / \ell^2) \eta_{\alpha\beta}$	<i>spinless</i> matter, a possible solution is $\lambda_{\alpha\beta} = (\gamma / \ell^2) \eta_{\alpha\beta}$, as will be shown below.
mielke-geometrodynamics_F00581	127	■ 1.000	$\sigma = \pm \mathrm{i} \gamma$	$\sigma = \pm i \gamma$	of BAEKLER et al. (1982); if $\sigma = \pm i \gamma$ is chosen for one of the free parameters there.
mielke-geometrodynamics_F00582	127	■ 1.000	$\theta_{\mathrm{T}} = \pm \mathrm{i}$	$\theta_{\mathrm{T}} = \pm i$	version (6.2.9) of Einstein’s GR for the choice $\theta_{\mathrm{T}} = \pm i$ into a <i>chiral gauge theory</i>
mielke-geometrodynamics_F00583	127	■ 0.984	$\stackrel{(\pm)}{T}^{\alpha} := \frac{1}{2} (T^{\alpha} \pm i T^{*\alpha})$	$\stackrel{(\pm)}{T}^{\alpha} := \frac{1}{2} (T^{\alpha} \pm i T^{*\alpha})$	where $\stackrel{(\pm)}{T}^{\alpha} := \frac{1}{2} (T^{\alpha} \pm i T^{*\alpha})$ denotes the self-dual or anti-self-dual torsion. The
mielke-geometrodynamics_F00584	128	■ 1.000	$\lambda_{\{\alpha \beta\}}^{\{\tau\}}$	$\lambda_{\alpha\beta}^{(\tau)}$	et al. 1989) an additional piece $\lambda_{\alpha\beta}^{(\tau)}$ in the ansatz for the Lagrangian multiplier,
mielke-geometrodynamics_F00585	128	■ 1.000	$\lambda_{\{\alpha \beta\}}^{\{\tau\}} = \tau_{\{\alpha \beta\}}$	$D^{(\tau)} \lambda_{\alpha\beta} = \tau_{\alpha\beta}$	provided it satisfies $D^{(\tau)} \lambda_{\alpha\beta} = \tau_{\alpha\beta}$.
mielke-geometrodynamics_F00586	128	■ 0.584	Π^{α}	Π^{α}	The complex translational field momenta Π^{α} replace the role of the torsion if we
mielke-geometrodynamics_F00587	128	■ 0.731	$\delta L_{ } / \delta \vartheta^{\alpha}$	$\delta L_{ } / \delta \vartheta^{\alpha}$	allow us to rederive the field Eq. (6.2.24) via $\delta L_{ } / \delta \vartheta^{\alpha}$ from a variational principle.
mielke-geometrodynamics_F00588	128	■ 1.000	$R_{\{\alpha \beta\}} = 0$	$R_{\alpha\beta} = 0$	Cartan spacetime with $R_{\alpha\beta} = 0$ and in view of the first Noether identity (6.8.11),
mielke-geometrodynamics_F00589	128	■ 0.999	Π_{α}	Π_{α}	terms of the momenta Π_{α} , it would be convenient to have a <i>relocalized</i> energy–momentum current
mielke-geometrodynamics_F00590	128	■ 0.121	Σ_{α}	Σ_{α}	Σ_{α} for which $D \Sigma_{\alpha} \cong 0$ holds.
mielke-geometrodynamics_F00591	128	■ 0.121	D	D	Σ_{α} for which $D \Sigma_{\alpha} \cong 0$ holds.

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mielke-geometrodynamics_F00592	129	0.821	$\stackrel{\scriptscriptstyle\text{F}}{\text{A}}_{\alpha}^{\beta}$	$A_{\alpha}^{(\mp)\beta}$	valued 1-form $A_{\alpha}^{(\mp)\beta}$ deforms our original Riemann–Cartan connection Γ_{α}^{β} similarly
mielke-geometrodynamics_F00593	129	0.946	$\underline{\Gamma}_{\parallel}^{\alpha\beta} \rightarrow 0$	$\underline{\Gamma}_{\parallel}^{\alpha\beta} \rightarrow 0$	(1973), THIRRING (1978), HEHL et al. (1980). In the teleparallelism limit $\underline{\Gamma}_{\parallel}^{\alpha\beta} \rightarrow 0$,
mielke-geometrodynamics_F00594	129	1.000	H_{α}	H_{α}	the “superpotential” H_{α} is the <i>Møller</i> or <i>Freud complex</i> (FREUD 1939)
mielke-geometrodynamics_F00595	129	0.700	$\overparen{D}\eta_{\alpha} := D\eta_{\alpha} -$	$\widehat{D}\eta_{\alpha} := D\eta_{\alpha} -$	³ The <i>transposed</i> connection, which has the property (MIELKE et al. 1989) that $\widehat{D}\eta_{\alpha} := D\eta_{\alpha} -$
mielke-geometrodynamics_F00596	129	0.521	$\left(e_{\alpha} \downarrow T^{\beta}\right) \wedge \eta_{\beta} \equiv 0$	$(e_{\alpha} \downarrow T^{\beta}) \wedge \eta_{\beta} \equiv 0$	$(e_{\alpha} \downarrow T^{\beta}) \wedge \eta_{\beta} \equiv 0$, may be regarded as a special real version of our Sen-type connection, for
mielke-geometrodynamics_F00597	129	0.593	$\stackrel{\scriptscriptstyle\text{F}}{D}\eta_{\alpha} = \pm(i/2)\ell^2\eta_{\alpha\beta} \wedge \Pi^{\beta}$	$D^{(\pm)}\eta_{\alpha} = \pm(i/2)\ell^2\eta_{\alpha\beta} \wedge \Pi^{(\pm)\beta}$	which $D^{(\pm)}\eta_{\alpha} = \pm(i/2)\ell^2\eta_{\alpha\beta} \wedge \Pi^{(\pm)\beta}$ holds.
mielke-geometrodynamics_F00598	130	1.000	E_{α}	E_{α}	et al. (2000) that E_{α} is a covariant conserved gravitational energy, i.e., that $DE_{\alpha} =$
mielke-geometrodynamics_F00599	130	1.000	$D E_{\alpha} =$	$DE_{\alpha} =$	et al. (2000) that E_{α} is a covariant conserved gravitational energy, i.e., that $DE_{\alpha} =$
mielke-geometrodynamics_F00600	130	1.000	$D D H_{\alpha} \equiv -R_{\alpha}^{\beta} \wedge H_{\beta} = 0$	$DDH_{\alpha} \equiv -R_{\alpha}^{\beta} \wedge H_{\beta} = 0$	$DDH_{\alpha} \equiv -R_{\alpha}^{\beta} \wedge H_{\beta} = 0$ due to the vanishing of the Ricci identity in a teleparallel
mielke-geometrodynamics_F00602	130	1.000	Σ_t	Σ_t	into a family of spacelike three-dimensional hypersurfaces Σ_t that are parametrized
mielke-geometrodynamics_F00603	130	1.000	$x^0 = t$	$x^0 = t$	by the coordinate time $x^0 = t$. Then there exists the future-directed <i>timelike</i> vector
mielke-geometrodynamics_F00604	130	0.496	$u := -N^2 dt, \Longleftrightarrow$	$u := -N^2 dt, \Longleftrightarrow$	which is uniquely linked to the hypersurface orthogonal 1-form $u := -N^2 dt, \Longleftrightarrow$
mielke-geometrodynamics_F00605	130	0.963	$du \wedge u = 0$	$du \wedge u = 0$	$du \wedge u = 0$ via $u = g(n, \cdot)$. The lapse function N and the shift vector N^A , which
mielke-geometrodynamics_F00606	130	0.963	$u = g(n,$	$u = g(n,$	$du \wedge u = 0$ via $u = g(n, \cdot)$. The lapse function N and the shift vector N^A , which
mielke-geometrodynamics_F00607	130	0.963	$)$. The lapse function N	$)$. The lapse function N	$du \wedge u = 0$ via $u = g(n, \cdot)$. The lapse function N and the shift vector N^A , which
mielke-geometrodynamics_F00608	130	0.963	N^A	N^A	$du \wedge u = 0$ via $u = g(n, \cdot)$. The lapse function N and the shift vector N^A , which
mielke-geometrodynamics_F00609	130	0.974	p	p	The <i>normal</i> part of a p -form Ψ is defined by
mielke-geometrodynamics_F00610	130	0.997	$(3+1)$	$(3+1)$	operators. (For more details on this very concise (3+1) decomposition of exterior
mielke-geometrodynamics_F00611	131	1.000	$L = L(\vartheta^{\alpha}, \Psi, D\Psi)$	$L = L(\vartheta^{\alpha}, \Psi, D\Psi)$	Let us consider a Lagrangian $L = L(\vartheta^{\alpha}, \Psi, D\Psi)$ which is of first differential
mielke-geometrodynamics_F00612	131	0.973	$\left.\mathscr{K} = L_{\perp} := n \rfloor L\right.$	$\mathscr{K} = L_{\perp} := n \rfloor L$	where $\mathscr{K} = L_{\perp} := n \rfloor L$ denotes the normal part of the Lagrangian. Since the “time
mielke-geometrodynamics_F00613	131	0.677	$\bullet$	$\bullet$	derivative” $\bullet$ is more precisely given by the Lie derivative $\ell_n := dn \rfloor + n \rfloor d$ along
mielke-geometrodynamics_F00614	131	0.677	$\left.\ell_n := dn \rfloor + n \rfloor d\right.$	$\ell_n := dn \rfloor + n \rfloor d$	derivative” $\bullet$ is more precisely given by the Lie derivative $\ell_n := dn \rfloor + n \rfloor d$ along
mielke-geometrodynamics_F00615	131	1.000	n	n	the timelike vector field n , we have
mielke-geometrodynamics_F00616	131	0.965	$n^{\alpha} := n \rfloor \vartheta^{\alpha}$	$n^{\alpha} := n \rfloor \vartheta^{\alpha}$	and transvect the resulting expression with the four-vector $n^{\alpha} := n \rfloor \vartheta^{\alpha}$ of <i>lapse and</i>
mielke-geometrodynamics_F00617	131	0.605	$n^{\alpha} e_{\alpha} \rfloor \Psi = n \rfloor \Psi$	$n^{\alpha} e_{\alpha} \rfloor \Psi = n \rfloor \Psi$	<i>shift</i> , for which the useful relation $n^{\alpha} e_{\alpha} \rfloor \Psi = n \rfloor \Psi$ can be derived. For the projection
mielke-geometrodynamics_F00618	132	0.919	$L_n = Dn \downarrow + n \downarrow D$	$L_n = Dn \downarrow + n \downarrow D$	In introducing the gauge-covariant Lie derivative $L_n = Dn \downarrow + n \downarrow D$, separating
mielke-geometrodynamics_F00619	132	0.651	$n \rfloor \Psi = n \downarrow (dt \wedge n \rfloor \Psi) = n \rfloor \Psi$	$n \rfloor \Psi = n \downarrow (dt \wedge n \rfloor \Psi) = n \rfloor \Psi$	sion is related to the Hamiltonian (6.4.5). Indeed, using $n \rfloor \Psi = n \downarrow (dt \wedge n \rfloor \Psi) = n \rfloor \Psi$

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mielke-geometrodynamics_F00620	132	■ 0.479	$\left.\Gamma_{\perp}^{\alpha\beta} := n\right\rfloor \Gamma^{\alpha\beta}$	$\Gamma_{\perp}^{\alpha\beta} := n\rfloor \Gamma^{\alpha\beta}$	where the normal vector $n^{\alpha} := n\rfloor \vartheta^{\alpha}$ comprises <i>lapse and shift</i> and $\Gamma_{\perp}^{\alpha\beta} := n\rfloor \Gamma^{\alpha\beta}$
mielke-geometrodynamics_F00621	132	■ 1.000	$\mathscr{G}_{\alpha}$	$\mathscr{G}_{\alpha}$	is the normal part of the Lorentz connection and $\mathscr{G}_{\alpha}$ and $\mathscr{G}_{\alpha\beta}$ represent the tangential
mielke-geometrodynamics_F00622	132	■ 1.000	$\mathscr{G}_{\alpha\beta}$	$\mathscr{G}_{\alpha\beta}$	is the normal part of the Lorentz connection and $\mathscr{G}_{\alpha}$ and $\mathscr{G}_{\alpha\beta}$ represent the tangential
mielke-geometrodynamics_F00623	133	■ 0.709	$D_{\alpha} := e_{\alpha}\rfloor D$	$D_{\alpha} := e_{\alpha}\rfloor D$	where $D_{\alpha} := e_{\alpha}\rfloor D$ are the anholonomic components of the exterior covariant
mielke-geometrodynamics_F00624	133	■ 0.927	$D\lambda_{\alpha\beta}$	$D\lambda_{\alpha\beta}$	arise the extra tangential term $\underline{D}\lambda_{\alpha\beta}$ in the <i>Lorentz</i> generator $\mathscr{G}_{\alpha\beta}$, where
mielke-geometrodynamics_F00625	133	■ 1.000	$\underline{\Psi} := \Psi - dt \wedge n\rfloor \Psi$	$\underline{\Psi} := \Psi - dt \wedge n\rfloor \Psi$	$\underline{\Psi} := \Psi - dt \wedge n\rfloor \Psi$ in the notation of differential forms. However, as suggested
mielke-geometrodynamics_F00626	133	■ 0.893	$\underline{\lambda}_{\alpha\beta} = \left(1/2\ell^2\right) * (\vartheta_{\alpha} \wedge \vartheta_{\beta}) = -\left(1/N\ell^2\right) n_{[\alpha} * \underline{\vartheta}_{\beta]}$	$\lambda_{\alpha\beta} = (1/2\ell^2) * (\vartheta_{\alpha} \wedge \vartheta_{\beta}) = -(1/N\ell^2) n_{[\alpha} * \underline{\vartheta}_{\beta]}$	by $\underline{\lambda}_{\alpha\beta} = (1/2\ell^2) * (\vartheta_{\alpha} \wedge \vartheta_{\beta}) = -(1/N\ell^2) n_{[\alpha} * \underline{\vartheta}_{\beta]}$. This is another manifestation of
mielke-geometrodynamics_F00627	133	■ 0.925	$d\stackrel{(\mp)}{C}_{\text{TT}}$	$d\stackrel{(\mp)}{C}_{\text{TT}}$	by the boundary term $d\stackrel{(\mp)}{C}_{\text{TT}}$.
mielke-geometrodynamics_F00628	133	■ 0.989	$(\alpha = \hat{0})$	$(\alpha = \hat{0})$	After this trivialization of the Lorentz constraints, the Hamiltonian $(\alpha = \hat{0})$ and
mielke-geometrodynamics_F00629	133	■ 1.000	$(\alpha = B)$	$(\alpha = B)$	diffeomorphism constraints $(\alpha = B)$ of complexified $\text{GR}_{\parallel}$, in terms of a complex
mielke-geometrodynamics_F00630	133	■ 0.464	$\stackrel{(\pm)}{\mathscr{A}}_{\alpha} := \stackrel{*}{\Pi}_{\alpha}$	$\stackrel{(\pm)}{\mathscr{A}}_{\alpha} := \stackrel{*}{\Pi}_{\alpha}$	(1994). Here the tangential Ashtekar-type connection $\stackrel{(\pm)}{\mathscr{A}}_{\alpha} := \stackrel{*}{\Pi}_{\alpha}$ is three-dual
mielke-geometrodynamics_F00631	133	■ 0.130	$\stackrel{*}{-}_{\alpha}$	$\stackrel{*}{-}_{\alpha}$	ties $\stackrel{*}{\vartheta}_{\alpha}$, where we note that the spatial Hodge dual is involutive, i.e., $\stackrel{*}{*} = +1$.
mielke-geometrodynamics_F00632	133	■ 0.130	$\stackrel{*}{*} = +1$	$\stackrel{*}{*} = +1$	ties $\stackrel{*}{\vartheta}_{\alpha}$, where we note that the spatial Hodge dual is involutive, i.e., $\stackrel{*}{*} = +1$.
mielke-geometrodynamics_F00633	133	■ 1.000	$\underline{\vartheta}^{\hat{0}} = 0$	$\underline{\vartheta}^{\hat{0}} = 0$	Moreover, by adopting tetrads in the temporal gauge $\underline{\vartheta}^{\hat{0}} = 0$ of SCHWINGER (1963)
mielke-geometrodynamics_F00634	133	■ 0.858	$\stackrel{(\pm)}{\mathscr{A}}_{\hat{0}} = d^{\pm}(\stackrel{(\pm)}{\phi})$	$\stackrel{(\pm)}{\mathscr{A}}_{\hat{0}} = d^{\pm}(\stackrel{(\pm)}{\phi})$	and the gauge $\stackrel{(\pm)}{\mathscr{A}}_{\hat{0}} = d^{\pm}(\stackrel{(\pm)}{\phi})$ of NESTER (1989), the Hamiltonian constraint
mielke-geometrodynamics_F00635	134	■ 1.000	$\alpha = \hat{0}$	$\alpha = \hat{0}$	i.e., (6.4.19) for $\alpha = \hat{0}$, vanishes identically.
mielke-geometrodynamics_F00636	134	■ 0.561	$\stackrel{*}{-}\vartheta_{\alpha}$	$\stackrel{*}{-}\vartheta_{\alpha}$	parts of the basic one-forms, i.e., the <i>triad densities</i> $\stackrel{*}{\vartheta}_{\alpha}$ and the tangential part of
mielke-geometrodynamics_F00637	134	■ 0.677	$\stackrel{(\pm)}{\mathscr{A}}_{\alpha} := \stackrel{*}{\Pi}_{\alpha}$	$\stackrel{(\pm)}{\mathscr{A}}_{\alpha} := \stackrel{*}{\Pi}_{\alpha}$	the self-dual or anti-self-dual connection $\stackrel{(\pm)}{\mathscr{A}}_{\alpha} := \stackrel{*}{\Pi}_{\alpha}$, the three-dual of the chiral
mielke-geometrodynamics_F00638	134	■ 0.995	$\stackrel{\pm}{p}$	$\stackrel{\pm}{p}$	momenta $\stackrel{(\pm)}{\Pi}_{\alpha}$, become the generalized coordinates q and momenta $\stackrel{\pm}{p}$ of the bosonic
mielke-geometrodynamics_F00639	134	■ 0.337	$\stackrel{*}{\Pi}_{\alpha}$	$\stackrel{*}{\Pi}_{\alpha}$	prevails (MIELKE 2002), where the complex field “momenta” $\stackrel{(\pm)}{\Pi}_{\alpha}$ become differ-
mielke-geometrodynamics_F00640	134	■ 0.316	$\stackrel{*}{\vartheta}^{\beta}$	$\stackrel{*}{\vartheta}^{\beta}$	ential operators, whereas the triad densities $\stackrel{*}{\vartheta}^{\beta}$ remain generalized coordinates q .
mielke-geometrodynamics_F00641	134	■ 1.000	$\alpha = B$	$\alpha = B$	For the remaining Gauss constraint (6.4.8) with $\alpha = B$, let us try the state vector
mielke-geometrodynamics_F00642	134	■ 0.731	$\underline{C}_{\text{TT}}$	$\underline{C}_{\text{TT}}$	Chern–Simons term $\stackrel{(\pm)}{\underline{C}}_{\text{TT}}$ in terms of the triads or triad densities and the tangential
mielke-geometrodynamics_F00643	134	■ 0.862	$\stackrel{*}{\vartheta}^B$	$\stackrel{*}{\vartheta}^B$	rewritten in terms of the triad densities $\stackrel{*}{\vartheta}^B$ with $B = 1, 2, 3$ and the tangential part

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mielke-geometrodynamics_F00644	134	■ 0.862	B=1,2,3	$B = 1, 2, 3$	rewritten in terms of the triad densities ${}^*\underline{\vartheta}^B$ with $B = 1, 2, 3$ and the tangential part
mielke-geometrodynamics_F00645	134	■ 0.637	$\stackrel{\scriptscriptstyle (\pm)}{\underline{T}}^B$	$\stackrel{\scriptscriptstyle (\pm)}{\underline{T}}^B$	momentum operator (6.4.22) returns the chiral torsion $\stackrel{\scriptscriptstyle (\pm)}{\underline{T}}^B$ as a prefactor:
mielke-geometrodynamics_F00646	134	■ 1.000	$\underline{\Gamma}_{\parallel}^{\alpha\beta} \stackrel{*}{=} 0$	$\Gamma_{\parallel}^{\alpha\beta} \stackrel{*}{=} 0$	Since the corresponding connection $\underline{\Gamma}_{\parallel}^{\alpha\beta} \stackrel{*}{=} 0$ is of pure gauge type, the state vector
mielke-geometrodynamics_F00647	135	■ 0.999	$\Psi \wedge {}^*\Phi = \Phi \wedge {}^*\Psi$	$\Psi \wedge {}^*\Phi = \Phi \wedge {}^*\Psi$	where the rule $\underline{\Psi} \wedge {}^*\Phi = \Phi \wedge {}^*\underline{\Psi}$ for differential forms of the same degree has
mielke-geometrodynamics_F00648	135	■ 0.994	$SU(2)$	$SU(2)$	borrowed from $SU(2)$ Chern–Simons field theory (WITTEN 1989; GUADAGNINI
mielke-geometrodynamics_F00649	135	■ 1.000	$\mathscr{H}_{\Lambda} \Psi_{\Lambda}(A) = 0$	$\mathscr{H}_{\Lambda} \Psi_{\Lambda}(A) = 0$	straint $\mathscr{H}_{\Lambda} \Psi_{\Lambda}(A) = 0$ of GR with cosmological constant in the nonperturbative
mielke-geometrodynamics_F00650	135	■ 1.000	$\ell \rightarrow 0$	$\ell \rightarrow 0$	ℓ , it is dimensionless and nonsingular even for $\ell \rightarrow 0$, due to $\dim \underline{\vartheta}^B = [\ell]$.
mielke-geometrodynamics_F00651	135	■ 1.000	$\operatorname{dim} \underline{\vartheta}^B = [\ell]$	$\dim \underline{\vartheta}^B = [\ell]$	ℓ , it is dimensionless and nonsingular even for $\ell \rightarrow 0$, due to $\dim \underline{\vartheta}^B = [\ell]$.
mielke-geometrodynamics_F00652	135	■ 0.937	$\stackrel{\scriptscriptstyle (\mp)}{\underline{T}}^B = 0$	$\underline{T}^{(\mp)B} = 0$	configurations with $\stackrel{\scriptscriptstyle (\mp)}{\underline{T}}^B = 0$, i.e., <i>self-dual or anti-self-dual</i> tangential torsion. Con-
mielke-geometrodynamics_F00653	135	■ 0.802	$\stackrel{\scriptscriptstyle (\pm)}{\underline{T}} = 0$	$\stackrel{\scriptscriptstyle (\pm)}{\underline{T}} = 0$	straint are dominated by <i>self-dual</i> torsion solutions satisfying $\stackrel{\scriptscriptstyle (\pm)}{\underline{T}} = 0$.
mielke-geometrodynamics_F00654	135	■ 1.000	$\mathbb{R}^4 \cup \infty = S^4$	$\mathbb{R}^4 \cup \infty = S^4$	$\mathbb{R}^4 \cup \infty = S^4$ with the spherically symmetric metric
mielke-geometrodynamics_F00655	136	■ 0.997	(r, ψ, θ, ϕ)	(r, ψ, θ, ϕ)	where (r, ψ, θ, ϕ) are the standard hyperspherical coordinates that parameterize the
mielke-geometrodynamics_F00656	136	■ 1.000	S^3	S^3	unit three-sphere S^3 . The function f carries physical dimension [length]. This space
mielke-geometrodynamics_F00657	136	■ 1.000	f	f	unit three-sphere S^3 . The function f carries physical dimension [length]. This space
mielke-geometrodynamics_F00658	136	■ 1.000	$df = \pm 2h dr$	$df = \pm 2h dr$	which is self-dual or anti-self-dual, provided that $df = \pm 2h dr$. Here the self-dual
mielke-geometrodynamics_F00659	136	■ 0.998	$\eta_{\mu\nu}^A := \eta_0^A{}_{\mu\nu} + \delta_{\mu}^A \delta_{\nu}^0 - \delta_{\nu}^A \delta_{\mu}^0$	$\eta_{\mu\nu}^A := \eta_0^A{}_{\mu\nu} + \delta_{\mu}^A \delta_{\nu}^0 - \delta_{\nu}^A \delta_{\mu}^0$	generalized Levi-Civita symbol $\eta_{\mu\nu}^A := \eta_0^A{}_{\mu\nu} + \delta_{\mu}^A \delta_{\nu}^0 - \delta_{\nu}^A \delta_{\mu}^0$ of 'T HOOFT (1991),
mielke-geometrodynamics_F00660	136	■ 1.000	$r \rightarrow \infty$	$r \rightarrow \infty$	radial infinity $r \rightarrow \infty$ yields
mielke-geometrodynamics_F00661	136	■ 0.954	$\stackrel{\scriptscriptstyle (\mp)}{\underline{T}}^B = 0$	$\stackrel{\scriptscriptstyle (\mp)}{\underline{T}}^B = 0$	self-dual, i.e., $\stackrel{\scriptscriptstyle (\mp)}{\underline{T}}^B = 0$, then integration over the chiral CS term $\stackrel{\scriptscriptstyle (\pm)}{\underline{C}}_{\text{TT}}$ yields the
mielke-geometrodynamics_F00662	136	■ 0.954	$\stackrel{\scriptscriptstyle (\pm)}{\underline{C}}_{\text{TT}}$	$\stackrel{\scriptscriptstyle (\pm)}{\underline{C}}_{\text{TT}}$	self-dual, i.e., $\stackrel{\scriptscriptstyle (\mp)}{\underline{T}}^B = 0$, then integration over the chiral CS term $\stackrel{\scriptscriptstyle (\pm)}{\underline{C}}_{\text{TT}}$ yields the
mielke-geometrodynamics_F00663	136	■ 1.000	$\Psi_{\parallel}(\vartheta) \propto e^{-C}$	$\Psi_{\parallel}(\vartheta) \propto e^{-C}$	Thus one can surmise that the state vector $\Psi_{\parallel}(\vartheta) \propto e^{-C}$ as a solution of the
mielke-geometrodynamics_F00664	136	■ 1.000	$\underline{\vartheta}^{\hat{0}} = 3\kappa d\theta_{\text{L}} = h dr = \pm df/2$	$\underline{\vartheta}^{\hat{0}} = 3\kappa d\theta_{\text{L}} = h dr = \pm df/2$	⁴ Interestingly, in the gauge $\underline{\vartheta}^{\hat{0}} = 3\kappa d\theta_{\text{L}} = h dr = \pm df/2$, such instantons are solutions to the
mielke-geometrodynamics_F00665	136	■ 0.991	$T^{\hat{0}} = 0$	$T^{\hat{0}} = 0$	topological Eq. (6.7.12), due to $T^{\hat{0}} = 0$.
mielke-geometrodynamics_F00666	137	■ 1.000	$A = A_{\mu}^I \lambda_I$	$A = A_{\mu}^I \lambda_I$	From the <i>holonomy</i> of a Lie-algebra-valued connection one-form $A = A_{\mu}^I \lambda_I$
mielke-geometrodynamics_F00667	137	■ 0.967	$dx^{\mu} \in \mathscr{C}$	$dx^{\mu} \in \mathscr{C}$	$dx^{\mu} \in \mathscr{C}$, one can construct a <i>gauge-invariant</i> operator, or character, along the loop
mielke-geometrodynamics_F00668	137	■ 1.000	$\gamma^{\mu}(s)$	$\gamma^{\mu}(s)$	with coordinates $\gamma^{\mu}(s)$. Then the Wilson loop can be written as

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mielke-geometrodynamics_F00669	137	0.826	$\Psi(A)$	$\Psi(A)$	The state vector $\Psi(A)$ for the gauge potential can be transformed via the Wilson
mielke-geometrodynamics_F00670	137	1.000	$\mathscr{C} / \mathscr{G}$	$\mathscr{C} / \mathscr{G}$	The integration is performed over a coset space $\mathscr{C} / \mathscr{G}$ in order to factor out gauge-
mielke-geometrodynamics_F00671	137	1.000	$G(\gamma)$	$G(\gamma)$	of the Wilson loop is the <i>self-linking number</i> $G(\gamma)$ of Gauss. It is interesting to see
mielke-geometrodynamics_F00672	137	1.000	$V(\gamma)$	$V(\gamma)$	of the Jones polynomial $V(\gamma)$. If we could formally transform these results of lattice
mielke-geometrodynamics_F00673	138	1.000	$ \Psi\rangle$	$ \Psi\rangle$	state vector $ \Psi\rangle$ of the universe would be characterized by knot invariants, such as
mielke-geometrodynamics_F00674	138	1.000	$\mathbb{R}^4$	$\mathbb{R}^4$	group $\mathbb{R}^4$ as structure group. Then, instead of Wilson-type loops, we encounter
mielke-geometrodynamics_F00675	138	0.979	$\left. n^\alpha \right _{\vartheta^\alpha}$	$n^\alpha = n _{\vartheta^\alpha}$	GUADAGNINI 1993) via one of the orthonormal legs $n^\alpha = n _{\vartheta^\alpha}$. In view of (6.4.25),
mielke-geometrodynamics_F00676	138	1.000	$(1/16\pi^2)$	$(1/16\pi^2)$	It is well known from differential geometry that the Euler number $(1/16\pi^2)$
mielke-geometrodynamics_F00677	138	1.000	$\int R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}$	$\int R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}$	$\int R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}$ and the Pontryagin integral $(1/8\pi^2) \int R_\alpha{}^\beta \wedge R_\beta{}^\alpha$ bear an intimate
mielke-geometrodynamics_F00678	138	1.000	$(1/8\pi^2) \int R_\alpha{}^\beta \wedge R_\beta{}^\alpha$	$(1/8\pi^2) \int R_\alpha{}^\beta \wedge R_\beta{}^\alpha$	$\int R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}$ and the Pontryagin integral $(1/8\pi^2) \int R_\alpha{}^\beta \wedge R_\beta{}^\alpha$ bear an intimate
mielke-geometrodynamics_F00679	138	1.000	ϵ	ϵ	to the following argument: For a real constant ϵ with $0 \leq \epsilon \leq 1$, the RC connec-
mielke-geometrodynamics_F00680	138	1.000	$0 \leq \epsilon \leq 1$	$0 \leq \epsilon \leq 1$	to the following argument: For a real constant ϵ with $0 \leq \epsilon \leq 1$, the RC connec-
mielke-geometrodynamics_F00681	138	0.973	$\Gamma_{\alpha\beta}(\epsilon) = \Gamma_{\alpha\beta}^{\{\}} - \epsilon K_{\alpha\beta}$	$\Gamma_{\alpha\beta}(\epsilon) = \Gamma_{\alpha\beta}^{\{\}} - \epsilon K_{\alpha\beta}$	to $\Gamma_{\alpha\beta}(\epsilon) = \Gamma_{\alpha\beta}^{\{\}} - \epsilon K_{\alpha\beta}$ such that $\Gamma_{\alpha\beta}(0) = \Gamma_{\alpha\beta}^{\{\}}$. Provided a topological quantity
mielke-geometrodynamics_F00682	138	0.973	$\Gamma_{\alpha\beta}(0) = \Gamma_{\alpha\beta}^{\{\}}$	$\Gamma_{\alpha\beta}(0) = \Gamma_{\alpha\beta}^{\{\}}$	to $\Gamma_{\alpha\beta}(\epsilon) = \Gamma_{\alpha\beta}^{\{\}} - \epsilon K_{\alpha\beta}$ such that $\Gamma_{\alpha\beta}(0) = \Gamma_{\alpha\beta}^{\{\}}$. Provided a topological quantity
mielke-geometrodynamics_F00683	138	1.000	$\Gamma_{\alpha\beta}$	$\Gamma_{\alpha\beta}$	depends on the connection $\Gamma_{\alpha\beta}$, it must be invariant under this continuous defor-
mielke-geometrodynamics_F00684	138	0.679	$T := e_\alpha T^\alpha = d\sigma$	$T := e_\alpha T^\alpha = d\sigma$	torsion $T := e_\alpha T^\alpha = d\sigma$ derived from a scalar field via a gradient (GREGORASH &
mielke-geometrodynamics_F00685	138	1.000	N	N	manifold, which may enclose the singularity line of the dislocation N times, yields
mielke-geometrodynamics_F00686	139	1.000	$\sigma =$	$\sigma =$	provided the scalar field is not single-valued, as, for instance, in the case of $\sigma =$
mielke-geometrodynamics_F00687	139	0.997	$\arcsin(x)$	$\arcsin(x)$	<u>$\arcsin(x)$</u> .
mielke-geometrodynamics_F00688	139	1.000	$\pi_3(SO(4)) =$	$\pi_3(SO(4)) =$	nontrivial winding number will occur as a result of the homotopy group <u>$\pi_3(SO(4)) \ni$</u>
mielke-geometrodynamics_F00689	139	1.000	$\mathbb{Z} \oplus \mathbb{Z}$	$\mathbb{Z} \oplus \mathbb{Z}$	<u>$\mathbb{Z} \oplus \mathbb{Z}$</u> . On the other hand, the 3D hypersurface admits the intriguing mapping $\ell \vartheta^A \rightarrow$
mielke-geometrodynamics_F00690	139	1.000	$\ell \vartheta^A \rightarrow$	$\ell \vartheta^A \rightarrow$	$\mathbb{Z} \oplus \mathbb{Z}$. On the other hand, the 3D hypersurface admits the intriguing mapping $\ell \vartheta^A \rightarrow$
mielke-geometrodynamics_F00691	139	0.999	$\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$	$\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$	$\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$ of the triads to the corresponding three-dual of the $\mathfrak{su}(2)$ -valued
mielke-geometrodynamics_F00692	139	0.999	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	$\underline{\Gamma}^{\star A} := \frac{1}{2} \eta^{ABC} \underline{\Gamma}_{BC}$ of the triads to the corresponding three-dual of the $\mathfrak{su}(2)$ -valued
mielke-geometrodynamics_F00693	139	0.718	$\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}} + \Lambda' \underline{\eta}$	$\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}} + \Lambda' \underline{\eta}$	connection (BAEKLER et al. 1992). Since this implies a mapping $\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}} + \Lambda' \underline{\eta}$
mielke-geometrodynamics_F00694	139	1.000	$\mathbb{R} \times W^3$	$\mathbb{R} \times W^3$	small scale of “quantum gravity,” the underlying spacetime manifold $\mathbb{R} \times W^3$ may
mielke-geometrodynamics_F00695	139	1.000	dj_5	dj_5	In classical EC theory, the net axial current production dj_5 seems (CHANG &

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mielke-geometrodynamics_F00696	139	■ 1.000	$d \, \mathscr{A} \wedge d \, \mathscr{A}$	$d\mathscr{A} \wedge d\mathscr{A}$	$d\mathscr{A} \wedge d\mathscr{A}$ involving the axial torsion. Since torsion instantons are characterized
mielke-geometrodynamics_F00697	140	■ 0.685	$\vartheta_{\mathrm{T}}=2/\gamma$	$\theta_{\mathrm{T}}=2/\gamma$	⁵ The translational angle $\theta_{\mathrm{T}}=2/\gamma$ is at times identified (FREIDEL et al. 2005) with the inverse
mielke-geometrodynamics_F00698	140	■ 0.985	$d \, C_{\mathrm{TT}}$	dC_{TT}	lational Chern–Simons term dC_{TT} were considered earlier by MIELKE (1992).
mielke-geometrodynamics_F00699	141	■ 1.000	$\vartheta_{[\beta} \wedge H_{\alpha]}$	$\vartheta_{[\beta} \wedge H_{\alpha]}$	Note that the angular momentum part $\vartheta_{[\beta} \wedge H_{\alpha]}$ of the gauge fields induced by
mielke-geometrodynamics_F00700	141	■ 1.000	$D \, H_{\alpha\beta}$	$DH_{\alpha\beta}$	the NY term cancels identically against one part of $DH_{\alpha\beta}$. In the case of Dirac fields
mielke-geometrodynamics_F00701	141	■ 1.000	μ_{α}	μ_{α}	(MIELKE 2004), the <i>spin–energy potential</i> μ_{α} , a two-form, is related to the axial
mielke-geometrodynamics_F00702	141	■ 1.000	$j_5=\bar{\psi}^*\gamma_5\psi$	$j_5=\bar{\psi}^*\gamma_5\psi$	current three-form $j_5=\bar{\psi}^*\gamma_5\psi$ via
mielke-geometrodynamics_F00703	141	■ 1.000	$D \, \vartheta_{\mathrm{L}} \neq 0$	$D\theta_{\mathrm{L}} \neq 0$	that (6.7.6) for $D\theta_{\mathrm{L}} \neq 0$ provides torsion with a dynamical coupling to RC curva-
mielke-geometrodynamics_F00704	141	■ 1.000	$\mu_{\alpha} \wedge \vartheta^{\alpha}=0$	$\mu_{\alpha} \wedge \vartheta^{\alpha}=0$	i.e., $\mu_{\alpha} \wedge \vartheta^{\alpha}=0$, in view of (6.7.8).
mielke-geometrodynamics_F00705	141	■ 0.726	$\left. T=e_{\alpha}\right\rfloor T^{\alpha}$	$T=e_{\alpha}\rfloor T^{\alpha}$	where the vector torsion one-form $T=e_{\alpha}\rfloor T^{\alpha}$ enters as an intermediate source. Due
mielke-geometrodynamics_F00706	141	■ 0.815	$D \, D \, \vartheta=d \, d \, \vartheta \equiv 0$	$DD\vartheta=dd\vartheta\equiv 0$	to the Poincaré lemma $DD\vartheta=dd\vartheta\equiv 0$ for a (pseudo) scalar field, Eq. (6.7.10) has
mielke-geometrodynamics_F00707	141	■ 1.000	ϑ^{β}	ϑ^{β}	as a first integral. After a contraction with ϑ^{β} , the dual of the axial torsion one-form
mielke-geometrodynamics_F00708	141	■ 0.996	$\mathscr{A}:=*\left(\vartheta^{\alpha}\wedge T_{\alpha}\right)$	$\mathscr{A}:=*(\vartheta^{\alpha}\wedge T_{\alpha})$	$\mathscr{A}:=*(\vartheta^{\alpha}\wedge T_{\alpha})$, i.e., the translational CS term, turns out to be related to the volume
mielke-geometrodynamics_F00709	141	■ 0.999	ϑ_{L}	θ_{L}	a coupling to a kinetic term arising from the pseudoscalar field θ_{L} rescaling the
mielke-geometrodynamics_F00710	141	■ 1.000	$T^{[\gamma}\wedge T^{\mu]}\eta_{\alpha\beta\gamma\mu}=0$	$T^{[\gamma}\wedge T^{\mu]}\eta_{\alpha\beta\gamma\mu}=0$	$T^{[\gamma}\wedge T^{\mu]}\eta_{\alpha\beta\gamma\mu}=0$. Again we end up with a first-order equation for torsion, where,
mielke-geometrodynamics_F00711	142	■ 1.000	$D \, H$	DH	Part of the sources for the Yang–Mills-type divergence terms “ DH ” are the mate-
mielke-geometrodynamics_F00712	143	■ 1.000	L_{g}	L_g	aid of the coframe ϑ^{α} , in general gives us back the gauge Lagrangian L_g amended
mielke-geometrodynamics_F00713	143	■ 1.000	$\left\{ \right\} ^{(i)}T^{\alpha}$	$^{(i)}T^{\alpha}$	Lagrangian, which is at most <i>quadratic</i> in the irreducible pieces $^{(i)}T^{\alpha}$ and $^{(j)}R^{\alpha\beta}$
mielke-geometrodynamics_F00714	143	■ 1.000	$\left\{ \right\} ^{(j)}R^{\alpha\beta}$	$^{(j)}R^{\alpha\beta}$	Lagrangian, which is at most <i>quadratic</i> in the irreducible pieces $^{(i)}T^{\alpha}$ and $^{(j)}R^{\alpha\beta}$
mielke-geometrodynamics_F00715	143	■ 1.000	$\ell\simeq 10^{-32}\,\mathrm{cm}$	$\ell\simeq 10^{-32}\,\mathrm{cm}$	The gravitational coupling constant is absorbed in the Planck length $\ell\simeq 10^{-32}\,\mathrm{cm}$,
mielke-geometrodynamics_F00716	143	■ 1.000	$a_{\mathrm{0}},\,a_{\mathrm{i}}(i=1,2,3),\,\kappa$	$a_0,a_i(i=1,2,3),\kappa$	whereas a_0 , a_i ($i=1,2,3$), κ , and b_j ($j=1,\dots,6$) are dimensionless coupling
mielke-geometrodynamics_F00717	143	■ 1.000	$b_{\mathrm{j}}(j=1,\,\ldots,\,6)$	$b_j(j=1,\dots,6)$	whereas a_0 , a_i ($i=1,2,3$), κ , and b_j ($j=1,\dots,6$) are dimensionless coupling
mielke-geometrodynamics_F00718	144	■ 1.000	$C:Q\rightarrow Q^{\prime}=-Q$	$C:Q\rightarrow Q'=-Q$	a charge conjugation $C:Q\rightarrow Q'=-Q$, one obtains the same gravitational field.
mielke-geometrodynamics_F00719	144	■ 1.000	C	C	Maxwell equations are invariant under CPT, charge conjugation C has to be accom-
mielke-geometrodynamics_F00720	144	■ 0.963	T	T	panied by an additional time reversal T (particles moving backward in time $\hat{=}$ antipar-
mielke-geometrodynamics_F00721	144	■ 0.963	$\stackrel{\wedge}{=}$	$\hat{=}$	panied by an additional time reversal T (particles moving backward in time $\hat{=}$ antipar-
mielke-geometrodynamics_F00722	144	■ 1.000	$a_{\mathrm{0}}=0,\,a_{\mathrm{i}}=1$	$a_0=0,a_i=1$	by the choice $a_0=0$, $a_i=1$, and $b_j=1$. In order to obtain its complexified version,

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mielke-geometrodynamics_F00723	144	■ 1.000	$b_{\{j\}}=1$	$b_j = 1$	by the choice $a_0 = 0$, $a_i = 1$, and $b_j = 1$. In order to obtain its complexified version,
mielke-geometrodynamics_F00724	144	■ 1.000	θ_{T}	θ_{T}	parameters θ_{T} and θ_{L} :
mielke-geometrodynamics_F00725	145	■ 1.000	$P: \vartheta^{\prime A} = -\vartheta^A$	$P: \vartheta^{\prime A} = -\vartheta^A$	are invariant under space reflections $\overline{P: \vartheta^{\prime A} = -\vartheta^A}$, where $A = 1, 2, 3$, due to the
mielke-geometrodynamics_F00726	145	■ 1.000	$A=1,2,3$	$A = 1, 2, 3$	are invariant under space reflections $P: \vartheta^{\prime A} = -\vartheta^A$, where $A = 1, 2, 3$, due to the
mielke-geometrodynamics_F00727	145	■ 1.000	$\eta_{\{\alpha \beta \gamma \delta\}}$	$\eta_{\alpha\beta\gamma\delta}$	occurrence of the determinant of the metric in $\eta_{\alpha\beta\gamma\delta}$. In contradistinction to these
mielke-geometrodynamics_F00728	145	■ 1.000	$R_{\{\alpha\}\ }^{\{\beta\}} \wedge R_{\{\beta\}\ }^{\{\alpha\}}$	$R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}$	The term $R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}$ is known to yield, after integration over spacetime, a topo-
mielke-geometrodynamics_F00729	145	■ 1.000	$\langle 0 R^{\alpha\beta} 0 \rangle = 0$	$\langle 0 R^{\alpha\beta} 0 \rangle = 0$	Even above the Planck scale, we can at most require $\langle 0 R^{\alpha\beta} 0 \rangle = \mathbf{0}$, but we could
mielke-geometrodynamics_F00730	151	■ 0.938	Ω_{C}	Ω_{C}	Let Ω_{C} be the 2-form of a generalized curvature tensor (NOMIZU 1972) that is
mielke-geometrodynamics_F00731	151	■ 0.929	$\operatorname{SL}(2, \mathbb{C}) \approx \widetilde{\operatorname{SO}}(3,1)$	$\operatorname{SL}(2, \mathbb{C}) \approx \widetilde{\operatorname{SO}}(3,1)$	¹ The bispinor representation of the covering group $\operatorname{SL}(2, \mathbb{C}) \approx \operatorname{SO}(3,1)$ is inapt for this purpose,
mielke-geometrodynamics_F00732	151	■ 1.000	$\sigma_{\{\alpha \beta\}} = \frac{1}{2} [\gamma_{\alpha}, \gamma_{\beta}]$	$\sigma_{\alpha\beta} = \frac{1}{2} [\gamma_{\alpha}, \gamma_{\beta}]$	since its infinitesimal generators $\sigma_{\alpha\beta} = \frac{1}{2} [\gamma_{\alpha}, \gamma_{\beta}]$ are neither self-dual nor anti-self-dual. As a direct
mielke-geometrodynamics_F00733	151	■ 1.000	$\sigma_{\{\alpha \beta\}}^{(\star)} = \gamma^5 \sigma_{\alpha\beta}$	$\sigma_{\alpha\beta}^{(\star)} = \gamma^5 \sigma_{\alpha\beta}$	sum of the spinor and its conjugate representation, they rather satisfy the relation $\sigma_{\alpha\beta}^{(\star)} = \gamma^5 \sigma_{\alpha\beta}$,
mielke-geometrodynamics_F00734	151	■ 0.829	$\gamma^5 := \frac{1}{4!} \varepsilon_{\mathrm{abcd}} \gamma^{\mathrm{a}} \gamma^{\mathrm{b}} \gamma^{\mathrm{c}} \gamma^{\mathrm{d}}$	$\gamma^5 := \frac{1}{4!} \varepsilon_{\mathrm{abcd}} \gamma^{\mathrm{a}} \gamma^{\mathrm{b}} \gamma^{\mathrm{c}} \gamma^{\mathrm{d}}$	whereby the matrix $\gamma^5 := \frac{1}{4!} \varepsilon_{\mathrm{abcd}} \gamma^{\mathrm{a}} \gamma^{\mathrm{b}} \gamma^{\mathrm{c}} \gamma^{\mathrm{d}}$ is, within the Clifford algebra, an axial Lorentz scalar.
mielke-geometrodynamics_F00735	152	■ 1.000	$s=0$	$s = 0$	In a four-dimensional Riemannian manifold with the Euclidean signature $s = 0$,
mielke-geometrodynamics_F00736	153	■ 1.000	$R_{\{\alpha \beta\}}$	$R_{\alpha\beta}$	where $R_{\alpha\beta}$ is the curvature two-form. In WEYL (1919), in the concluding chapter
mielke-geometrodynamics_F00737	153	■ 0.813	$\Gamma^{\{\alpha \beta\}} := \Gamma_i^{\alpha\beta} dx^i$	$\Gamma^{\alpha\beta} := \Gamma_i^{\alpha\beta} dx^i$	$\Gamma^{\alpha\beta} := \Gamma_i^{\alpha\beta} dx^i$. In his purely affine approach, Yang did not consider the canonical
mielke-geometrodynamics_F00738	153	■ 0.977	$\vartheta^{\{\alpha\}} = E_{\{j\}}^{\{\alpha\}} dx^{\{j\}}$	$\vartheta^{\alpha} = E_j^{\alpha} dx^j$	with respect to the coframe $\vartheta^{\alpha} = E_j^{\alpha} dx^j$ dual to the local anholonomic frame e_{α} .
mielke-geometrodynamics_F00739	153	■ 0.977	$e_{\{\alpha\}}$	e_{α}	with respect to the coframe $\vartheta^{\alpha} = E_j^{\alpha} dx^j$ dual to the local anholonomic frame e_{α} .
mielke-geometrodynamics_F00740	154	■ 1.000	$G_{\{i \ j\}}$	G_{ij}	which is dual to the usual Einstein tensor G_{ij} in Riemannian spacetime. In exterior
mielke-geometrodynamics_F00741	154	■ 0.494	$\operatorname{Ric}_{\{\alpha \beta\}} := (-1)^s * \left(R_{(\alpha}^{\delta} \wedge \eta_{\delta \beta)} \right)$	$\operatorname{Ric}_{\alpha\beta} := (-1)^s * \left(R_{(\alpha}^{\delta} \wedge \eta_{\delta \beta)} \right)$	$\operatorname{Ric}_{\alpha\beta} := (-1)^s * (R_{(\alpha}^{\delta} \wedge \eta_{\delta \beta)})$.
mielke-geometrodynamics_F00742	154	■ 1.000	$\tau_{\{\alpha \beta\}} = \vartheta_{[\alpha} \wedge$	$\tau_{\alpha\beta} = \vartheta_{[\alpha} \wedge$	Macroscopically, the matter coupling ² to the spin current three-form $\tau_{\alpha\beta} = \vartheta_{[\alpha} \wedge$
mielke-geometrodynamics_F00743	154	■ 0.998	$\mu_{\{\beta\}}$	$\mu_{\beta}]$	$\mu_{\beta}]$ seems to disfavor such models, a criticism which has already raised by FAIRCHILD
mielke-geometrodynamics_F00744	154	■ 0.589	$\Gamma^{\{\beta \alpha \beta\}} \propto \partial g$	$\Gamma^{\{\beta \alpha \beta\}} \propto \partial g$	In Riemannian spacetime, with its Levi–Civita connection $\Gamma^{ \alpha\beta} \propto \partial g$, these
mielke-geometrodynamics_F00745	154	■ 1.000	$\{ \}^{(\star)}$	$(\star)$	and the Lie dual $^{(\star)}$ of the curvature $R_{\alpha\beta}$ in a space(time) of signature s .
mielke-geometrodynamics_F00746	154	■ 1.000	$L_{\{\text{SKY}\}}^{(\star)} = 2 L_{\{\text{SKY}\}}$	$L_{\mathrm{SKY}}^{(\star)} = 2 L_{\mathrm{SKY}}$	i.e., they lead to $L_{\mathrm{SKY}}^{(\star)} = 2 L_{\mathrm{SKY}}$. Both second-order duality conditions satisfy Yang’s
mielke-geometrodynamics_F00747	154	■ 1.000	$j_{\{5\}} := \bar{\psi}^{\{\star\}} \gamma_{\{5\}} \psi$	$j_5 := \bar{\psi}^{\star} \gamma_5 \psi$	current $\overline{j_5} := \bar{\psi}^{\star} \gamma_5 \psi$.
mielke-geometrodynamics_F00748	155	■ 1.000	$D \eta_{\{\alpha \beta\}} = 0$	$D \eta_{\alpha\beta} = 0$	its Lie dual, due to $D \eta_{\alpha\beta} = 0$ in a Riemannian spacetime. More on this “vacuum

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mielke-geometrodynamics_F00749	155	■ 1.000	$\backslash\mathrm{zeta}=\backslash\mathrm{pm}\ 1$	$\zeta = \pm 1$	whereby only $\boxed{\xi = \pm 1}$ (and zero) are admitted.
mielke-geometrodynamics_F00750	155	■ 1.000	$\backslash\mathrm{zeta}=+1$	$\zeta = +1$	$\boxed{\xi = +1}$: <i>self-double-dual curvature</i>
mielke-geometrodynamics_F00751	155	■ 1.000	$\backslash\mathrm{zeta}=-1$	$\zeta = -1$	$\boxed{\xi = -1}$: <i>anti-self-double-dual curvature</i>
mielke-geometrodynamics_F00752	155	■ 0.999	$\backslash\mathrm{mathscr{A}}:={\ }^*\left(\vartheta_{\alpha}\wedge T^{\alpha}\right)$	$\mathscr{A} := {}^*(\vartheta_{\alpha} \wedge T^{\alpha})$	Riemannian curvature $R_{\alpha\beta}^{()}$ and the axial torsion one-form $\mathscr{A} := {}^*(\vartheta_{\alpha} \wedge \dot{T}^{\alpha})$ is moti-
mielke-geometrodynamics_F00753	155	■ 0.638	$\backslash\mathrm{left}\backslash\mathrm{angle}\ d\ j_5\backslash\mathrm{right}\backslash\mathrm{rangle}=2\ i^{*}\ m\backslash\mathrm{left}\backslash\mathrm{angle}\ \backslash\mathrm{bar}\{\psi\}\ \backslash\mathrm{gamma}_5\ \backslash\mathrm{psi}\backslash\mathrm{right}\backslash\mathrm{rangle}-\backslash\mathrm{left}(R_{\alpha}\backslash\mathrm{beta}\wedge\{\ \}\}\ \backslash\mathrm{wedge}\ R^{\{\backslash j\ \backslash\alpha\ \backslash\mathrm{beta}\}+\backslash\mathrm{frac}\{1\}\{2\}}\ d\ \backslash\mathrm{mathscr{A}}\ \backslash\mathrm{wedge}\ d\ \backslash\mathrm{mathscr{A}}\backslash\mathrm{right})\ /\$	$\langle dj_5 \rangle = 2i^*m \langle \bar{\psi} \gamma_5 \psi \rangle - \left(R_{\alpha\beta}^{()} \wedge R^{\{j\alpha\beta} + \frac{1}{2} d\mathscr{A} \wedge d\mathscr{A} \right) /$	vated by the <i>axial anomaly</i> $\langle dj_5 \rangle = 2i^*m \langle \bar{\psi} \gamma_5 \psi \rangle - (R_{\alpha\beta}^{()} \wedge R^{() \alpha\beta} + \frac{1}{2} d\mathscr{A} \wedge d\mathscr{A}) /$
mielke-geometrodynamics_F00754	155	■ 1.000	$48\ \backslash\mathrm{pi}^{\{2\}}$	$48\pi^2$	$48\pi^2$ in the coupling to Dirac fields ψ .
mielke-geometrodynamics_F00755	156	■ 1.000	$d^{\{4\}}\ x$	d^4x	arises, where the four-dimensional integration over d^4x is already implied in our
mielke-geometrodynamics_F00756	156	■ 0.996	${\ }^*\mathrm{R}_{\alpha\beta}=\theta_{\mathrm{L}}^*\backslash\mathrm{theta}_{\mathrm{L}}^{\{\star\}}\ \mathrm{R}_{\alpha\beta}^{\{\backslash\star\}}+\backslash\mathrm{left}(\backslash\mathrm{theta}_{\mathrm{T}}^*\backslash\mathrm{theta}_{\mathrm{T}}^{\{\star\}}\ \backslash\mathrm{Lambda}\ /\ 6\backslash\mathrm{right})\ \mathrm{eta}_{\alpha\beta}$	${}^*R_{\alpha\beta} = \theta_{\mathrm{L}}^* R_{\alpha\beta}^{(*)} + (\theta_{\mathrm{T}}^* \Lambda / 6) \eta_{\alpha\beta}$	1994) via a modified duality ${}^*R_{\alpha\beta} = \theta_{\mathrm{L}}^* R_{\alpha\beta}^{(*)} + (\theta_{\mathrm{T}}^* \Lambda / 6) \eta_{\alpha\beta}$, which, however, explic-
mielke-geometrodynamics_F00757	156	■ 0.998	$\backslash\mathrm{sigma}_{\alpha}:=\backslash\mathrm{Sigma}_{\alpha}\cdot D^{\{\ \backslash\ \}}\ \backslash\mathrm{mu}_{\alpha}$	$\sigma_{\alpha} := \Sigma_{\alpha} - D^{\{ \} } \mu_{\alpha}$	& MAGGIOLO 2005) symmetrized energy–momentum current $\sigma_{\alpha} := \Sigma_{\alpha} - D^{()}\mu_{\alpha}.$)
mielke-geometrodynamics_F00758	156	■ 0.682	$\backslash\mathrm{left}.\backslash\mathrm{left}.\mathrm{R}:=e_{\alpha}\backslash\mathrm{right}\backslash\mathrm{rfloor}\ e_{\beta}\backslash\mathrm{right}\backslash\mathrm{rfloor}\ R^{\{\alpha\beta\}}$	$R := e_{\alpha} \rfloor e_{\beta} \rfloor R^{\alpha\beta}$	trace with $R := e_{\alpha} \rfloor e_{\beta} \rfloor R^{\alpha\beta}$ as irreducible pieces. Then the double-dual curvature
mielke-geometrodynamics_F00759	157	■ 1.000	$o_{ij}^{\{\mathrm{Mink}\}}$	o_{ij}^{Mink}	with o_{ij}^{Mink} denoting the (pseudo)orthogonal Minkowski metric. Earlier, EINSTEIN &
mielke-geometrodynamics_F00760	157	■ 0.529	$D^{\{\backslash\ \backslash\ \}}\ G_{\alpha}^{\{\backslash\ \backslash\ \}}\ \backslash\mathrm{equiv}\ 0$	$D^{\{ \} } G_{\alpha}^{\{ \} } \equiv 0$	Due to the contracted Bianchi identity $D^{() } G_{\alpha}^{() } \equiv 0$ in Riemannian spacetime, the
mielke-geometrodynamics_F00761	157	■ 0.999	$\mathrm{R}=-4\ \backslash\mathrm{Lambda}$	$R = -4\Lambda$	curvature scalar is constant, i.e., more precisely, $R = -4\Lambda$. Thus, (7.4.6) is equivalent
mielke-geometrodynamics_F00762	157	■ 1.000	$\backslash\mathrm{Lambda}\ \backslash\mathrm{simeq}\ 1\ /\ \backslash\mathrm{kappa}$	$\Lambda \simeq 1/\kappa$	with cosmological constant $\Lambda \simeq 1/\kappa$. It is crucial to note that the latter carries phys-
mielke-geometrodynamics_F00763	157	■ 1.000	$\mathrm{R}=0$	$R = 0$	sical Einstein–Fokker background due to $R = 0$ in double-dual Thompson spaces.
mielke-geometrodynamics_F00764	157	■ 1.000	$h_{\{i\ j\}}:=g_{\{i\ j\}}-g_{\{i\ j\}}^{\{\mathrm{dS}\}}$	$h_{ij} := g_{ij} - g_{ij}^{\mathrm{dS}}$	On the other hand, the linearized Einstein equations for $h_{ij} := g_{ij} - g_{ij}^{\mathrm{dS}}$ in a de
mielke-geometrodynamics_F00765	157	■ 1.000	$g_{\{i\ j\}}^{\{\mathrm{dS}\}}$	g_{ij}^{dS}	Sitter background g_{ij}^{dS} with a positive cosmological constant are
mielke-geometrodynamics_F00766	158	■ 1.000	$m\ /\ H$	m/H	de Sitter space if m/H is kept finite in the massless limit. For gravitons, the mass
mielke-geometrodynamics_F00767	158	■ 1.000	$m\ \backslash\mathrm{leq}\ 7\ \backslash\mathrm{times}\ 10^{\{-32\}}\ \backslash\mathrm{mathrm{eV}}$	$m \leq 7 \times 10^{-32} \mathrm{eV}$	(GOLDHABER & NIETO 2010) of $m \leq 7 \times 10^{-32} \mathrm{eV}$, but larger than the Hubble
mielke-geometrodynamics_F00768	158	■ 1.000	$H\ \backslash\mathrm{simeq}\ 10^{\{-33\}}\ \backslash\mathrm{mathrm{eV}}$	$H \simeq 10^{-33} \mathrm{eV}$	scale of $\boxed{H \sim 10^{-33} \mathrm{eV}}$.
mielke-geometrodynamics_F00769	158	■ 0.994	$\backslash\mathrm{Lambda}_{\{\mathrm{text}\ \mathrm{obs}\ \}}>0$	$\Lambda_{\mathrm{obs}} > 0$	is induced via the tiny cosmological constant $\Lambda_{\mathrm{obs}} > 0$ of our expanding universe.
mielke-geometrodynamics_F00770	158	■ 0.994	$1\ /\ g^{\{2\}}$	$1/g^2$	constant $1/g^2$ is ‘traded’ for a dimensional one. This ‘dimensional transmutation’ à
mielke-geometrodynamics_F00771	159	■ 0.662	$\backslash\mathrm{operatorname{SU}}(2)$	$SU(2)$	Originally, Yang and Mills formulated their model as an $SU(2)$ gauge theory with
mielke-geometrodynamics_F00772	159	■ 0.968	1974)	1974)	1974) is an $SO(1, 3)$ gauge theory, whose covering gauge group, after a Wick rotation
mielke-geometrodynamics_F00773	159	■ 1.000	$\backslash\mathrm{overline}\{S\ 0\}(4)\ \backslash\mathrm{cong}$	$\overline{SO}(4) \cong$	to Euclidean space in the path integral quantization, is locally isomorphic to $\overline{SO}(4) \cong$

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mielke-geometrodynamics_F00774	159	■ 1.000	$S U(2) \otimes S U(2)$	$SU(2) \otimes SU(2)$	$SU(2) \otimes SU(2)$. This doubling of the (internal) unitary structure group $SU(2)$ was
mielke-geometrodynamics_F00775	159	■ 0.719	$\kappa \Lambda_{\text{obs}} \simeq 10^{-123}$	$\kappa \Lambda_{\text{obs}} \simeq 10^{-123}$	occur at the tiny <i>dimensionless scale</i> of $\kappa \Lambda_{\text{obs}} \simeq 10^{-123}$, as suggested by observa-
mielke-geometrodynamics_F00776	159	■ 0.999	ΔL	ΔL	In perturbative quantum gravity (ALVAREZ 1989), there arise counterterms ΔL of
mielke-geometrodynamics_F00777	159	■ 0.995	$\operatorname{Ric}_{\alpha\beta} := \left(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)} \right)$	$\operatorname{Ric}_{\alpha\beta} := \left(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)} \right)$	version of the zero-form $\operatorname{Ric}_{\alpha\beta} := \left(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)} \right)$.
mielke-geometrodynamics_F00778	160	■ 1.000	$H_{\alpha\beta}$	$H_{\alpha\beta}$	Here the field momenta H_{α} and $H_{\alpha\beta}$ are understood as arising from a generating
mielke-geometrodynamics_F00779	160	■ 1.000	L_{eff}	L_{eff}	n-form G as part of some effective gauge Lagrangian L_{eff} that includes the countert-
mielke-geometrodynamics_F00780	160	■ 0.637	$n-2$	$n-2$	In our dynamical approach, the $(n-2)$ -forms H_{α} and $H_{\alpha\beta}$ will be gauge field
mielke-geometrodynamics_F00781	160	■ 1.000	$P := \mathbb{R}^4 \otimes$	$P := \mathbb{R}^4 \otimes$	tively. Due to the <i>semidirect product</i> structure of the Poincaré group $P := \mathbb{R}^4 \ltimes$
mielke-geometrodynamics_F00782	160	■ 0.945	$H^{\alpha\beta} \rightarrow \vartheta^{\alpha}$	$H^{\alpha\beta} \rightarrow \vartheta^{\alpha}$	$SO(1, 3)$, the gauge field momenta contribute to the gauge potentials via $H^{\alpha\beta} \rightarrow \vartheta^{\alpha}$
mielke-geometrodynamics_F00783	160	■ 1.000	$H^{\alpha} \rightarrow \Gamma^{\alpha\beta}$	$H^{\alpha} \rightarrow \Gamma^{\alpha\beta}$	and $H^{\alpha} \rightarrow \Gamma^{\alpha\beta}$ in the FR (7.7.2, 7.7.3) just in an <i>intertwined</i> manner.
mielke-geometrodynamics_F00784	160	■ 0.991	$\{ \}^* \mathbb{D} H_{\alpha\beta}$	$*DH_{\alpha\beta}$	similarly as for Yang–Mills fields, a term proportional to $*DH_{\alpha\beta}$. However, “on
mielke-geometrodynamics_F00785	160	■ 0.388	$\{ \}^* \left(H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$	$* \left(H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$	$* \left(H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$ due to an algebraic identity. In the FR (7.7.2) of the coframe,
mielke-geometrodynamics_F00786	160	■ 0.388	H^{β}	H^{β}	$* \left(H^{\beta} \wedge \vartheta_{\alpha} \right) \equiv e_{\alpha}$ due to an algebraic identity. In the FR (7.7.2) of the coframe,
mielke-geometrodynamics_F00787	160	■ 0.644	$\{ \}^* \mathbb{D} H^{\alpha} \cong E^{\alpha}$	$*DH^{\alpha} \cong *E^{\alpha}$	the same situation arises, with the modification that the “on shell” term $*DH^{\alpha} \cong *E^{\alpha}$
mielke-geometrodynamics_F00788	160	■ 1.000	$\widetilde{G}$	$\widetilde{G}$	order generation functional $\widetilde{G}$. When coupling to matter, FRs have to be handled with
mielke-geometrodynamics_F00789	161	■ 1.000	$\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$	$\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$	momenta $\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$ can be expanded as $\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$. (For the
mielke-geometrodynamics_F00790	161	■ 1.000	$\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$	$\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$	momenta $\widetilde{H}_{\alpha\beta} := -\partial \widetilde{L} / \partial R^{\alpha\beta}$ can be expanded as $\widetilde{H}_{\alpha\beta} = \overset{(2)}{H}_{\alpha\beta} + \overset{(4)}{H}_{\alpha\beta} + \dots$. (For the
mielke-geometrodynamics_F00791	161	■ 1.000	$\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$	$\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$	time being, the field momentum $\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$ conjugate to the torsion $T^{\alpha} :=$
mielke-geometrodynamics_F00792	161	■ 1.000	$T^{\alpha} :=$	$T^{\alpha} :=$	time being, the field momentum $\widetilde{H}_{\alpha} := -\partial \widetilde{L} / \partial T^{\alpha}$ conjugate to the torsion $T^{\alpha} :=$
mielke-geometrodynamics_F00793	161	■ 1.000	$\mathbb{D} \vartheta^{\alpha}$	$D\vartheta^{\alpha}$	$D\vartheta^{\alpha}$ will be set to zero.)
mielke-geometrodynamics_F00794	161	■ 1.000	$\widetilde{H}_{\alpha\beta\mu\nu}$	$\widetilde{H}_{\alpha\beta\mu\nu}$	For a <i>fixed point</i> of the transformation (7.7.9), the Hessian $\widetilde{H}_{\alpha\beta\mu\nu}$ obviously has
mielke-geometrodynamics_F00795	161	■ 1.000	$\partial \widetilde{H}_{\mu\nu} / \partial R^{\alpha\beta} = 0$	$\partial \widetilde{H}_{\mu\nu} / \partial R^{\alpha\beta} = 0$	to vanish. This condition, i.e., $\partial \widetilde{H}_{\mu\nu} / \partial R^{\alpha\beta} = 0$, can be readily solved.
mielke-geometrodynamics_F00796	161	■ 1.000	$\theta_{\text{T}} R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}$	$\theta_{\text{T}} R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}$	If parity-violating terms such as $\theta_{\text{T}} R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}$ arising from the Nieh–Yan
mielke-geometrodynamics_F00797	162	■ 1.000	$\widetilde{L}$	$\widetilde{L}$	Lagrangian $\widetilde{L}$.
mielke-geometrodynamics_F00798	162	■ 1.000	$\eta_{\alpha\beta}$	$\eta_{\alpha\beta}$	Accordingly, $\eta_{\alpha\beta}$ and $\widetilde{H}_{\alpha\beta}$ interchange their roles as generalized coordinates and
mielke-geometrodynamics_F00799	162	■ 1.000	$\widetilde{H}_{\alpha\beta}$	$\widetilde{H}_{\alpha\beta}$	Accordingly, $\eta_{\alpha\beta}$ and $\widetilde{H}_{\alpha\beta}$ interchange their roles as generalized coordinates and

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mielke-geometrodynamics_F00822	163	■ 1.000	$\operatorname{Tr}(\left(\gamma_{\alpha} \gamma_{\beta}\right)=4 o_{\alpha \beta}$	$\operatorname{Tr}\left(\gamma_{\alpha} \gamma_{\beta}\right)=4 o_{\alpha \beta}$	They are normalized by the traces $\operatorname{Tr}\left(\gamma_{\alpha} \gamma_{\beta}\right)=4 o_{\alpha \beta}$ and $\operatorname{Tr}\left(\sigma_{\alpha \beta} \sigma^{\gamma \delta}\right)=8 \delta_{[\alpha}^{\gamma} \delta_{\beta]}^{\delta}$,
mielke-geometrodynamics_F00823	163	■ 1.000	$\operatorname{Tr}(\left(\sigma_{\alpha \beta} \sigma^{\gamma \delta}\right)=8 \delta_{[\alpha}^{\gamma} \delta_{\beta]}^{\delta}$	$\operatorname{Tr}\left(\sigma_{\alpha \beta} \sigma^{\gamma \delta}\right)=8 \delta_{[\alpha}^{\gamma} \delta_{\beta]}^{\delta}$	They are normalized by the traces $\operatorname{Tr}\left(\gamma_{\alpha} \gamma_{\beta}\right)=4 o_{\alpha \beta}$ and $\operatorname{Tr}\left(\sigma_{\alpha \beta} \sigma^{\gamma \delta}\right)=8 \delta_{[\alpha}^{\gamma} \delta_{\beta]}^{\delta}$,
mielke-geometrodynamics_F00824	163	■ 1.000	$[\alpha \beta]=\frac{1}{2}(\alpha \beta-\beta \alpha)$	$[\alpha \beta]=\frac{1}{2}(\alpha \beta-\beta \alpha)$	where $[\alpha \beta]=\frac{1}{2}(\alpha \beta-\beta \alpha)$ denotes the antisymmetrization of indices. For the unit
mielke-geometrodynamics_F00825	164	■ 0.988	Lie	Lie	whereas for the <i>Lie</i> (or right) dual $\sigma_{\alpha \beta}^{(*)}:=\frac{1}{2} \sigma^{\gamma \delta} \eta_{\alpha \beta \gamma \delta}=i \gamma_5 \sigma_{\alpha \beta}$, we obtain the asso-
mielke-geometrodynamics_F00826	164	■ 0.988	$\sigma_{\alpha \beta}^{(*)}:=\frac{1}{2} \sigma^{\gamma \delta} \eta_{\alpha \beta \gamma \delta}=i \gamma_5 \sigma_{\alpha \beta}$	$\sigma_{\alpha \beta}^{(*)}:=\frac{1}{2} \sigma^{\gamma \delta} \eta_{\alpha \beta \gamma \delta}=i \gamma_5 \sigma_{\alpha \beta}$	whereas for the <i>Lie</i> (or right) dual $\sigma_{\alpha \beta}^{(*)}:=\frac{1}{2} \sigma^{\gamma \delta} \eta_{\alpha \beta \gamma \delta}=i \gamma_5 \sigma_{\alpha \beta}$, we obtain the asso-
mielke-geometrodynamics_F00827	164	■ 0.998	$\sigma^{(*)}$	$\sigma^{(*)}$	$\sigma^{(*)}$ solely by multiplication by γ_5 , which here is regarded as just an anticommuting
mielke-geometrodynamics_F00828	164	■ 0.998	γ_5	γ_5	$\sigma^{(*)}$ solely by multiplication by γ_5 , which here is regarded as just an anticommuting
mielke-geometrodynamics_F00829	164	■ 0.995	$12 i^{*} \gamma$	$12 i^{*} \gamma$	relates this to $12 i^{*} \gamma$, the Hodge dual ³ for the basis of Clifforms has been recovered.
mielke-geometrodynamics_F00830	164	■ 1.000	$P_{\pm}=\frac{1}{2}\left(1 \pm \gamma_5\right)$	$P_{\pm}=\frac{1}{2}\left(1 \pm \gamma_5\right)$	representation involves explicitly the <i>chirality projector</i> $P_{\pm}=\frac{1}{2}\left(1 \pm \gamma_5\right)$, obeying
mielke-geometrodynamics_F00831	164	■ 1.000	$P_{\pm} P_{\pm}=P_{\pm}$	$P_{\pm} P_{\pm}=P_{\pm}$	$P_{\pm} P_{\pm}=P_{\pm}$.
mielke-geometrodynamics_F00832	164	■ 0.981	$\gamma \rightarrow \widetilde{\gamma}=e^{\varphi} \gamma$	$\gamma \rightarrow \widetilde{\gamma}=e^{\varphi} \gamma$	(ATIYAH et al. 1978) under the Weyl rescaling $\gamma \rightarrow \widetilde{\gamma}=e^{\varphi} \gamma$, where φ can be viewed
mielke-geometrodynamics_F00833	164	■ 1.000	$\{ \}^{\#}$	$\#$	$\#$ on <i>arbitrary</i> two-forms allows us to <i>reconstruct</i> a spacetime metric h , which is,
mielke-geometrodynamics_F00834	164	■ 1.000	h	h	$\#$ on <i>arbitrary</i> two-forms allows us to <i>reconstruct</i> a spacetime metric h , which is,
mielke-geometrodynamics_F00835	164	■ 0.524	$\{ \}^{* *}=-1$	$^{* *}=-1$	³ For Minkowski signature (sign = 1), the Hodge dual satisfies * * = -1 , and therefore, i^{*} is used
mielke-geometrodynamics_F00836	164	■ 0.524	i^{*}	i^{*}	³ For Minkowski signature (sign = 1), the Hodge dual satisfies $^{* *}=-1$, and therefore, i^{*} is used
mielke-geometrodynamics_F00837	165	■ 1.000	$\Gamma:=\frac{i}{4} \Gamma^{\alpha \beta} \sigma_{\alpha \beta}$	$\Gamma:=\frac{i}{4} \Gamma^{\alpha \beta} \sigma_{\alpha \beta}$	of gravity. In terms of the Clifford-algebra-valued <i>connection</i> $\Gamma:=\frac{i}{4} \Gamma^{\alpha \beta} \sigma_{\alpha \beta}$, the
mielke-geometrodynamics_F00838	165	■ 1.000	$\overline{S O}_0(1,3) \cong S L(2, C)$	$\overline{S O}_o(1,3) \cong S L(2, C)$	$\overline{S O}_o(1,3) \cong S L(2, C)$ -covariant exterior derivative
mielke-geometrodynamics_F00839	165	■ 0.975	$[\Psi, \Phi]:=\Psi \wedge \Phi-(-1)^{p q}$	$[\Psi, \Phi]:=\Psi \wedge \Phi-(-1)^{p q}$	of p-forms employs the algebra-valued <i>form commutator</i> $[\Psi, \Phi]:=\Psi \wedge \Phi-(-1)^{p q}$
mielke-geometrodynamics_F00840	165	■ 0.755	$\Phi \wedge \Psi$	$\Phi \wedge \Psi$	$\Phi \wedge \Psi$.
mielke-geometrodynamics_F00841	165	■ 1.000	$K:=\frac{i}{4} K^{\alpha \beta} \sigma_{\alpha \beta}$	$K:=\frac{i}{4} K^{\alpha \beta} \sigma_{\alpha \beta}$	involves $K:=\frac{i}{4} K^{\alpha \beta} \sigma_{\alpha \beta}$, the contortion one-form.
mielke-geometrodynamics_F00842	166	■ 1.000	$\{ \}^{(2)} \Theta:=$	$^{(2)} \Theta:=$	The torsion two-form can be irreducibly decomposed into the trace part $^{(2)} \Theta:=$
mielke-geometrodynamics_F00843	166	■ 1.000	$\frac{1}{3} \gamma \wedge T$	$\frac{1}{3} \gamma \wedge T$	$\frac{1}{3} \gamma \wedge T$, the axial torsion $^{(3)} \Theta:=-\frac{1}{3} *\left(\gamma \wedge \mathscr{A}\right)$, and the tensor torsion $^{(1)} \Theta:=$
mielke-geometrodynamics_F00844	166	■ 1.000	$\{ \}^{(3)} \Theta:=-\frac{1}{3} *\left(\gamma \wedge \mathscr{A}\right)$	$^{(3)} \Theta:=-\frac{1}{3} *\left(\gamma \wedge \mathscr{A}\right)$	$\frac{1}{3} \gamma \wedge T$, the axial torsion $^{(3)} \Theta:=-\frac{1}{3} *\left(\gamma \wedge \mathscr{A}\right)$, and the tensor torsion $^{(1)} \Theta:=$
mielke-geometrodynamics_F00845	166	■ 1.000	$\{ \}^{(1)} \Theta:=$	$^{(1)} \Theta:=$	$\frac{1}{3} \gamma \wedge T$, the axial torsion $^{(3)} \Theta:=-\frac{1}{3} *\left(\gamma \wedge \mathscr{A}\right)$, and the tensor torsion $^{(1)} \Theta:=$
mielke-geometrodynamics_F00846	166	■ 1.000	$\Theta-^{(2)} \Theta-^{(3)} \Theta$	$\Theta-^{(2)} \Theta-^{(3)} \Theta$	$\Theta-^{(2)} \Theta-^{(3)} \Theta$, where the one-forms of the trace and axial vector torsion, respec-

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mielke-geometrodynamics_F00847	166	■ 0.984	$\widetilde{\mathrm{A}}=\mathrm{A}$	$\widetilde{\mathcal{A}} = \mathcal{A}$	Under a Weyl rescaling, the axial torsion remains invariant, i.e., $\widetilde{\mathcal{A}} = \mathcal{A}$. Both
mielke-geometrodynamics_F00848	167	■ 1.000	$\mathrm{D} \, \Omega^{\mathrm{g}(\star)} \equiv 0$	$D\Omega^{g(\star)} \equiv 0$	i.e., $D\Omega^{g(\star)} \equiv 0$, provided that $D\eta_{\alpha\beta\gamma\delta} = 0$ holds as in spaces with vanishing Weyl
mielke-geometrodynamics_F00849	167	■ 1.000	$\mathrm{D} \, \eta_{\{\alpha \, \beta \, \gamma \, \delta\}}=0$	$D\eta_{\alpha\beta\gamma\delta} = 0$	i.e., $D\Omega^{g(\star)} \equiv 0$, provided that $D\eta_{\alpha\beta\gamma\delta} = 0$ holds as in spaces with vanishing Weyl
mielke-geometrodynamics_F00850	167	■ 0.995	$\mathrm{A}:=\{ \, \}^*\!\!\left(\vartheta_{\alpha} \, \wedge \, \mathrm{T}^{\alpha} \right)\!\!\left(\mathrm{R}_{\{\alpha\}}\right)=\mathrm{A}_{\mathrm{i}} \, \mathrm{d} \, x^{\mathrm{i}}$	$\mathcal{A} := \,^*(\vartheta_{\alpha} \wedge T^{\alpha}) = \mathcal{A}_i dx^i$	where $\mathcal{A} := \,^*(\vartheta_{\alpha} \wedge T^{\alpha}) = \mathcal{A}_i dx^i$ is the <i>axial torsion</i> one-form. The translational
mielke-geometrodynamics_F00851	168	■ 0.471	$\mathrm{R}^{\{\backslash \, \backslash \}}:=(-1)^{\text{sig}+1} \sim *\!\!\left(\mathrm{R}^{\{\backslash \, \backslash \}}_{\beta} \, \wedge \, \eta_{\beta \alpha} \right)$	$R^{\{\}} := (-1)^{\mathrm{sig}+1} \,^*(R^{\{\}j\beta} \wedge \eta_{\beta\alpha})$	vature scalar $R^{[\!]} := (-1)^{\mathrm{sig}+1} \,^*(R^{[\!]\alpha\beta} \wedge \eta_{\beta\alpha})$ and the axial torsion piece $d\mathcal{A} \wedge d\mathcal{A}$
mielke-geometrodynamics_F00852	168	■ 0.995	L_{SKY}	L_{SKY}	Lagrangian L_{SKY} as well as a Ricci-squared term and a scalar-curvature-squared
mielke-geometrodynamics_F00853	168	■ 0.827	$\operatorname{Ric}_{\{\alpha \, \beta \}}:=(-1)^{\text{sig} \, *} \!\!\left(\mathrm{R}_{\{\alpha \}}^{\{\backslash \, \backslash \}} \, \wedge \, \eta_{\delta \, \mid \, \beta \}} \right)$	$\mathrm{Ric}_{\alpha\beta} := (-1)^{\mathrm{sig}} \,^*(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)})$	$\mathrm{Ric}_{\alpha\beta} := (-1)^{\mathrm{sig}} \,^*(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \beta)})$, is also known as the Gauss–Bonnet term.
mielke-geometrodynamics_F00854	169	■ 1.000	$\mathrm{G}_{\alpha}:=\mathrm{R}^{\{\beta \, \gamma \}} \, \wedge \, \eta_{\alpha \, \beta \, \gamma} \, / \, 2$	$G_{\alpha} := R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma}/2$	$G_{\alpha} := R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma}/2$ and the curvature scalar R . The latter is the zero-form
mielke-geometrodynamics_F00855	169	■ 0.894	$\mathrm{G}^{\{\backslash \, \backslash \}}=\mathrm{G}_{\alpha}^{\{\backslash \, \backslash \}}_{\beta} \, \eta_{\beta \gamma} \, \gamma^{\alpha}$	$G^{\{\}} = G_{\alpha}^{\{\}\beta} \eta_{\beta} \gamma^{\alpha}$	decomposes into the Einstein three-form $G^{[\!]} = G_{\alpha}^{[\!]\beta} \eta_{\beta} \gamma^{\alpha}$ with respect to the
mielke-geometrodynamics_F00856	170	■ 0.983	2+1	2 + 1	Feynman RP (1981) The qualitative behavior of Yang-Mills theory in 2+1 dimensions. Nuclear
mielke-geometrodynamics_F00857	171	■ 0.998	$\mathrm{m} \rightarrow 0$	$\mathrm{m} \rightarrow 0$	Kogan II, Mouslopoulos S, Papazoglou A (2001) The m→0 limit for massive graviton in dS ₄ and
mielke-geometrodynamics_F00858	171	■ 0.998	dS_4	dS_4	Kogan II, Mouslopoulos S, Papazoglou A (2001) The m→0 limit for massive graviton in dS ₄ and
mielke-geometrodynamics_F00859	171	■ 0.999	AdS_4	AdS_4	AdS ₄ . How to circumvent the van Dam-Veltman-Zakharov discontinuity. Phys Lett B 503(1):173–
mielke-geometrodynamics_F00860	171	■ 0.878	5, $\mathbb{R}$	$5, \mathbb{R}$	Mielke EW (2012) Einstein-Weyl gravity from a topological $\mathrm{SL}(5, \mathbb{R})$ gauge invariant action. Adv
mielke-geometrodynamics_F00861	172	■ 1.000	$R+R^2+Q^2$	$R + R^2 + Q^2$	Zhytnikov V (1994) Wavelike exact solutions of $R + R^2 + Q^2$ gravity. J Math Phys 35(11):6001–
mielke-geometrodynamics_F00862	173	■ 0.999	(2014,2006)	(2014, 2006)	“with flying colors”; see WILL (2014, 2006) for recent reviews. However, Einstein’s
mielke-geometrodynamics_F00863	173	■ 0.999	$\lambda_{\mathrm{DE}}=\left(\hbar \, c \, / \, \rho_{\mathrm{DE}}\right)^{1 \, / \, 4} \, \mathrm{simeq} \, 85 \, \mu \, \mathrm{m}$	$\lambda_{\mathrm{DE}} = (\hbar c/\rho_{\mathrm{DE}})^{1/4} \simeq 85 \mu \, \mathrm{m}$	cal dark energy (DE) scale of $\lambda_{\mathrm{DE}} = (\hbar c/\rho_{\mathrm{DE}})^{1/4} \simeq 85 \, \mu\mathrm{m}$. Thus this “window” of
mielke-geometrodynamics_F00864	174	■ 1.000	$[\hbar]=[p][q]$	$[\hbar] = [p][q]$	dimensional transmutation due to the physical dimension $[\hbar] = [p][q]$ of Planck’s
mielke-geometrodynamics_F00865	174	■ 1.000	$\hbar$	$\hbar$	(reduced) constant $\hbar$.
mielke-geometrodynamics_F00866	174	■ 1.000	$C:=\frac{i}{4} \, C^{\alpha \beta} \, \sigma_{\alpha \beta}$	$C := \frac{i}{4} C^{\alpha\beta} \sigma_{\alpha\beta}$	ghosts, into operator transformations. Let $C:=\frac{i}{4} C^{\alpha\beta} \sigma_{\alpha\beta}$ denote the zero-form of the
mielke-geometrodynamics_F00867	174	■ 0.998	$\Psi:=\frac{i}{4} \, \Psi_i^{\alpha \beta} \, \sigma_{\alpha \beta} \, \mathrm{d} \, x^{\mathrm{j}}$	$\Psi := \frac{i}{4} \Psi_j^{\alpha\beta} \sigma_{\alpha\beta} dx^j$	$\Psi:=\frac{i}{4} \Psi_i^{\alpha\beta} \sigma_{\alpha\beta} dx^j$ the topological ghost one-form, and $\Phi:=\frac{i}{4} \Phi^{\alpha\beta} \sigma_{\alpha\beta}$ the correspond-
mielke-geometrodynamics_F00868	174	■ 0.998	$\Phi:=\frac{i}{4} \, \Phi^{\alpha \beta} \, \sigma_{\alpha \beta}$	$\Phi := \frac{i}{4} \Phi^{\alpha\beta} \sigma_{\alpha\beta}$	$\Psi:=\frac{i}{4} \Psi_i^{\alpha\beta} \sigma_{\alpha\beta} dx^j$ the topological ghost one-form, and $\Phi:=\frac{i}{4} \Phi^{\alpha\beta} \sigma_{\alpha\beta}$ the correspond-
mielke-geometrodynamics_F00869	174	■ 1.000	$\sigma_{\alpha \beta}$	$\sigma_{\alpha\beta}$	of the generator $\sigma_{\alpha\beta}$ of the linear (or Lorentz) group.
mielke-geometrodynamics_F00870	175	■ 1.000	Γ	Γ	This is consistent with the interpretation of Γ and Ω^g as connection one-form and
mielke-geometrodynamics_F00871	175	■ 1.000	Ω^g	Ω^g	This is consistent with the interpretation of Γ and Ω^g as connection one-form and
mielke-geometrodynamics_F00872	175	■ 1.000	$C, \, \Psi$	C, Ψ	number of degrees of freedom carried by the ghosts C , Ψ , and Φ is equal to that of
mielke-geometrodynamics_F00873	175	■ 1.000	$s^2=0$	$s^2 = 0$	variables, i.e., $s^2 = 0$.

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mielke-geometrodynamics_F00874	175	■ 0.948	$\wedge^2$	2	(1988) by introducing the <i>graded</i> ² connection and curvature forms
mielke-geometrodynamics_F00875	175	■ 1.000	$s \ s=0$	$ss = 0$	Moreover, a straightforward proof of the nilpotency $ss = 0$ of the BRST operator
mielke-geometrodynamics_F00876	175	■ 1.000	$(d \ \oplus s)(d \ \oplus s) \ \equiv 0$	$(d \oplus s)(d \oplus s) \equiv 0$	s follows simply from $\boxed{d \oplus s)(d \oplus s) \equiv 0}$ as a result of the graded Bianchi identity
mielke-geometrodynamics_F00877	175	■ 1.000	$[s, d] := s \ d + d \ s = 0$	$[s, d] := sd + ds = 0$	(8.2.3), the anticommutation of the graded commutator $[s, d] := sd + ds = 0$, and
mielke-geometrodynamics_F00878	175	■ 0.615	$\bar{C}, \ \bar{\chi}$	$\bar{C}, \bar{\chi}$	ism, where the Lorentz-algebra-valued antighosts $\bar{C}$, $\bar{\chi}$, and $\bar{\Phi}$ obey the following
mielke-geometrodynamics_F00879	175	■ 0.615	$\bar{\Phi}$	$\bar{\Phi}$	ism, where the Lorentz-algebra-valued antighosts $\bar{C}$, $\bar{\chi}$, and $\bar{\Phi}$ obey the following
mielke-geometrodynamics_F00880	175	■ 1.000	$[\Psi, \Phi] := \Psi \wedge \Phi -$	$[\Psi, \Phi] := \Psi \wedge \Phi -$	ghost number g , such that the graded commutator is more generally defined by $[\psi, \phi] := \psi \wedge \phi -$
mielke-geometrodynamics_F00881	175	■ 0.510	$(-1)^{\left(\left(p_1+g_1\right)\left(p_2+g_2\right)\right)} \Phi \wedge \Psi$	$(-1)^{(p_1+g_1)(p_2+g_2)} \Phi \wedge \Psi$	$\boxed{(-1)^{(p_1+g_1)(p_2+g_2)} \phi \wedge \psi}$, (CHANG & SOO 1992).
mielke-geometrodynamics_F00882	176	■ 1.000	$\bar{\eta}$	$\bar{\eta}$	By construction, s is nilpotent for the antighosts, since the Lagrange multiplier $\bar{\eta}$,
mielke-geometrodynamics_F00883	176	■ 1.000	$\mathbb{B}$	$\mathbb{B}$	the self-dual two-forms $\mathbb{B}$, and β are auxiliary fields introduced as trivial pairs.
mielke-geometrodynamics_F00884	176	■ 0.998	$\alpha = \alpha_i dx^i$	$\alpha = \alpha_i dx^i$	By introducing a BRST gauge field $\alpha = \alpha_i dx^i$ with ghost number -1 and a
mielke-geometrodynamics_F00885	176	■ 0.999	λ	λ	commuting ghost λ of α , one can promote (DE CARVALHO & BAULIEU 1992) the
mielke-geometrodynamics_F00886	176	■ 0.999	α	α	commuting ghost λ of α , one can promote (DE CARVALHO & BAULIEU 1992) the
mielke-geometrodynamics_F00887	176	■ 1.000	$C \rightarrow (\alpha +$	$C \rightarrow (\alpha +$	promotion, which likewise, can be generated via the field redefinitions $C \rightarrow (\alpha +$
mielke-geometrodynamics_F00888	176	■ 0.712	$\lambda) C, \Psi \rightarrow (\alpha + \lambda) \Psi$	$\lambda) C, \Psi \rightarrow (\alpha + \lambda) \Psi$	$\lambda) C, \Psi \rightarrow (\alpha + \lambda) \Psi$ as well as $\Phi \rightarrow (\alpha + \lambda)^2 \Phi$ of the ghosts. Thus, local BRST
mielke-geometrodynamics_F00889	176	■ 0.712	$\Phi \rightarrow (\alpha + \lambda)^2 \Phi$	$\Phi \rightarrow (\alpha + \lambda)^2 \Phi$	$\lambda) C, \Psi \rightarrow (\alpha + \lambda) \Psi$ as well as $\Phi \rightarrow (\alpha + \lambda)^2 \Phi$ of the ghosts. Thus, local BRST
mielke-geometrodynamics_F00890	176	■ 1.000	$\gamma = \vartheta^\alpha \gamma_\alpha$	$\gamma = \vartheta^\alpha \gamma_\alpha$	In the affine or (broken) Poincaré gauge theory, the coframe $\gamma = \vartheta^\alpha \gamma_\alpha$ (locally
mielke-geometrodynamics_F00891	176	■ 1.000	$\widetilde{\Theta} := \Theta \oplus \psi$	$\widetilde{\Theta} := \Theta \oplus \psi$	respectively. In the graded torsion two-form $\widetilde{\Theta} := \Theta \oplus \psi$, there occurs the transla-
mielke-geometrodynamics_F00892	176	■ 0.999	$\psi := \psi_j^\alpha \gamma_\alpha dx^j$	$\psi := \psi_j^\alpha \gamma_\alpha dx^j$	tional “Clifform” $\psi := \psi_j^\alpha \gamma_\alpha dx^j$ of the corresponding topological ghost.
mielke-geometrodynamics_F00893	177	■ 0.885	$s \rightarrow \widetilde{s} = s + \mathbb{L}_\zeta$	$s \rightarrow \widetilde{s} = s + \mathbb{L}_\zeta$	alizing the BRST transformations s via $s \rightarrow \widetilde{s} = s + \mathbb{L}_\zeta$ involving the covariant Lie
mielke-geometrodynamics_F00894	177	■ 0.566	$\left.\left.\mathrm{L}_{\zeta}:=\zeta \lrcorner D-D \zeta\right\rfloor\right\rfloor$	$\mathbb{L}_\zeta := \zeta \lrcorner D - D\zeta$	derivative $\mathbb{L}_\zeta := \zeta \lrcorner D - D\zeta$ along an anticommuting ghost vector field $\zeta = \zeta^i \partial_i$.
mielke-geometrodynamics_F00895	177	■ 0.566	$\zeta = \zeta^i \partial_i$	$\zeta = \zeta^i \partial_i$	derivative $\mathbb{L}_\zeta := \zeta \lrcorner D - D\zeta$ along an anticommuting ghost vector field $\zeta = \zeta^i \partial_i$.
mielke-geometrodynamics_F00896	177	■ 0.339	$\mathbb{L}_\zeta$	$\mathbb{L}_\zeta$	of $\mathbb{L}_\zeta$, since ζ has ghost number one. Its BRST transformation is $s\zeta = \phi + \mathbb{L}_\zeta \zeta$, such
mielke-geometrodynamics_F00897	177	■ 0.339	ζ	ζ	of $\mathbb{L}_\zeta$, since ζ has ghost number one. Its BRST transformation is $s\zeta = \phi + \mathbb{L}_\zeta \zeta$, such
mielke-geometrodynamics_F00898	177	■ 0.339	$s \zeta = \phi + \mathbb{L}_\zeta \zeta$	$s\zeta = \phi + \mathbb{L}_\zeta \zeta$	of $\mathbb{L}_\zeta$, since ζ has ghost number one. Its BRST transformation is $\boxed{s\zeta = \phi + \mathbb{L}_\zeta \zeta}$, such
mielke-geometrodynamics_F00899	177	■ 1.000	$p+1$	$p + 1$	Its action on a p -form, reduces to a finite series with up to $p + 1$ terms. In effect, the
mielke-geometrodynamics_F00900	177	■ 0.497	$\exp (\zeta \rfloor) \widetilde{\Omega}^g$	$\exp(\zeta \rfloor) \widetilde{\Omega}^g$	by $\exp(\zeta \rfloor) \widetilde{\Omega}^g$ and $\exp(\zeta \rfloor)(\widetilde{\Theta} \oplus \phi)$, respectively, where $\phi := \phi^\alpha \gamma_\alpha$ are the ghosts of

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mielke-geometrodynamics_F00901	177	■ 0.497	$\left.\exp\left(\zeta\right)\right _{\Theta+\phi}$	$\exp(\zeta)(\tilde{\Theta} \oplus \phi)$	by $\exp(\zeta)\tilde{\Omega}^g$ and $\exp(\zeta)(\tilde{\Theta} \oplus \phi)$, respectively, where $\phi:=\phi^\alpha\gamma_\alpha$ are the ghosts of
mielke-geometrodynamics_F00902	177	■ 0.497	$\phi:=\phi^\alpha\gamma_\alpha$	$\phi:=\phi^\alpha\gamma_\alpha$	by $\exp(\zeta)\tilde{\Omega}^g$ and $\exp(\zeta)(\tilde{\Theta} \oplus \phi)$, respectively, where $\phi:=\phi^\alpha\gamma_\alpha$ are the ghosts of
mielke-geometrodynamics_F00903	177	■ 1.000	d	d	(BAULIEU 1987), the exterior derivative d suffers from the similarity transformation
mielke-geometrodynamics_F00904	177	■ 1.000	$\Gamma^{\mathrm{T}}=(\gamma-D\xi)/\ell$	$\Gamma^{(\mathrm{T})}=(\gamma-D\xi)/\ell$	<i>translational</i> connection $\Gamma^{(\mathrm{T})}=(\gamma-D\xi)/\ell$ and corresponding curvature $\Omega^{(\mathrm{T})}=$
mielke-geometrodynamics_F00905	177	■ 1.000	$\Omega^{\mathrm{T}}=$	$\Omega^{(\mathrm{T})}=$	<i>translational</i> connection $\Gamma^{(\mathrm{T})}=(\gamma-D\xi)/\ell$ and corresponding curvature $\Omega^{(\mathrm{T})}=$
mielke-geometrodynamics_F00906	177	■ 1.000	$D\Gamma^{\mathrm{T}}=(\Theta-DD\xi)/\ell$	$D\Gamma^{(\mathrm{T})}=(\Theta-DD\xi)/\ell$	$D\Gamma^{(\mathrm{T})}=(\Theta-DD\xi)/\ell$. Then the translational connection $\Gamma^{(\mathrm{T})}$ should be graded
mielke-geometrodynamics_F00907	177	■ 1.000	Γ^{T}	$\Gamma^{(\mathrm{T})}$	$D\Gamma^{(\mathrm{T})}=(\Theta-DD\xi)/\ell$. Then the translational connection $\Gamma^{(\mathrm{T})}$ should be graded
mielke-geometrodynamics_F00908	177	■ 1.000	$\gamma\rightarrow\tilde{\gamma}:=\ell\left(\Gamma^{\mathrm{T}}\oplus c\right)$	$\gamma\rightarrow\tilde{\gamma}:=\ell\left(\Gamma^{(\mathrm{T})}\oplus c\right)$	as well, e.g., by the substitution $\gamma\rightarrow\tilde{\gamma}:=\ell(\Gamma^{(\mathrm{T})}\oplus c)$ in the first structure equation.
mielke-geometrodynamics_F00909	177	■ 1.000	$\xi:=\xi^\alpha\gamma_\alpha$	$\xi:=\xi^\alpha\gamma_\alpha$	The “quartet” of scalars $\xi:=\xi^\alpha\gamma_\alpha$ corresponds to the “generalized radius vectors”
mielke-geometrodynamics_F00910	177	■ 1.000	$A(4,\mathbb{R})/GL(4,\mathbb{R})\approx\mathbb{R}^4$	$A(4,\mathbb{R})/GL(4,\mathbb{R})\approx\mathbb{R}^4$	of Cartan and lives on the coset space $A(4,\mathbb{R})/GL(4,\mathbb{R})\approx\mathbb{R}^4$ of the affine group.
mielke-geometrodynamics_F00911	178	■ 1.000	$\Gamma^{\mathrm{T}}\stackrel{*}{=}0$	$\Gamma^{(\mathrm{T})}\stackrel{*}{=}0$	naturally arises in the affine gauge theory after, locally, the gauge $\Gamma^{(\mathrm{T})}\stackrel{*}{=}0$ is imposed
mielke-geometrodynamics_F00912	178	■ 0.995	$SU(N)$	$SU(N)$	and replace the internal $SU(N)$ group by the linear group $SL(4,\mathbb{R})$ of the tangent
mielke-geometrodynamics_F00913	178	■ 0.995	$SL(4,\mathbb{R})$	$SL(4,\mathbb{R})$	and replace the internal $SU(N)$ group by the linear group $SL(4,\mathbb{R})$ of the tangent
mielke-geometrodynamics_F00914	178	■ 0.899	$SO(1,3)$	$SO(1,3)$	space embracing the Lorentz group $SO(1,3)$ (or $SO(4)$ in Euclidean space with
mielke-geometrodynamics_F00915	178	■ 0.467	$=0$	$=0$	signature $\mathrm{sig}=0$) as subgroup. Then one starts from the gravitational Pontryagin
mielke-geometrodynamics_F00916	178	■ 0.999	$s\{\cdots\}$	$s\{\cdots\}$	This purely topological action can be amended by any s -exact four-form $s\{\cdots\}$,
mielke-geometrodynamics_F00917	178	■ 1.000	$d^*\Gamma=0, D^*\Psi=0$	$d^*\Gamma=0, D^*\Psi=0$	(1988), we may choose the Lorentz-type conditions $d^*\Gamma=0, D^*\Psi=0$ on the
mielke-geometrodynamics_F00918	178	■ 1.000	$\Omega^{g(\pm)}=0$	$\Omega^{g(\pm)}=0$	self-duality condition $\Omega^{g(\pm)}=0$ on the curvature as <i>gauge constraints</i> consistently
mielke-geometrodynamics_F00919	178	■ 1.000	CP	CP	³ For the CP -invariant Euler term, a similar result would hold, i.e., $sL_{\mathrm{Euler}}=$
mielke-geometrodynamics_F00920	178	■ 1.000	$sL_{\mathrm{Euler}}=$	$sL_{\mathrm{Euler}}=$	³ For the CP -invariant Euler term, a similar result would hold, i.e., $sL_{\mathrm{Euler}}=$
mielke-geometrodynamics_F00921	178	■ 0.998	$(-1)^{\mathrm{sig}+1}d\left(\Psi^{\alpha\beta}\wedge R_{\alpha\beta}^{(*)}\right)$	$(-1)^{\mathrm{sig}+1}d\left(\Psi^{\alpha\beta}\wedge R_{\alpha\beta}^{(*)}\right)$	$(-1)^{\mathrm{sig}+1}d\left(\Psi^{\alpha\beta}\wedge R_{\alpha\beta}^{(*)}\right)$. However, the latter is only partially metric-free, since it involves
mielke-geometrodynamics_F00922	179	■ 1.000	$\bar{\chi}$	$\bar{\chi}$	with the four-form character of a Lagrangian, the auxiliary fields $\bar{\chi}$ and β have
mielke-geometrodynamics_F00923	179	■ 0.993	$\tilde{\Psi}_{\mathrm{F}}:=\bar{C}_{\alpha\beta}\left(d^*\Gamma^{\alpha\beta}+\frac{1}{2}*\mathbb{B}^{\alpha\beta}\right)$	$\tilde{\Psi}_{\mathrm{F}}:=\bar{C}_{\alpha\beta}\left(d^*\Gamma^{\alpha\beta}+\frac{1}{2}*\mathbb{B}^{\alpha\beta}\right)$	$\tilde{\Psi}_{\mathrm{F}}:=\bar{C}_{\alpha\beta}\left(d^*\Gamma^{\alpha\beta}+\frac{1}{2}*\mathbb{B}^{\alpha\beta}\right)$.
mielke-geometrodynamics_F00924	179	■ 0.835	$\{\cdots\}$	$\{\cdots\}$	four-form $\{\cdots\}$ is able to induce a dependence on the metric that in is not present in
mielke-geometrodynamics_F00925	179	■ 0.998	$d^*\Gamma=*\mathbb{B}$	$d^*\Gamma=*\mathbb{B}$	$d^*\Gamma=*\mathbb{B}$ on the linear connection. Moreover, for vanishing real <i>gauge parameter</i>
mielke-geometrodynamics_F00926	179	■ 1.000	$\rho=0$	$\rho=0$	$\rho=0$, the equation of motion for the auxiliary Lagrange-multiplier-type two-form

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mielke-geometrodynamics_F00927	180	■ 1.000	$\rho=1$	$\rho = 1$	In the case of the choice $\rho = 1$ of the real gauge parameter, the two-form β is present
mielke-geometrodynamics_F00928	180	■ 1.000	$D\,\omega^g\equiv 0$	$D\Omega^g\equiv 0$	Due to the second Bianchi identity $D\Omega^g\equiv 0$, the dual field equation
mielke-geometrodynamics_F00929	180	■ 0.960	$T_{\mu\nu}:=2^*\left(\partial L/\partial g_{\mu\nu}\right)=0$	$T_{\mu\nu}:=2^*(\partial L/\partial g_{\mu\nu})=0$	i.e., $T_{\mu\nu}:=2^*(\partial L/\partial g_{\mu\nu})=0$. In turn, this implies that the BRST quantization is
mielke-geometrodynamics_F00930	181	■ 0.999	$\mathscr{A}:=\left(\vartheta^*\right)\left(\vartheta_{\alpha}\wedge\right)$	$\mathscr{A}:=*(\vartheta_{\alpha}\wedge$	respect to the Riemannian curvature $R_{\alpha\beta}^{\parallel}$ and the <i>axial torsion</i> one-form $\mathscr{A}:=*(\vartheta_{\alpha}\wedge$
mielke-geometrodynamics_F00931	181	■ 0.999	T^{α}	T^{α}	$T^{\alpha})$ is rather well motivated by the <i>axial anomaly</i> in RC spacetime
mielke-geometrodynamics_F00932	181	■ 1.000	$\gamma:=\vartheta^{\alpha}\gamma_{\alpha}$	$\gamma:=\vartheta^{\alpha}\gamma_{\alpha}$	In addition, the coframe $\gamma:=\vartheta^{\alpha}\gamma_{\alpha}$ is constrained by the Hilbert–de Donder gauge
mielke-geometrodynamics_F00933	181	■ 1.000	$d^*\gamma=0$	$d^*\gamma=0$	condition $d^*\gamma=0$ complying with the BRST algebra of translations.
mielke-geometrodynamics_F00934	181	■ 1.000	$\widetilde{L}_{\mathrm{FP}}:=L_{\mathrm{FP}}+\Delta L_{\mathrm{FP}}$	$\widetilde{L}_{\mathrm{FP}}:=L_{\mathrm{FP}}+\Delta L_{\mathrm{FP}}$	Then for $\rho=0$ there arises from $\widetilde{L}_{\mathrm{FP}}:=L_{\mathrm{FP}}+\Delta L_{\mathrm{FP}}$ the <i>modified double duality</i>
mielke-geometrodynamics_F00935	181	■ 1.000	i	i	on the curvature two-form Ω^g , which is real due to the extra imaginary unit i .
mielke-geometrodynamics_F00936	181	■ 0.692	$\stackrel{\pm}{\Omega}^g$	$\stackrel{\pm}{\Omega}^g$	only one sign for the dual curvature $\stackrel{\pm}{\Omega}^g$ leads to a self-consistent constraint. This
mielke-geometrodynamics_F00937	182	■ 0.811	$\theta=\pi/2$	$\theta=\pi/2$	of the curvature, which for $\theta=\pi/2$ specializes to the self-dual curvature $\stackrel{+}{\Omega}^g$, and for
mielke-geometrodynamics_F00938	182	■ 0.811	$\stackrel{+}{\Omega}^g$	$\stackrel{+}{\Omega}^g$	of the curvature, which for $\theta=\pi/2$ specializes to the self-dual curvature $\stackrel{+}{\Omega}^g$, and for
mielke-geometrodynamics_F00939	182	■ 1.000	$\theta=3\pi/2$	$\theta=3\pi/2$	$\theta=3\pi/2$ to the anti-self-dual curvature $\bar{\Omega}^g$. Since the “unit”-curvature two-form σ
mielke-geometrodynamics_F00940	182	■ 1.000	$\bar{\Omega}^g$	$\bar{\Omega}^g$	$\theta=3\pi/2$ to the anti-self-dual curvature $\bar{\Omega}^g$. Since the “unit”-curvature two-form σ
mielke-geometrodynamics_F00941	182	■ 1.000	$\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$	$\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$	a special case of the Ansatz (8.6.4) for the choice $\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$ and $\theta_{\mathrm{L}}^*=\mp(-1)^{\mathrm{sig}}$. Interestingly
mielke-geometrodynamics_F00942	182	■ 1.000	$\theta_{\mathrm{L}}^*=\mp(-1)^{\mathrm{sig}}$	$\theta_{\mathrm{L}}^*=\mp(-1)^{\mathrm{sig}}$	a special case of the Ansatz (8.6.4) for the choice $\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$ and $\theta_{\mathrm{L}}^*=\mp(-1)^{\mathrm{sig}}$. Interestingly
mielke-geometrodynamics_F00943	183	■ 0.983	L_{qPG}	L_{qPG}	tion parts of the gravitational Lagrangian L_{qPG} . In the field momenta, each of the
mielke-geometrodynamics_F00944	183	■ 1.000	$a_{(M)}$	$a_{(M)}$	the Lagrangian with individual dimensionless weights $a_{(M)}$ and $b_{(N)}$, respectively.
mielke-geometrodynamics_F00945	183	■ 1.000	$b_{(N)}$	$b_{(N)}$	the Lagrangian with individual dimensionless weights $a_{(M)}$ and $b_{(N)}$, respectively.
mielke-geometrodynamics_F00946	183	■ 1.000	$\theta_{\mathrm{T}},\theta_{\mathrm{T}}^*$	$\theta_{\mathrm{T}},\theta_{\mathrm{T}}^*$	for the rotational field momenta (MIELKE 1992; ZHYTNIKOV 1994), where $\theta_{\mathrm{T}},\theta_{\mathrm{T}}^*$,
mielke-geometrodynamics_F00947	183	■ 1.000	θ_{L}^*	θ_{L}^*	θ_{L} , and θ_{L}^* are dimensionless constants. The constraint (8.5.3) corresponds to the
mielke-geometrodynamics_F00948	183	■ 1.000	$a_0=0, b_{(N)}=1$	$a_0=0, b_{(N)}=1$	SKY Lagrangian with $a_0=0, b_{(N)}=1$, and the choice $\theta_{\mathrm{L}}^*=\mp(-1)^{\mathrm{sig}}$ as well as
mielke-geometrodynamics_F00949	183	■ 0.359	$dC_{\mathrm{TT}}^*:=d(\vartheta^{\alpha}\wedge T_{\alpha})/2\ell^2$	$dC_{\mathrm{TT}}^*:=d(\vartheta^{\alpha}\wedge T_{\alpha})/2\ell^2$	$dC_{\mathrm{TT}}^*:=d(\vartheta^{\alpha}\wedge T_{\alpha})/2\ell^2$ inducing the teleparallelism equivalence, as well as the
mielke-geometrodynamics_F00950	183	■ 1.000	$\frac{1}{2}s\bar{\chi}^{\alpha\beta}\wedge(H_{\alpha\beta}+\partial L_{\theta}/\partial R^{\alpha\beta})$	$\frac{1}{2}s\bar{\chi}^{\alpha\beta}\wedge(H_{\alpha\beta}+\partial L_{\theta}/\partial R^{\alpha\beta})$	BRST gauge constraint $\frac{1}{2}s\bar{\chi}^{\alpha\beta}\wedge(H_{\alpha\beta}+\partial L_{\theta}/\partial R^{\alpha\beta})$ generalizing the corresponding
mielke-geometrodynamics_F00951	183	■ 0.680	$\widetilde{L}_{\mathrm{FP}}$	$\widetilde{L}_{\mathrm{FP}}$	terms in the Faddeev–Popov Lagrangian $\widetilde{L}_{\mathrm{FP}}$ modified by (8.5.1).

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mielke-geometrodynamics_F00952	184	■ 1.000	D	D	The covariant derivative D of the “unit”-curvature pieces proportional to $\eta_{\alpha\beta}$ and
mielke-geometrodynamics_F00953	184	■ 1.000	$\vartheta_{\alpha} \wedge \vartheta_{\beta}$	$\vartheta_{\alpha} \wedge \vartheta_{\beta}$	$\vartheta_{\alpha} \wedge \vartheta_{\beta}$ in the DD Ansatz is responsible for the two additional torsion terms. For
mielke-geometrodynamics_F00954	184	■ 0.999	$\tau_{\alpha\beta}=0$	$\tau_{\alpha\beta} = 0$	spinless matter, i.e., $\tau_{\alpha\beta} = 0$, one can algebraically resolve (8.6.9) using Eq. (A.1.26)
mielke-geometrodynamics_F00955	184	■ 1.000	$H_{\alpha}(**)$	$H_{\alpha}(**)$	of HEHL et al. (1995) for the translational momenta $H_{\alpha}(**)$ constrained by double
mielke-geometrodynamics_F00956	185	■ 1.000	$H_{\alpha\beta}(**) \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}$	$H_{\alpha\beta}(**) \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}$	The contraction $H_{\alpha\beta}(**) \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}$ of the DD Ansatz (8.6.4) together with the linear
mielke-geometrodynamics_F00957	185	■ 1.000	$\theta_L \neq 0$	$\theta_L \neq 0$	expansion (8.6.3) implies, for $\theta_L \neq 0$, that the pseudoscalar curvature vanishes for
mielke-geometrodynamics_F00958	185	■ 1.000	$R_{\alpha\beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta} = 0$	$R_{\alpha\beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta} = 0$	consistency, i.e., $R_{\alpha\beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta} = 0$. Then the scalar curvature
mielke-geometrodynamics_F00959	185	■ 1.000	$\theta_L = 0$	$\theta_L = 0$	⁵ There occurs an interesting modification in the case that $\theta_L = 0$, since then the pseudoscalar cur-
mielke-geometrodynamics_F00960	185	■ 1.000	$\theta_T R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} / 4 \ell^2$	$\theta_T R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} / 4 \ell^2$	vature four-form $\theta_T R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} / 4 \ell^2$ needs to be subtracted from (8.6.12), which would induce
mielke-geometrodynamics_F00961	186	■ 1.000	$T^{\alpha} \simeq 0$	$T^{\alpha} \simeq 0$	vanishing torsion $T^{\alpha} \simeq 0$ via Lagrange multipliers in the gravitational gauge field
mielke-geometrodynamics_F00962	186	■ 1.000	$T^{\alpha} = 0$	$T^{\alpha} = 0$	Then obviously, $T^{\alpha} = 0$ will emerge by varying the Lagrange multiplier, and the
mielke-geometrodynamics_F00963	186	■ 1.000	λ_{α}	λ_{α}	second field equation (8.6.7) amounts to an algebraic equation for the two-form λ_{α} .
mielke-geometrodynamics_F00964	186	■ 0.382	$\Gamma^{j\alpha\beta}$	$\Gamma^{j\alpha\beta}$	which, generically, is of third order in the Levi-Civita connection $\Gamma^{(\cdot)\alpha\beta}$ of Rie-
mielke-geometrodynamics_F00965	186	■ 0.803	$\mu_{\alpha} = \frac{1}{4} \vartheta_{\alpha} \wedge * j_5$	$\mu_{\alpha} = \frac{1}{4} \vartheta_{\alpha} \wedge * j_5$	In the case of Dirac spinors, the <i>spin-energy potential</i> $\mu_{\alpha} = \frac{1}{4} \vartheta_{\alpha} \wedge * j_5$ is dual to
mielke-geometrodynamics_F00966	186	■ 1.000	$\mu_{\alpha} \wedge \delta T^{\alpha}$	$\mu_{\alpha} \wedge \delta T^{\alpha}$	can likewise be obtained from the variation $\mu_{\alpha} \wedge \delta T^{\alpha}$.
mielke-geometrodynamics_F00967	186	■ 1.000	$DR_{\alpha\beta} \equiv 0, DR_{\alpha\beta}^{(*)} \equiv$	$DR_{\alpha\beta} \equiv 0, DR_{\alpha\beta}^{(*)} \equiv$	coupling constants in the θ -type boundary term (8.6.5): since $DR_{\alpha\beta} \equiv 0$, $DR_{\alpha\beta}^{(*)} \equiv$
mielke-geometrodynamics_F00968	186	■ 0.420	$0, D\eta_{\alpha\beta}=0$	$0, D\eta_{\alpha\beta} = 0$	0, $D\eta_{\alpha\beta} = 0$, and $D(\vartheta_{\alpha} \wedge \vartheta_{\beta}) = 2T_{[\alpha} \wedge \vartheta_{\beta]} = 0$ in a Riemannian spacetime, the
mielke-geometrodynamics_F00969	186	■ 0.420	$D(\vartheta_{\alpha} \wedge \vartheta_{\beta}) = 2T_{[\alpha} \wedge \vartheta_{\beta]} = 0$	$D(\vartheta_{\alpha} \wedge \vartheta_{\beta}) = 2T_{[\alpha} \wedge \vartheta_{\beta]} = 0$	0, $D\eta_{\alpha\beta} = 0$, and $D(\vartheta_{\alpha} \wedge \vartheta_{\beta}) = 2T_{[\alpha} \wedge \vartheta_{\beta]} = 0$ in a Riemannian spacetime, the
mielke-geometrodynamics_F00970	186	■ 1.000	$R_{\alpha\beta}^{(*)}$	$R_{\alpha\beta}^{(*)}$	the Lie dual $R_{\alpha\beta}^{(*)}$ of the curvature does not contribute in (8.6.8), due to the Bach–
mielke-geometrodynamics_F00971	186	■ 0.428	$G^{(\cdot)} := R^{(\cdot)\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \gamma^{\alpha} / 2$	$G^{(\cdot)} := R^{(\cdot)\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \gamma^{\alpha} / 2$	Here $G^{(\cdot)} := R^{(\cdot)\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \gamma^{\alpha} / 2$ is the Einstein three-form dual to the usual Einstein
mielke-geometrodynamics_F00972	186	■ 0.997	$G_{ij} := \text{Ric}_{ij} - \frac{1}{2} g_{ij}$	$G_{ij} := \text{Ric}_{ij} - \frac{1}{2} g_{ij}$	tensor $G_{ij} := \text{Ric}_{ij} - \frac{1}{2} g_{ij}$ written in terms of the Riemannian connection $\Gamma^{(\cdot)}$. Matter
mielke-geometrodynamics_F00973	186	■ 0.993	$\kappa_{\text{eff}} = \ell^2 / \theta_T^*$	$\kappa_{\text{eff}} = \ell^2 / \theta_T^*$	whereas the effective gravitational coupling constant $\kappa_{\text{eff}} = \ell^2 / \theta_T^*$ depends on a
mielke-geometrodynamics_F00974	186	■ 1.000	θ_T^*	θ_T^*	“bare” length scale ℓ “renormalized” by the vacuum angle θ_T^* .
mielke-geometrodynamics_F00975	187	■ 1.000	$\widehat{\Sigma}_{\alpha}$	$\widehat{\Sigma}_{\alpha}$	<i>Einstein equation</i> (8.6.18) to the symmetrized energy–momentum current $\widehat{\Sigma}_{\alpha}$ of
mielke-geometrodynamics_F00976	187	■ 0.987	$\kappa_{\text{eff}} = 8\pi G_N / c^4$	$\kappa_{\text{eff}} = 8\pi G_N / c^4$	matter requires, at least locally, the macroscopic Newtonian value $\kappa_{\text{eff}} = 8\pi G_N / c^4$
mielke-geometrodynamics_F00977	187	■ 0.894	L_{qPG}	L_{qPG}	most quadratic Lagrangian L_{qPG} with ten ⁶ parameters $a_{(M)}$ and $b_{(N)}$.
mielke-geometrodynamics_F00978	187	■ 1.000	$\langle \varphi \rangle \propto 1 / \ell$	$\langle \varphi \rangle \propto 1 / \ell$	tive to associate this with a (spontaneous) <i>symmetry breaking</i> $\langle \varphi \rangle \propto 1 / \ell$ of the scale

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mielke-geometrodynamics_F00979	188	■ 1.000	a	a	When these covectors are induced by an axion a and dilaton φ as respective potentials,
mielke-geometrodynamics_F00980	188	■ 1.000	$\Phi = a + i f_{\varphi} \exp(-\varphi / f_{\varphi})$	$\Phi = a + i f_{\varphi} \exp(-\varphi / f_{\varphi})$	even combine into a single complex scalar, the <i>axidilaton</i> $\Phi = a + i f_{\varphi} \exp(-\varphi / f_{\varphi})$;
mielke-geometrodynamics_F00981	192	■ 0.954	$\Lambda_{\text{eff}} > 0$	$\Lambda_{\text{eff}} > 0$	(here generalized to an n-dimensional manifold) and second for $\Lambda_{\text{eff}} > 0$ to the Nariai
mielke-geometrodynamics_F00982	193	■ 0.989	$\mathrm{N} = \mathrm{N}(\mathrm{r} \pm \mathrm{t})$	$N = N(r \pm t)$	with $N = N(r \pm t)$ denoting an arbitrary harmonic function of r and t ; cf. Ni (1975).
mielke-geometrodynamics_F00983	193	■ 0.989	r	r	with $N = N(r \pm t)$ denoting an arbitrary harmonic function of r and t ; cf. Ni (1975).
mielke-geometrodynamics_F00984	193	■ 0.998	$\Lambda_{\text{eff}} = 0$	$\Lambda_{\text{eff}} = 0$	spacetime, however, the Schwarzschild solution (9.1.1, 2) with $\Lambda_{\text{eff}} = 0$ is of para-
mielke-geometrodynamics_F00985	194	■ 0.358	$\{ \}^{\{ * \}}$	$*$	nonsingular one, we first pass to a different coordinate system. Let r^* be the “tortoise”
mielke-geometrodynamics_F00986	194	■ 0.996	$\mathrm{u}^2 - \mathrm{v}^2 = 1$	$u^2 - v^2 = 1$	$u^2 - v^2 = 1$, that is, for $r = r^* = 0$, this geometry becomes singular in real space-
mielke-geometrodynamics_F00987	194	■ 0.996	$\mathrm{r} = \mathrm{r}^{\{ * \}} = \mathrm{o}$	$r = r^* = 0$	$u^2 - v^2 = 1$, that is, for $r = r^* = 0$, this geometry becomes singular in real space-
mielke-geometrodynamics_F00988	194	■ 1.000	$\tilde{\mathrm{v}} := \mathrm{i} \mathrm{v}$	$\tilde{v} := i v$	be avoided by making use of a purely “imaginary” coordinate $\tilde{v} := i v$. The sphere
mielke-geometrodynamics_F00989	194	■ 0.531	S^2	S^2	S^2 given by $e^v = 0$ deteriorates to a rotational axis, while the imaginary time axis
mielke-geometrodynamics_F00990	194	■ 0.531	$\mathrm{e}^{\mathrm{v}} = 0$	$e^v = 0$	S^2 given by $e^v = 0$ deteriorates to a rotational axis, while the imaginary time axis
mielke-geometrodynamics_F00991	194	■ 1.000	$\tau :=$	$\tau :=$	$\tau :=$ it represents an angle variable about this axis in Euclidean spacetime. (Corre-
mielke-geometrodynamics_F00992	194	■ 1.000	K_3	K_3	K_3 -surface represents the only compact manifold that is allowed by a metric with
mielke-geometrodynamics_F00993	195	■ 0.629	ω^g	ω^g	connection ω^g into a Christoffel-type connection ω^{I} and the remaining contortion
mielke-geometrodynamics_F00994	195	■ 0.975	$R \ C$	RC	2-form in an RC spacetime. Its Riemannian part may be decomposed further into
mielke-geometrodynamics_F00995	195	■ 0.917	$\Omega_{\mathrm{C}}^{\text{I}}$	Ω_C^{I}	where Ω_C^{I} denotes Weyl’s conformal curvature 2-form in a Riemannian spacetime.
mielke-geometrodynamics_F00996	195	■ 1.000	$\mathrm{f}_{\mathrm{i}} = \mathrm{o}$	$f_i = 0$	choice $f_i = 0$ and $\zeta = -1$, as discussed by BAEKLER et al. (1982), the duality ansatz
mielke-geometrodynamics_F00997	196	■ 1.000	$K \wedge K$	$K \wedge K$	fact that $K \wedge K$ is already self-double-dual.
mielke-geometrodynamics_F00998	196	■ 1.000	$\mu = 0$	$\mu = 0$	cosmos, vanishes without Schwarzschild’s source term, i.e., for $\mu = 0$. This is in
mielke-geometrodynamics_F00999	196	■ 1.000	$m \ u$	mu	parameter mu vanishes in the metric ground form. Here the radial functions occurring
mielke-geometrodynamics_F01000	197	■ 1.000	$\kappa < 0$	$\kappa < 0$	For a real solution, $\kappa < 0$ is necessary. In this case, torsion can exist only in an anti-de
mielke-geometrodynamics_F01001	197	■ 1.000	$\mathrm{t} \rightarrow -\mathrm{i} \tau$	$t \rightarrow -i\tau$	Weyl’s trick, i.e., the formal substitution $t \rightarrow -i\tau$ of the timelike variable. It follows
mielke-geometrodynamics_F01002	197	■ 0.687	$\mathrm{f} = \mathrm{T}_{\hat{0}\hat{1}}^{\hat{0}}$	$f = T_{\hat{0}\hat{1}}^{\hat{0}}$	from the matrix representation (9.1.15) that the anholonomic components $f = T_{\hat{0}\hat{1}}^{\hat{0}}$
mielke-geometrodynamics_F01003	197	■ 0.790	$\mathrm{g} = \mathrm{T}_{\hat{1}\hat{2}}^{\hat{2}} = \mathrm{T}_{\hat{1}\hat{3}}^{\hat{3}}$	$g = T_{\hat{1}\hat{2}}^{\hat{2}} = T_{\hat{1}\hat{3}}^{\hat{3}}$	and $g = T_{\hat{1}\hat{2}}^{\hat{2}} = T_{\hat{1}\hat{3}}^{\hat{3}}$ are the only ones that are then allowed to be nonzero. A solution
mielke-geometrodynamics_F01004	198	■ 1.000	$\mathrm{n} = 4$	$n = 4$	ourselves to a manifold of dimension $n = 4$. The previous classification of the source-
mielke-geometrodynamics_F01005	198	■ 0.998	R^6	R^6	space R^6 . For then

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mielke-geometrodynamics_F01006	199	■ 0.995	γ_2^g	γ_2^g	the form γ_2^g over the basis manifold, which provides for a topological invariant. It is
mielke-geometrodynamics_F01007	199	■ 1.000	$\mathrm{H}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$	$H_p(M, \mathbb{R})$	rank of the pth homology group $H_p(M, \mathbb{R})$, i.e., from the so-called Betti numbers
mielke-geometrodynamics_F01008	199	■ 1.000	$\mathrm{b} := \operatorname{dim} \mathrm{H}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$	$b := \dim H_p(M, \mathbb{R})$	$b := \dim H_p(M, \mathbb{R})$, which were introduced by Riemann. Similarly as in the case of
mielke-geometrodynamics_F01009	199	■ 1.000	$\chi(\mathrm{M})$	$\chi(M)$	the vanishing of $\chi(M)$. In particular, it is known that a compact even-dimensional
mielke-geometrodynamics_F01010	199	■ 1.000	$\mathrm{s}=1$	$s = 1$	manifold admits a pseudo-Riemannian metric of signature $\mathbb{S}=\mathbb{I}$ (“Lorentz metric”)
mielke-geometrodynamics_F01011	199	■ 0.996	$\alpha: \omega^g - \omega^g$	$\alpha: \omega^g - \omega^g$	$\alpha: \omega^g - \omega^g$ of the embedding of the boundary ∂M in M . Although it is exactly
mielke-geometrodynamics_F01012	200	■ 1.000	B^n	B^n	an open ball B^n is to be cut out of M and N in order to identify the then respectively
mielke-geometrodynamics_F01013	200	■ 1.000	$\partial \mathrm{B}^n = \mathrm{S}^{n-1}$	$\partial B^n = S^{n-1}$	existing boundary spheres $\partial B^n = S^{n-1}$ with each other via a diffeomorphism that
mielke-geometrodynamics_F01014	200	■ 0.736	$\mathrm{W}^n := \mathrm{S}^1 \times \mathrm{S}^{n-1}$	$W^n := S^1 \times S^{n-1}$	$W^n := S^1 \times S^{n-1}$ (WHEELER 1955), the manifold $M\#(S^1 \times S^{n-1})$ comes into exis-
mielke-geometrodynamics_F01015	200	■ 0.736	$(\mathrm{S}^1 \times \mathrm{S}^{n-1})$	$(S^1 \times S^{n-1})$	$W^n := S^1 \times S^{n-1}$ (WHEELER 1955), the manifold $M\#(S^1 \times S^{n-1})$ comes into exis-
mielke-geometrodynamics_F01016	200	■ 0.980	$\mathbb{C} P^2$	$\mathbb{C}P^2$	number” of a surface (cf. SEIFERT & THRE FALL 1934), and $\mathbb{C}P^2$ denotes the complex
mielke-geometrodynamics_F01017	200	■ 1.000	E^2	E^2	projective two-space in E^2 . For spaces with uneven dimensions,
mielke-geometrodynamics_F01018	201	■ 1.000	$\mathrm{T}^4 = \mathrm{S}^1 \times \mathrm{S}^1 \times \mathrm{S}^1 \times \mathrm{S}^1$	$T^4 = S^1 \times S^1 \times S^1 \times S^1$	by the flat torus $T^4 = S^1 \times S^1 \times S^1 \times S^1$. Solutions with anti-self-double-dual cur-
mielke-geometrodynamics_F01019	202	■ 0.982	$\mathcal{S}_*(\mathrm{M}) := \mathcal{M} / \tilde{\mathcal{D}}$	$\mathcal{S}_*(M) := \mathcal{M} / \tilde{\mathcal{D}}$	space” $\mathcal{S}_*(M) := \mathcal{M} / \tilde{\mathcal{D}}$ of all Riemannian metrics modulo the diffeomorphisms
mielke-geometrodynamics_F01020	203	■ 1.000	S^2	S^2	Euler–Poincaré topological invariant (9.3.1). The two-sphere part S^2 of these $\mathbb{R} \times$
mielke-geometrodynamics_F01021	203	■ 1.000	$\mathbb{R} \times$	$\mathbb{R} \times$	Euler–Poincaré topological invariant (9.3.1). The two-sphere part S^2 of these $\mathbb{R} \times$
mielke-geometrodynamics_F01022	203	■ 1.000	$\mathrm{S}^1 \times \mathrm{S}^2$	$S^1 \times S^2$	$S^1 \times S^2$ topological configurations may even be knotted (MIELKE 1977a), and within
mielke-geometrodynamics_F01023	203	■ 1.000	$\mathbf{R}^4$	$\mathbf{R}^4$	piecewise linear structures (MILNOR 2011) on a noncompact $\mathbf{R}^4$. However, mani-
mielke-geometrodynamics_F01024	203	■ 1.000	U_4	U_4	Baekler P (1980) The unique spherically symmetric solution of the U_4 -theory of gravity in the
mielke-geometrodynamics_F01025	205	■ 0.999	$\mathrm{R} + \mathrm{R}^2$	$R + R^2$	Neville DE (1980) Birkhoff theorems for $R + R^2$ gravity theories with torsion. Phys Rev D
mielke-geometrodynamics_F01026	208	■ 0.837	C_{TL}	C_{TL}	term C_{TL} , which simulates to some extent Einstein’s theory in 3D, allowing us to
mielke-geometrodynamics_F01027	208	■ 0.994	$\varepsilon \rightarrow 1 / \varepsilon$	$\varepsilon \rightarrow 1/\varepsilon$	$\varepsilon \rightarrow 1/\varepsilon$.
mielke-geometrodynamics_F01028	208	■ 0.674	$\Gamma^{\beta \gamma} = \Gamma_j^{\beta \gamma} d x^j$	$\Gamma^{\beta \gamma} = \Gamma_j^{\beta \gamma} dx^j$	connection $\Gamma^{\beta \gamma} = \Gamma_j^{\beta \gamma} dx^j$, i.e.,
mielke-geometrodynamics_F01029	209	■ 1.000	$\Gamma_{\alpha}^{\star}$	$\Gamma_{\alpha}^{\star}$	Since the coframe ϑ_{α} has physical dimensions [length] and the connection $\Gamma_{\alpha}^{\star}$ is
mielke-geometrodynamics_F01030	209	■ 1.000	$\vartheta_{\alpha} / \ell$	$\vartheta_{\alpha} / \ell$	dimensionless, one can use $\vartheta_{\alpha} / \ell$ and $\Gamma_{\alpha}^{\star}$ as <i>dimensionless</i> gauge potentials of local
mielke-geometrodynamics_F01031	209	■ 1.000	$C = \operatorname{Tr}\{A \wedge F\}$	$C = \operatorname{Tr}\{A \wedge F\}$	The corresponding Chern–Simons three-forms of gauge type $C = \operatorname{Tr}\{A \wedge F\}$ are the
mielke-geometrodynamics_F01032	209	■ 1.000	T_{α}	T_{α}	are uniquely related to the torsion T_{α} , the curvature $R_{\alpha}^{\star}$, and a cosmological term η_{α} ,

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mielke-geometrodynamics_F01033	209	■ 1.000	$R_{\alpha}^{\star}$	$R_{\alpha}^{\star}$	are uniquely related to the torsion T_{α} , the curvature $R_{\alpha}^{\star}$, and a cosmological term η_{α} ,
mielke-geometrodynamics_F01034	209	■ 1.000	η_{α}	η_{α}	are uniquely related to the torsion T_{α} , the curvature $R_{\alpha}^{\star}$, and a cosmological term η_{α} ,
mielke-geometrodynamics_F01041	210	■ 0.872	$\mathrm{SO}(1,2)$	$SO(1,2)$	dimensional Lorentz group $SO(1,2)$ as a $9 = 5 \otimes 3 \otimes 1$ multiplet after bivectors are
mielke-geometrodynamics_F01043	211	■ 1.000	$H = -\partial L / \partial F$	$H = -\partial L / \partial F$	are the translational and rotational field momenta (or the excitation $H = -\partial L / \partial F$
mielke-geometrodynamics_F01044	211	■ 1.000	$H_{\alpha} = H_{[\alpha\beta]} \vartheta^{\beta}$	$H_{\alpha} = H_{[\alpha\beta]} \vartheta^{\beta}$	(In the particular case of an antisymmetric field $H_{\alpha} = H_{[\alpha\beta]} \vartheta^{\beta}$, this is dual to the
mielke-geometrodynamics_F01045	211	■ 1.000	$S_{\alpha} = -{}^* H_{\alpha}$	$S_{\alpha} = -{}^* H_{\alpha}$	translational field strength, i.e., $S_{\alpha} = -{}^* H_{\alpha}$.) The sources for the gravitational field
mielke-geometrodynamics_F01046	211	■ 0.957	$\mathbb{R}^3 \otimes S_0(1,2)$	$\mathbb{R}^3 \otimes SO(1,2)$	$\mathbb{R}^3 \otimes SO(1,2)$, we arrive at the MIELKE & BAEKLER (1991) (MB) model, which is
mielke-geometrodynamics_F01047	211	■ 1.000	θ_T, θ_L	θ_T, θ_L	Allowing for arbitrary “vacuum angles” θ_T , θ_L , and $\theta_{TL} = -\chi$, the most general
mielke-geometrodynamics_F01048	211	■ 1.000	$\theta_{TL} = -\chi$	$\theta_{TL} = -\chi$	Allowing for arbitrary “vacuum angles” θ_T , θ_L , and $\theta_{TL} = -\chi$, the most general
mielke-geometrodynamics_F01049	212	■ 0.997	C_T, C_L	C_T, C_L	where C_T , C_L , and C_{TL} are respectively the translational, rotational, and “mixed” CS
mielke-geometrodynamics_F01050	212	■ 1.000	C_T	C_T	tional term C_T is covariant, it appears that the MB model is only semi-topological,
mielke-geometrodynamics_F01051	212	■ 1.000	$\Gamma^{\star\alpha}$	$\Gamma^{\star\alpha}$	Consequently, varying the Lagrangian (10.2.2) with respect to ϑ^{α} and $\Gamma^{\star\alpha}$ yields
mielke-geometrodynamics_F01052	212	■ 1.000	$\theta_T / 2\ell$	$\theta_T / 2\ell$	$\theta_T / 2\ell$, resembling a cosmological constant familiar from 4D gravity.
mielke-geometrodynamics_F01053	212	■ 1.000	$\varepsilon = \theta_{TL} \theta_T / A$	$\varepsilon = \theta_{TL} \theta_T / A$	where the contortional constants $\varepsilon = \theta_{TL} \theta_T / A$ and $\rho = -\theta_T^2 / A$ are related to the
mielke-geometrodynamics_F01054	212	■ 1.000	$\rho = -\theta_T^2 / A$	$\rho = -\theta_T^2 / A$	where the contortional constants $\varepsilon = \theta_{TL} \theta_T / A$ and $\rho = -\theta_T^2 / A$ are related to the
mielke-geometrodynamics_F01055	212	■ 1.000	$A := -(-1)^s \theta_{TL}^2 + 2\theta_T \theta_L \neq 0$	$A := -(-1)^s \theta_{TL}^2 + 2\theta_T \theta_L \neq 0$	$A := -(-1)^s \theta_{TL}^2 + 2\theta_T \theta_L \neq 0$ is nonvanishing.
mielke-geometrodynamics_F01056	213	■ 0.994	$K_{\alpha}^{\star} = -{}^* T_{\alpha} / 2 = -(-1)^s (\varepsilon / 2\ell) \vartheta_{\alpha}$	$K_{\alpha}^{\star} = -{}^* T_{\alpha} / 2 = -(-1)^s (\varepsilon / 2\ell) \vartheta_{\alpha}$	for the RC curvature. From the relation $K_{\alpha}^{\star} = -{}^* T_{\alpha} / 2 = -(-1)^s (\varepsilon / 2\ell) \vartheta_{\alpha}$ for the
mielke-geometrodynamics_F01057	213	■ 1.000	$\varepsilon = 0$	$\varepsilon = 0$	$\varepsilon = 0$, i.e., in a purely Riemannian spacetime. Alternatively, for $\rho = 0$, i.e., in the
mielke-geometrodynamics_F01058	213	■ 1.000	Λ_{eff}	Λ_{eff}	cosmological term Λ_{eff} has the anti-de Sitter (AdS) metric
mielke-geometrodynamics_F01059	214	■ 1.000	J	J	respectively. Observe that the shift is proportional to the angular momentum J of
mielke-geometrodynamics_F01060	214	■ 1.000	$J^2 \leq M^2$	$J^2 \leq M^2$	the solution, which allows for $J^2 \leq M^2$ to interpret this configuration as a <i>rotating</i>
mielke-geometrodynamics_F01061	214	■ 1.000	$M = -1$	$M = -1$	normalization $M = -1$ and $J = 0$, it reduces to the AdS metric.
mielke-geometrodynamics_F01062	214	■ 1.000	$J = 0$	$J = 0$	normalization $M = -1$ and $J = 0$, it reduces to the AdS metric.
mielke-geometrodynamics_F01063 [conf 0.080]	214	■ 0.080	$\Gamma_{\alpha}^{\{\star\}} \rightarrow \Gamma_{\alpha}^{\star} = \Gamma_{\alpha}^{\{\star\}} - K_{\alpha}^{\star}$	$\Gamma_{\alpha}^{\{\star\}} \rightarrow \Gamma_{\alpha}^{\star} = \Gamma_{\alpha}^{\{\star\}} - K_{\alpha}^{\star}$	rest essentially on a “prolongation” $\Gamma_{\alpha}^{\{\star\}} \rightarrow \Gamma_{\alpha}^{\star} = \Gamma_{\alpha}^{\{\star\}} - K_{\alpha}^{\star}$ of the Riemannian to
mielke-geometrodynamics_F01064	214	■ 1.000	ε	ε	effective cosmological constant Λ_{eff} is related to ε and ρ via (10.2.10), the black hole
mielke-geometrodynamics_F01065	214	■ 1.000	$\Gamma_{\alpha}^{\star} = \frac{1}{2} \eta_{\alpha\beta\gamma} \Gamma^{\beta\gamma}$	$\Gamma_{\alpha}^{\star} = \frac{1}{2} \eta_{\alpha\beta\gamma} \Gamma^{\beta\gamma}$	where ε is a continuous parameter, involves the Lie dual $\Gamma_{\alpha}^{\star} = \frac{1}{2} \eta_{\alpha\beta\gamma} \Gamma^{\beta\gamma}$ of the
mielke-geometrodynamics_F01066	214	■ 1.000	$\tilde{T}_{\alpha} = 0$	$\tilde{T}_{\alpha} = 0$	Riemannian spacetime with $\tilde{T}_{\alpha} = 0$ and deformed <i>teleparallelism</i> in the local gauge

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mielke-geometrodynamics_F01067	214	■ 0.999	$\widetilde{\Gamma}_{\alpha}^{\star} \stackrel{*}{=} 0$	$\widetilde{\Gamma}_{\alpha}^{\star} \stackrel{*}{=} 0$	$\widetilde{\Gamma}_{\alpha}^{\star} \stackrel{*}{=} 0$, equivalent to the covariant constraint of vanishing modified RC curvature,
mielke-geometrodynamics_F01068	214	■ 1.000	$\widetilde{R}_{\alpha}^{\star} = 0$	$\widetilde{R}_{\alpha}^{\star} = 0$	i.e., $\widetilde{R}_{\alpha}^{\star} = 0$. In the latter case, coframe and connection are locally <i>Lie dual</i> to each
mielke-geometrodynamics_F01069	215	■ 1.000	$1 / \varepsilon$	$1/\varepsilon$	large rotational momentum proportional to $1/\varepsilon$ and conversely, resembling, to some
mielke-geometrodynamics_F01070	215	■ 0.915	$\rho = (-1)^{\text{sign}} \varepsilon^2 / 4$	$\rho = (-1)^{\text{sign}} \varepsilon^2 / 4$	as originally envisioned by CARTAN (1924). Here $\rho = (-1)^{\text{sign}} \varepsilon^2 / 4$ depends quadrat-
mielke-geometrodynamics_F01071	215	■ 0.518	$R_{\alpha}^{\star} = (-1)^{\text{sign}} \varepsilon T_{\alpha} / 4 \ell$	$R_{\alpha}^{\star} = (-1)^{\text{sign}} \varepsilon T_{\alpha} / 4 \ell$	should not ignore that $R_{\alpha}^{\star} = (-1)^{\text{sign}} \varepsilon T_{\alpha} / 4 \ell$ induces, for $\varepsilon \neq 0$, a constant-curvature
mielke-geometrodynamics_F01072	215	■ 0.518	$\varepsilon \neq 0$	$\varepsilon \neq 0$	should not ignore that $R_{\alpha}^{\star} = (-1)^{\text{sign}} \varepsilon T_{\alpha} / 4 \ell$ induces, for $\varepsilon \neq 0$, a constant-curvature
mielke-geometrodynamics_F01073	216	■ 1.000	$\varepsilon \rightarrow -\varepsilon$	$\varepsilon \rightarrow -\varepsilon$	symmetry” $\varepsilon \rightarrow -\varepsilon$ in (10.3.9) and (10.3.10), we see that the limit $\gamma = (1 - \delta\varepsilon) \rightarrow$
mielke-geometrodynamics_F01074	216	■ 1.000	$\gamma = (1 - \delta\varepsilon) \rightarrow$	$\gamma = (1 - \delta\varepsilon) \rightarrow$	symmetry” $\varepsilon \rightarrow -\varepsilon$ in (10.3.9) and (10.3.10), we see that the limit $\gamma = (1 - \delta\varepsilon) \rightarrow$
mielke-geometrodynamics_F01075	216	■ 1.000	$\gamma \neq 0$	$\gamma \neq 0$	(10.2.1). For $\gamma \neq 0$, the vacuum solution (10.3.9) yields
mielke-geometrodynamics_F01076	216	■ 1.000	$\varepsilon \rightarrow 0$	$\varepsilon \rightarrow 0$	in the limit $\varepsilon \rightarrow 0$ of <i>weak</i> axial torsion coupling, the translational field momenta
mielke-geometrodynamics_F01077	216	■ 1.000	$H_{\alpha}^{\star} = \gamma \vartheta_{\alpha} / \ell$	$H_{\alpha}^{\star} = \gamma \vartheta_{\alpha} / \ell$	i.e., $H_{\alpha}^{\star} = \gamma \vartheta_{\alpha} / \ell$, similar to the case of the EC action. Conversely, in the limit
mielke-geometrodynamics_F01078	216	■ 1.000	$\varepsilon \rightarrow \infty$	$\varepsilon \rightarrow \infty$	$\varepsilon \rightarrow \infty$ of <i>strong</i> axial torsion coupling, the translational momenta (10.3.9) become
mielke-geometrodynamics_F01079	216	■ 1.000	$H_{\alpha} = -\gamma \vartheta_{\alpha} / \delta \ell^2$	$H_{\alpha} = -\gamma \vartheta_{\alpha} / \delta \ell^2$	$H_{\alpha} = -\gamma \vartheta_{\alpha} / \delta \ell^2$, and the rotational momenta turn out to be $H_{\alpha}^{\star} = 0$. This implies
mielke-geometrodynamics_F01080	216	■ 1.000	$H_{\alpha}^{\star} = 0$	$H_{\alpha}^{\star} = 0$	$H_{\alpha} = -\gamma \vartheta_{\alpha} / \delta \ell^2$, and the rotational momenta turn out to be $H_{\alpha}^{\star} = 0$. This implies
mielke-geometrodynamics_F01081	217	■ 1.000	$\{H, dt, d\mathscr{A}, dh\}$	$\{H, dt, d\mathscr{A}, dh\}$	a basis $\{H, dt, d\mathscr{A}, dh\}$ of one-forms can be constructed that, for nonzero torsion
mielke-geometrodynamics_F01082	217	■ 1.000	$H = 0$	$H = 0$	For $H = 0$, the three one-forms on the right-hand side become linearly dependent,
mielke-geometrodynamics_F01083	217	■ 0.998	$H_{\alpha} \neq 0$	$H_{\alpha} \neq 0$	thereby sending us back to the exact solution (10.3.9). However, for $H_{\alpha} \neq 0$, an exact
mielke-geometrodynamics_F01084	217	■ 1.000	$d\mathscr{A}$	$d\mathscr{A}$	can be proposed in which one coordinate “leg,” for instance $d\mathscr{A}$, needs to be con-
mielke-geometrodynamics_F01085	218	■ 1.000	$d^* \vartheta_{\alpha} = 0$	$d^* \vartheta_{\alpha} = 0$	generalizes the Hilbert–de Donder or <i>transversality condition</i> $d^* \vartheta_{\alpha} = 0$ for the
mielke-geometrodynamics_F01086	218	■ 1.000	$\eta_{\alpha} = {}^* \vartheta_{\alpha}$	$\eta_{\alpha} = {}^* \vartheta_{\alpha}$	To this end, we will employ the relation $\eta_{\alpha} = {}^* \vartheta_{\alpha}$ for the η_{α} basis and iterate the
mielke-geometrodynamics_F01087	219	■ 1.000	$m = \varepsilon / \ell$	$m = \varepsilon / \ell$	where $m = \varepsilon / \ell$, results in vacuum and the gauge-covariant Hilbert–de Donder or
mielke-geometrodynamics_F01088	219	■ 0.782	$\vartheta^{\alpha} = E_i^{\alpha} dx^i$	$\vartheta^{\alpha} = E_i^{\alpha} dx^i$	is as follows: In 3D, the coframe $\vartheta^{\alpha} = E_i^{\alpha} dx^i$ has $3 \times 3 = 9$ components, of which
mielke-geometrodynamics_F01089	219	■ 0.782	$3 \times 3 = 9$	$3 \times 3 = 9$	is as follows: In 3D, the coframe $\vartheta^{\alpha} = E_i^{\alpha} dx^i$ has $3 \times 3 = 9$ components, of which
mielke-geometrodynamics_F01090	219	■ 0.924	$SO(1, 2)$	$SO(1, 2)$	3 are pure gauge due to the $SO(1, 2)$ Lorentz transformation $\vartheta'^{\alpha} = \Lambda_{\beta}^{\alpha}(x) \vartheta^{\beta}$. The
mielke-geometrodynamics_F01091	219	■ 0.924	$\vartheta'^{\alpha} = \Lambda_{\beta}^{\alpha}(x) \vartheta^{\beta}$	$\vartheta'^{\alpha} = \Lambda_{\beta}^{\alpha}(x) \vartheta^{\beta}$	3 are pure gauge due to the $SO(1, 2)$ Lorentz transformation $\vartheta'^{\alpha} = \Lambda_{\beta}^{\alpha}(x) \vartheta^{\beta}$. The
mielke-geometrodynamics_F01092	220	■ 0.992	$2 / \theta_{\text{L}}$	$2/\theta_{\text{L}}$	contributes to the mass with an “anyon”-type g-factor of $2/\theta_{\text{L}}$; cf. JACKIW & NAIR
mielke-geometrodynamics_F01093	220	■ 1.000	$\theta_{\text{T}} \theta_{\text{L}} / 2 \chi = 1$	$\theta_{\text{T}} \theta_{\text{L}} / 2 \chi = 1$	to be constrained to $\theta_{\text{T}} \theta_{\text{L}} / 2 \chi = 1$, or at least to be positive in order to avoid nega-

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mielke-geometrodynamics_F01094	220	■ 1.000	$\theta_{\mathrm{T}} \neq 0$	$\theta_{\mathrm{T}} \neq 0$	$\theta_{\mathrm{T}} \neq 0$, the vacuum angle of the translational Chern–Simons term C_{T} . In any case,
mielke-geometrodynamics_F01095	221	■ 1.000	$\varepsilon \neq \tilde{\varepsilon}$	$\varepsilon \neq \tilde{\varepsilon}$	Then, for <i>different</i> continuous parameters $\varepsilon \neq \tilde{\varepsilon}$, we obtain
mielke-geometrodynamics_F01096	221	■ 1.000	$\tilde{\theta}_{\mathrm{L}}$	$\tilde{\theta}_{\mathrm{L}}$	parent. Consequently, the constants θ_{L} and $\tilde{\theta}_{\mathrm{L}}$ give rise to the two central charges
mielke-geometrodynamics_F01097	222	■ 1.000	$C^{\mathrm{k\ l}}=C^{\mathrm{l\ k}}$	$C^{kl} = C^{lk}$	is the one-form associated with the symmetric Cotton tensor $C^{kl} = C^{lk}$. In general,
mielke-geometrodynamics_F01098	222	■ 0.940	$\Gamma^{\{\lambda\alpha\beta}$	$\Gamma^{\{\lambda\alpha\beta}$	this is a third-order equation in the Levi-Civita connection $\Gamma^{(\cdot)\alpha\beta}$, i.e., of fourth order
mielke-geometrodynamics_F01099	222	■ 1.000	$T_{\alpha}=D\vartheta_{\alpha}=0$	$T_{\alpha} = D\vartheta_{\alpha} = 0$	Due to $T_{\alpha} = D\vartheta_{\alpha} = 0$ in Riemannian spacetime, implying $H_{\alpha} = 0$ in 3D, the mod-
mielke-geometrodynamics_F01100	222	■ 1.000	$H_{\alpha}=0$	$H_{\alpha} = 0$	Due to $T_{\alpha} = D\vartheta_{\alpha} = 0$ in Riemannian spacetime, implying $H_{\alpha} = 0$ in 3D, the mod-
mielke-geometrodynamics_F01101	222	■ 0.891	$D\,H_{\alpha}^{\star}=\delta\ell\,D\,H_{\alpha}+(\gamma/\ell)\,D\vartheta_{\alpha}\equiv 0$	$DH_{\alpha}^{\star} = \delta\ell DH_{\alpha} + (\gamma/\ell)D\vartheta_{\alpha} \equiv 0$	ified S -duality leads to <u>$DH_{\alpha}^{\star} = \delta\ell DH_{\alpha} + (\gamma/\ell)D\vartheta_{\alpha} \equiv 0$</u> , and the two-form $D^{(\cdot)}H_{[\alpha\beta]}$
mielke-geometrodynamics_F01102	222	■ 0.891	$D^{[4]}H_{[\alpha\beta]}$	$D^{[4]}H_{[\alpha\beta]}$	ified S -duality leads to $DH_{\alpha}^{\star} = \delta\ell DH_{\alpha} + (\gamma/\ell)D\vartheta_{\alpha} \equiv 0$, and the two-form $D^{(\cdot)}H_{[\alpha\beta]}$
mielke-geometrodynamics_F01103	222	■ 0.539	$G_{\alpha}=R_{\alpha}^{\{\cdot\}^{\star}}$	$G_{\alpha} = R_{\alpha}^{\{\cdot\}^{\star}}$	in Riemannian spacetime. Since $G_{\alpha} = R_{\alpha}^{(\cdot)\star}$ is the Einstein current two-form in 3D,
mielke-geometrodynamics_F01104	222	■ 0.990	$\widetilde{\Sigma}_{\alpha}:=\Sigma_{\alpha}-D^{\{\cdot\}}\mu_{\alpha}$	$\widetilde{\Sigma}_{\alpha} := \Sigma_{\alpha} - D^{\{\cdot\}}\mu_{\alpha}$	<i>symmetric</i> Belinfante–Rosenfeld two-form $\widetilde{\Sigma}_{\alpha} := \Sigma_{\alpha} - D^{(\cdot)}\mu_{\alpha}$ as source.
mielke-geometrodynamics_F01105	223	■ 1.000	$E=p\,c_{\mathrm{F}}$	$E = pc_{\mathrm{F}}$	spectrum is $E = pc_{\mathrm{F}}$, where c_{F} is the effective speed of fermions in the 2D carbon
mielke-geometrodynamics_F01106	223	■ 1.000	c_{F}	c_{F}	spectrum is $E = pc_{\mathrm{F}}$, where c_{F} is the effective speed of fermions in the 2D carbon
mielke-geometrodynamics_F01107	223	■ 0.825	$\mathrm{SO}(3)$	$\mathrm{SO}(3)$	space(time): in the Euclidean case, the covering group of the rotation group $\mathrm{SO}(3)$
mielke-geometrodynamics_F01108	224	■ 0.902	$s=1$	$s = 1$	On the other hand, for Lorentzian signature $s = 1$, the covering group of $\mathrm{SO}(1,2)$ is
mielke-geometrodynamics_F01109	224	■ 0.942	$S\,L(2,\,\mathbb{R})$	$SL(2, \mathbb{R})$	isomorphic to the real group $SL(2, \mathbb{R})$, or as well to the symplectic algebra $\mathfrak{sp}(2, \mathbb{R})$.
mielke-geometrodynamics_F01110	224	■ 0.942	$\mathfrak{sp}(2,\,\mathbb{R})$	$\mathfrak{sp}(2, \mathbb{R})$	isomorphic to the real group $SL(2, \mathbb{R})$, or as well to the symplectic algebra $\mathfrak{sp}(2, \mathbb{R})$.
mielke-geometrodynamics_F01111	224	■ 1.000	$\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2g_{\alpha\beta}$	$\gamma_{\alpha}\gamma_{\beta} + \gamma_{\beta}\gamma_{\alpha} = 2g_{\alpha\beta}$	thus providing a realization of the Clifford algebra $\gamma_{\alpha}\gamma_{\beta} + \gamma_{\beta}\gamma_{\alpha} = 2g_{\alpha\beta}$ in 3D.
mielke-geometrodynamics_F01112	224	■ 1.000	$D\,\psi:=d\,\psi-\frac{1}{2}\gamma_{\alpha}\wedge\Gamma^{\star\alpha}\psi$	$D\psi := d\psi - \frac{1}{2}\gamma_{\alpha} \wedge \Gamma^{\star\alpha}\psi$	where $D\psi := d\psi - \frac{1}{2}\gamma_{\alpha} \wedge \Gamma^{\star\alpha}\psi$ denotes the covariant spinor derivative in 3D.
mielke-geometrodynamics_F01113	224	■ 1.000	L_{D}	L_{D}	is obtained by varying L_{D} with respect to $\overline{\Psi}$. Since
mielke-geometrodynamics_F01114	224	■ 1.000	$\bar{\Psi}$	$\bar{\Psi}$	is obtained by varying L_{D} with respect to $\overline{\Psi}$. Since
mielke-geometrodynamics_F01115	224	■ 1.000	$\hat{D}$	$\hat{D}$	The curvature associated with the deformed covariant derivative $\hat{D}$ is
mielke-geometrodynamics_F01116	225	■ 0.999	$F:=d\,A$	$F := dA$	field strength reads $F := dA$, i.e., it remains an exact two-form. In 3D, the Maxwell
mielke-geometrodynamics_F01117	225	■ 0.539	$d\left(A\wedge^{\star}F\right)/2$	$d\left(A\wedge^{\star}F\right)/2$	However, this three-form is gauge-invariant only modulo an exact form $d(A\wedge^{\star}F)/2$.
mielke-geometrodynamics_F01118	225	■ 0.528	$\square:=d^{\star}d^{\star}-^{\star}d^{\star}d$	$\square := d^{\star}d^{\star} - ^{\star}d^{\star}d$	where $\square := d^{\star}d^{\star} - ^{\star}d^{\star}d$ is the d’Alembertian in 2 + 1 dimensions. Since (10.6.3) is a
mielke-geometrodynamics_F01119	225	■ 1.000	$\lambda=\hbar/c\,m_{\mathrm{photon}}$	$\lambda = \hbar/cm_{\mathrm{photon}}$	reasons, via the Compton wavelength $\lambda = \hbar/cm_{\mathrm{photon}}$ for massive photons.
mielke-geometrodynamics_F01120	226	■ 0.987	x^{α}	x^{α}	vention that x^{α} and y^{α} are spacelike orthogonal vectors that span the (x, y) -plane

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mielke-geometrodynamics_F01121	226	■ 0.987	y^{α}	y^{α}	vention that x^{α} and y^{α} are spacelike orthogonal vectors that span the (x, y) -plane
mielke-geometrodynamics_F01122	226	■ 0.987	(x, y)	(x, y)	vention that x^{α} and y^{α} are spacelike orthogonal vectors that span the (x, y) -plane
mielke-geometrodynamics_F01123	226	■ 1.000	n^{α}	n^{α}	Moreover, the vector n^{α} is a timelike unit vector normal to the hypersurface with
mielke-geometrodynamics_F01124	226	■ 1.000	$n^{\alpha} n_{\alpha}=s$	$n^{\alpha} n_{\alpha} = s$	$n^{\alpha} n_{\alpha} = s$, the signature s of our 3D spacetime. Following SOLENG (1992), ANANDAN
mielke-geometrodynamics_F01125	226	■ 1.000	$\tau_{\alpha}^{\star}$	$\tau_{\alpha}^{\star}$	(1994), and BAKKE et al. (2009), we assume that the two-forms Σ_{α} and $\tau_{\alpha}^{\star}$ of the
mielke-geometrodynamics_F01126	227	■ 1.000	X	X	of the defect; cf. Fig. 10.2. From the specification (10.7.1) of the one-forms X and
mielke-geometrodynamics_F01127	227	■ 1.000	Y	Y	Y , it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}} \neq 0$ and $\tau_{\hat{1}\hat{2}} =$
mielke-geometrodynamics_F01128	227	■ 1.000	$\Sigma_{\hat{0}} \neq 0$	$\Sigma_{\hat{0}} \neq 0$	Y , it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}} \neq 0$ and $\tau_{\hat{1}\hat{2}} =$
mielke-geometrodynamics_F01129	227	■ 1.000	$\tau_{\hat{1}\hat{2}} =$	$\tau_{\hat{1}\hat{2}} =$	Y , it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}} \neq 0$ and $\tau_{\hat{1}\hat{2}} =$
mielke-geometrodynamics_F01130	227	■ 0.975	$-\tau_{\hat{2}\hat{1}} \neq 0$	$-\tau_{\hat{2}\hat{1}} \neq 0$	$-\tau_{\hat{2}\hat{1}} \neq 0$.
mielke-geometrodynamics_F01131	227	■ 1.000	$x^{[\alpha} y^{\beta]} R_{\alpha\beta} = R_{\hat{1}\hat{2}} = -R_{\hat{2}\hat{1}} \neq 0$	$x^{[\alpha} y^{\beta]} R_{\alpha\beta} = R_{\hat{1}\hat{2}} = -R_{\hat{2}\hat{1}} \neq 0$	with x^{α} and y^{α} reveal that $x^{[\alpha} y^{\beta]} R_{\alpha\beta} = R_{\hat{1}\hat{2}} = -R_{\hat{2}\hat{1}} \neq 0$ are the only nonvanishing
mielke-geometrodynamics_F01132	227	■ 0.511	$N^{\alpha} = n \rfloor \vartheta^{\alpha}$	$N^{\alpha} = n \rfloor \vartheta^{\alpha}$	We recall that $N^{\alpha} = n \rfloor \vartheta^{\alpha}$ is the lapse and shift vector in the (2+1) decomposition à
mielke-geometrodynamics_F01133	228	■ 1.000	$d(\mathscr{A} \wedge d\mathscr{A})$	$d(\mathscr{A} \wedge d\mathscr{A})$	Pointryagin-type term $d(\mathscr{A} \wedge d\mathscr{A})$ from the axial torsion. Moreover, the Nieh–Yan
mielke-geometrodynamics_F01134	228	■ 1.000	dC_{T}	dC_{T}	term dC_{T} proportional to $d^* \mathscr{A}$ vanishes identically for this example of a spinning
mielke-geometrodynamics_F01135	228	■ 1.000	$d^* \mathscr{A}$	$d^* \mathscr{A}$	term dC_{T} proportional to $d^* \mathscr{A}$ vanishes identically for this example of a spinning
mielke-geometrodynamics_F01136	228	■ 1.000	$\mathcal{N}=1$	$\mathcal{N} = 1$	try generator, i.e., $\mathcal{N} = 1$, represents the simplest consistent coupling of a Rarita–
mielke-geometrodynamics_F01137	228	■ 0.961	$\Psi_{\alpha} := e_{\alpha} \rfloor \Psi$	$\Psi_{\alpha} := e_{\alpha} \rfloor \Psi$	as $\Psi_{\alpha} := e_{\alpha} \rfloor \Psi$ involves the tetrad that is inverse to the frame in the anholonomic
mielke-geometrodynamics_F01138	228	■ 1.000	$i, j, k, \ldots = 0, 1, 2$	$i, j, k, \ldots = 0, 1, 2$	run over $i, j, k, \ldots = 0, 1, 2$, whereas $\alpha, \beta, \ldots = \hat{0}, \hat{1}, \hat{2}$ for the anholonomic indices.
mielke-geometrodynamics_F01139	228	■ 1.000	$\alpha, \beta, \ldots = \hat{0}, \hat{1}, \hat{2}$	$\alpha, \beta, \ldots = \hat{0}, \hat{1}, \hat{2}$	run over $i, j, k, \ldots = 0, 1, 2$, whereas $\alpha, \beta, \ldots = \hat{0}, \hat{1}, \hat{2}$ for the anholonomic indices.
mielke-geometrodynamics_F01140	228	■ 1.000	$\bar{\Psi} := \Psi^{\dagger} \gamma^0$	$\bar{\Psi} := \Psi^{\dagger} \gamma^0$	$\bar{\Psi} := \Psi^{\dagger} \gamma^0$.
mielke-geometrodynamics_F01141	228	■ 1.000	$\Psi := \Psi_{\alpha} \vartheta^{\alpha}$	$\Psi := \Psi_{\alpha} \vartheta^{\alpha}$	³ In four dimensions (4D), the RS field $\Psi := \Psi_{\alpha} \vartheta^{\alpha}$ entering (10.8.1) is a <i>Majorana-spinor</i> -valued
mielke-geometrodynamics_F01142	228	■ 0.700	$\Psi = C \bar{\Psi}^{\mathrm{T}}$	$\Psi = C \bar{\Psi}^{\mathrm{T}}$	i.e., $\Psi = C \bar{\Psi}^{\mathrm{T}}$, where C is the charge conjugation matrix given by $C = -i\gamma_0$ satisfying $C^{\dagger} =$
mielke-geometrodynamics_F01143	228	■ 0.700	$C = -i \gamma_0$	$C = -i\gamma_0$	i.e., $\Psi = C \bar{\Psi}^{\mathrm{T}}$, where C is the charge conjugation matrix given by $C = -i\gamma_0$ satisfying $C^{\dagger} =$
mielke-geometrodynamics_F01144	228	■ 0.700	$C^{\dagger} =$	$C^{\dagger} =$	i.e., $\Psi = C \bar{\Psi}^{\mathrm{T}}$, where C is the charge conjugation matrix given by $C = -i\gamma_0$ satisfying $C^{\dagger} =$
mielke-geometrodynamics_F01145	228	■ 1.000	$C^{-1}, \quad C^{\mathrm{T}} = -C$	$C^{-1}, \quad C^{\mathrm{T}} = -C$	$C^{-1}, \quad C^{\mathrm{T}} = -C$ and $C^{-1} \gamma^{\alpha} C = -(\gamma^{\alpha})^{\mathrm{T}}$. Consequently,
mielke-geometrodynamics_F01146	228	■ 1.000	$C^{-1} \gamma^{\alpha} C = -(\gamma^{\alpha})^{\mathrm{T}}$	$C^{-1} \gamma^{\alpha} C = -(\gamma^{\alpha})^{\mathrm{T}}$	$C^{-1}, \quad C^{\mathrm{T}} = -C$ and $C^{-1} \gamma^{\alpha} C = -(\gamma^{\alpha})^{\mathrm{T}}$. Consequently,
mielke-geometrodynamics_F01147	228	■ 1.000	γ^{α}	γ^{α}	For the real Majorana representation, all γ^{α} are purely imaginary, and the components of the

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mielke-geometrodynamics_F01148	229	0.803	L_{RS}	L_{RS}	The generalization of L_{RS} analyzed by MIELKE & MAGGIOLO (2012) exists only
mielke-geometrodynamics_F01149	230	0.995	$\vartheta^\alpha, \Gamma_\alpha^\star, \Psi$	$\vartheta^\alpha, \Gamma_\alpha^\star, \Psi$	independent variables $(\vartheta^\alpha, \Gamma_\alpha^\star, \Psi)$ yields
mielke-geometrodynamics_F01150	230	1.000	c	c	where σ stands in for a spinor-valued zero-form and c is a real constant. Let us probe
mielke-geometrodynamics_F01151	230	1.000	$\overline{c \, \gamma \, \sigma} = c \, \overline{\sigma} \, \gamma$	$\overline{c \, \gamma \, \sigma} = c \, \overline{\sigma} \, \gamma$	where $\overline{c \, \gamma \, \sigma} = c \, \overline{\sigma} \, \gamma$ holds for the Dirac adjoint.
mielke-geometrodynamics_F01152	230	1.000	$\delta L / \delta \Gamma_\alpha^\star \cong 0$	$\delta L / \delta \Gamma_\alpha^\star \cong 0$	In the following, let us assume that the second field equation $\delta L / \delta \Gamma_\alpha^\star \cong 0$ satisfies,
mielke-geometrodynamics_F01153	231	1.000	$c = -m$	$c = -m$	$\tau_\alpha^\star$, vanish. Moreover, we require $c = -m$, in order to eliminate kinetic terms like
mielke-geometrodynamics_F01154	231	1.000	$\gamma \wedge D \Psi$	$\gamma \wedge D \Psi$	$\gamma \wedge D \Psi$. Then, using the 3D formula
mielke-geometrodynamics_F01155	232	0.999	L_∞	L_∞	for the coupling constants of the bosonic part of L_∞ . Consequently, massless RS
mielke-geometrodynamics_F01156	232	1.000	$\mathcal{N} = 2$	$\mathcal{N} = 2$	More recently, ALVAREZ et al. (2015) considered an $\mathcal{N} = 2$ supersymmetry with-
mielke-geometrodynamics_F01157	233	0.988	$D = 3$	$D = 3$	Deser S, McCarthy J (1990) Self-dual formulations of $D = 3$ gravity theories. Phys Lett B
mielke-geometrodynamics_F01158	233	0.666	$(2+1)$	$(2+1)$	Garcia-Compean H, Obregon O, Ramirez C, Sabido M (2001) On S duality in (2+1) Chern–Simons
mielke-geometrodynamics_F01159	237	0.772	$1 / 2$	$1/2$	$1/2$ are described by bispinor representations $D^{(-\frac{1}{2}, \frac{3}{2})} \otimes D^{(\frac{1}{2}, \frac{3}{2})}(\widetilde{SO}_o(1, 3))$ of the
mielke-geometrodynamics_F01160	237	0.772	$D^{\left(-\frac{1}{2}, \frac{3}{2}\right)} \otimes D^{\left(\frac{1}{2}, \frac{3}{2}\right)} \left(\widetilde{SO}_o(1, 3)\right)$	$D^{(-\frac{1}{2}, \frac{3}{2})} \otimes D^{(\frac{1}{2}, \frac{3}{2})} \left(\widetilde{SO}_o(1, 3)\right)$	$1/2$ are described by bispinor representations $D^{(-\frac{1}{2}, \frac{3}{2})} \otimes D^{(\frac{1}{2}, \frac{3}{2})}(\widetilde{SO}_o(1, 3))$ of the
mielke-geometrodynamics_F01161	237	1.000	$SO_o(1, 3)$	$SO_o(1, 3)$	proper Lorentz group $SO_o(1, 3)$ (GEL’FAND et al. 1963). ² These representations
mielke-geometrodynamics_F01162	237	0.983	$P_o := \mathbb{R}^4 \otimes SO_o(1, 3)$	$P_o := \mathbb{R}^4 \otimes SO_o(1, 3)$	resentations of the (restricted) Poincaré group $P_o := \mathbb{R}^4 \otimes SO_o(1, 3)$ of flat
mielke-geometrodynamics_F01163	237	1.000	$\psi_\infty := \psi_{1/2} \otimes \psi_{5/2} \otimes \psi_{9/2} \otimes \dots$	$\psi_\infty := \psi_{1/2} \otimes \psi_{5/2} \otimes \psi_{9/2} \otimes \dots$	$\psi_\infty := \psi_{1/2} \otimes \psi_{5/2} \otimes \psi_{9/2} \otimes \dots$.
mielke-geometrodynamics_F01164	237	0.706	$SO(s, n - s)$	$SO(s, n - s)$	(pseudo-orthogonal) structure group $SO(s, n - s)$ of the tangent bundle of an
mielke-geometrodynamics_F01165	237	1.000	$s \neq 0$	$s \neq 0$	n-dimensional manifold ³ of signature s . Provided $s \neq 0$, this topological space
mielke-geometrodynamics_F01166	237	1.000	$SO_o(s, n - s)$	$SO_o(s, n - s)$	ing to homotopy theory. Concerning the connected component $SO_o(s, n - s)$ of the
mielke-geometrodynamics_F01167	237	0.929	$\widetilde{SO}_o(s, n - s) (= \operatorname{Spin}(n))$	$\widetilde{SO}_o(s, n - s) (= \operatorname{Spin}(n))$	$\widetilde{SO}_o(s, n - s) (= \operatorname{Spin}(n))$ for $s = 0$), whose irreducible representations provide
mielke-geometrodynamics_F01168	237	0.929	$s = 0$	$s = 0$	$\widetilde{SO}_o(s, n - s) (= \operatorname{Spin}(n))$ for $s = 0$), whose irreducible representations provide
mielke-geometrodynamics_F01169	238	1.000	$\Lambda : \tilde{G} \rightarrow G$	$\Lambda : \tilde{G} \rightarrow G$	This transition is achieved by a group homomorphism $\Lambda : \tilde{G} \rightarrow G$, which is
mielke-geometrodynamics_F01170	238	1.000	$C(s, n - s)$	$C(s, n - s)$	based on the following explicit construction: Let $C(s, n - s)$ denote the so-called
mielke-geometrodynamics_F01171	238	1.000	γ_α	γ_α	Clifford algebra, which is spanned by <i>generalized</i> complex Dirac matrices γ_α . These
mielke-geometrodynamics_F01172	238	1.000	$N \times N$	$N \times N$	and admit realizations by $N \times N$ matrices for which $N = 2^{[n/2]}$ corresponds to the
mielke-geometrodynamics_F01173	238	1.000	$N = 2^{[n/2]}$	$N = 2^{[n/2]}$	and admit realizations by $N \times N$ matrices for which $N = 2^{[n/2]}$ corresponds to the
mielke-geometrodynamics_F01174	238	1.000	$\mathbf{E}^n$	$\mathbf{E}^n$	Let the transformations of the Euclidean vector space $\mathbf{E}^n$ with coordinates x^α be

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mielke-geometrodynamics_F01175	238	■ 0.963	$\mathrm{g} \in S O(s, n-s)$	$g \in SO(s, n-s)$	determined by the (Lorentz) rotations $g \in SO(s, n-s)$ on behalf of the group action
mielke-geometrodynamics_F01176	238	■ 0.983	$2^{\{N\}}$	2^N	to a 2^N -dimensional bundle, the so-called Clifford bundle CV. For details, see ATIYAH
mielke-geometrodynamics_F01177	239	■ 0.976	$\widetilde{S O}_{\circ}(1,3) \approx S L(2, \mathbb{C})$	$\widetilde{SO}_{\circ}(1,3) \approx SL(2, \mathbb{C})$	orthochronous Lorentz group $SO_{\circ}(1,3)$ is represented by $\widetilde{SO}_{\circ}(1,3) \approx SL(2, \mathbb{C})$
mielke-geometrodynamics_F01178	239	■ 1.000	$\widetilde{S O}_{\circ}(s, n-s)$	$\widetilde{SO}_{\circ}(s, n-s)$	infinitesimal generators of the group $\widetilde{SO}_{\circ}(s, n-s)$ are
mielke-geometrodynamics_F01179	239	■ 0.588	$\mathrm{P}_0 := \mathbb{R}^n \otimes \widetilde{S O}_{\circ}$	$P_0 := \mathbb{R}^n \otimes \widetilde{SO}_{\circ}$	as induced representations of the “covering” Poincaré group $P_0 := \mathbb{R}^n \otimes \widetilde{SO}_{\circ}$.
mielke-geometrodynamics_F01180	239	■ 1.000	$(s, n-s)$	$(s, n-s)$	$(s, n-s)$ in a curved spacetime. It is necessary to obtain from a more general
mielke-geometrodynamics_F01181	239	■ 1.000	$L^{\mathfrak{g}}(M)$	$L^{\mathfrak{g}}(M)$	“geometric arena” the bundle $L^{\mathfrak{g}}(M)$ of orthogonal frames concerning the foundation
mielke-geometrodynamics_F01182	239	■ 0.974	$L_{\circ}^{\mathfrak{g}}(M)$	$L_{\circ}^{\mathfrak{g}}(M)$	a <i>spin structure</i> is imposed on a base space M as follows: Besides $L_{\circ}^{\mathfrak{g}}(M)$, the
mielke-geometrodynamics_F01183	239	■ 0.998	$\mathrm{f}: \tilde{L}(M) \rightarrow L_{\circ}^{\mathfrak{g}}(M)$	$f: \tilde{L}(M) \rightarrow L_{\circ}^{\mathfrak{g}}(M)$	is introduced (MILNOR 1963) together with the bundle map $f: \tilde{L}(M) \rightarrow L_{\circ}^{\mathfrak{g}}(M)$,
mielke-geometrodynamics_F01184	239	■ 0.950	$L_{\circ}^{\mathfrak{g}}(M)$	$L_{\circ}^{\mathfrak{g}}(M)$	A spin structure over M is equivalent to a double covering of $L_{\circ}^{\mathfrak{g}}(M)$. This spin
mielke-geometrodynamics_F01185	239	■ 0.949	$\tilde{L}^{\prime}(M), f^{\prime}$	$\tilde{L}^{\prime}(M), f^{\prime}$	structure is identical to a second one that is represented by $(\tilde{L}^{\prime}(M), f^{\prime})$ if the iso-
mielke-geometrodynamics_F01186	239	■ 1.000	$f^{\prime}$	$f^{\prime}$	bundle mapping $f^{\prime}$. Additionally, a <i>spin manifold</i> is understood as a Riemannian
mielke-geometrodynamics_F01187	239	■ 1.000	$\mathrm{w}_2 \in \mathrm{H}^2(M, \mathbb{Z}_2)$	$w_2 \in H^2(M, \mathbb{Z}_2)$	it is necessary and sufficient that the Stiefel–Whitney class $w_2 \in H^2(M, \mathbb{Z}_2)$ of
mielke-geometrodynamics_F01188	239	■ 1.000	g^{th}	g^{th}	the base space M be zero. Here the g^{th} Stiefel–Whitney class is considered as the
mielke-geometrodynamics_F01189	240	■ 1.000	$\mathrm{H}^{\mathfrak{g}}(M, \mathbb{Z}_2)$	$H^{\mathfrak{g}}(M, \mathbb{Z}_2)$	characteristic class of the g^{th} relative cohomology group $H^{\mathfrak{g}}(M, \mathbb{Z}_2)$ of the manifold
mielke-geometrodynamics_F01190	240	■ 1.000	$\tilde{s} = \dim \mathrm{H}^1(M, \mathbb{Z}_2)$	$\tilde{s} = \dim H^1(M, \mathbb{Z}_2)$	defined by the dimension $\tilde{s} = \dim H^1(M, \mathbb{Z}_2)$ of the first relative cohomology group
mielke-geometrodynamics_F01191	240	■ 0.813	$\tilde{s} = 2^v$	$\tilde{s} = 2^v$	(MILNOR 1963). Accordingly, there are exactly $\tilde{s} = 2^v$ spin structures possible
mielke-geometrodynamics_F01192	240	■ 0.997	ν	ν	hypersurface consists of the connected sum of v handlebodies (“wormholes”) $W^3 :=$
mielke-geometrodynamics_F01193	240	■ 0.997	$W^3 :=$	$W^3 :=$	hypersurface consists of the connected sum of v handlebodies (“wormholes”) $W^3 :=$
mielke-geometrodynamics_F01194	240	■ 0.999	$\mathrm{w}_1 = 0$	$w_1 = 0$	$s)$ bundles. The last bundle can be oriented if $w_1 = 0$ is valid, while the number of
mielke-geometrodynamics_F01195	240	■ 0.993	$\mathrm{H}^{\circ}(M, \mathbb{Z}_2)$	$H^{\circ}(M, \mathbb{Z}_2)$	distinguishable orientations has to be processed out of the dimension of $H^{\circ}(M, \mathbb{Z}_2)$
mielke-geometrodynamics_F01196	240	■ 1.000	S^7	S^7	dimensional sphere S^7 (MILNOR 1965). Attention can be drawn to other topologically
mielke-geometrodynamics_F01197	240	■ 0.998	(1968, 1970)	(1968, 1970)	However, it was GEROCH (1968, 1970) who pointed out an intuitive and in the
mielke-geometrodynamics_F01198	240	■ 0.969	$T(M)$	$T(M)$	structures on $T(M)$ is dependent only on the mapping f (ISHAM 1978).
mielke-geometrodynamics_F01199	240	■ 0.925	$S U(3)$	$SU(3)$	invariance or Gell-Mann’s $SU(3)$ -group as well into these global constructions, the
mielke-geometrodynamics_F01200	241	■ 1.000	$\tilde{L}(M)$	$\tilde{L}(M)$	<i>associated</i> to the spin bundle $\tilde{L}(M)$. For this reason, it is also called an <i>associ-</i>
mielke-geometrodynamics_F01201	241	■ 0.614	D^{F}	D^{F}	plex) representation D^{F} of dimension $N = 2^{\lfloor n/2 \rfloor}$ of the structure group $\widetilde{SO}_{\circ}$.

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mielke-geometrodynamics_F01282	241	0.614	$\widetilde{S_0}_{\circ}$	$\widetilde{SO}_\circ$	plex) representation D^F of dimension $N = 2^{[n/2]}$ of the structure group $\widetilde{SO}_\circ$.
mielke-geometrodynamics_F01283	241	0.986	$\widetilde{S_0}_{\circ}(1,3)$	$\widetilde{SO}_\circ(1,3)$	$(s, n - s)$, which contains the physical Lorentz group $\widetilde{SO}_\circ(1,3)$ as a subgroup.
mielke-geometrodynamics_F01284	241	0.996	$SL(2, \mathbb{C}) \approx \widetilde{S_0}_{\circ}(1,3)$	$SL(2, \mathbb{C}) \approx \widetilde{SO}_\circ(1,3)$	denotes the bispinor representation of the covering group $SL(2, \mathbb{C}) \approx \widetilde{SO}_\circ(1,3)$ of
mielke-geometrodynamics_F01285	241	1.000	$\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus \left(\frac{1}{2}, \frac{3}{2}\right)$	$\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus \left(\frac{1}{2}, \frac{3}{2}\right)$	the proper Lorentz group, which itself is the direct sum $\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus \left(\frac{1}{2}, \frac{3}{2}\right)$ of spin-
mielke-geometrodynamics_F01286	241	0.999	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$ representations labeled according to GEL'FAND et al. (1963). This formulation
mielke-geometrodynamics_F01287	241	1.000	$\tilde{0}(1,3)$	$\tilde{O}(1,3)$	group $\tilde{O}(1,3)$ if required. Concerning the spinor basis $\tilde{b}_{(1)}, i = 1, \dots, N$, their cross
mielke-geometrodynamics_F01288	241	1.000	$\tilde{b}_{(1)}, i=1, \ldots, N$	$\tilde{b}_{(1)}, i = 1, \dots, N$	group $\tilde{O}(1,3)$ if required. Concerning the spinor basis $\tilde{b}_{(1)}, i = 1, \dots, N$, their cross
mielke-geometrodynamics_F01289	242	0.999	$\mathscr{G}_S$	$\mathscr{G}_S$	of the group $\mathscr{G}_S$ of (local) spin gauge transformations. This group results from $\mathscr{G}_p$, a
mielke-geometrodynamics_F01210	242	0.999	$\mathscr{G}_p$	$\mathscr{G}_p$	of the group $\mathscr{G}_S$ of (local) spin gauge transformations. This group results from $\mathscr{G}_p$, a
mielke-geometrodynamics_F01211	242	1.000	$\tilde{L}(M)$	$\tilde{L}(M)$	The spin bundle $\tilde{L}(M)$ can, analogously to the bundle $L^g(M)$ of the orthonormal
mielke-geometrodynamics_F01212	242	1.000	γ^{n+1}	γ^{n+1}	The matrix γ^{n+1} that is thereby implicitly defined is an axial Lorentz scalar and can
mielke-geometrodynamics_F01213	243	1.000	γ^{n+1}	γ^{n+1}	If one realizes the Dirac matrices in the “Cartan basis,” then γ^{n+1} can be regarded as
mielke-geometrodynamics_F01214	243	0.999	$\tilde{\omega}$	$\tilde{\omega}$	The introduction of a $\widetilde{SO}_\circ(s, n - s)$ -valued 1-form $\tilde{\omega}$ of the connection in the asso-
mielke-geometrodynamics_F01215	243	0.790	$\Gamma_i^{\alpha\beta}$	$\Gamma_i^{\alpha\beta}$	The local components of the spin connection, i.e., Ricci’s rotation coefficients $\Gamma_i^{\alpha\beta}$,
mielke-geometrodynamics_F01216	243	0.958	$\Gamma_{ij}^{\cdot k}$	$\Gamma_{ij}^{\cdot k}$	can be generated from the holonomic connection coefficients $\Gamma_{ij}^{\cdot k}$ by means of
mielke-geometrodynamics_F01217	244	0.995	$\mathbb{R}^n \otimes SO$	$\mathbb{R}^n \otimes SO$	If we had started from a spinor representation of the Poincaré group $\mathbb{R}^n \otimes SO_\circ$.
mielke-geometrodynamics_F01218	244	0.996	$e_{\alpha}^{-i}(m)$	$e_{\alpha}^{-i}(m)$	strived for an extension by inserting the tetrad fields $e_{\alpha}^i(m)$, i.e., the local bundle coor-
mielke-geometrodynamics_F01219	245	1.000	$\mathcal{D}(M)$	$\mathcal{D}(M)$	with respect to the group $\mathcal{D}(M)$ of differentiable coordinate transformations is again
mielke-geometrodynamics_F01220	245	1.000	$\delta L_{\mathrm{D}} / \delta \bar{\psi}$	$\delta L_{\mathrm{D}} / \delta \bar{\psi}$	By variation of (11.2.1) for $\delta L_{\mathrm{D}} / \delta \bar{\psi}$ or for $\delta L_{\mathrm{D}} / \delta \psi$, respectively, the generally
mielke-geometrodynamics_F01221	245	1.000	$\delta L_{\mathrm{D}} / \delta \psi$	$\delta L_{\mathrm{D}} / \delta \psi$	By variation of (11.2.1) for $\delta L_{\mathrm{D}} / \delta \bar{\psi}$ or for $\delta L_{\mathrm{D}} / \delta \psi$, respectively, the generally
mielke-geometrodynamics_F01222	246	1.000	$\mathrm{D}\gamma = 0$	$\mathrm{D}\gamma = 0$	for a covariant constant γ , i.e., for $\mathrm{D}\gamma = 0$. However, the latter condition is globally
mielke-geometrodynamics_F01223	246	1.000	$\tau := \delta L / \delta \omega$	$\tau := \delta L / \delta \omega$	using the rules concerning exterior forms. According to the definitions $\tau := \delta L / \delta \omega$
mielke-geometrodynamics_F01224	246	1.000	$\tau = i \bar{\psi} \gamma \psi$	$\tau = i \bar{\psi} \gamma \psi$	$\tau = i \bar{\psi} \gamma \psi$ of its charge current, but also on the (Lorentz) rotational gauge fields of
mielke-geometrodynamics_F01225	246	0.992	τ_s	τ_s	respect to the current τ or τ_s , and this according to whether the “internal” Yang–
mielke-geometrodynamics_F01226	246	0.999	$j_{n+1} := i \bar{\psi}^* \gamma \gamma^{n+1} \psi$	$j_{n+1} := i \bar{\psi}^* \gamma \gamma^{n+1} \psi$	⁶ A corresponding conservation law is also valid for the axial vector current $j_{n+1} := i \bar{\psi}^* \gamma \gamma^{n+1} \psi$
mielke-geometrodynamics_F01227	247	1.000	H^{n-1}	H^{n-1}	hypersurfaces H^{n-1} , this is not only independent of time, but even <i>positive definite</i> ,

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mielke-geometrodynamics_F01228	247	■ 1.000	$\tau_{\mathrm{s}}^{\mathrm{\{0\}}}$	τ_s^0	Nevertheless, one could not interpret τ_s^0 as the probability density in a non quantized
mielke-geometrodynamics_F01229	247	■ 0.999	$\tilde{\omega}^{\mathrm{\{\{ \}\}}}$	$\tilde{\omega}^{\{\}}$	into the Christoffel-like connection $\tilde{\omega}^{\{\}}$ and the spin contortion that is projected by
mielke-geometrodynamics_F01230	247	■ 0.826	$\mathrm{D}^{\mathrm{\{\{ \}\}}}$	$D^{\{\}}$	Here $D^{\{\}}$ denotes—similarly as in HEHL et al. (1976)—the covariant derivative with
mielke-geometrodynamics_F01231	247	■ 0.993	$\Theta=\mathrm{[K, \vartheta]}$	$\Theta = [K, \vartheta]$	occurs in RC spacetime. Considering the resolution of the relation $\Theta = [K, \vartheta]$ for
mielke-geometrodynamics_F01232	248	■ 1.000	$[\Theta, \vartheta]$	$[\Theta, \vartheta]$	it follows that the modified torsion $[\Theta, \vartheta]$ couples <i>algebraically</i> to the canonical
mielke-geometrodynamics_F01233	248	■ 1.000	$\delta L_{\mathrm{D}} / \delta \bar{\psi}$	$\delta L_D / \delta \bar{\psi}$	$\delta L_D / \delta \bar{\psi}$. This equation will be referred to as the <i>nonlinear Heisenberg–Pauli–Weyl</i>
mielke-geometrodynamics_F01234	249	■ 1.000	$D=\vartheta^{\alpha} \nabla_{\alpha}$	$D = \vartheta^{\alpha} \nabla_{\alpha}$	with $D = \vartheta^{\alpha} \nabla_{\alpha}$ taken into considerations, as follows:
mielke-geometrodynamics_F01235	249	■ 1.000	(1938,1957)	(1938, 1957)	was IVANENKO (1938, 1957) who, for quantum-theoretic reasons, pointed out the
mielke-geometrodynamics_F01236	249	■ 1.000	P C	PC	reflection and charge conjugation PC , and the time reflection T . To be added is the
mielke-geometrodynamics_F01237	249	■ 0.520	$\psi:=\mathrm{binom}\{\psi_{\mathrm{p}}\}\{\psi_{\mathrm{N}}\}$	$\psi := \begin{pmatrix} \psi_p \\ \psi_N \end{pmatrix}$	fundamental equation for the isotopic doublet $\psi := \begin{pmatrix} \psi_p \\ \psi_N \end{pmatrix}$, this equation had to
mielke-geometrodynamics_F01238	249	■ 1.000	S L(2, $\mathbb{C}$)	$SL(2, \mathbb{C})$	group $SL(2, \mathbb{C})$ of Weyl to a gauge theory with $\overline{G} = SL(2\mathrm{f}, \mathbb{C})$ as a structure group,
mielke-geometrodynamics_F01239	249	■ 1.000	$\overline{\mathrm{G}}=\mathrm{S\,L}(2\,\mathrm{f},\,\mathbb{C})$	$\overline{G} = SL(2\mathrm{f}, \mathbb{C})$	group $SL(2, \mathbb{C})$ of Weyl to a gauge theory with $\overline{G} = SL(2\mathrm{f}, \mathbb{C})$ as a structure group,
mielke-geometrodynamics_F01240	249	■ 0.994	$\overline{\mathrm{G}}$	$\overline{G}$	results in a $\overline{G}$ -invariant nonlinear Heisenberg–Pauli–Weyl spinor equation (MIELKE
mielke-geometrodynamics_F01241	251	■ 1.000	$1 / r^{\mathrm{\{2\}}}$	$1/r^2$	and thus replaces Coulomb’s $1/r^2$ -dependence on the distance $r := \vec{\mathbf{x}} $ from the cen-
mielke-geometrodynamics_F01242	251	■ 1.000	$r:= \overrightarrow{\mathrm{\mathbf{x}}} $	$r := \vec{\mathbf{x}} $	and thus replaces Coulomb’s $1/r^2$ -dependence on the distance $r := \vec{\mathbf{x}}$ from the cen-
mielke-geometrodynamics_F01243	251	■ 1.000	(1934,1935)	(1934, 1935)	context. Theoretically, it was coined by BORN & INFELD (1934, 1935) and, as far as a quantum-
mielke-geometrodynamics_F01244	252	■ 0.979	n=4	$n = 4$	that is given by (11.3.7) exclusively in the physically(!) relevant case of $n = 4$ dimen-
mielke-geometrodynamics_F01245	252	■ 1.000	$\varepsilon=\pm 1, \ell$	$\varepsilon = \pm 1, \ell$	the following analysis. On the other hand, the constants $\varepsilon = \pm 1, \ell$, and a are to be
mielke-geometrodynamics_F01246	253	■ 0.999	$v=v(\rho)$	$v = v(\rho)$	Concerning <i>static</i> spinor fields, the functions $v = v(\rho)$ and $\mu = \mu(\rho)$ depend only
mielke-geometrodynamics_F01247	253	■ 0.999	$\mu=\mu(\rho)$	$\mu = \mu(\rho)$	Concerning <i>static</i> spinor fields, the functions $v = v(\rho)$ and $\mu = \mu(\rho)$ depend only
mielke-geometrodynamics_F01248	253	■ 1.000	$\mathrm{r}:= \overrightarrow{\mathrm{\mathbf{x}}} , \vartheta$	$r := \vec{\mathbf{x}} , \vartheta$	and the metric becomes static itself. Thereby the spherical coordinates $r := \vec{\mathbf{x}} , \vartheta$,
mielke-geometrodynamics_F01249	253	■ 1.000	$\overrightarrow{\mathrm{\mathbf{x}}}$	$\vec{\mathbf{x}}$	and ϕ are related to the Cartesian coordinates $\vec{\mathbf{x}}$ in the familiar manner. The search for
mielke-geometrodynamics_F01250	253	■ 1.000	g_{kl}	g_{kl}	can be simplified by conformally relating the components g_{kl} of the metric with those
mielke-geometrodynamics_F01251	253	■ 0.919	$\Gamma_{\kappa}^{\{f \cdot \alpha \beta\}}$	$\Gamma_{\kappa}^{\{f \cdot \alpha \beta$	between the Christoffel symbols and the torsion-free rotation coefficients $\Gamma_{\kappa}^{\{\} \cdot \alpha \beta}$, the
mielke-geometrodynamics_F01252	254	■ 0.742	$\mathrm{R}_{\circ}$	$R_{\circ}$	$R_{\circ}$. For this purpose, the de Sitter space is to be covered with the so-called horospher-
mielke-geometrodynamics_F01253	254	■ 0.990	$\lambda, \mathbf{y}$	$\lambda, \mathbf{y}$	ical coordinate system $(\lambda, \mathbf{y})$. Concerning these coordinates, the line element of the

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mielke-geometrodynamics_F01254	254	■ 0.984	$S_{0_{\circ}(1,4)}$	$SO_{\circ}(1,4)$	de Sitter group $SO_{\circ}(1,4)$ (see MIELKE (1977c), Eqs. (1.23) and (4.5) in the case of
mielke-geometrodynamics_F01255	254	■ 1.000	$\bar{\Gamma}_k$	$\bar{\Gamma}_k$	For the sake of completeness, the conformally related component $\bar{\Gamma}_k$ of the spinor
mielke-geometrodynamics_F01256	254	■ 1.000	$i \vec{\gamma} \cdot \vec{\partial}$	$i \vec{\gamma} \cdot \vec{\partial}$	spacelike Dirac operator $i \vec{\gamma} \cdot \vec{\partial}$ as being written again in Cartesian coordinates.
mielke-geometrodynamics_F01257	255	■ 0.999	$\exp(-\mu/2 - \nu/4)$	$\exp(-\mu/2 - \nu/4)$	$\exp(-\mu/2 - \nu/4)$, as stated, into the separation ansatz
mielke-geometrodynamics_F01258	255	■ 0.643	χ_k^m	χ_k^m	Following the notation of ROSE (1961), the spin-weighted spherical harmonics χ_k^m
mielke-geometrodynamics_F01259	255	■ 1.000	$\mathrm{P}=(-1)^l$	$P = (-1)^l$	of parity $\boxed{P = (-1)}$ are given by
mielke-geometrodynamics_F01260	255	■ 1.000	l	l	Here the parameter κ is related to the quantum numbers j and l of the total angular
mielke-geometrodynamics_F01261	255	■ 1.000	$\mathrm{C}\left(l\frac{1}{2}j; m - \bar{m}, \bar{m}\right)$	$C\left(l\frac{1}{2}j; m - \bar{m}, \bar{m}\right)$	Here $C(l\frac{1}{2}j; m - \bar{m}, \bar{m})$ denote the Clebsch–Gordan coefficients of the rotation group
mielke-geometrodynamics_F01262	255	■ 0.996	$\mathrm{Y}_{l\{m\}}(\vartheta, \phi)$	$Y_l^m(\vartheta, \phi)$	SU(2), and $Y_l^m(\vartheta, \phi)$ are the spherical harmonics. Concerning the operators $\vec{J}$ and
mielke-geometrodynamics_F01263	255	■ 0.996	$\vec{J}$	$\vec{J}$	SU(2), and $Y_l^m(\vartheta, \phi)$ are the spherical harmonics. Concerning the operators $\vec{J}$ and
mielke-geometrodynamics_F01264	255	■ 1.000	$\vec{L}$	$\vec{L}$	$\vec{L}$ of the total spin and the orbital angular momentum, these spinors satisfy the
mielke-geometrodynamics_F01265	256	■ 1.000	$ \kappa =1$	$ \kappa = 1$	follows from the properties of the spherical harmonics that this is the case for $\boxed{ \kappa = 1}$
mielke-geometrodynamics_F01266	256	■ 1.000	$j=\frac{1}{2}, l=0,1$	$j = \frac{1}{2}, l = 0, 1$	to the values $j = \frac{1}{2}, l = 0, 1$, and $m = \pm\frac{1}{2}$, respectively. States of higher quantum
mielke-geometrodynamics_F01267	256	■ 1.000	$\mathrm{m}= \pm \frac{1}{2}$	$m = \pm\frac{1}{2}$	to the values $j = \frac{1}{2}, l = 0, 1$, and $m = \pm\frac{1}{2}$, respectively. States of higher quantum
mielke-geometrodynamics_F01268	256	■ 1.000	$\int L_{D'}$	$\int L_{D'}$	functional $\int L_{D'}$ belonging to (11.4.3). After integrating over the angular variables
mielke-geometrodynamics_F01269	256	■ 1.000	ρ^*	ρ^*	schild solution, an utmost slow tending of ρ^* to the “coordinate singularity” is
mielke-geometrodynamics_F01270	257	■ 1.000	M^*	M^*	mass M^* of the (possibly “strong”) gravitational field is absorbed in the coefficients
mielke-geometrodynamics_F01271	257	■ 0.999	$j=\frac{1}{2}$	$j = \frac{1}{2}$	localized and regular solutions. For a soliton with total spin $j = \frac{1}{2}$, i.e., for $\kappa = -1$,
mielke-geometrodynamics_F01272	257	■ 0.999	$\kappa=-1$	$\kappa = -1$	localized and regular solutions. For a soliton with total spin $j = \frac{1}{2}$, i.e., for $\boxed{\kappa = -1}$,
mielke-geometrodynamics_F01273	257	■ 1.000	$H(\rho)$	$H(\rho)$	For a soliton with total spin $j = \frac{1}{2}$ and $\kappa = -1$, the radial functions $H(\rho)$ in the
mielke-geometrodynamics_F01274	257	■ 1.000	$l=0$	$l = 0$	ansatz (11.4.15) account for a spherically symmetric distribution with $l = 0$, while
mielke-geometrodynamics_F01275	257	■ 1.000	$l=1$	$l = 1$	according to (11.4.17), the quantum number $l = 1$ of angular momentum can be
mielke-geometrodynamics_F01276	257	■ 1.000	$F(\rho)$	$F(\rho)$	ascribed to $F(\rho)$. Let us compare this with the situation in a nonlocal version of the
mielke-geometrodynamics_F01277	258	■ 0.943	$ \varphi ^6$	$ \varphi ^6$	$\boxed{\varphi ^6}$ -model of ROSEN (1965). With reference to an expansion in terms of spherical
mielke-geometrodynamics_F01278	258	■ 1.000	$\mathrm{Y}_{l\{m\}}(\vartheta, \phi)$	$Y_l^m(\vartheta, \phi)$	harmonics $Y_l^m(\vartheta, \phi)$, the occurring <i>exact</i> radial solutions
mielke-geometrodynamics_F01279	259	■ 0.726	$ \omega <1$	$ \omega < 1$	ansatz (11.4.15) satisfies $\boxed{ \omega < 1}$. On the other hand, it is known (VÁZQUEZ 1977)

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mielke-geometrodynamics_F01280	259	■ 1.000	$ \omega >1$	$ \omega > 1$	that bounded and square-integrable radial solutions <i>do not</i> exist for $\boxed{\omega > 1}$.
mielke-geometrodynamics_F01281	259	■ 1.000	$ \psi \sim e^{-m_{\pi} r}$	$ \psi \sim e^{-m_{\pi} r}$	$ \psi \sim e^{-m_{\pi} r}$ between the centers of charge of the solitons. This is the case if the
mielke-geometrodynamics_F01282	259	■ 1.000	m_{s}	m_s	The total mass m_s of a spinor soliton is obtained by the integration
mielke-geometrodynamics_F01283	259	■ 1.000	Σ_{D}	Σ_D	of the canonical energy momentum current Σ_D over a spacelike hypersurface H^3 .
mielke-geometrodynamics_F01284	259	■ 1.000	m_{s}	m_s	(11.4.40) is inversely proportional to the “fundamental” length ℓ^* and that m_s would
mielke-geometrodynamics_F01285	260	■ 0.979	(1951,1956)	(1951,1956)	Following the preceding studies of FINKELSTEIN et al. (1951, 1956) and SOLER
mielke-geometrodynamics_F01286	260	■ 0.356	$a=0$	$a = 0$	(1970), which were mainly concerned with a model (11.4.3) with $a = 0$, RAÑADA
mielke-geometrodynamics_F01287	260	■ 0.988	$a=1$	$a = 1$	equation (11.3.7), i.e., on the Eq.(11.4.3) with $a = 1$ but with the opposite
mielke-geometrodynamics_F01288	260	■ 1.000	$\varepsilon=+1$	$\varepsilon = +1$	sign $\varepsilon = +1$. For solitons corresponding to an $S_{1/2}$ -bound state with normalized
mielke-geometrodynamics_F01289	260	■ 1.000	$S_{1 / 2}$	$S_{1/2}$	sign $\varepsilon = +1$. For solitons corresponding to an $S_{1/2}$ -bound state with normalized
mielke-geometrodynamics_F01290	260	■ 1.000	$Q=e$	$Q = e$	charge $Q = e$, only nodeless radial solutions whose total rest mass shows an
mielke-geometrodynamics_F01291	263	■ 1.000	$D_{\mathrm{F}}(x)$	$D_F(x)$	tudes are not yet subjected to field (“second”) quantization. Here $D_F(x)$ denotes
mielke-geometrodynamics_F01292	263	■ 1.000	(1957,1958)	(1957,1958)	A further reason for IVANENKO (1957, 1958) to consider the necessity of nonlin-
mielke-geometrodynamics_F01293	263	■ 1.000	$\ell^*=8.1 \times 10^{-33} \mathrm{~cm}$	$\ell^* = 8.1 \times 10^{-33} \text{ cm}$	$\ell^* = 8.1 \times 10^{-33} \text{ cm}$ to the far larger amount of the Compton wavelength $\ell = 10^{-13}$
mielke-geometrodynamics_F01294	263	■ 1.000	$\ell=10^{-13}$	$\ell = 10^{-13}$	$\ell^* = 8.1 \times 10^{-33} \text{ cm}$ to the far larger amount of the Compton wavelength $\ell = 10^{-13}$
mielke-geometrodynamics_F01295	263	■ 1.000	$\ell=10^{-13} \mathrm{~cm}$	$\ell = 10^{-13} \text{ cm}$	field theory (1967), which starts from $\ell = 10^{-13} \text{ cm}$, can refrain from using a “bare”
mielke-geometrodynamics_F01296	269	■ 1.000	$\langle dj \rangle=0$	$\langle dj \rangle = 0$	charge current j can be conserved, i.e., $\langle dj \rangle = 0$, whereas the conservation of the
mielke-geometrodynamics_F01297	269	■ 1.000	$\langle dj_5 \rangle \neq 0$	$\langle dj_5 \rangle \neq 0$	axial current j_5 is broken, $\boxed{dj_5 \neq 0}$, in the most common convention.
mielke-geometrodynamics_F01298	269	■ 0.999	$j_5(x ; \varepsilon):=\bar{\psi}(x) \gamma_5 \gamma \psi(x+\varepsilon)$	$j_5(x ; \varepsilon) := \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon)$	sidering the point-split current $\boxed{j_5(x ; \varepsilon) := \psi(x) \gamma_5^* \gamma \psi(x + \varepsilon)}$, where ε is an infin-
mielke-geometrodynamics_F01299	269	■ 1.000	$\delta / \delta A$	$\delta / \delta A$	The variation $\delta / \delta A$ of the current $j_5(x ; \varepsilon)$ is compensated by the variation of the
mielke-geometrodynamics_F01300	269	■ 1.000	$j_5(x ; \varepsilon)$	$j_5(x ; \varepsilon)$	The variation $\delta / \delta A$ of the current $j_5(x ; \varepsilon)$ is compensated by the variation of the
mielke-geometrodynamics_F01301	269	■ 0.999	$x^i \rightarrow x^i+\varepsilon^i$	$x^i \rightarrow x^i + \varepsilon^i$	exponential. Since the parallel transport from $\boxed{x^i \rightarrow x^i + \varepsilon^i}$ along the infinitesimal
mielke-geometrodynamics_F01302	270	■ 1.000	$P=\bar{\psi} \gamma_5 \psi$	$P = \bar{\psi} \gamma_5 \psi$	for massive fermions, where $P = \bar{\psi} \gamma_5 \psi$ is the pseudoscalar current and $F := dA$
mielke-geometrodynamics_F01303	270	■ 1.000	$C:=A \wedge d A$	$C := A \wedge dA$	Simons (CS) term $C := A \wedge dA$ corresponds to the spin or helicity of the photon,
mielke-geometrodynamics_F01304	270	■ 0.760	$\mathbf{A} \cdot \mathbf{B}$	$\mathbf{A} \cdot \mathbf{B}$	with its spacelike part $\mathbf{A} \cdot \mathbf{B}$ known as magnetic helicity (JACKIW & PI 2000). Since
mielke-geometrodynamics_F01305	270	■ 1.000	$\mathcal{D}:=D+i A \wedge$	$\mathcal{D} := D + iA \wedge$	potential $A = A_i dx^i$ is accounted for via $\mathcal{D} := D + iA \wedge$, where $D \psi := d\psi + \Gamma \wedge$
mielke-geometrodynamics_F01306	270	■ 1.000	$D \psi:=d \psi+\Gamma \wedge$	$D \psi := d\psi + \Gamma \wedge$	potential $A = A_i dx^i$ is accounted for via $\mathcal{D} := D + iA \wedge$, where $D \psi := d\psi + \Gamma \wedge$
mielke-geometrodynamics_F01307	270	■ 0.566	$\Gamma^{\alpha \beta}=\Gamma_i^{\alpha \beta} d x^i$	$\Gamma^{\alpha \beta} = \Gamma_i^{\alpha \beta} dx^i$	nection one-form $\Gamma^{\alpha \beta} = \Gamma_i^{\alpha \beta} dx^i$.

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mielke-geometrodynamics_F01308	270	■ 1.000	$\Theta := D \gamma$	$\Theta := D\gamma$	with respect to $\overline{\psi}$ and ψ . Making use of the torsion $\Theta := D\gamma$ and of the properties
mielke-geometrodynamics_F01309	271	■ 0.652	$\left.\left.\text{Tr}\left(\frac{1}{4}\check{\gamma}\right)\right)\right)=e_{\alpha}\right\rfloor T^{\alpha}$	$T := \frac{1}{4} \text{Tr}(\check{\gamma} \rfloor \Theta) = e_{\alpha} \rfloor T^{\alpha}$	where $T := \frac{1}{4} Tr(\check{\gamma} \rfloor \Theta) = e_{\alpha} \rfloor T^{\alpha}$ is the one-form of the trace (or vector) torsion.
mielke-geometrodynamics_F01310	271	■ 1.000	$K=\frac{i}{4} K^{\alpha \beta} \sigma_{\alpha \beta}$	$K = \frac{i}{4} K^{\alpha\beta} \sigma_{\alpha\beta}$	(or Christoffel) connection $\Gamma^{(\cdot)}$ and the <i>contortion</i> one-form $K = \frac{i}{4} K^{\alpha\beta} \sigma_{\alpha\beta}$, sat-
mielke-geometrodynamics_F01311	271	■ 1.000	$D \gamma=[\gamma, K]=\gamma_{\alpha} T^{\alpha}$	$D\gamma = [\gamma, K] = \gamma_{\alpha} T^{\alpha}$	isfying $D\gamma = [\gamma, K] = \gamma_{\alpha} T^{\alpha}$. Accordingly, the Dirac Lagrangian (12.2.1) splits
mielke-geometrodynamics_F01312	271	■ 0.509	$D^{(t)} \gamma=0$	$D^{(t)}\gamma = 0$	$D^{(\cdot)}\gamma = 0$. Hence, in an RC spacetime, a Dirac spinor feels only the <i>axial torsion</i> ,
mielke-geometrodynamics_F01313	271	■ 1.000	$\gamma \rightarrow \gamma^{\beta}=$	$\gamma \rightarrow \gamma^{\beta} =$	which is invariant under Weyl rescalings and <i>chiral transformations</i> $\gamma \rightarrow \gamma^{\beta} =$
mielke-geometrodynamics_F01314	271	■ 1.000	$e^{i \gamma^5 \beta} \gamma e^{-i \gamma^5 \beta}$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P: \vartheta^{\check{B}} \rightarrow -\vartheta^B$, where $B =$
mielke-geometrodynamics_F01315	271	■ 1.000	$P: \vartheta^B \rightarrow -\vartheta^B$	$P: \vartheta^B \rightarrow -\vartheta^B$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P: \vartheta^{\check{B}} \rightarrow -\vartheta^B$, where $B =$
mielke-geometrodynamics_F01316	271	■ 1.000	$B=$	$B =$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P: \vartheta^{\check{B}} \rightarrow -\vartheta^B$, where $B =$
mielke-geometrodynamics_F01317	271	■ 0.992	1, 2, 3	1, 2, 3	<u>1, 2, 3</u> .
mielke-geometrodynamics_F01318	271	■ 1.000	$d j \simeq 0$	$dj \simeq 0$	From the Dirac equation (12.2.2) and its adjoint one can readily deduce that $dj \simeq 0$
mielke-geometrodynamics_F01319	271	■ 1.000	$m \rightarrow 0$	$m \rightarrow 0$	spacetime. If we restore chiral symmetry in the limit $m \rightarrow 0$, this leads to the classical
mielke-geometrodynamics_F01320	272	■ 1.000	$d j_5=0$	$dj_5 = 0$	conservation law $dj_5 = 0$ of the axial current for massless Weyl spinors, or since
mielke-geometrodynamics_F01321	272	■ 1.000	$\mathscr{A}=2 d \theta$	$\mathscr{A} = 2d\theta$	torsion via $\mathscr{A} = 2d\theta$. Then there arises in (12.2.3) a derivative coupling of the
mielke-geometrodynamics_F01322	272	■ 1.000	$a=\theta f_a$	$a = \theta f_a$	would-be axion $a = \theta f_a$ to two fermions via the CPT-invariant term
mielke-geometrodynamics_F01323	272	■ 0.972	$U(1)_{\mathrm{PQ}}$	$U(1)_{\text{PQ}}$	associated with a spontaneously broken Peccei–Quinn symmetry $U(1)_{\text{PQ}}$; cf. KIM &
mielke-geometrodynamics_F01324	273	■ 1.000	$\eta^{\alpha \beta}:=\left(\vartheta^{\alpha} \wedge \vartheta^{\beta}\right)$	$\eta^{\alpha\beta} := *(\vartheta^{\alpha} \wedge \vartheta^{\beta})$	where $\eta^{\alpha\beta} := *(\vartheta^{\alpha} \wedge \vartheta^{\beta})$ is dual to the unit two-form.
mielke-geometrodynamics_F01325	273	■ 1.000	$\left\langle d j_5\right\rangle$	$\langle dj_5 \rangle$	quantization, where the vacuum expectation value $\langle dj_5 \rangle$ of the axial current picks up
mielke-geometrodynamics_F01326	273	■ 0.999	$d C_{\mathrm{TT}} \cong(1 / 4) d j_5 \rightarrow 0$	$dC_{\text{TT}} \cong (1/4) dj_5 \rightarrow 0$	NY four-form tends to zero “on shell,” i.e. <u>$dC_{\text{TT}} \cong (1/4) dj_5 \rightarrow 0$</u> . This is consistent
mielke-geometrodynamics_F01327	273	■ 0.997	${ }^* j_5$	${ }^* j_5$	Here we implicitly assume that the light-cone structure of the axial covector ${ }^* j_5$
mielke-geometrodynamics_F01328	273	■ 1.000	$n=4 k+2$	$n = 4k + 2$	$n = 4k + 2$ dimensions (LEUTWYLER 1986).
mielke-geometrodynamics_F01329	274	■ 0.999	$d^{*} \mathscr{A} \sim d C_{\mathrm{TT}}$	$d^{*}\mathscr{A} \sim dC_{\text{TT}}$	arises in the axial anomaly, but <i>not</i> the NY-type term $d^{*}\mathscr{A} \sim dC_{\text{TT}}$ as was claimed
mielke-geometrodynamics_F01330	274	■ 1.000	$d \mathscr{A} \wedge d \mathscr{A}=-2 \mathcal{E} \cdot \mathcal{B} \eta$	$d\mathscr{A} \wedge d\mathscr{A} = -2\mathcal{E} \cdot \mathcal{B}\eta$	arises in the anomaly. Thus only the Weyl invariant term <u>$d\mathscr{A} \wedge d\mathscr{A} = -2\mathcal{E} \cdot \mathcal{B}\eta$</u>
mielke-geometrodynamics_F01331	274	■ 1.000	$F \wedge F=d A \wedge d A$	$F \wedge F = dA \wedge dA$	<u>$F \wedge F = dA \wedge dA$</u> of the Pontryagin term (12.4.8). Torsion terms like $d\mathscr{A} \wedge d\mathscr{A}$
mielke-geometrodynamics_F01332	274	■ 0.835	$d^{*} \mathscr{A} \wedge *\left(d^{*} \mathscr{A}\right)=4 \ell^4 V_{\mathrm{NY}} \wedge{ }^* V_{\mathrm{NY}}$	$d^{*}\mathscr{A} \wedge *(d^{*}\mathscr{A}) = 4\ell^4 V_{\text{NY}} \wedge { }^* V_{\text{NY}}$	and $d^{*}\mathscr{A} \wedge *(d^{*}\mathscr{A}) = 4\ell^4 V_{\text{NY}} \wedge { }^* V_{\text{NY}}$ have been considered previously, as part of
mielke-geometrodynamics_F01333	274	■ 0.984	$d^{*} \mathscr{A}=2 \ell^2 d C_{\mathrm{TT}}=2 \ell^2 V_{\mathrm{NY}}$	$d^{*}\mathscr{A} = 2\ell^2 dC_{\text{TT}} = 2\ell^2 V_{\text{NY}}$	identity (12.4.10) for the NY term <u>$d^{*}\mathscr{A} = 2\ell^2 dC_{\text{TT}} = 2\ell^2 V_{\text{NY}}$</u> , the second term is

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mielke-geometrodynamics_F01334	274	■ 1.000	Z	Z	Z-factor, which would creep into the definition of the NY term. This is in sharp con-
mielke-geometrodynamics_F01335	274	■ 0.588	$\Psi=\Psi_{\mathrm{i}}\,\mathrm{d}\,x^{\mathrm{i}}$	$\Psi = \Psi_i dx^i$	Schwinger spinor-valued one-form $\Psi = \Psi_i dx^i$; cf. URRUTIA & VERGARA (1991),
mielke-geometrodynamics_F01336	275	■ 0.999	$J_5:=i\,\bar{\Psi}\,\wedge\,\gamma\,\wedge\,\Psi$	$J_5 := i\bar{\Psi} \wedge \gamma \wedge \Psi$	The anomaly for the corresponding <i>axial current</i> $J_5 := i\bar{\Psi} \wedge \gamma \wedge \Psi$ is −21 times
mielke-geometrodynamics_F01337	275	■ 1.000	$t\rightarrow +0$	$t \rightarrow +0$	In the heat kernel approach, there exists for small $t \rightarrow +0$ the asymptotic expansion
mielke-geometrodynamics_F01338	275	■ 0.988	$\mathbb{A}:=$	$\mathbb{A} :=$	of the kernel in n dimensions, where the usual Feynman “dagger” convention $\mathbb{A} :=$
mielke-geometrodynamics_F01339	275	■ 0.794	$\check{\gamma}\rfloor A=\gamma^\alpha e_\alpha\rfloor A=(-1)^{s+1}\ast[\gamma\wedge A]$ $\downharpoonleft A=(-1)^{s+1}\{\}^*\left[\{\}^*\wedge A\right]$	$\check{\gamma}\rfloor A = \gamma^\alpha e_\alpha \rfloor A = (-1)^{s+1} \ast [\gamma \wedge A]$	$\check{\gamma}\rfloor A = \gamma^\alpha e_\alpha \rfloor A = (-1)^{s+1} \ast [\gamma \wedge A]$ for one-forms is used.
mielke-geometrodynamics_F01340	275	■ 0.996	$\square:=$	$\square :=$	terms additional to the generally covariant Riemannian d’Alembertian operator $\square :=$
mielke-geometrodynamics_F01341	275	■ 0.980	$\partial_i\left(\sqrt{ g }g^{ij}\partial_j\right)/\sqrt{ g }$	$\partial_i \left(\sqrt{ g } g^{ij} \partial_j \right) / \sqrt{ g }$	$\partial_i \left(\sqrt{ g } g^{ij} \partial_j \right) / \sqrt{ g }$ can be explicitly identified (MIELKE & KREIMER 1998). Not
mielke-geometrodynamics_F01342	275	■ 0.998	$K_{\mathrm{k}}\left(x,\,\mathrm{not}\,D^2\right)$	$K_k \left(x, \not{D}^2 \right)$	The coefficients $K_k(x, \not{D}^2)$ are completely determined by the form of the second-
mielke-geometrodynamics_F01343	275	■ 0.952	$\mathrm{not}\,D^2$	$\not{D}^2$	order differential operator $\not{D}^2$, which is positive for Euclidean signature diag $\mathrm{o}_{\alpha\beta} =$
mielke-geometrodynamics_F01344	275	■ 0.952	$\mathrm{no}_{\alpha\beta}:=$	$\mathrm{o}_{\alpha\beta} =$	order differential operator $\not{D}^2$, which is positive for Euclidean signature diag $\mathrm{o}_{\alpha\beta} =$
mielke-geometrodynamics_F01345	276	■ 0.950	$(-1,\,\mathrm{ldots},-1)$	$(-1, \dots, -1)$	$(-1, \dots, -1)$. For odd $k = 1, 3, \dots$, these coefficients are zero, while the first non-
mielke-geometrodynamics_F01346	276	■ 0.950	$k=1,3,\,\mathrm{ldots}$	$k = 1, 3, \dots$	$(-1, \dots, -1)$. For odd $k \equiv 1, 3, \dots$, these coefficients are zero, while the first non-
mielke-geometrodynamics_F01347	276	■ 0.483	$\mathrm{d}\,\mathscr{K}=\mathrm{d}^*\,\stackrel{\smile}{D}\,\wedge^*\,\breve{D}\,\mathscr{A}$	$d\mathcal{K} = d^* \widetilde{D} \wedge^* \breve{D}^* \mathcal{A}$	where the higher-order term $d\mathcal{K} = d^* \widetilde{D} \wedge^* \breve{D}^* \mathcal{A}$ involves the covariant derivative
mielke-geometrodynamics_F01348	276	■ 0.507	$D=\mathrm{D}^{\{\}}+i\,\mathscr{A}\,\gamma_5/4$	$D = D^{\{\}} + i\mathcal{A}\gamma_5/4$	$\widetilde{D} = D^{\{\}} + i\mathcal{A}\gamma_5/4$ modified by the axial torsion.
mielke-geometrodynamics_F01349	276	■ 1.000	K_{2}	K_2	terms K_2 and K_4 . Whereas in $n = 4$ dimensions, the Pontryagin-type term K_4 is
mielke-geometrodynamics_F01350	276	■ 1.000	K_{4}	K_4	terms K_2 and K_4 . Whereas in $n = 4$ dimensions, the Pontryagin-type term K_4 is
mielke-geometrodynamics_F01351	276	■ 1.000	$k=4$	$k = 4$	dimensionless and thus for $k = 4$ gets multiplied by $t^{(k-4)/2} = 1$, the term $K_2 \sim$
mielke-geometrodynamics_F01352	276	■ 1.000	$t^{\{(k-4)\,/\,2\}}=1$	$t^{(k-4)/2} = 1$	dimensionless and thus for $k = 4$ gets multiplied by $t^{(k-4)/2} = 1$, the term $K_2 \sim$
mielke-geometrodynamics_F01353	276	■ 1.000	$K_{\mathrm{2}}\,\mathrm{sim}$	$K_2 \sim$	dimensionless and thus for $k = 4$ gets multiplied by $t^{(k-4)/2} = 1$, the term $K_2 \sim$
mielke-geometrodynamics_F01354	276	■ 0.960	$\mathrm{d}^*\,\mathscr{A}=2\,\ell^2\,\mathrm{d}\,C_{\mathrm{TT}}$	$d^* \mathcal{A} = 2\ell^2 dC_{\mathrm{TT}}$	$d^* \mathcal{A} = 2\ell^2 dC_{\mathrm{TT}}$ carries dimensions. Since a <i>massive</i> Dirac spinor has canonical
mielke-geometrodynamics_F01355	276	■ 0.986	$[\mathrm{length}]^{-3\,/\,2}$	$[\mathrm{length}]^{-3/2}$	dimension $[length]^{-3/2}$, it scales as $\psi \sim m^{3/2}$. Moreover, the term $t = 1/M^2$ is
mielke-geometrodynamics_F01356	276	■ 0.986	$\psi\,\mathrm{sim}\,m^{3\,/\,2}$	$\psi \sim m^{3/2}$	dimension $[length]^{-3/2}$, it scales as $\psi \sim m^{3/2}$. Moreover, the term $t = 1/M^2$ is
mielke-geometrodynamics_F01357	276	■ 0.986	$t=1\,/\,M^2$	$t = 1/M^2$	dimension $[length]^{-3/2}$, it scales as $\psi \sim m^{3/2}$. Moreover, the term $t = 1/M^2$ is
mielke-geometrodynamics_F01358	276	■ 1.000	$M\rightarrow\infty$	$M \rightarrow \infty$	related to the regulator mass $M \rightarrow \infty$ in the FUJIKAWA (1979) method. Then the
mielke-geometrodynamics_F01359	276	■ 1.000	$m\,\mathrm{neq}\,0$	$m \neq 0$	heat kernel expansion will tend to zero in the chiral limit $m \rightarrow 0$. In the case $m \neq 0$,

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mielke-geometrodynamics_F01360	276	■ 1.000	$\vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha} = M \vartheta^{\alpha}$	$\vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha} = M \vartheta^{\alpha}$	To rescale the coframe by $\vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha} = M \vartheta^{\alpha}$ does not help, since that would also
mielke-geometrodynamics_F01361	276	■ 1.000	$[\hbar]$	$[\hbar]$	change the dimension of the Dirac field in order to retain the physical dimension $[\hbar]$
mielke-geometrodynamics_F01362	276	■ 1.000	$(n=2)$	$(n = 2)$	sions, neither classically nor in QFT. One would surmise that in $(n = 2)$ -dimensional
mielke-geometrodynamics_F01363	276	■ 1.000	$\sigma_{\alpha} := \Sigma_{\alpha} - D \mu_{\alpha}$	$\sigma_{\alpha} := \Sigma_{\alpha} - D \mu_{\alpha}$	where $\sigma_{\alpha} := \Sigma_{\alpha} - D \mu_{\alpha}$ is the energy–momentum current Belinfante symmetrized
mielke-geometrodynamics_F01364	277	■ 0.996	$\stackrel{(\pm)}{\Gamma}$	$\stackrel{(\pm)}{\Gamma}$	anti-self-dual variables $\stackrel{(\pm)}{\Gamma}$ in EC theory as well as in simple supergravity (MIELKE
mielke-geometrodynamics_F01365	277	■ 0.839	$d C_{\mathrm{RR}}^{\{\}}$	$d C_{\mathrm{RR}}^{\{\}}$	The appearance of the Riemannian Pontryagin term $d C_{\mathrm{RR}}^{\{\}}$ in the anomaly (12.3.1)
mielke-geometrodynamics_F01366	277	■ 1.000	$\Lambda \neq 0$	$\Lambda \neq 0$	essary condition $\Lambda \neq 0$ could even be the dominating configurations, one gets a net
mielke-geometrodynamics_F01367	277	■ 0.570	$\underline{C}_{\mathrm{RR}}^{\{\}}$	$\underline{C}_{\mathrm{RR}}^{\{\}}$	to gravity, the tangential complexified CS term $\underline{C}_{\mathrm{RR}}^{\{\}}$ is known (KODAMA 1990) to
mielke-geometrodynamics_F01368	277	■ 0.936	$\stackrel{(\pm)}{\underline{\Gamma}}$	$\stackrel{(\pm)}{\underline{\Gamma}}$	of gravity with cosmological constant, where the complex Ashtekar variable $\stackrel{(\pm)}{\underline{\Gamma}}$
mielke-geometrodynamics_F01369	277	■ 1.000	$\mathscr{C} := \Gamma / \mathscr{G}$	$\mathscr{C} := \Gamma / \mathscr{G}$	theory on a three-dimensional hypersurface, with $\mathscr{C} := \Gamma / \mathscr{G}$ of inequivalent gauge
mielke-geometrodynamics_F01370	278	■ 0.423	$\mathbb{L}_n := n \downarrow D + D n$	$\mathbb{L}_n := n \downarrow D + D n$	where $\mathbb{L}_n := n \downarrow D + D n$ is the gauge-covariant Lie derivative along the normal
mielke-geometrodynamics_F01371	278	■ 0.523	τ^A	τ^A	for the matter current τ^A on the spacelike hypersurface (JIANG 1991). Only when
mielke-geometrodynamics_F01372	278	■ 1.000	$\tau^A := (1/2) \eta^{0A\beta\gamma} \tau_{\beta\gamma}$	$\tau^A := (1/2) \eta^{0A\beta\gamma} \tau_{\beta\gamma}$	field, it is the canonical spin $\tau^A := (1/2) \eta^{0A\beta\gamma} \tau_{\beta\gamma}$ that appears on the right-hand
mielke-geometrodynamics_F01373	278	■ 1.000	$\gamma_{\alpha} \gamma_{\beta} + \gamma_{\beta} \gamma_{\alpha} = 2 o_{\alpha\beta}$	$\gamma_{\alpha} \gamma_{\beta} + \gamma_{\beta} \gamma_{\alpha} = 2 o_{\alpha\beta}$	When the constant Dirac matrices γ_{α} satisfying $\gamma_{\alpha} \gamma_{\beta} + \gamma_{\beta} \gamma_{\alpha} = 2 o_{\alpha\beta}$ saturate the
mielke-geometrodynamics_F01374	278	■ 1.000	$\eta^{\alpha} :=$	$\eta^{\alpha} :=$	index of the orthonormal coframe one-form $\vartheta^{\alpha} = E_j^{\alpha} dx^j$ and its Hodge dual $\eta^{\alpha} :=$
mielke-geometrodynamics_F01375	278	■ 0.996	$* \vartheta^{\alpha}$	$* \vartheta^{\alpha}$	$* \vartheta^{\alpha}$, we obtain a basis of Clifford-algebra-valued exterior forms via
mielke-geometrodynamics_F01376	278	■ 0.500	$\Gamma := \frac{i}{4} \Gamma_i^{\alpha\beta} \sigma_{\alpha\beta} dx^i$	$\Gamma := \frac{i}{4} \Gamma_i^{\alpha\beta} \sigma_{\alpha\beta} dx^i$	In terms of the Clifford-algebra-valued <i>connection</i> $\Gamma := \frac{i}{4} \Gamma_i^{\alpha\beta} \sigma_{\alpha\beta} dx^i$, the $SL(2, \mathbb{C})$ -
mielke-geometrodynamics_F01377	278	■ 1.000	$D = d + \Gamma \wedge$	$D = d + \Gamma \wedge$	covariant exterior derivative is given by $D = d + \Gamma \wedge$, where $\sigma_{\alpha\beta} = \frac{i}{2} (\gamma_{\alpha} \gamma_{\beta} - \gamma_{\beta} \gamma_{\alpha})$
mielke-geometrodynamics_F01378	278	■ 1.000	$\sigma_{\alpha\beta} = \frac{i}{2} (\gamma_{\alpha} \gamma_{\beta} - \gamma_{\beta} \gamma_{\alpha})$	$\sigma_{\alpha\beta} = \frac{i}{2} (\gamma_{\alpha} \gamma_{\beta} - \gamma_{\beta} \gamma_{\alpha})$	covariant exterior derivative is given by $D = d + \Gamma \wedge$, where $\sigma_{\alpha\beta} = \frac{i}{2} (\gamma_{\alpha} \gamma_{\beta} - \gamma_{\beta} \gamma_{\alpha})$
mielke-geometrodynamics_F01379	278	■ 1.000	$\sigma := \frac{i}{2} \gamma \wedge \gamma = \frac{1}{2} \sigma_{\alpha\beta} \vartheta^{\alpha} \wedge \vartheta^{\beta}$	$\sigma := \frac{i}{2} \gamma \wedge \gamma = \frac{1}{2} \sigma_{\alpha\beta} \vartheta^{\alpha} \wedge \vartheta^{\beta}$	are the Lorentz generators entering in the two-form $\sigma := \frac{i}{2} \gamma \wedge \gamma = \frac{1}{2} \sigma_{\alpha\beta} \vartheta^{\alpha} \wedge \vartheta^{\beta}$.
mielke-geometrodynamics_F01380	279	■ 1.000	$R := * (R^{\alpha\beta} \wedge \eta_{\beta\alpha})$	$R := * (R^{\alpha\beta} \wedge \eta_{\beta\alpha})$	to the curvature scalar $R := * (R^{\alpha\beta} \wedge \eta_{\beta\alpha})$ and the axial torsion piece $d \mathscr{A} \wedge d \mathscr{A}$ of
mielke-geometrodynamics_F01381	279	■ 1.000	$\mathfrak{sl}(5, \mathbb{R})$	$\mathfrak{sl}(5, \mathbb{R})$	the $\mathfrak{sl}(5, \mathbb{R})$ -valued connection $\hat{F} = \bar{F} + (\vartheta^{\alpha} L^4_{\alpha} + \vartheta_{\beta} L^{\bar{\beta}}_{\bar{4}}) / \ell$ is expanded into the
mielke-geometrodynamics_F01382	279	■ 1.000	$\hat{\Gamma} = \Gamma + (\vartheta^{\alpha} L^4_{\alpha} + \vartheta_{\beta} L^{\beta}_{\bar{4}}) / \ell$	$\hat{\Gamma} = \Gamma + (\vartheta^{\alpha} L^4_{\alpha} + \vartheta_{\beta} L^{\beta}_{\bar{4}}) / \ell$	the $\mathfrak{sl}(5, \mathbb{R})$ -valued connection $\hat{F} = \bar{F} + (\vartheta^{\alpha} L^4_{\alpha} + \vartheta_{\beta} L^{\bar{\beta}}_{\bar{4}}) / \ell$ is expanded into the
mielke-geometrodynamics_F01383	279	■ 0.763	$\hat{C}_{\mathrm{RR}}$	$\hat{C}_{\mathrm{RR}}$	canonical dimension $[length]$. The corresponding Pontryagin term $\hat{C}_{\mathrm{RR}}$ splits via
mielke-geometrodynamics_F01384	280	■ 1.000	$\pi^0 \rightarrow 2 \gamma$	$\pi^0 \rightarrow 2 \gamma$	Bell JS, Jackiw R (1969) A PCAC puzzle: $\pi^0 \rightarrow 2 \gamma$ in the σ model. Il Nuovo Cimento A 60(1):

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mielke-geometrodynamics_F01385	282	0.775	$\mathrm{R}^4 \otimes \mathrm{SO}(1,3)$	$\mathbb{R}^4 \otimes \mathrm{SO}(1,3)$	Lorentz or Poincaré group $\mathbb{R}^4 \ltimes \mathrm{SO}(1,3)$ to a metilinear group $\mathrm{SL}(5, \mathbb{R})$ as a gauge
mielke-geometrodynamics_F01386	283	1.000	$d\,B=0$	$dB = 0$	Independent variations with respect to A and B lead to $dB = 0$ and to the <i>constraint</i>
mielke-geometrodynamics_F01387	284	0.999	$\sigma=\sigma_i\,d\,x^i$	$\sigma = \sigma_i dx^i$	where $\sigma = \sigma_i dx^i$ is a one-form or a covector; cf. LUCCHESI et al. (1993). Since Bianchi
mielke-geometrodynamics_F01388	284	0.672	$\mathrm{SL}(5,\mathrm{R})$	$\mathrm{SL}(5, \mathbb{R})$	“semi-topological” BF scheme based on the metric-free structure group $\mathrm{SL}(5, \mathbb{R})$
mielke-geometrodynamics_F01389	284	1.000	$\mathrm{SL}(n+1,\mathrm{R})$	$\mathrm{SL}(n+1, \mathbb{R})$	In general, the metilinear group $\mathrm{SL}(n+1, \mathbb{R})$ with $\det g = 1$ is generated by
mielke-geometrodynamics_F01390	284	1.000	$\mathrm{det}\,g=1$	$\det g = 1$	In general, the metilinear group $\mathrm{SL}(n+1, \mathbb{R})$ with $\det g = 1$ is generated by
mielke-geometrodynamics_F01391	284	1.000	$n(n+2)$	$n(n+2)$	$n(n+2)$ trace-free generators
mielke-geometrodynamics_F01392	284	1.000	$\mathfrak{gl}(n,\mathrm{R})$	$\mathfrak{gl}(n, \mathbb{R})$	comprise those of a $\mathfrak{gl}(n, \mathbb{R})$ subalgebra
mielke-geometrodynamics_F01393	284	1.000	$\alpha,\ldots,\delta=0,\ldots,n-1$	$\alpha, \dots, \delta = 0, \dots, n-1$	where $\alpha, \dots, \delta = 0, \dots, n-1$.
mielke-geometrodynamics_F01394	284	0.367	$\ell\,P_{\alpha}:=\mathbb{L}^{n+1}_{\alpha}=L^{n+1}_{\alpha}$	$\ell P_{\alpha} := \mathbb{L}^{n+1}_{\alpha} = L^{n+1}_{\alpha}$	The generators $\ell P_{\alpha} := \mathbb{L}^{n+1}_{\alpha} = L^{n+1}_{\alpha}$ and $\ell P^{\beta}_{*} := \mathbb{L}^{\beta}_{n+1} = L^{\beta}_{n+1}$ transform
mielke-geometrodynamics_F01395	284	0.367	$\ell\,P^{\beta}_{*}:=\swarrow^{\beta}_{n+1}=L^{\beta}_{n+1}$	$\ell P^{\beta}_{*} := \swarrow^{\beta}_{n+1} = L^{\beta}_{n+1}$	The generators $\ell P_{\alpha} := \mathbb{L}^{n+1}_{\alpha} = L^{n+1}_{\alpha}$ and $\ell P^{\beta}_{*} := \mathbb{L}^{\beta}_{n+1} = L^{\beta}_{n+1}$ transform
mielke-geometrodynamics_F01396	285	0.998	P_{*}^{α}	P^{α}_{*}	This reveals that P^{α}_{*} is a vector and P_{β} is a covector with respect to the $\mathrm{GL}(n, \mathbb{R})$
mielke-geometrodynamics_F01397	285	0.998	P_{β}	P_{β}	This reveals that P^{α}_{*} is a vector and P_{β} is a covector with respect to the $\mathrm{GL}(n, \mathbb{R})$
mielke-geometrodynamics_F01398	285	0.998	$\mathrm{GL}(n,\mathrm{R})$	$\mathrm{GL}(n, \mathbb{R})$	This reveals that P^{α}_{*} is a vector and P_{β} is a covector with respect to the $\mathrm{GL}(n, \mathbb{R})$
mielke-geometrodynamics_F01399	285	0.996	$\mathrm{A}_{*}(n,\mathrm{R})$	$A_{*}(n, \mathbb{R})$	to the semisimple <i>graded affine</i> group $A_{*}(n, \mathbb{R})$ of the same rank:
mielke-geometrodynamics_F01400	285	0.998	$\mathrm{SO}(n+1)$	$\mathrm{SO}(n+1)$	(Pseudo)orthogonal groups like $\mathrm{SO}(n+1)$ involve a Cartan–Killing metric with
mielke-geometrodynamics_F01401	285	1.000	$\hat{g}_{AB}$	$\hat{g}_{AB}$	components $\hat{g}_{AB}$. Then a basis of the Lie generators
mielke-geometrodynamics_F01402	285	0.998	$n(n+1)/2$	$n(n+1)/2$	is only $n(n+1)/2$ -dimensional. The same would apply to the respective connections
mielke-geometrodynamics_F01403	285	0.999	$A(x)\in\mathrm{A}_{*}(n,\mathrm{R})$	$A(x) \in A_{*}(n, R)$	where $A(x) \in A_{*}(n, R)$ and the $\Lambda(x) \in \mathrm{GL}(n, \mathbb{R})$ are linear gauge transformations,
mielke-geometrodynamics_F01404	285	0.999	$\Lambda(x)\in\mathrm{GL}(n,\mathrm{R})$	$\Lambda(x) \in \mathrm{GL}(n, \mathbb{R})$	where $A(x) \in A_{*}(n, R)$ and the $\Lambda(x) \in \mathrm{GL}(n, \mathbb{R})$ are linear gauge transformations,
mielke-geometrodynamics_F01405	285	1.000	$\tau(x):=\exp\left[\xi^{\alpha}\,P_{\alpha}\right]\in\mathrm{mathscr{T}}(n,\mathrm{R})$	$\tau(x) := \exp[\xi^{\alpha} P_{\alpha}] \in \mathcal{T}(n, R)$	while $\tau(x) := \exp[\xi^{\alpha} P_{\alpha}] \in \mathcal{T}(n, R)$ and $\tau_{*}(x) := \exp[\xi^{\beta}_{*} P^{\beta}_{*}] \in \mathcal{T}_{*}(n, R)$ represent
mielke-geometrodynamics_F01406	285	1.000	$\tau_{*}(x):=\exp\left[\xi^{\alpha}_{*}\,P_{\alpha}\right]\in\mathrm{mathscr{T}}_{*}(n,\mathrm{R})$	$\tau_{*}(x) := \exp[\xi^{\alpha}_{*} P^{\alpha}_{*}] \in \mathcal{T}_{*}(n, R)$	while $\tau(x) := \exp[\xi^{\alpha} P_{\alpha}] \in \mathcal{T}(n, R)$ and $\tau_{*}(x) := \exp[\xi^{\beta}_{*} P^{\beta}_{*}] \in \mathcal{T}_{*}(n, R)$ represent
mielke-geometrodynamics_F01407	286	1.000	$\ell\rightarrow\infty$	$\ell \rightarrow \infty$	For $\ell \rightarrow \infty$, this holds automatically due to the degeneracy of the commutation
mielke-geometrodynamics_F01408	286	1.000	$\mathrm{Ad}\,A(B)=[A,\,B]$	$\mathrm{Ad}\,A(B) = [A, B]$	the adjoint representation $\mathrm{Ad}A(B) = [A, B]$. The curvature
mielke-geometrodynamics_F01409	286	1.000	$\hat{\Gamma}$	$\hat{\Gamma}$	The inhomogeneous transformation law (13.4.3) for the graded affine connection $\hat{\Gamma}$
mielke-geometrodynamics_F01410	286	1.000	$\Gamma^{(L)}$	$\Gamma^{(L)}$	can be split into that of the linear connection $\Gamma^{(L)}$, which acquires the conventional
mielke-geometrodynamics_F01411	286	0.990	$\mathrm{GL}(n,\mathrm{R})$	$\mathrm{GL}(n, \mathbb{R})$	of a Yang–Mills-type connection for the gauge group $\mathrm{GL}(n, \mathbb{R})$. The remaining trans-

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mielke-geometrodynamics_F01412	287	■ 0.998	$D \tau(x) := d \tau(x) + \Gamma^{\{L\}} \tau(x)$	$D\tau(x) := d\tau(x) + \Gamma^{(L)}\tau(x)$	derivative $D\tau(x) := d\tau(x) + \Gamma^{(L)}\tau(x)$, the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not
mielke-geometrodynamics_F01413	287	■ 0.998	$\Gamma^{\{T\}}$	$\Gamma^{(T)}$	derivative $D\tau(x) := d\tau(x) + \Gamma^{(L)}\tau(x)$, the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not
mielke-geometrodynamics_F01414	287	■ 0.998	$\Gamma_*^{\{T\}}$	$\Gamma_*^{(T)}$	derivative $D\tau(x) := d\tau(x) + \Gamma^{(L)}\tau(x)$, the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not
mielke-geometrodynamics_F01415	287	■ 1.000	$\xi = \xi^\alpha P_\alpha$	$\xi = \xi^\alpha P_\alpha$	Since $\xi = \xi^\alpha P_\alpha$ and $\xi_* = \xi_\beta^* P_\beta^*$ acquire their values in the coset space
mielke-geometrodynamics_F01416	287	■ 1.000	$\xi_* = \xi_\beta^* P_\beta^*$	$\xi_* = \xi_\beta^* P_\beta^*$	Since $\xi = \xi^\alpha P_\alpha$ and $\xi_* = \xi_\beta^* P_\beta^*$ acquire their values in the coset space
mielke-geometrodynamics_F01417	287	■ 1.000	$\mathscr{T}(n, R) \otimes \mathscr{T}_*(n, R)$	$\mathscr{T}(n, R) \otimes \mathscr{T}_*(n, R)$	“hide” the action of the local pseudotranslations $\mathscr{T}(n, R) \otimes \mathscr{T}_*(n, R)$ on ϑ and θ .
mielke-geometrodynamics_F01418	287	■ 1.000	$\xi = \xi^* = 0$	$\xi = \xi^* = 0$	If we postulate even stronger constraints $\xi = \xi^* = 0$ of “zero section” vector
mielke-geometrodynamics_F01419	287	■ 1.000	$\theta_\alpha = g_{\alpha\beta} \vartheta^\beta$	$\theta_\alpha = g_{\alpha\beta} \vartheta^\beta$	mutually transposed one-forms, i.e., $\theta_\alpha = g_{\alpha\beta} \vartheta^\beta$.
mielke-geometrodynamics_F01420	287	■ 1.000	$D\xi^\alpha \stackrel{*}{=} \vartheta^\alpha$	$D\xi^\alpha \stackrel{*}{=} \vartheta^\alpha$	When $D\xi^\alpha \stackrel{*}{=} \vartheta^\alpha$, a Cartan transport of a radius vector ξ^α occurs via “rolling
mielke-geometrodynamics_F01421	287	■ 1.000	ξ^α	ξ^α	When $D\xi^\alpha \stackrel{*}{=} \vartheta^\alpha$, a Cartan transport of a radius vector ξ^α occurs via “rolling
mielke-geometrodynamics_F01422	288	■ 0.593	$1 / [$	$1 / [$	In view of the canonical dimension 1/[length] of the generators P_α and P_β^* of pseudo-
mielke-geometrodynamics_F01423	288	■ 0.593	$]$	$]$	In view of the canonical dimension 1/[length] of the generators P_α and P_β^* of pseudo-
mielke-geometrodynamics_F01424	288	■ 0.593	P_*^β	P_*^β	In view of the canonical dimension 1/[length] of the generators P_α and P_β^* of pseudo-
mielke-geometrodynamics_F01425	288	■ 0.647	$\vartheta^\alpha = \ell \Gamma_4^\alpha$	$\vartheta^\alpha = \ell \Gamma_4^\alpha$	i.e., it becomes a Cartan connection, where $\vartheta^\alpha = \ell \Gamma_4^\alpha$ and $\theta_\beta = \ell \Gamma_\beta^{\hat{4}}$ are two inde-
mielke-geometrodynamics_F01426	288	■ 0.647	$\theta_\beta = \ell \Gamma_\beta^{\hat{4}}$	$\theta_\beta = \ell \Gamma_\beta^{\hat{4}}$	i.e., it becomes a Cartan connection, where $\vartheta^\alpha = \ell \Gamma_4^\alpha$ and $\theta_\beta = \ell \Gamma_\beta^{\hat{4}}$ are two inde-
mielke-geometrodynamics_F01427	288	■ 1.000	$\Lambda = 3 / \ell^2$	$\Lambda = 3 / \ell^2$	where $\Lambda = 3 / \ell^2$ will later be identified with the cosmological constant.
mielke-geometrodynamics_F01428	289	■ 0.998	$d \hat{C}$	$d\hat{C}$	$d\hat{C}$ provides us, after variation with respect to Γ_A^B , with the Bianchi identity
mielke-geometrodynamics_F01429	289	■ 0.998	$\Gamma_A^{\{B\}}$	Γ_A^B	$d\hat{C}$ provides us, after variation with respect to Γ_A^B , with the Bianchi identity
mielke-geometrodynamics_F01430	289	■ 1.000	$\hat{B}$	$\hat{B}$	The variation with respect to $\hat{B}$ would lead to $\hat{R} = 0$, a physically too strong constraint.
mielke-geometrodynamics_F01431	289	■ 1.000	$\hat{R} = 0$	$\hat{R} = 0$	The variation with respect to $\hat{B}$ would lead to $\hat{R} = 0$, a physically too strong constraint.
mielke-geometrodynamics_F01432	289	■ 0.994	$\eta_{AB C D E} := \sqrt{\hat{g}} \varepsilon_{AB C D E}$	$\eta_{ABCDE} := \sqrt{\hat{g}} \varepsilon_{ABCDE}$	Here, the antisymmetric unit tensor $\eta_{ABCDE} := \sqrt{\hat{g}} \varepsilon_{ABCDE}$ is constructed from the
mielke-geometrodynamics_F01433	289	■ 1.000	$\varepsilon_{AB C D E}$	ε_{ABCDE}	<i>metric-free</i> Levi-Civita symbol ε_{ABCDE} , being invariant under the five-dimensional
mielke-geometrodynamics_F01434	289	■ 0.913	$\hat{g}$	$\hat{g}$	structure group $\text{SL}(5, \mathbb{R})$, together with the determinant $\hat{g}$ of the invertible group
mielke-geometrodynamics_F01435	289	■ 1.000	$B^{AB} := \hat{g}^{AC} B_C^B$	$B^{AB} := \hat{g}^{AC} B_C^B$	$B^{AB} := \hat{g}^{AC} B_C^B$. Thus, the additional quadratic term provides a <i>dynamical symmetry</i> -
mielke-geometrodynamics_F01436	289	■ 0.853	$\text{operatorname{SO}}(2, 3)$	$\text{SO}(2, 3)$	or $\text{SO}(2, 3)$, respectively. Moreover, the five-index structure of η_{ABCDE} has forced us
mielke-geometrodynamics_F01437	289	■ 0.853	$\eta_{AB C D E}$	η_{ABCDE}	or $\text{SO}(2, 3)$, respectively. Moreover, the five-index structure of η_{ABCDE} has forced us

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mielke-geometrodynamics_F01438	289	■ 0.997	$\backslash\mathrm{Phi}^{\mathrm{A}}=\backslash\mathrm{left}\{\backslash\mathrm{phi}^{\mathrm{\alpha}},\ \backslash\mathrm{phi}^{\mathrm{\hat{4}}}\}\backslash\mathrm{right}\{\}$	$\Phi^A = \left\{ \phi^\alpha, \phi^{\hat{4}} \right\}$	to complement the construction by a quintet of pseudoscalars $\Phi^A = \{ \phi^\alpha, \phi^{\hat{4}} \}$, our
mielke-geometrodynamics_F01439	289	■ 0.988	$\mathrm{U}\backslash\mathrm{left}(\backslash\mathrm{Phi}^{\mathrm{B}}\ \backslash\mathrm{Phi}_{\mathrm{B}})\backslash\mathrm{right}()$	$U\left(\Phi^B\Phi_B\right)$	arise from a minimum of a Higgs-type potential $U(\Phi^B\Phi_B)$; cf. PAGELS (1984) and
mielke-geometrodynamics_F01440	290	■ 1.000	$\backslash\mathrm{Phi}=\backslash\mathrm{Re}\ \backslash\mathrm{Phi}+i\ \backslash\mathrm{Im}\ \backslash\mathrm{Phi}$	$\Phi = \Re\Phi + i\Im\Phi$	$\Phi = \Re\Phi + i\Im\Phi$ is invariant
mielke-geometrodynamics_F01441	290	■ 0.995	$\mathrm{d}\ \mathrm{C}_{\mathrm{\mathrm{RR}}}$	dC_{RR}	where the resulting Pontryagin term dC_{RR} serves merely to reproduce the Bianchi
mielke-geometrodynamics_F01442	290	■ 0.989	$\backslash\mathrm{stackrel{\circ}{rel}}\{\backslash\mathrm{circ}\}\{\backslash\mathrm{Gamma}\}^{\mathrm{\alpha}}\backslash\mathrm{alpha}\ \backslash\mathrm{beta}\}=\backslash\mathrm{stackrel{\circ}{rel}}\{\backslash\mathrm{circ}\}\{\backslash\mathrm{Gamma}\}^{\mathrm{\alpha}}\backslash\mathrm{beta}\ \backslash\mathrm{alpha}\}$	$\overset{\circ}{\Gamma}^{\alpha\beta} = - \overset{\circ}{\Gamma}^{\beta\alpha}$	identity (13.5.5) for the curvature, now in terms of the Lorentz connection $\overset{\circ}{\Gamma}^{\alpha\beta} = - \overset{\circ}{\Gamma}^{\beta\alpha}$. The variation with respect to the Lorentz-valued two-form $B^{\alpha\beta}$ leads to
mielke-geometrodynamics_F01443	290	■ 1.000	$\backslash\mathrm{stackrel{\circ}{rel}}\{\backslash\mathrm{circ}\}\{\backslash\mathrm{Gamma}\}^{\mathrm{\alpha}}\backslash\mathrm{beta}\ \backslash\mathrm{alpha}\}$	$B^{\alpha\beta}$	$- \overset{\circ}{\Gamma}^{\beta\alpha}$. The variation with respect to the Lorentz-valued two-form $B^{\alpha\beta}$ leads to
mielke-geometrodynamics_F01444	290	■ 1.000	$\mathrm{B}^{\mathrm{\alpha}}\backslash\mathrm{alpha}\ \backslash\mathrm{beta}\}$	$B^{\alpha\beta}$	$- \overset{\circ}{\Gamma}^{\beta\alpha}$. The variation with respect to the Lorentz-valued two-form $B^{\alpha\beta}$ leads to
mielke-geometrodynamics_F01445	290	■ 1.000	$\backslash\mathrm{mu}\ \backslash\mathrm{neq}\ 0$	$\mu \neq 0$	as an equation of motion. Thus, for $\mu \neq 0$, our Lagrangian is “on shell” equivalent
mielke-geometrodynamics_F01446	291	■ 1.000	$\backslash\mathrm{vartheta}^{\mathrm{\alpha}}=\mathrm{g}^{\mathrm{\alpha}}\backslash\mathrm{alpha}\ \backslash\mathrm{beta}\}\ \backslash\mathrm{theta}_{\mathrm{\beta}}\}$	$\vartheta^\alpha = g^{\alpha\beta}\theta_\beta$	For a nondegenerate coframe $\vartheta^\alpha = g^{\alpha\beta}\theta_\beta$ and vanishing torsion, the covariant
mielke-geometrodynamics_F01447	291	■ 1.000	$\backslash\mathrm{kappa}=\backslash\mathrm{stackrel{\circ}{rel}}\{\backslash\mathrm{circ}\}\{\backslash\mathrm{Gamma}\}^{\mathrm{\alpha}}\backslash\mathrm{beta}\ \backslash\mathrm{alpha}\}$	$\kappa =$	Pontryagin four-forms (here suppressed). Einstein’s gravitational constant $\kappa =$
mielke-geometrodynamics_F01448	291	■ 1.000	$8\ \backslash\mathrm{pi}\ \mathrm{G}_{\mathrm{\mathrm{N}}}\}=8\ \backslash\mathrm{pi}\ \backslash\mathrm{ell}_{\mathrm{\mathrm{Planck}}}\}^{\mathrm{2}}$	$8\pi G_{\mathrm{N}} = 8\pi\ell_{\mathrm{Planck}}^2$	$8\pi G_{\mathrm{N}} = 8\pi\ell_{\mathrm{Planck}}^2$ and the density $\rho_\Lambda = \Lambda/\kappa = \mathcal{Q}_\Lambda\rho_{\mathrm{crit}} < \rho_{\mathrm{crit}}$ corresponding to
mielke-geometrodynamics_F01449	291	■ 1.000	$\backslash\mathrm{rho}_{\mathrm{\backslash\mathrm{Lambda}}}\}= \backslash\mathrm{Lambda}\ /\ \backslash\mathrm{kappa}=\backslash\mathrm{Omega}_{\mathrm{\backslash\mathrm{Lambda}}}\}\ \backslash\mathrm{rho}_{\mathrm{\backslash\mathrm{crit}}}\}\<\backslash\mathrm{rho}_{\mathrm{\backslash\mathrm{crit}}}\}\}$	$\rho_\Lambda = \Lambda/\kappa = \Omega_\Lambda\rho_{\mathrm{crit}} < \rho_{\mathrm{crit}}$	$8\pi G_{\mathrm{N}} = 8\pi\ell_{\mathrm{Planck}}^2$ and the density $\rho_\Lambda = \Lambda/\kappa = \mathcal{Q}_\Lambda\rho_{\mathrm{crit}} < \rho_{\mathrm{crit}}$ corresponding to
mielke-geometrodynamics_F01450	291	■ 1.000	$\backslash\mathrm{kappa}=\backslash\mathrm{mu}\ \backslash\mathrm{ell}^{\mathrm{2}}\ /\ 4$	$\kappa = \mu\ell^2/4$	implies that these constants are related via $\kappa = \mu\ell^2/4$ and $\Lambda = 12\kappa/\mu\ell^4 = 3/\ell^2$ to
mielke-geometrodynamics_F01451	291	■ 1.000	$\backslash\mathrm{Lambda}=12\ \backslash\mathrm{kappa}\ /\ \backslash\mathrm{mu}\ \backslash\mathrm{ell}^{\mathrm{4}}=3\ /\ \backslash\mathrm{ell}^{\mathrm{2}}$	$\Lambda = 12\kappa/\mu\ell^4 = 3/\ell^2$	implies that these constants are related via $\kappa = \mu\ell^2/4$ and $\Lambda = 12\kappa/\mu\ell^4 = 3/\ell^2$ to
mielke-geometrodynamics_F01452	291	■ 1.000	$\backslash\mathrm{mu}\ \backslash\mathrm{simeq}\ 10^{\mathrm{-123}}$	$\mu \simeq 10^{-123}$	The symmetry-breaking parameter $\mu \simeq 10^{-123}$ is observationally an extremely small
mielke-geometrodynamics_F01453	291	■ 1.000	$\backslash\mathrm{hat{B}}\ \backslash\mathrm{wedge}\ \backslash\mathrm{hat{B}}$	$\hat{B} \wedge \hat{B}$	breaking $\hat{B} \wedge \hat{B}$ term, as anticipated by SMOLIN (2000) in the restricted case of
mielke-geometrodynamics_F01454	291	■ 1.000	$\mathrm{G}_{\mathrm{\mathrm{N}}}\}$	G_{N}	gravitational constant G_{N} has its origin in SSB, as suggested already by ENGLERT et
mielke-geometrodynamics_F01455	292	■ 0.997	$0(5)$	$O(5)$	Since our $\mathrm{SL}(5, \mathbb{R})$ gauge model is broken into two steps from $O(5)$ down to the
mielke-geometrodynamics_F01456	292	■ 0.466	$\backslash\mathrm{operatorname{SO}}(1,3)$	$\mathrm{SO}(1,3)$	Lorentz group $\mathrm{SO}(1,3)$ as an exact subgroup, a local holonomic spacetime metric
mielke-geometrodynamics_F01457	292	■ 1.000	$\mathrm{g}_{\mathrm{\mathrm{i}}\ \mathrm{j}}$	g_{ij}	g_{ij} is then induced. In the last step, we assumed that the translational connections
mielke-geometrodynamics_F01458	292	■ 0.999	$\backslash\mathrm{Gamma}_{\mathrm{\backslash\mathrm{4}}}\{\ \ \}^{\mathrm{\alpha}}\backslash\mathrm{alpha}\}\ \backslash\mathrm{otimes}_{\mathrm{\mathrm{s}}}\backslash\mathrm{Gamma}_{\mathrm{\backslash\mathrm{alpha}}}\{\ \ \}^{\mathrm{4}}$	$\Gamma_4{}^\alpha \otimes_{\mathrm{s}} \Gamma_\alpha{}^4$	$\Gamma_4{}^\alpha \otimes_{\mathrm{s}} \Gamma_\alpha{}^4$ is dimensionless, we had to multiply the vacuum expectation (13.6.1) of
mielke-geometrodynamics_F01459	292	■ 1.000	$\backslash\mathrm{ell}^{\mathrm{2}}\ /\ \backslash\mathrm{mu}^{\mathrm{2}}=\backslash\mathrm{left}(\backslash\mathrm{kappa}\ /\ 4\ \backslash\mathrm{Lambda}_{\mathrm{\mathrm{H}}}\}\ \backslash\mathrm{right})$	$\ell^2/\mu^2 = (\kappa/4\Lambda_{\mathrm{H}})$	the derivative of the Higgs field by a <i>huge</i> dimensional factor $\ell^2/\mu^2 = (\kappa/4\Lambda_{\mathrm{H}})$. This
mielke-geometrodynamics_F01460	294	■ 1.000	$\mathrm{W}^{\mathrm{+}}$	W^+	W^+ and W^- gauge bosons or even “heavy photons” Z , again via pair creation. These
mielke-geometrodynamics_F01461	294	■ 1.000	$\mathrm{W}^{\mathrm{-}}$	W^-	W^+ and W^- gauge bosons or even “heavy photons” Z , again via pair creation. These
mielke-geometrodynamics_F01462	294	■ 0.633	$\backslash\mathrm{mu}\ \backslash\mathrm{rightarrow}\ 0$	$\mu \rightarrow 0$	In the limit $\mu \rightarrow 0$ of $\mathrm{SL}(5, \mathbb{R})$ symmetry restoration, concepts like macroscopic
mielke-geometrodynamics_F01463	294	■ 1.000	$\mathrm{d}\ \mathrm{s}$	ds	logical field theory does a line element ds emerge as a necessary means for measuring
mielke-geometrodynamics_F01464	294	■ 1.000	$1\ /\ \backslash\mathrm{alpha}:=$	$1/\alpha :=$	slowly decay (ZEE 2004). The inverse <i>dimensionless</i> coupling constants $1/\alpha :=$

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mielke-geometrodynamics_F01465	295	■ 1.000	$\hbar c / e^2 \simeq 137$	$\hbar c / e^2 \simeq 137$	<u>$\hbar c / e^2 \simeq 137$</u> of Sommerfeld and $1/\mu \simeq 10^{123}$ of gravity seem to be on a par except
mielke-geometrodynamics_F01466	295	■ 1.000	$1 / \mu \simeq 10^{123}$	$1/\mu \simeq 10^{123}$	$\hbar c / e^2 \simeq 137$ of Sommerfeld and $1/\mu \simeq 10^{123}$ of gravity seem to be on a par except
mielke-geometrodynamics_F01467	295	■ 0.991	C_{TT}	C_{TT}	contains a translational CS three-form C_{TT} that gives rise to the <i>parity-violating</i>
mielke-geometrodynamics_F01468	295	■ 1.000	$\mathfrak{gl}(4, \mathbb{R})$	$\mathfrak{gl}(4, \mathbb{R})$	of Nieh and Yan (NY) (NIEH 2007). On the other hand, the $\mathfrak{gl}(4, \mathbb{R})$ -valued Pontryagin
mielke-geometrodynamics_F01469	296	■ 0.865	$\operatorname{Ric}_{\alpha\beta} := (-1)^{\text{\texttt{sig}}}$	$\operatorname{Ric}_{\alpha\beta} := (-1)^{\text{\texttt{sig}}}$	sion involving the symmetric Ricci tensor, i.e., the zero-form $\operatorname{Ric}_{\alpha\beta} := (-1)^{\text{\texttt{sig}}}$
mielke-geometrodynamics_F01470	296	■ 0.998	$\left(R_{(\alpha} \wedge \eta_{\delta \beta)}\right)$	$(R_{(\alpha}{}^\delta \wedge \eta_{\delta \beta)})$	$(R_{(\alpha}{}^\delta \wedge \eta_{\delta \beta)})$, is known as the <i>Gauss-Bonnet identity</i> .
mielke-geometrodynamics_F01471	300	■ 1.000	$\mathscr{S} := \mathscr{M} / \mathscr{D}$	$\mathscr{S} := \mathscr{M} / \mathscr{D}$	² The quotient space $\mathscr{S} := \mathscr{M} / \mathscr{D}$ of all Riemannian metrics on a 3-dimensional manifold modulo the
mielke-geometrodynamics_F01472	300	■ 1.000	$\mathscr{D}$	$\mathscr{D}$	group $\mathscr{D}$ of the diffeomorphisms of M^3 is called the “superspace”. Accordingly, it has a completely
mielke-geometrodynamics_F01473	300	■ 1.000	M^3	M^3	group $\mathscr{D}$ of the diffeomorphisms of M^3 is called the “superspace”. Accordingly, it has a completely
mielke-geometrodynamics_F01474	301	■ 1.000	F_{ij}	F_{ij}	ponents F_{ij} of the electromagnetic field strengths. Such a theory was developed not
mielke-geometrodynamics_F01475	301	■ 1.000	$g_{[ij]} g^{[ij]}$	$g_{[ij]} g^{[ij]}$	term proportional to $g_{[ij]} g^{[ij]}$ occurs side by side with Hilbert’s Lagrangian 4-form,
mielke-geometrodynamics_F01476	301	■ 1.000	$\operatorname{g}_{[ij]} := \ell^{*2} F_{ij}$	$g_{[ij]} := \ell^{*2} F_{ij}$	FAT (1979a), in a more recent study, has given up the identification $g_{[ij]} := \ell^{*2} F_{ij}$.
mielke-geometrodynamics_F01477	301	■ 1.000	$m \in M$	$m \in M$	be dependent not only on the point $m \in M$ but also on locally distinguished direc-
mielke-geometrodynamics_F01478	303	■ 1.000	Σ_ω	Σ_ω	mathematical structure of the canonical energy–momentum current Σ_ω of the Yang–
mielke-geometrodynamics_F01479	304	■ 0.995	$\Sigma_{\text{\texttt{mat}}} = 0$	$\Sigma_{\text{\texttt{mat}}} = 0$	$\Sigma_{\text{\texttt{mat}}} = 0$.
mielke-geometrodynamics_F01480	304	■ 1.000	$\delta = \pi / 2$	$\delta = \pi / 2$	$\delta = \pi / 2$ is exactly equal to the operation of taking the Hodge dual ³
mielke-geometrodynamics_F01481	305	■ 1.000	Σ_{YM}	Σ_{YM}	transformation). The <i>invariance</i> of Σ_{YM} with respect to duality rotations can easily
mielke-geometrodynamics_F01482	305	■ 1.000	$\cos^2 \delta +$	$\cos^2 \delta +$	be proved by writing out and applying the elementary functional relation $\cos^2 \delta +$
mielke-geometrodynamics_F01483	305	■ 1.000	$\sin^2 \delta = 1$	$\sin^2 \delta = 1$	$\sin^2 \delta = 1$ for trigonometric functions.
mielke-geometrodynamics_F01484	307	■ 0.999	$R_{ij}^{\{}$	$R_{ij}^{\{}$	for the holonomically written components $R_{ij}^{\{}$ or the Ricci tensor. Combined with
mielke-geometrodynamics_F01485	307	■ 0.993	$\bar{\tau}_e$	$\bar{\tau}_e$	out of the matter currents $\bar{\tau}_e$ and $\bar{\tau}_p$ of a theory (SCHWINGER 1968; JULIA &
mielke-geometrodynamics_F01486	307	■ 0.993	$\bar{\tau}_p$	$\bar{\tau}_p$	out of the matter currents $\bar{\tau}_e$ and $\bar{\tau}_p$ of a theory (SCHWINGER 1968; JULIA &
mielke-geometrodynamics_F01487	308	■ 1.000	$\delta = 0$	$\delta = 0$	traditional electrodynamics, which is defined by the choice $\delta = 0$ of a “dual gauge”
mielke-geometrodynamics_F01488	308	■ 0.562	$\xi \wedge \vartheta$	$\xi \wedge \vartheta$	After the exterior multiplication of (14.1.32) by $\xi \wedge \vartheta$ and (14.1.33) by ${}^* \xi \wedge \vartheta$ and
mielke-geometrodynamics_F01489	308	■ 0.562	${}^* \xi \wedge \vartheta$	${}^* \xi \wedge \vartheta$	After the exterior multiplication of (14.1.32) by $\xi \wedge \vartheta$ and (14.1.33) by ${}^* \xi \wedge \vartheta$ and
mielke-geometrodynamics_F01490	308	■ 0.966	$d\delta$	$d\delta$	$d\delta$, Eq.(14.1.37) has to be contracted by ${}^* \Sigma_{\mathrm{YM}}(\xi)$. On account of the invariance
mielke-geometrodynamics_F01491	308	■ 0.966	${}^* \Sigma_{\mathrm{YM}}(\xi)$	${}^* \Sigma_{\mathrm{YM}}(\xi)$	$d\delta$, Eq.(14.1.37) has to be contracted by ${}^* \Sigma_{\mathrm{YM}}(\xi)$. On account of the invariance

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mielke-geometrodynamics_F01492	309	■ 1.000	$d \, d \, \delta \equiv 0$	$dd\delta \equiv 0$	Due to the identity $dd\delta \equiv 0$, this follows in an unequivocal way from the exactness
mielke-geometrodynamics_F01493	309	■ 0.928	$b_1 = \operatorname{dim} H^1(M) = \nu$	$b_1 = \dim H^1(M) = \nu$	of the form $d\delta$. For manifolds with Betti number $b_1 = \dim H^1(M) = \nu$, e.g., for
mielke-geometrodynamics_F01494	309	■ 1.000	ν	ν	those that exhibit ν -fold global connectivity, however,
mielke-geometrodynamics_F01495	310	■ 1.000	(1962, 1969)	(1962, 1969)	Encouraged by this remarkable result, WHEELER (1962, 1969) was inclined not
mielke-geometrodynamics_F01496	311	■ 1.000	$\operatorname{n}=4+\operatorname{K}$	$n = 4 + K$	(or supergravity) in $\boxed{n = 4 + K}$ dimensions, which is then shown or assumed to
mielke-geometrodynamics_F01497	311	■ 0.844	(1926, 1928)	(1926, 1928)	(1926, 1928) proved that this is possible after proceeding to 5-dimensional ⁵ man-
mielke-geometrodynamics_F01498	312	■ 1.000	M^{4+K}	M^{4+K}	M^{4+K} that is endowed with a bundle $\hat{L}(M^{4+K})$ of linear frames and is furthermore
mielke-geometrodynamics_F01499	312	■ 1.000	$\hat{L}(\hat{M}^{4+K})$	$\hat{L}(M^{4+K})$	M^{4+K} that is endowed with a bundle $\hat{L}(M^{4+K})$ of linear frames and is furthermore
mielke-geometrodynamics_F01500	312	■ 0.998	$\hat{\vartheta}$	$\hat{\vartheta}$	equipped with a canonical 1-form $\hat{\vartheta}$ associated with $\hat{L}(M^{4+K})$. Only later does one
mielke-geometrodynamics_F01501	312	■ 1.000	$\hat{E} \approx M^4 \times B$	$\hat{E} \approx M^4 \times B$	such that M^{4+K} takes on the form $\hat{E} \approx M^4 \times B$, where B is a (non)compact K-
mielke-geometrodynamics_F01502	312	■ 1.000	G / H	G/H	dimensional homogeneous space, G/H . Here G denotes a semisimple Lie group of
mielke-geometrodynamics_F01503	312	■ 0.951	$\operatorname{K}=\operatorname{dim} \operatorname{G}$	$K = \dim G$	dimension $K = \dim G$, and H an isotropic subgroup. As shown by COQUEREAUX &
mielke-geometrodynamics_F01504	312	■ 0.600	$P(\hat{M}^4, G, \pi, \delta)$	$P(M^4, G, \pi, \delta)$	JADCZYK (1983), the “geometric arena” is not the fiber bundle $P(M^4, G, \pi, \delta)$ but
mielke-geometrodynamics_F01505	312	■ 0.983	$P(\hat{M}^4, N / H, \pi, \delta)$	$P(M^4, N/H, \pi, \delta)$	$P(M^4, N/H, \pi, \delta)$, where N denotes the normalizer of H in G. The extended space
mielke-geometrodynamics_F01506	312	■ 1.000	$\hat{E}$	$\hat{E}$	$\hat{E}$, which is locally isomorphic to $M^4 \times G/H$, is globally the associated bundle
mielke-geometrodynamics_F01507	312	■ 1.000	$M^4 \times G / H$	$M^4 \times G/H$	$\hat{E}$, which is locally isomorphic to $M^4 \times G/H$, is globally the associated bundle
mielke-geometrodynamics_F01508	312	■ 0.996	$\hat{E}(\hat{M}^4, F=G / H, N / H, P)$	$\hat{E}(M^4, F = G/H, N/H, P)$	$\hat{E}(M^4, F = G/H, N/H, P)$ with N/H as structure group.
mielke-geometrodynamics_F01509	312	■ 0.996	N / H	N/H	$\hat{E}(M^4, F = G/H, N/H, P)$ with N/H as structure group.
mielke-geometrodynamics_F01510	312	■ 0.858	$\omega=\omega^j{}_{\alpha} L_j \otimes \vartheta^{\alpha}$	$\omega = \omega^j{}_{\alpha} L_j \otimes \vartheta^{\alpha}$	and $\omega = \omega^j{}_{\alpha} L_j \otimes \vartheta^{\alpha}$ is a 1-form with values in the Lie algebra of G or N/H ,
mielke-geometrodynamics_F01511	312	■ 0.760	$\stackrel{\circ}{\vartheta}$	$\overset{\circ}{\vartheta}$	Mills-type gauge fields. The 1-form $\overset{\circ}{\vartheta}$ is defined on the cotangent bundle $T_e^*(G)$ of
mielke-geometrodynamics_F01512	312	■ 0.760	$T_e{}^*(G)$	$T_e^*(G)$	Mills-type gauge fields. The 1-form $\overset{\circ}{\vartheta}$ is defined on the cotangent bundle $T_e^*(G)$ of
mielke-geometrodynamics_F01513	312	■ 1.000	ξ^j	ξ^j	the group G, being parametrized by the coordinates ξ^j . In its expansion
mielke-geometrodynamics_F01514	312	■ 1.000	φ_j	φ_j	the coefficients φ_j may be regarded as a multiplet of (Higgs-type) scalar fields trans-
mielke-geometrodynamics_F01515	312	■ 1.000	$d \, \xi^j$	$d\xi^j$	(holonomic) basis $d\xi^j$ is “dual” to the infinitesimal generators I_j of G. Moreover,
mielke-geometrodynamics_F01516	312	■ 1.000	I_j	I_j	(holonomic) basis $d\xi^j$ is “dual” to the infinitesimal generators I_j of G. Moreover,
mielke-geometrodynamics_F01517	312	■ 0.881	$V=V^j I_j, \stackrel{\circ}{\vartheta}$	$V = V^j I_j, \overset{\circ}{\vartheta}$	as a frame “dual” to the Lie vector field $V = V^j I_j, \overset{\circ}{\vartheta}$ satisfies the Maurer–Cartan

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mielke-geometrodynamics_F01570	320	■ 1.000	$\hat{\psi}$	$\hat{\psi}$	field $\hat{\psi}$ with $2^{[2+K/2]}$ components is adopted:
mielke-geometrodynamics_F01571	320	■ 1.000	$2^{\{[2+\mathrm{K} / 2]\}}$	$2^{[2+K/2]}$	field $\hat{\psi}$ with $2^{[2+K/2]}$ components is adopted:
mielke-geometrodynamics_F01572	320	■ 1.000	$\chi_{\rho F}^{\pm}(\xi)$	$\chi_{\rho F}^{\pm}(\xi)$	Here the spinorial harmonics $\chi_{\rho F}^{\pm}(\xi)$ furnish a representation ρ of the symmetry
mielke-geometrodynamics_F01573	320	■ 0.989	$\stackrel{\circ}{\gamma}^{K+1}$	$\stackrel{\circ}{\gamma}^{K+1}$	of the internal helicity operator $\stackrel{\circ}{\gamma}^{K+1}$. After projection π to four-dimensional compo-
mielke-geometrodynamics_F01574	321	■ 1.000	$\tilde{\omega}^g$	$\tilde{\omega}^g$	the (spinor) connections $\tilde{\omega}^g$ and ω , respectively, but that in addition, inherits a gauge
mielke-geometrodynamics_F01575	321	■ 1.000	$\gamma \wedge \gamma \wedge {}^* \Omega \psi$	$\gamma \wedge \gamma \wedge {}^* \Omega \psi$	covariant term $\gamma \wedge \gamma \wedge {}^* \Omega \psi$ of electric dipole type. This term in the Lagrangian
mielke-geometrodynamics_F01576	321	■ 0.527	$\hat{\mathrm{m}}=0$	$\hat{\mathrm{m}}=0$	which follows from the Lagrangian $(4+K)$ -form in the case $\hat{\mathrm{m}}=0$. This equation
mielke-geometrodynamics_F01577	322	■ 1.000	$\stackrel{\circ}{D}$	$\stackrel{\circ}{D}$	depends via the covariant derivative $\stackrel{\circ}{D}$ on the spinor connection $\stackrel{\circ}{\omega}$ and therefore acts
mielke-geometrodynamics_F01578	322	■ 1.000	$\stackrel{\circ}{\omega}$	$\stackrel{\circ}{\omega}$	depends via the covariant derivative $\stackrel{\circ}{D}$ on the spinor connection $\stackrel{\circ}{\omega}$ and therefore acts
mielke-geometrodynamics_F01579	322	■ 0.978	$i \not{D}^{\text{(int)}}$	$i \not{D}^{\text{(int)}}$	only on the internal space B. Provided B is compact, $i \not{D}^{\text{(int)}}$ has a discrete spectrum
mielke-geometrodynamics_F01580	322	■ 1.000	$\gamma^5 \psi = (\mp) \psi$	$\gamma^5 \psi = (\mp) \psi$	obeying $\gamma^5 \psi = (\mp) \psi$, transform according to inequivalent representations of the
mielke-geometrodynamics_F01581	322	■ 1.000	$n=4+K$	$n=4+K$	dimension of the total space. For $\underline{n} \equiv 4+K$, the spinor fields transform accord-
mielke-geometrodynamics_F01582	322	■ 0.999	$\widetilde{SO}(1, 3+K)$	$\widetilde{SO}(1, 3+K)$	ing to the fundamental representation of the covering tangent group $\widetilde{SO}(1, 3+K)$.
mielke-geometrodynamics_F01583	322	■ 0.995	$\hat{\gamma}^{n+1}$	$\hat{\gamma}^{n+1}$	eigenvalues of the n-dimensional chirality operator $\hat{\gamma}^{n+1}$. Under the subgroup
mielke-geometrodynamics_F01584	322	■ 0.615	$\widehat{SO}(1, 3) \otimes \widetilde{SO}(K)$	$\widehat{SO}(1, 3) \otimes \widetilde{SO}(K)$	$\widetilde{SO}(1, 3) \otimes \widetilde{SO}(K)$, such a representation branches according to
mielke-geometrodynamics_F01585	322	■ 1.000	$n=2p+1$	$n=2p+1$	(i) $\underline{n} \equiv 2p+1$: The transformation $\hat{\gamma}^A \rightarrow -\hat{\gamma}^A$ does not affect the infinitesimal
mielke-geometrodynamics_F01586	322	■ 1.000	$\hat{\gamma}^A \rightarrow -\hat{\gamma}^A$	$\hat{\gamma}^A \rightarrow -\hat{\gamma}^A$	(i) $n=2p+1$: The transformation $\hat{\gamma}^A \rightarrow -\hat{\gamma}^A$ does not affect the infinitesimal
mielke-geometrodynamics_F01587	322	■ 1.000	$\hat{\sigma}^{AB}$	$\hat{\sigma}^{AB}$	generators $\hat{\sigma}^{AB}$ but $\hat{\gamma}^{n+1} \rightarrow -\hat{\gamma}^{n+1}$. Since G-transformations are
mielke-geometrodynamics_F01588	322	■ 1.000	$\hat{\gamma}^{n+1} \rightarrow -\hat{\gamma}^{n+1}$	$\hat{\gamma}^{n+1} \rightarrow -\hat{\gamma}^{n+1}$	generators $\hat{\sigma}^{AB}$ but $\hat{\gamma}^{n+1} \rightarrow -\hat{\gamma}^{n+1}$. Since G-transformation s are
mielke-geometrodynamics_F01589	322	■ 1.000	$n=4p$	$n=4p$	(ii) $n=4p$: This implies that
mielke-geometrodynamics_F01590	322	■ 1.000	$n=4p+2$	$n=4p+2$	(iii) $n=4p+2$: Then it follows that
mielke-geometrodynamics_F01591	323	■ 0.543	$i \not{D}^{\text{(int)}}$	$i \not{D}^{\text{(int)}}$	Then it remains to reveal the conditions under which $i \not{D}^{\text{(int)}}$ has zero eigenmodes.
mielke-geometrodynamics_F01592	323	■ 1.000	$\mathrm{D}^* = {}^* \mathrm{D}$	$\mathrm{D}^* = {}^* \mathrm{D}$	Since $\underline{\mathrm{D}}^* = {}^* \underline{\mathrm{D}}$ holds for a metric-compatible (spinor) connection,
mielke-geometrodynamics_F01593	323	■ 0.922	$\stackrel{\circ}{\Theta}=0$	$\stackrel{\circ}{\Theta}=0$	for vanishing torsion, i.e., $\stackrel{\circ}{\Theta}=0$, Eq.(14.3.18) is the sum of a nonnegative and a
mielke-geometrodynamics_F01594	324	■ 1.000	$n>2$	$n>2$	izable in $n>2$ dimensions according to perturbation theory. For supergravity with
mielke-geometrodynamics_F01595	324	■ 0.999	$n=11$	$n=11$	turbative expansion. Moreover, simple ($\mathcal{N}=1$) supergravity in $n=11$ dimensions

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mielke-geometrodynamics_F01596	324	■ 0.999	$G=\mathrm{SO}(32)$	$G = \mathrm{SO}(32)$	ical strings. These <i>superstring theories</i> with $G = \mathrm{SO}(32)$ or the exceptional group
mielke-geometrodynamics_F01597	324	■ 1.000	$\mathrm{E}_8 \times \mathrm{E}_8$	$E_8 \times E_8$	$E_8 \times E_8$ as structure group seem to provide a unification of all physical interactions
mielke-geometrodynamics_F01598	324	■ 1.000	$n=10+\operatorname{dim} G=506$	$n = 10 + \dim G = 506$	in $n = 10 + \dim G = 506$ dimensions (DUFF et al. 1985) and a consistent and <i>finite</i>
mielke-geometrodynamics_F01599	324	■ 0.993	$\{ \}^{\wedge 10} J_3 \hbar$	$^{10} J_3 \hbar$	by the mass M, the z-component ¹⁰ $J_3 \hbar$ of the angular momentum, and the total angu-
mielke-geometrodynamics_F01600	324	■ 1.000	$\sqrt{J(J+1)} \hbar$	$\sqrt{J(J+1)} \hbar$	lar momentum $\sqrt{J(J+1)} \hbar$. Furthermore, it may carry the generalized “internal”
mielke-geometrodynamics_F01601	324	■ 1.000	$Q^{\{i\}}$	$Q^{(i)}$	Here $Q^{(i)}$ and $P^{(i)}$ denote the “electric”- and “magnetic”-type charges of a “dually”
mielke-geometrodynamics_F01602	324	■ 1.000	$P^{\{i\}}$	$P^{(i)}$	Here $Q^{(i)}$ and $P^{(i)}$ denote the “electric”- and “magnetic”-type charges of a “dually”
mielke-geometrodynamics_F01603	324	■ 1.000	$J_3 \hat{=} J$	$J_3 \hat{=} J$	rotation group. As for our still purely <i>classical</i> considerations, $J_3 \hat{=} J$ remains valid.
mielke-geometrodynamics_F01604	325	■ 0.798	$r^2 e^v$	$r^2 e^v$	substitutes the function $r^2 e^v$ in the Schwarzschild metric. The generalization of
mielke-geometrodynamics_F01605	326	■ 0.976	$\bar{Q}^{\{o\}}$	$\bar{Q}^{(o)}$	mass M, generalized charge $\bar{Q}^{(o)}$, and (classical) angular momentum component $J_3 \hbar$
mielke-geometrodynamics_F01606	326	■ 0.976	$J_3 \hbar$	$J_3 \hbar$	mass M, generalized charge $\bar{Q}^{(o)}$, and (classical) angular momentum component $J_3 \hbar$
mielke-geometrodynamics_F01607	326	■ 1.000	$\mu_m = \frac{Q_e}{M} J$	$\mu_m = \frac{Q_e}{M} J$	with respect to the axis of rotation, but it also has the magnetic moment $\mu_m = \frac{Q_e}{M} J$,
mielke-geometrodynamics_F01608	326	■ 0.823	M_{ir}	M_{ir}	The total mass M of a black hole depends on its geometric remnant mass M_{ir} ,
mielke-geometrodynamics_F01609	326	■ 0.938	$M_{\mathrm{ir}}, \bar{Q}^{\{o\}}$	$M_{\mathrm{ir}}, \bar{Q}^{(o)}$	dependence of M on the characteristic values M_{ir} , $\bar{Q}^{(o)}$, and J is nonlinear. More
mielke-geometrodynamics_F01610	326	■ 0.811	$J \hat{=} J_3 = 0$	$J \hat{=} J_3 = 0$	formulated by REISSNER (1916) and NORDSTRÖM (1918) if $J \hat{=} J_3 = 0$ holds.
mielke-geometrodynamics_F01611	327	■ 1.000	λ_3	λ_3	The generalized Gell-Mann matrix λ_3 , which is identical to the Pauli matrix σ_3 for
mielke-geometrodynamics_F01612	327	■ 1.000	σ_3	σ_3	The generalized Gell-Mann matrix λ_3 , which is identical to the Pauli matrix σ_3 for
mielke-geometrodynamics_F01613	327	■ 1.000	Δ	Δ	Here Δ is to be substituted by the term (14.4.5) with $J = 0$. Apart from the horizon
mielke-geometrodynamics_F01614	327	■ 0.993	$r=r_{\mathrm{o}}$	$r = r_{\mathrm{o}}$	given by (14.4.6), this spacetime geometry has obviously still a further one at $r = r_{\mathrm{o}}$,
mielke-geometrodynamics_F01615	327	■ 1.000	$r=0$	$r = 0$	away from the essential singularity at $r = 0$.
mielke-geometrodynamics_F01616	328	■ 0.672	$Q=-\bar{Q} \cos \delta$	$Q = -\bar{Q} \cos \delta$	a particle of mass M with an electric charge $Q = -Q \cos \delta$ and a magnetic monopole
mielke-geometrodynamics_F01617	328	■ 1.000	$P=-\bar{Q} \sin \delta$	$P = -\bar{Q} \sin \delta$	strength $P = -\bar{Q} \sin \delta$. If the surface area S_A of the event horizon, which is defined
mielke-geometrodynamics_F01618	328	■ 1.000	S_{A}	S_A	strength $P = -\bar{Q} \sin \delta$. If the surface area S_A of the event horizon, which is defined
mielke-geometrodynamics_F01619	328	■ 0.546	$\bar{Q}$	$\bar{Q}$	and a large dual charge $\bar{Q}$, the charge–mass relation and the total mass M of the
mielke-geometrodynamics_F01620	328	■ 0.990	$1 / \kappa$	$1/\kappa$	necessarily fixed by the coupling constant $1/\kappa$ of the Lorentz “rotons”.
mielke-geometrodynamics_F01621	330	■ 1.000	$\mathrm{d}=10$	$d = 10$	Green MB, Schwarz JH (1984) Anomaly cancellations in supersymmetric $d = 10$ gauge theory and
mielke-geometrodynamics_F01622	335	■ 1.000	$1 / \alpha_e$	$1/\alpha_e$	than electromagnetic forces by the factor $1/\alpha_e$. On the other hand, all experiments

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mielke-geometrodynamics_F01623	335	■ 0.967	$m_{\pi}=2\ m_e\ /\ \alpha_e$	$m_{\pi} = 2m_e/\alpha_e$	(1964), is explained by a virtual exchange of scalar π -mesons of mass $m_{\pi} = 2\ m_e/\alpha_e$.
mielke-geometrodynamics_F01624	335	■ 0.829	$S\ U(3)^c$	$SU(3)^c$	ory with an exact $SU(3)^c$ structure group. The introduction of these additional color
mielke-geometrodynamics_F01625	336	■ 0.924	$f_{i\ j}$	f_{ij}	tensor fields f_{ij} and g_{ij} or via the canonical 1-forms ϑ^f and ϑ^g as follows:
mielke-geometrodynamics_F01626	336	■ 0.924	$\vartheta^{\{f\}}$	ϑ^f	tensor fields f_{ij} and g_{ij} or via the canonical 1-forms ϑ^f and ϑ^g as follows:
mielke-geometrodynamics_F01627	336	■ 0.924	$\vartheta^{\{g\}}$	ϑ^g	tensor fields f_{ij} and g_{ij} or via the canonical 1-forms ϑ^f and ϑ^g as follows:
mielke-geometrodynamics_F01628	336	■ 1.000	L_{fg}	L_{fg}	necessitates the introduction of an additional mixing term L_{fg} . Were it not for this
mielke-geometrodynamics_F01629	336	■ 1.000	ϑ^{f}	ϑ^{f}	mixing term is understood to have only a polynomial dependence on ϑ^{f} and ϑ^{g} of
mielke-geometrodynamics_F01630	336	■ 1.000	$b-4$	$b-4$	degree a or $b-4$,
mielke-geometrodynamics_F01631	337	■ 1.000	$\sum^{\{f\}}$	$\sum^f$	Let $\sum^f$ and $\sum^g$ be the canonical energy–momentum currents that are generated
mielke-geometrodynamics_F01632	337	■ 1.000	$\sum^{\{g\}}$	$\sum^g$	Let $\sum^f$ and $\sum^g$ be the canonical energy–momentum currents that are generated
mielke-geometrodynamics_F01633	338	■ 0.921	G_{N}	G_{N}	of the (weak) Newtonian constant G_{N} of gravity to the coupling constant G_{S} of strong
mielke-geometrodynamics_F01634	338	■ 0.921	G_{S}	G_{S}	of the (weak) Newtonian constant G_{N} of gravity to the coupling constant G_{S} of strong
mielke-geometrodynamics_F01635	338	■ 1.000	$=10^{-13}\ \mathrm{cm}$	$=10^{-13}\ \mathrm{cm}$	spacetime structure beyond hadronic distances of the size of one fermi = $10^{-13}\ \mathrm{cm}$.
mielke-geometrodynamics_F01636	338	■ 1.000	$\mathrm{f}^{\{0\}}$	f^0	f^0 -meson and the massless graviton in a first approximation.
mielke-geometrodynamics_F01637	339	■ 1.000	$\mu^{\{g\}},\ \mu^{\{f\}}$	μ^g,μ^f	is the result for the f-metric. The integration constants are μ^g,μ^f , and γ . For $u = 1/2$,
mielke-geometrodynamics_F01638	339	■ 1.000	$\mathrm{u}=1\ /\ 2$	$u = 1/2$	is the result for the f-metric. The integration constants are μ^g,μ^f , and γ . For $u = 1/2$,
mielke-geometrodynamics_F01639	339	■ 0.992	$\mu^{\{g\}}=\Lambda=0$	$\mu^g = \Lambda = 0$	<u>$\mu^g = \Lambda = 0$</u> , this solution reduces to the configuration that was given by SALAM &
mielke-geometrodynamics_F01640	339	■ 1.000	$m_{\{f\}}=1-2\ \mathrm{GeV}\ /\ c^2$	$m_f = 1-2\mathrm{GeV}/c^2$	f-mesons of mass $m_f = 1-2\ \mathrm{GeV}/c^2$. This additional property of the tensor dom-
mielke-geometrodynamics_F01641	340	■ 1.000	$E_{\{j\}}^{\{\beta\}}$	E_j^{β}	the Poincaré gauge theory. These are given by the tetrads E_j^{β} and Ricci’s rotation
mielke-geometrodynamics_F01642	340	■ 1.000	$\Gamma_{i}^{\{\alpha\}\{\beta\}}$	$\Gamma_i^{\alpha\beta}$	coefficients $\Gamma_i^{\alpha\beta}$, respectively. With respect to anholonomic frame, these potentials
mielke-geometrodynamics_F01643	340	■ 1.000	$\left(\frac{1}{2},\ \frac{1}{2}\right)$	$\left(\frac{1}{2},\ \frac{1}{2}\right)$	transform under the Lorentz group according to the representation $\left(\frac{1}{2},\ \frac{1}{2}\right)$ or $(1, 0) \oplus$
mielke-geometrodynamics_F01644	340	■ 1.000	$(1,0)\ \oplus$	$(1,0)\oplus$	transform under the Lorentz group according to the representation $\left(\frac{1}{2},\ \frac{1}{2}\right)$ or $(1, 0) \oplus$
mielke-geometrodynamics_F01645	340	■ 0.609	$(0,1)$	$(0,1)$	(0, 1), respectively, and are therefore have an holonomic spin $\overline{s}_a = 1$. After the tran-
mielke-geometrodynamics_F01646	340	■ 0.609	$\operatorname{spin}\ \overline{s}_{\{a\}}=1$	$\operatorname{spin}\ \overline{s}_a = 1$	(0, 1), respectively, and are therefore have an holonomic spin $\overline{s}_a = 1$. After the tran-
mielke-geometrodynamics_F01647	340	■ 1.000	$\overline{s}=2$	$\overline{s} = 2$	induces gauge potentials of maximal (reducible) spin or helicity $\overline{s} = 2$ (HEHL et al.
mielke-geometrodynamics_F01648	340	■ 1.000	$\Lambda_{\mathrm{eff}}\neq 0$	$\Lambda_{\mathrm{eff}} \neq 0$	nonlinear theory with $\Lambda_{\mathrm{eff}} \neq 0$. Other quantization procedures are needed in order to isolate the

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mielke-geometrodynamics_F01649	341	■ 1.000	$\ell=\ell^* / \sqrt{\kappa}$	$\ell = \ell^* / \sqrt{\kappa}$	proportional to $\ell = \ell^* / \sqrt{\kappa}$ is superposed in the metric background of the solutions.
mielke-geometrodynamics_F01650	341	■ 1.000	$1 / \sqrt{\kappa}$	$1 / \sqrt{\kappa}$	ℓ^* scaled by $1 / \sqrt{\kappa}$ is identical to the Compton wavelength of a proton according to
mielke-geometrodynamics_F01651	341	■ 1.000	$10^{13} \mathrm{~cm}$	$10^{13} \mathrm{~cm}$	minimum radius would be increased to about $10^{13} \mathrm{~cm}$ as a result of the hypothesis of
mielke-geometrodynamics_F01652	341	■ 0.999	ℓ^2	ℓ^2	“strong” gravity. The relative strength of the coupling constants ℓ^2 and ℓ^{*2} , which
mielke-geometrodynamics_F01653	341	■ 0.999	ℓ^{*2}	ℓ^{*2}	“strong” gravity. The relative strength of the coupling constants ℓ^2 and ℓ^{*2} , which
mielke-geometrodynamics_F01654	341	■ 1.000	$\kappa=0,38 \times 10^{-38}$	$\kappa = 0,38 \times 10^{-38}$	small number of order of magnitude $\kappa = 0,38 \times 10^{-38}$. It was again EINSTEIN in
mielke-geometrodynamics_F01655	342	■ 1.000	10^{20}	10^{20}	there would be a fundamental length ℓ being enlarged by about 10^{20} . This hypothe-
mielke-geometrodynamics_F01656	342	■ 1.000	M_{U}	M_U	between the total mass M_U of the universe and the mass M_P of the proton while
mielke-geometrodynamics_F01657	342	■ 1.000	M_{P}	M_P	between the total mass M_U of the universe and the mass M_P of the proton while
mielke-geometrodynamics_F01658	342	■ 1.000	ℓ_{H}	ℓ_H	relating it to the product of the relative Hubble radius ℓ_H and the relative Compton
mielke-geometrodynamics_F01659	342	■ 1.000	$S U(f)$	$SU(f)$	resentations of the unitary groups $SU(f)$. Concerning this aspect, particular success
mielke-geometrodynamics_F01660	342	■ 0.995	$S U(4)$	$SU(4)$	to a far larger extent for the enlarged group $SU(4)$ that is postulated in the context of the charm
mielke-geometrodynamics_F01661	342	■ 0.997	$S U(5)$	$SU(5)$	hypothesis, and also for the grand unified theories (GUTs) that embarks upon $SU(5)$ as internal
mielke-geometrodynamics_F01662	343	■ 0.999	$S O(3) \approx S U(2)$	$SO(3) \approx SU(2)$	transform according to spinor representations of the group $SO(3) \approx SU(2)$ of the
mielke-geometrodynamics_F01663	343	■ 1.000	$S U(4) \approx S O(6) \supset$	$SU(4) \approx SO(6) \supset$	nal spin within a common group-theoretic scheme, the group $SU(4) \approx SO(6) \supset$
mielke-geometrodynamics_F01664	343	■ 1.000	$S O(4) \approx S U(2) \times S U(2)$	$SO(4) \approx SU(2) \times SU(2)$	$SO(4) \approx SU(2) \times SU(2)$ was proposed by Wigner. Later on, the group $SU(6)$ was
mielke-geometrodynamics_F01665	343	■ 1.000	$S U(6)$	$SU(6)$	$SO(4) \approx SU(2) \times SU(2)$ was proposed by Wigner. Later on, the group $SU(6)$ was
mielke-geometrodynamics_F01666	343	■ 1.000	$J=1 / 2$	$J = 1/2$	$J = 1/2$ and $J = 3/2$ of total angular momentum are thought to fill out a 56-fold
mielke-geometrodynamics_F01667	343	■ 1.000	$J=3 / 2$	$J = 3/2$	$J = 1/2$ and $J = 3/2$ of total angular momentum are thought to fill out a 56-fold
mielke-geometrodynamics_F01668	343	■ 1.000	$S L(6, \mathbb{C}) \supset S L(2, \mathbb{C}) \times S U(3)$	$SL(6, \mathbb{C}) \supset SL(2, \mathbb{C}) \times SU(3)$	this problem is to extend $SU(6)$ to the linear group $SL(6, \mathbb{C}) \supset SL(2, \mathbb{C}) \times SU(3)$.
mielke-geometrodynamics_F01669	343	■ 0.962	$\mathrm{SU}(3)$	$SU(3)$	Thereby a relativistic union of the internal symmetry group $SU(3)$ with the (simply
mielke-geometrodynamics_F01670	343	■ 1.000	$S L(6, \mathbb{C})$	$SL(6, \mathbb{C})$	spacetime, is achieved. If the symmetry is regarded only as a local one, an $SL(6, \mathbb{C})$
mielke-geometrodynamics_F01671	343	■ 0.933	$\mathrm{P}(\mathrm{M}, \overline{\mathrm{G}}, \pi, \delta)$	$P(M, \overline{G}, \pi, \delta)$	fiber bundle $P(M, \overline{G}, \pi, \delta)$ as a geometric arena. But this time, it is
mielke-geometrodynamics_F01672	343	■ 1.000	$[S U(4) \otimes]^4$	$[SU(4) \otimes]^4$	chiral $[SU(4) \otimes]^4$ model of ELIAS et al. (1978) ready to hand for the case $f = c = 4$.
mielke-geometrodynamics_F01673	343	■ 1.000	$f=c=4$	$f = c = 4$	chiral $[SU(4) \otimes]^4$ model of ELIAS et al. (1978) ready to hand for the case $f = c = 4$.
mielke-geometrodynamics_F01674	344	■ 0.993	$\overline{\mathrm{L}}(\mathrm{M})$	$\overline{L}(M)$	with $\overline{L}(M)$. This 1-form is supposed to have holonomic components in the local
mielke-geometrodynamics_F01675	344	■ 0.993	A_2	A_2	fields—as is, for instance, realized in nature via the $SU(3)$ -nonet f, f', A_2 , and K^*
mielke-geometrodynamics_F01676	344	■ 0.995	$S L(2 N, \mathbb{C}) \supset S L(2, \mathbb{C}) \times U(N)$	$SL(2N, \mathbb{C}) \supset SL(2, \mathbb{C}) \times U(N)$	erators of $SL(2N, \mathbb{C}) \supset SL(2, \mathbb{C}) \times U(N)$ can be realized by the matrices

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mielke-geometrodynamics_F01677	344	■ 1.000	λ_{i}	λ_i	where the generalized Gell-Mann matrices λ_i are the N^2-1 vector operators of the
mielke-geometrodynamics_F01678	344	■ 1.000	$\mathrm{N}^{\wedge{2}}-1$	$N^2 - 1$	where the generalized Gell-Mann matrices λ_i are the N^2-1 vector operators of the
mielke-geometrodynamics_F01679	344	■ 0.983	$\operatorname{SU}(N)$	$SU(N)$	$SU(N)$ subgroup. If the gauge potentials of the unitary groups—which are contained
mielke-geometrodynamics_F01680	345	■ 1.000	A_{i}	A_i	The Yang–Mills-like gauge potentials A_i aside, this total connection 1-form com-
mielke-geometrodynamics_F01681	345	■ 0.998	$\bar{\gamma}$	$\bar{\gamma}$	field strengths that belong to the 1-forms $\overline{\gamma}$ and $\overline{\omega}$ are the 2-form
mielke-geometrodynamics_F01682	345	■ 1.000	$L_{\{\mathrm{W}\}}(\mathrm{f})$	$L_{\mathrm{W}}(\mathrm{f})$	freedom, this is achieved by replacing the Weyl term $L_{\mathrm{W}}(\mathrm{f})$ in the total Lagrangian
mielke-geometrodynamics_F01683	345	■ 1.000	$\mathrm{f}=\mathrm{c}=0$	$f = c = 0$	If the internal degrees of freedom are suppressed, i.e., for the case $f = c = 0$, the
mielke-geometrodynamics_F01684	345	■ 1.000	$\mathrm{f}=3$	$f = 3$	as a structure group. In the framework of an $SL(6, \mathbb{C})$ gauge theory, i.e., for $f = 3$
mielke-geometrodynamics_F01685	345	■ 0.987	$\mathrm{c}=0$	$c = 0$	and $c = 0$, they also proposed a mixing term $\overline{L}_{\mathrm{fg}}$ that is responsible for the medi-
mielke-geometrodynamics_F01686	345	■ 0.987	$\bar{L}_{\{\mathrm{fg}\}}$	$\bar{L}_{\mathrm{fg}}$	and $c = 0$, they also proposed a mixing term $\overline{L}_{\mathrm{fg}}$ that is responsible for the medi-
mielke-geometrodynamics_F01687	346	■ 1.000	$\mathrm{G}=\mathrm{S}\ \mathrm{U}(2)$	$G = SU(2)$	made allowance only for the isospin group, i.e., $G = SU(2)$. If we followed that
mielke-geometrodynamics_F01688	346	■ 0.989	$\mathrm{S}\ \mathrm{U}(4)^{\mathrm{f}}\ \otimes\ \mathrm{S}\ \mathrm{U}(4)^{\mathrm{c}}$	$SU(4)^f \otimes SU(4)^c$	theories. Within the scheme A of this $SU(4)^f \otimes SU(4)^c$ -gauge theory, the quarklike
mielke-geometrodynamics_F01689	346	■ 1.000	$\mathrm{S}\ \mathrm{L}(8,\ \mathbb{C})^{\mathrm{f}}\ \otimes\ \mathrm{S}\ \mathrm{U}(8,\ \mathbb{C})^{\mathrm{c}}$	$SL(8, \mathbb{C})^f \otimes SU(8, \mathbb{C})^c$	$SL(8, \mathbb{C})^f \otimes SU(8, \mathbb{C})^c$ -gauge theory, however, is rather more related—following
mielke-geometrodynamics_F01690	347	■ 1.000	$\mathrm{M}=\mathbb{R}\ \times\ S^1\ \times\ S^2$	$M = \mathbb{R} \times S^1 \times S^2$	fields within the nontrivial topology of the space $M = \mathbb{R} \times S^1 \times S^2$. Its apparent
mielke-geometrodynamics_F01691	348	■ 0.994	$\mathrm{G}\ \mathrm{L}(8,\ \mathbb{C})^{\mathrm{f}}\ \otimes$	$GL(8, \mathbb{C})^f \otimes$	1980) as indicated previously; see also CASTRO (2012). Thereby, a $GL(8, \mathbb{C})^f \otimes$
mielke-geometrodynamics_F01692	348	■ 1.000	$\mathrm{G}\ \mathrm{L}(8,\ \mathbb{C})^{\mathrm{c}}$	$GL(8, \mathbb{C})^c$	$GL(8, \mathbb{C})^c$ gauge unification of all basic particle forces emerged into a rather spec-
mielke-geometrodynamics_F01693	348	■ 1.000	$\mathrm{U}(5)$	$U(5)$	Since $SU(4)$ and $U(5)$ together with the local Lorentz group can also be embedded
mielke-geometrodynamics_F01694	348	■ 0.987	$\mathrm{S}\ \mathrm{O}(1,9)$	$SO(1, 9)$	(FADDEEV 2011) such as $SO(1, 9)$ and the embedding of the $SO(10)$ GUT into the
mielke-geometrodynamics_F01695	348	■ 0.987	$\mathrm{S}\ \mathrm{O}(10)$	$SO(10)$	(FADDEEV 2011) such as $SO(1, 9)$ and the embedding of the $SO(10)$ GUT into the
mielke-geometrodynamics_F01696	348	■ 0.994	$\mathrm{S}\ \mathrm{O}(1,13)$	$SO(1, 13)$	group $SO(1, 13)$ have been proposed (PERCACCI 1991). Even “graviweak” groups
mielke-geometrodynamics_F01697	348	■ 0.998	$\mathrm{S}\ \mathrm{O}(4,\ \mathbb{C})$	$SO(4, \mathbb{C})$	such as $SO(4, \mathbb{C})$ have been considered by NESTI & PERCACCI (2010). Again, in
mielke-geometrodynamics_F01698	348	■ 0.995	$\mathrm{E}(8)$	$E(8)$	such as $E(8)$ and hyperbolic Kac–Moody algebras (KLEINSCHMIDT & NICOLAI
mielke-geometrodynamics_F01699	348	■ 0.995	$\mathrm{E}(10)$	$E(10)$	2010) like $E(10)$ have been suggested to achieve this. However, these approaches
mielke-geometrodynamics_F01700	349	■ 0.984	$(5,\ \mathbb{C})$	$(5, \mathbb{C})$	Castro C (2012) A Clifford Cl $(5, \mathbb{C})$ unified gauge field theory of conformal gravity, Maxwell and
mielke-geometrodynamics_F01701	349	■ 0.992	$\operatorname{SL}(2\ \mathrm{n},\ \mathbb{C})\ \otimes\ \operatorname{SL}(2,\ \mathbb{C})$	$SL(2n, \mathbb{C}) \otimes SL(2, \mathbb{C})$	Dennis PW, Huang JC (1977) Spontaneous symmetry breakdown in an $SL(2n, \mathbb{C}) \otimes SL(2, \mathbb{C})$ -
mielke-geometrodynamics_F01702	349	■ 0.998	$1000\ \mathrm{m}_{\mathrm{W}}\ \mathrm{W}$	1000 m _W	1000 m _W . Phys Rev Lett 40(14):920
mielke-geometrodynamics_F01703	350	■ 0.473	$2\ \mathrm{n},\ \mathbb{C}$	$2n, \mathbb{C}$	Huang JC, Dennis PW (1981a) Spin-1 and spin-2 gauge fields in unified $SL(2n, \mathbb{C})$ model. Phys
mielke-geometrodynamics_F01704	350	■ 0.805	$2^{\mathrm{+}}$	2^+	Isham C, Salam A, Strathdee J (1973a) 2^+ nonet as gauge particles for $SL(6, \mathbb{C})$ symmetry. Phys

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mielke-geometrodynamics_F01705	354	■ 0.952	$\mathrm{R}-\mathrm{R}_{\mathrm{o}}$	$R - R_o$	$R - R_o$) whose gradient balances the electrodynamic force. ¹
mielke-geometrodynamics_F01706	354	■ 1.000	(1966,1967)	(1966,1967)	for instance, PRASAD (1966, 1967) and ROMAN & KOH (1966). These notions have
mielke-geometrodynamics_F01707	354	■ 1.000	$\kappa>0$	$\kappa > 0$	responsible for the negative (for $\kappa > 0$) pressure term
mielke-geometrodynamics_F01708	354	■ 0.983	$2\ \mu\ /\ \mathrm{r}$	$2\mu/r$	sented by a Schwarzschild–de Sitter microuniverse, whereby the $2\mu/r$ -singularity
mielke-geometrodynamics_F01709	355	■ 1.000	ℓ_{m}	ℓ_m	coordinate. The (dimensionless) ratio of the Compton wavelength ℓ_m of the test
mielke-geometrodynamics_F01710	355	■ 0.998	$\mathrm{F}=\mathrm{F}_{\mathrm{nm}}^{\iota}\left(\rho^*\right)$	$F = F_{\mathrm{nm}}^{\iota}(\rho^*)$	with reference to the still to be determined functions $F = F_{\mathrm{nm}}^{\iota}(\rho^*)$.
mielke-geometrodynamics_F01711	356	■ 0.877	$\iota=0$	$\iota = 0$	For the sake of illustration (Fig. 16.1), the simplest case $\iota = 0$ for $\kappa > 0$, i.e., the
mielke-geometrodynamics_F01712	357	■ 1.000	$\rho^*=\frac{\pi}{2}$	$\rho^* = \frac{\pi}{2}$	At $\rho^* = \frac{\pi}{2}$ there arises for $\beta^2 < 2$ a steeply rising potential wall that can never be
mielke-geometrodynamics_F01713	357	■ 1.000	$\beta^2<2$	$\beta^2 < 2$	At $\rho^* = \frac{\pi}{2}$ there arises for $\beta^2 < 2$ a steeply rising potential wall that can never be
mielke-geometrodynamics_F01714	357	■ 1.000	$\mathrm{V}(\mathrm{r})=2\ \mu\ /\ \mathrm{r}+\mathrm{cr}$	$V(r) = 2\mu/r + cr$	In the latter, it is usually the potential $V(r) = 2\mu/r + cr$ that is postulated for the
mielke-geometrodynamics_F01715	358	■ 0.997	$\overline{\mathrm{Q}}$	$\bar{Q}$	quantum number J of total angular momentum, and generalized charge $\bar{Q}$ is (almost)
mielke-geometrodynamics_F01716	358	■ 1.000	Q_{e}	Q_e	while the electric charge Q_e is determined via the generalized Gell-Mann–Nishijima
mielke-geometrodynamics_F01717	358	■ 0.522	M_{ir}	M_{ir}	irreducible mass M_{ir} ; concerning the tensor dominance model, it is thus suggestive
mielke-geometrodynamics_F01718	358	■ 0.630	$\mathrm{M}^*:=\hbar\sqrt{8\pi}\ /\ \ell^*\mathrm{c}$	$M^* := \hbar\sqrt{8\pi}/\ell^*c$	to propose that M_{ir} as well as the modified Planck mass $M^* := \hbar\sqrt{8\pi}/\ell^*c$ are
mielke-geometrodynamics_F01719	358	■ 0.966	$\mathrm{M}_{\mathrm{ir}}=\mathrm{M}^*=1\ \mathrm{GeV}/\mathrm{c}^2$	$M_{\mathrm{ir}} = M^* = 1\mathrm{GeV}/c^2$	approximately identical to the mass of a proton, i.e., $M_{\mathrm{ir}} = M^* = 1\mathrm{GeV}/c^2$. For
mielke-geometrodynamics_F01720	359	■ 0.999	$M,\ J$	M, J	M , J , and I are the characteristics of particles that are here defined <i>canonically</i> in
mielke-geometrodynamics_F01721	359	■ 0.999	I	I	M , J , and I are the characteristics of particles that are here defined <i>canonically</i> in
mielke-geometrodynamics_F01722	359	■ 1.000	$\mathrm{J}=0$	$J = 0$	(However, if $J=0$, this leads precisely back to the linear Gell-Mann–Okubo formula!)
mielke-geometrodynamics_F01723	359	■ 0.999	$\mathrm{b}=\frac{1}{5}$	$b = \frac{1}{5}$	surprisingly well represented by (16.3.3). (In Fig. 16.2, $\bar{b} = \frac{1}{3}$ has been chosen, while
mielke-geometrodynamics_F01724	359	■ 0.998	$\Lambda(1116)$	$\Lambda(1116)$	M^* has been accommodated to the state $\Lambda(1116)$ of excited baryons.) A detailed
mielke-geometrodynamics_F01725	360	■ 1.000	$k_{\mathrm{B}}=8.6\ \times 10^{-5}\ \mathrm{eV}/\mathrm{K}$	$k_B = 8.6 \times 10^{-5}\mathrm{eV}/\mathrm{K}$	Here $\bar{k}_B = 8.6 \times 10^{-5}\ \mathrm{eV}/\mathrm{K}$ denotes the Boltzmann constant. Under the assump-
mielke-geometrodynamics_F01726	362	■ 0.877	$S\ U(8)$	$SU(8)$	Okubo S (1975) $SU(4)$ and $SU(8)$ mass formulas and weak interactions. Phys Rev D 11(11):3261
mielke-geometrodynamics_F01727	362	■ 0.824	$(6,\ \mathbb{C})$	$(6, \mathbb{C})$	Salam A (1973). On SL (6, $\mathbb{C}$) gauge invariance, In: Fundamental Interactions in Physics, 1973 Coral
mielke-geometrodynamics_F01738	365	■ 0.998	$2\ \times 2$	2×2	2×2 Pauli matrices
mielke-geometrodynamics_F01739	365	■ 0.984	$\mathrm{C}(0,2)=\mathbb{H}$	$C(0,2) = \mathbb{H}$	of the Clifford algebra $\mathbb{C}(0,2) = \mathbb{H}$ (field of quaternions).
mielke-geometrodynamics_F01740	365	■ 0.989	$4\ \times 4$	4×4	4×4 Dirac matrices in the so-called Pauli realization:
mielke-geometrodynamics_F01741	366	■ 1.000	$4<2\ \mathrm{p}+1<2\mathrm{p}+2\leq n=2\ l+2$	$4 < 2p + 1 < 2p + 2 \leq n = 2l + 2$	where $\bar{4} < 2p + 1 < 2p + 2 \leq n = 2l + 2$; cf. KALINOWSKI (1984).

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mielke-geometrodynamics_F01742	366	■ 0.997	<code>\emptyset</code>	$\emptyset$	(iii) Both X itself and the empty set $\emptyset$ are open.
mielke-geometrodynamics_F01743	366	■ 1.000	<code>\mathrm{T}_{1}</code>	T_1	T_1 : For each two distinct points $m_1, m_2 \in X$, there exist neighborhoods $U_i \ni m_i$ of
mielke-geometrodynamics_F01744	366	■ 1.000	<code>\mathrm{m}_{1}, \mathrm{~m}_{2} \in \mathrm{X}</code>	$m_1, m_2 \in X$	T_1 : For each two distinct points $m_1, m_2 \in X$, there exist neighborhoods $U_i \ni m_i$ of
mielke-geometrodynamics_F01745	366	■ 1.000	<code>\mathrm{U}_{\{\mathrm{i}\}} \ni \mathrm{m}_{\{\mathrm{i}\}}</code>	$U_i \ni m_i$	T_1 : For each two distinct points $m_1, m_2 \in X$, there exist neighborhoods $U_i \ni m_i$ of
mielke-geometrodynamics_F01746	366	■ 1.000	<code>\mathrm{T}_{2}</code>	T_2	T_2 : For any two distinct points $m_1, m_2 \in X$ there exist neighborhoods $U_i \ni m_i$ that
mielke-geometrodynamics_F01747	366	■ 0.566	<code>\mathrm{U}_{1} \bigcap \mathrm{U}_{2}=\varnothing</code>	$U_1 \bigcap U_2 = \emptyset$	are disjoint, namely $U_1 \bigcap U_2 = \emptyset$.
mielke-geometrodynamics_F01748	367	■ 1.000	<code>\sigma: \mathrm{M} \rightarrow \mathrm{N}</code>	$\sigma : M \rightarrow N$	A <u>diffeomorphism</u> $\sigma : M \rightarrow N$ between manifolds of the same dimension (e.g., $M \approx$
mielke-geometrodynamics_F01749	367	■ 1.000	<code>\mathrm{M} \approx</code>	$M \approx$	A <u>diffeomorphism</u> $\sigma : M \rightarrow N$ between manifolds of the same dimension (e.g., $M \approx$
mielke-geometrodynamics_F01750	367	■ 1.000	<code>\mathrm{N} \approx \mathbb{R}^{\mathrm{n}}</code>	$N \approx \mathbb{R}^n$	$N \approx \mathbb{R}^n$) is an injective mapping for which σ and the corresponding inverse mapping
mielke-geometrodynamics_F01751	367	■ 0.980	<code>{ }^{-1}</code>	-1	σ^{-1} are C^∞ functions, i.e., every partial derivative of σ exists and is continuous in
mielke-geometrodynamics_F01752	367	■ 0.990	<code>\sigma_{*}: \mathrm{T}_{\{\mathrm{m}\}}(\mathrm{M}) \rightarrow \mathrm{T}_{\{\sigma(\mathrm{m})\}}(\mathrm{N})</code>	$\sigma_* : T_m(M) \rightarrow T_{\sigma(m)}(N)$	m gives rise to a mapping $\sigma_* : T_m(M) \rightarrow T_{\sigma(m)}(N)$ within the space of tangent
mielke-geometrodynamics_F01753	367	■ 0.999	<code>\sigma(\mathrm{s}(\mathrm{t}))</code>	$\sigma(s(t))$	vectors along the curve $s(t)$ or $\sigma(s(t))$ if constructed according to the directional
mielke-geometrodynamics_F01754	367	■ 1.000	<code>e^f</code>	e^f	derivative e^f . On the other hand, the transposed mapping of σ_* defines a linear
mielke-geometrodynamics_F01755	367	■ 1.000	<code>\sigma_{*}</code>	σ_*	derivative e^f . On the other hand, the transposed mapping of σ_* defines a linear
mielke-geometrodynamics_F01756	367	■ 1.000	<code>\sigma_{*}^T: T_{\{\sigma(\mathrm{m})\}}^{*}(\mathrm{N}) \rightarrow T_{\{\mathrm{m}\}}^{*}(\mathrm{M})</code>	$\sigma_*^T : T_{\sigma(m)}^*(N) \rightarrow T_M^*(M)$	mapping $\sigma_*^T : T_{\sigma(m)}^*(N) \rightarrow T_M^*(M)$ between the “dual” cotangent spaces. As such,
mielke-geometrodynamics_F01757	367	■ 1.000	<code>{ }^{*} \alpha</code>	$^*\alpha$	Consequently, $^*\alpha$ denotes the inverse image of the differential form α . For cross
mielke-geometrodynamics_F01758	368	■ 0.858	<code>(p, 0)</code>	$(p, 0)$	Completely <i>antisymmetric</i> covariant tensor fields of degree $(p, 0)$ are of prime impor-
mielke-geometrodynamics_F01759	369	■ 0.982	<code>{ }^1 \mathrm{d} f=\vartheta^{\alpha} \partial_{\alpha} f</code>	${}^1df = \vartheta^\alpha \partial_\alpha f$	f , it is nothing but the total differential ¹ $df = \vartheta^\alpha \partial_\alpha f$. This action is extended to
mielke-geometrodynamics_F01760	369	■ 0.937	<code>\sigma_{*} \alpha^{(0)}:=\alpha^{(0)} \circ \sigma</code>	$\sigma_* \alpha^{(0)} := \alpha^{(0)} \circ \sigma$	that for a 0-form is obtained by composition the $\sigma_* \alpha^{(0)} := \alpha^{(0)} \circ \sigma$.
mielke-geometrodynamics_F01761	369	■ 1.000	<code>\varepsilon_{\alpha_1 \ldots \alpha_n}</code>	$\varepsilon_{\alpha_1 \dots \alpha_n}$	Let $\varepsilon_{\alpha_1 \dots \alpha_n}$ be the totally antisymmetric unit tensor (Levi-Civita tensor; see MTW,
mielke-geometrodynamics_F01762	369	■ 0.999	<code>\varepsilon_{\hat{1} \ldots \tilde{n}}=+1</code>	$\varepsilon_{\hat{1} \dots \tilde{n}} = +1$	p. 87) with the conventional choice of signs $\varepsilon_{\hat{1} \dots \tilde{n}} = +1$. In a pseudo-Riemannian
mielke-geometrodynamics_F01763	369	■ 1.000	<code>g_{\{\mathrm{i} j\}}</code>	g_{ij}	manifold endowed with the metric tensor g_{ij} , this unit tensor allows one to define the
mielke-geometrodynamics_F01764	369	■ 0.997	<code>\mathrm{d} f=\left(\partial f / \partial x^i\right) d x^i</code>	$df = \left(\partial f / \partial x^i\right) dx^i$	to a coordinate basis, it would read $df = (\partial f / \partial x^i) dx^i$.
mielke-geometrodynamics_F01765	371	■ 1.000	<code>\delta \alpha=0</code>	$\delta \alpha = 0$	if and only if it is closed as well as <i>coclosed</i> , i.e., $\delta \alpha = 0$.
mielke-geometrodynamics_F01766	371	■ 0.901	<code>\mathrm{B}(\mathrm{M}):=\mathrm{d} \mathrm{D}(\mathrm{M})</code>	$B(M) := dD(M)$	$B(M) := dD(M)$ of <i>exact</i> forms
mielke-geometrodynamics_F01767	371	■ 1.000	<code>\mathrm{B}^0(\mathrm{M})=0</code>	$B^0(M) = 0$	On the other hand, we have $B^0(M) = 0$. Consequently,
mielke-geometrodynamics_F01768	372	■ 0.995	<code>\mathrm{H}^{\{\mathrm{p}\}}(\mathrm{M})</code>	$H^p(M)$	manifolds), the dimensions of the spaces $H^p(M)$ are called the <i>Betti numbers</i>

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mielke-geometrodynamics_F01769	372	■ 1.000	$\mathrm{t}=-1$	$t = -1$	with the Betti numbers as coefficients, gives for $\overline{f} = -1$ the alternating sum
mielke-geometrodynamics_F01776	372	■ 0.998	η_{i}	η_i	with a countable basis η_i , a p-dimensional standard simplex is defined by the set
mielke-geometrodynamics_F01771	373	■ 0.843	$\mathrm{c}:\Delta_{\mathrm{p}}\rightarrow \mathrm{M}$	$c : \Delta_p \rightarrow M$	the C^∞ -mapping $c : \Delta_p \rightarrow M$. These simplices form a free $\mathbb{R}$ -module $C_p(M, \mathbb{R})$,
mielke-geometrodynamics_F01772	373	■ 0.843	$\mathrm{C}_{\mathrm{p}}(\mathrm{M},\mathbb{R})$	$C_p(M, \mathbb{R})$	the C^∞ -mapping $c : \Delta_p \rightarrow M$. These simplices form a free $\mathbb{R}$ -module $C_p(M, \mathbb{R})$,
mielke-geometrodynamics_F01773	373	■ 1.000	$\partial \mathrm{c}$	∂c	whose elements are called p-chains. The boundary ∂c of a chain will be generated
mielke-geometrodynamics_F01774	373	■ 0.997	$\mathrm{Z}_{\mathrm{p}}(\mathrm{M},\mathbb{R})=(\mathrm{Ker}\partial)_{\mathrm{p}}, \mathrm{B}_{\mathrm{p}}(\mathrm{M},\mathbb{R}):=(\mathrm{Im}\partial)_{\mathrm{p}}$	$Z_p(M, \mathbb{R}) = (\mathrm{Ker} \partial)_p, B_p(M, \mathbb{R}) := (\mathrm{Im} \partial)_p$	tically.” The spaces $Z_p(M, \mathbb{R}) = (\mathrm{Ker} \partial)_p, B_p(M, \mathbb{R}) := (\mathrm{Im} \partial)_p$ as well as
mielke-geometrodynamics_F01775	373	■ 1.000	$\mathrm{N}_{\mathrm{(p)}}\subset \mathrm{M}$	$N_{(p)} \subset M$	dimensional (compact) submanifold $N_{(p)} \subset M$ of M. Consequently, they can be
mielke-geometrodynamics_F01776	373	■ 1.000	$\mathrm{N}_{\mathrm{(p)}}$	$N_{(p)}$	integrated, provided $N_{(p)}$ admits a simplicial covering by the chain $c_p \in C_p(M, \mathbb{R})$.
mielke-geometrodynamics_F01777	373	■ 1.000	$\mathrm{c}_{\mathrm{p}}\in \mathrm{C}_{\mathrm{p}}(\mathrm{M},\mathbb{R})$	$c_p \in C_p(M, \mathbb{R})$	integrated, provided $N_{(p)}$ admits a simplicial covering by the chain $c_p \in C_p(M, \mathbb{R})$.
mielke-geometrodynamics_F01778	374	■ 1.000	$\mathrm{H}^{\mathrm{P}}(\mathrm{M})\times \mathrm{H}_{\mathrm{p}}(\mathrm{M},\mathbb{R})\rightarrow \mathbb{R}$	$H^P(M) \times H_p(M, \mathbb{R}) \rightarrow \mathbb{R}$	ping $H^P(M) \times H_p(M, \mathbb{R}) \rightarrow \mathbb{R}$ from the Cartesian product of the cohomology and
mielke-geometrodynamics_F01779	375	■ 1.000	$\mathrm{g}\in \mathrm{G}$	$g \in G$	(i) the elements $g \in G$ form an (analytic) manifold,
mielke-geometrodynamics_F01780	375	■ 1.000	$\mathrm{G}\times \mathrm{G}\rightarrow \mathrm{G}$	$G \times G \rightarrow G$	(ii) the mapping $G \times G \rightarrow G$ originating from the pointwise multiplication
mielke-geometrodynamics_F01781	375	■ 0.999	$\left(\mathrm{g}_1,\mathrm{g}_2\right)\rightarrow \mathrm{g}_1\mathrm{g}_2^{-1}$	$(g_1, g_2) \rightarrow g_1 g_2^{-1}$	$(g_1, g_2) \rightarrow g_1 g_2^{-1}$ of (group) elements is analytic as well.
mielke-geometrodynamics_F01782	375	■ 1.000	$\mathrm{G}=\mathrm{GL}(\mathrm{N},\mathbb{C})$	$G = \mathrm{GL}(N, \mathbb{C})$	The most general example is given by $G = \mathrm{GL}(N, \mathbb{C})$, i.e., by the general linear
mielke-geometrodynamics_F01783	375	■ 0.999	$\mathrm{T}_{\mathrm{e}}(\mathrm{G})$	$T_e(G)$	The tangent space $T_e(G)$ attached to the unit element $e \in G$ of the group possesses
mielke-geometrodynamics_F01784	375	■ 0.999	$\mathrm{e}\in \mathrm{G}$	$e \in G$	The tangent space $T_e(G)$ attached to the unit element $e \in G$ of the group possesses
mielke-geometrodynamics_F01785	375	■ 0.952	$\xi=\left\{\xi^{\mathrm{j}},\mathrm{j}=1,\ldots,\mathrm{K}\right\}$	$\xi = \{\xi^j, j = 1, \dots, K\}$	the <i>Lie algebra</i> of $\mathfrak{g}$ of G (KN I, p. 38). Let $\xi = \{\xi^j, j = 1, \dots, K\}$ be an appropriate
mielke-geometrodynamics_F01786	375	■ 0.943	$\mathrm{c}_{\mathrm{ij}}^{\mathrm{k}}$	c_{ij}^k	Here the real <i>structure constants</i> c_{ij}^k are characteristic of the group G. They are always
mielke-geometrodynamics_F01787	376	■ 1.000	$\mathrm{A}=\mathrm{A}^{\mathrm{i}}\mathrm{I}_{\mathrm{i}}$	$A = A^i I_i$	defined for generic Lie-algebra-valued vectors $A = A^i I_i$ and $B = B^j I_j$ is linear and
mielke-geometrodynamics_F01788	376	■ 1.000	$\mathrm{B}=\mathrm{B}^{\mathrm{j}}\mathrm{I}_{\mathrm{j}}$	$B = B^j I_j$	defined for generic Lie-algebra-valued vectors $A = A^i I_i$ and $B = B^j I_j$ is linear and
mielke-geometrodynamics_F01789	376	■ 1.000	$\stackrel{\circ}{g}_{\mathrm{I}s}$	$\overset{\circ}{g}_{Is}$	may be introduced that is determined by the <i>Cartan–Killing metric</i> $\overset{\circ}{g}_{Is}$ of the Lie
mielke-geometrodynamics_F01790	376	■ 0.863	$\mathfrak{gl}(\mathrm{N},\mathbb{C})$	$\mathfrak{gl}(N, \mathbb{C})$	of $\mathfrak{g}$. In the case of the Lie algebra $\mathfrak{gl}(N, \mathbb{C})$ of the general linear group, this product
mielke-geometrodynamics_F01791	376	■ 0.737	$\mathrm{G}=\exp^{\mathrm{Trg}}$	$G = \exp^{\mathrm{Trg}}$	$\det G = \exp^{\mathrm{Trg}}$. The scalar product (C.12) is nondegenerate if and only if G is
mielke-geometrodynamics_F01792	377	■ 0.995	ξ^{j}	ξ^j	algebra. In terms of a real parametrization ξ^j , an arbitrary group element is generated
mielke-geometrodynamics_F01793	377	■ 1.000	λ_{j}	λ_j	Here the λ_j denote the generalized Gell-Mann matrices, which are Hermitian and